A finite proof of the all-prime W_p congruence

wmy's research note

Theorem 1. *For every natural prime $p \geq 7$, the p-integral rational number*

$$W_p = \sum_{n=0}^{p-1} \frac{(6n)!}{(n!)^6\, 10^{6n}} (532n^2 + 126n + 9)$$

satisfies $W_p \equiv 9p^2 \pmod{p^5}$.

The proof uses finite rational polynomials, coefficient residues and finite sums in characteristic p. The coefficient calculations and their complete witnesses are included below. A congruence in $\mathbb{Z}_{(p)}$ means divisibility of the difference by the indicated power of p; reductions are made only after proving integrality. Letters are local to their sections. In particular integer Fermat quotients and their characteristic-p reductions are distinguished. For $p \geq 19$, the argument first derives $W_p \equiv 3375p^2 c_p M_p \pmod{p^5}$ and $c_p \equiv 120/1728^p \pmod{p^3}$. The required moment value is then equivalent to the equality $S = P$ for the original reduced scalar. The finite-field calculation attaches that same scalar to its complete sources and proves $S = P$ by their coefficient identities. The four primes below 19 have separate exact integer certificates. The final section combines these results.

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1 The literal sum, conductor and normalization

1.1 The literal sum and the complete native moment

For a prime p, put $R_p = \{a/b \in \mathbb{Q} : p \nmid b\}$, with fractions in lowest terms. For nonzero $q \in \mathbb{Q}$, $v_p(q)$ is the exponent of p in its numerator minus that in its denominator. We write $q = O(p^k)$ when $q = 0$ or $v_p(q) \geq k$, and $a \equiv b \pmod{p^k}$ when $a - b = O(p^k)$. Thus an exactly zero error satisfies every precision bound. Finite sums preserve a common precision; multiplication by an element of R_p preserves precision; multiplying $O(p^k)$ by $O(p^l)$ gives $O(p^{k+l})$. These statements follow from $v_p(ab) = v_p(a) + v_p(b)$ for nonzero factors and the ultrametric inequality, with zero factors and zero sums treated separately.

The objects to be related are

$$w(n) = 532n^2 + 126n + 9, \qquad W_p = \sum_{n=0}^{p-1} \frac{(6n)!}{(n!)^6\, 10^{6n}} w(n), \tag{1}$$

$$c_n = \frac{(6n)!}{(3n)!(n!)^3\,1728^n}, \qquad A_p = (p-\tfrac{1}{2})(p-\tfrac{5}{6})(p-\tfrac{1}{6})c_{p-1} = p^3 c_p. \tag{2}$$

For $0 \le n < p$ every factor of $n!$ is prime to p; for $p \ge 7$ the number 10 is also prime to p. Hence every literal summand defining W_p, and therefore W_p itself, belongs to R_p. The equality $A_p = p^3 c_p$ follows by cancelling factorials in

$$(n+1)^3 c_{n+1} = (n+\tfrac{1}{2})(n+\tfrac{1}{6})(n+\tfrac{5}{6})c_n. \tag{3}$$

Define, for $0 \le n < p$,

$$N_{p,n}(X) = (4X-1)^{2p+n-1}(4-X)^{p+2n-1}(11-4X), \tag{4}$$

$$d_{n,a} = 2a - 6n + 1, \tag{5}$$

$$m_{p,n} = \frac{2}{3\,125^{p+n}} \sum_{a=0}^{3(p+n)-1} [X^a]N_{p,n}(X)\frac{p(2^{d_{n,a}} - 2^{-d_{n,a}})}{d_{n,a}}, \tag{6}$$

$$M_p = \sum_{n=0}^{p-1} c_n m_{p,n}. \tag{7}$$

Integer powers of 2 in this formula include negative powers. The degree of $N_{p,n}$ is at most $3(p+n)-1$, so its whole coefficient range is present. The two signed terms are the two actual endpoints $s = 2$ and $s = 1/2$.

For $p \ge 7$, every $m_{p,n}$ is p-integral. Indeed, $d_{n,a}$ is odd and nonzero, and

$$-6p < d_{n,a} < 6p < p^2.$$

Hence $v_p(d_{n,a}) \le 1$ and $p/d_{n,a} \in R_p$. The two powers of 2, the polynomial coefficients and $2/(3\,125^{p+n})$ also belong to R_p. No possible resonance denominator costs more than the digit supplied by the explicit factor p in the primitive. Legendre's formula, with all factorial arguments below p^2, gives

$$v_p(c_n) = \left\lfloor \frac{6n}{p} \right\rfloor - \left\lfloor \frac{3n}{p} \right\rfloor \ge 0 \quad (0 \le n < p). \tag{8}$$

Thus $M_p \in R_p$. Also $v_p(c_p) = 6 - 3 - 3 = 0$ and $v_p(A_p) = 3$.

Put $e = p - 1 - \lfloor p/6 \rfloor$ and $C_p = \sum_{n=0}^{e} c_n m_{p,n}$. For $e < n < p$, the two quotients in (8) are 5 and 2. Consequently

$$c_n = O(p^3) \quad (e < n < p), \qquad M_p - C_p = O(p^3). \tag{9}$$

The same valuation-three conclusion holds for $p - 3 \le n < p$ when $p \ge 19$.

1.2 The factorial normalization

The normalization constant is the actual coefficient c_p in (2). Its value modulo p^3 is derived as follows. In $\mathbb{Z}/p^3\mathbb{Z}$, define

$$H_k = \sum_{r=1}^{p-1} r^{-k}, \qquad B_j = \prod_{r=1}^{p-1}(jp + r).$$

All inverses here exist. Reduction modulo p and the finite-field sum of inverse squares give $H_2 \equiv 0 \pmod{p}$. One elementary justification of that sum is to choose $u \in \mathbb{F}_p^\times$ with $u^2 \ne 1$ (there are more than two nonzero elements), permute the sum by multiplication by u, and cancel $1 - u^{-2}$. Thus $p^2 H_2 = 0$ modulo p^3. Pairing r with $p - r$ gives the exact identity modulo p^3

$$p\big(r^{-1} + (p-r)^{-1}\big) = -p^2 r^{-2}.$$

Summing it and cancelling the unit 2 yields $pH_1 = 0$ modulo p^3. For any finite family f_r, expansion of the product through degree two in a variable z with $z^3 = 0$ gives

$$\prod_r (1 + zf_r) = 1 + z\sum_r f_r + \frac{z^2}{2}\left(\left(\sum_r f_r\right)^2 - \sum_r f_r^2\right).$$

Apply this with $z = jp$ and $f_r = r^{-1}$. The two harmonic annihilators show $B_j \equiv B_0 \pmod{p^3}$ for every j.

Partitioning a factorial into its multiples of p and its unit blocks gives the exact integer equation

$$(kp)! = p^k k! \prod_{j=0}^{k-1} B_j.$$

Let $D_n = \binom{6n}{3n}\binom{3n}{n}\binom{2n}{n}$, so $D_n = (6n)!/((3n)!(n!)^3)$ and $c_n = D_n/1728^n$. The factorial equation for $k = 1, 3, 6$, with exact cancellation in the integers before reduction, becomes

$$D_p B_0^3 = 120 B_3 B_4 B_5.$$

Here $120 = 6!/3!$. Since B_0 is a unit modulo p^3, the block congruences give $D_p \equiv 120 \pmod{p^3}$. The denominator 1728^p is a unit for $p \geq 7$, and therefore

$$c_p \equiv \frac{120}{1728^p} \pmod{p^3}. \tag{10}$$

The argument takes place in a ring with $p^3 = 0$, not in characteristic p after prematurely dividing by p.

1.3 The finite Gamma functional and literal normalization

For $k, L \geq 0$ and a rational coefficient sequence (f_k), define

$$\gamma_k = \frac{4^k (k!)^2}{(2k+1)!}, \qquad \Gamma_L(f) = \sum_{k=0}^{L} \gamma_k f_k, \qquad \beta(T) = (1-x)T' - \tfrac{1}{2}T.$$

For a Laurent expansion $T = \sum t_k x^k$,

$$[x^k]\beta(T) = (k+1)t_{k+1} - (k+\tfrac{1}{2})t_k.$$

Cancellation of factorials proves $(k+1)\gamma_k = (k+3/2)\gamma_{k+1}$ and $\gamma_0 = 1$. Telescoping, with both endpoint terms retained, now gives

$$\Gamma_L([x^k]\beta(T)) = -\frac{t_0}{2} + (L+1)\gamma_L t_{L+1}. \tag{11}$$

This is valid for every rational Laurent sequence; no infinite integration or convergence assertion is used.

Set $a = 729/15625$, $\lambda = ax^3$, and

$$G_p(x) = \sum_{n=0}^{p-1} w(n) c_n^2 a^n x^{3n}.$$

The exact identity

$$(6n+1)\gamma_{3n} c_n^2 a^n = \frac{(6n)!}{(n!)^6\, 10^{6n}}$$

follows by cancelling $(3n)!$ and using $4^{3n}a^n/1728^{2n} = 10^{-6n}$. In particular,

$$\Gamma_{3p-1}([x^k](2xG_p' + G_p)) = W_p. \tag{12}$$

For $0 \le k \le 3p-1$ and $p \ge 7$, all relevant factorial arguments are below p^2, and

$$v_p(\gamma_k) = 2\left\lfloor \frac{k}{p} \right\rfloor - \left\lfloor \frac{2k+1}{p} \right\rfloor \in \{0, -1\}. \tag{13}$$

Thus applying Γ_{3p-1} to coefficientwise $O(p^6)$ errors gives $O(p^5)$, and a finite sum loses no further digit. At the final weight, $v_p(\gamma_{3p-1}) = 2 \cdot 2 - 5 = -1$.

1.4 The actual native connection and conductor residue

Here are explicit finite definitions of the rational objects used in the adapter. The integer witness tables in Section 1.9 are part of these definitions. They involve no prime-dependent choices. Put $F_p(z) = \sum_{n=0}^{p-1} c_n z^n$, $H_p(z) = \sum_{n=0}^{p-1} c_n^2 z^n$, $\theta = z\,d/dz$, and

$$b(t) = t^3 + \tfrac{3}{2}t^2 + \tfrac{23}{36}t + \tfrac{5}{72}, \qquad Y = (-\theta^3 H_p, -\theta^4 H_p, -\theta^5 H_p, b(\theta)H_p, \theta b(\theta)H_p, \theta^2 b(\theta)H_p).$$

Write $Y_i(x) = Y_i(\lambda(x))$. Define the constant matrix

$$C = \begin{pmatrix} 49456/125 & 153648/125 & 113472/125 & 72/5 & -3312/25 & 113472/125 \\ -1576/25 & -4608/25 & -3312/25 & 0 & 72/5 & -3312/25 \\ 46/5 & 108/5 & 72/5 & 0 & 0 & 72/5 \\ -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 \end{pmatrix}.$$

Then $CY = (H_p, \theta H_p, \dots, \theta^5 H_p)^t$ by the six polynomial identities in the variable t obtained from the displayed definition of Y: explicitly,

$$C(-t^3, -t^4, -t^5, b(t), tb(t), t^2 b(t))^t = (1, t, t^2, t^3, t^4, t^5)^t.$$

Let $w_i = 9C_{0i} + 126C_{1i} + 532C_{2i}$ and $w_i^+ = 9C_{1i} + 126C_{2i} + 532C_{3i}$; consequently $G_p = \sum_i w_i Y_i(x)$. The six-state connection matrix is

$$L = \frac{1}{1-\lambda} \begin{pmatrix} 0 & 1-\lambda & 0 & 0 & 0 & 0 \\ 0 & 0 & 1-\lambda & 0 & 0 & 0 \\ 5\lambda/72 & 23\lambda/36 & 3\lambda/2 & -5\lambda/72 & -23\lambda/36 & -3\lambda/2 \\ 0 & 0 & 0 & 0 & 1-\lambda & 0 \\ 0 & 0 & 0 & 0 & 0 & 1-\lambda \\ -5/72 & -23/36 & -3/2 & 5\lambda/72 & 23\lambda/36 & 3\lambda/2 \end{pmatrix}.$$

Squaring (3) gives

$$\big(\theta^6 - z(\theta + \tfrac{1}{2})^2(\theta + \tfrac{1}{6})^2(\theta + \tfrac{5}{6})^2\big)H_p = -A_p^2 z^p.$$

Taking its coefficients and using $d\lambda/dx = 3\lambda/x$ proves

$$Y_i' = \frac{3}{x}\sum_j L_{ij}Y_j - \frac{3A_p^2\lambda^p}{x(1-\lambda)}\nu_i, \qquad \nu = (0, 0, -1, 0, 0, 1). \tag{14}$$

Also $w\nu = 0$ and $wL = w^+$, by direct matrix multiplication.

Let $e_x = 99x - 50$. The tables define integer polynomials $P_i(x)$ and positive integers d_i, with $r = (1, 1, 4, 1, 1, 1)$. Define the actual primitive row and its associated source row by

$$U_i(x) = \frac{P_i(x)}{d_i x^{r_i} e_x^2}, \tag{15}$$

$$V_i(x)=\frac{w_i+6w_i^{+}}{3375}-2\left((1-x)U_i'+\frac{3(1-x)}{x}\sum_j U_jL_{ji}-\frac{U_i}{2}\right). \tag{16}$$

The second formula is an explicit rational definition, equivalent by clearing denominators to the actual six source fractions. It does not choose a source value or assume a differential equation. Put

$$T_p^{\mathrm{nat}}=\sum_i U_iY_i(x),\qquad J_p=\frac{20250A_p^2\lambda^p(1-x)}{x(1-\lambda)}(U_5-U_2).$$

Substitution of (14) into the product rule gives

$$2xG_p'+G_p=6750\beta(T_p^{\mathrm{nat}})+3375\sum_i V_iY_i(x)+J_p. \tag{17}$$

The term J_p comes from the actual square residual in (14) and remains present throughout the calculation.

For the conductor calculation put

$$\begin{aligned} m&=\frac{27}{1000}x(80-99x+27x^2), & \ell&=\frac{27}{1000}x(9x-16),\\ h&=25-9x, & f(s,x)&=s^2-2ms+\lambda,\\ D&=x^4he_x^2. \end{aligned}$$

Set $A=81x^2-198x+80$, $B=243x^3-1053x^2+1266x-400$, $c=1953125/209952$, $d=244140625/944784$, and

$$Z=c(59049x^6-511758x^5+1724085x^4-2860596x^3+2441988x^2-1011200x+160000).$$

Define all nine entries of the two ordinary matrices by

$$P=h\begin{pmatrix}0&0&cAB\\0&0&-dA\\-cAB&dA&0\end{pmatrix},\qquad Q=\begin{pmatrix}0&0&Z\\0&3906250xe_x/6561&-dB\\Z&-dB&30517578125/8503056\end{pmatrix}.$$

The ordinary rational rows are

$$R_{ij}^{\mathrm{ord}}=\frac{2(P_{ij}(s-m)+\ell hQ_{ij})}{Df}, \tag{18}$$

$$U_{ij}^{\mathrm{ord}}=\frac{-2(1-x)P_{ij}\partial_x f+4(1-x)\ell hQ_{ij}m'-2Q_{ij}(s-m)((\ell-2(1-x)\ell')h+9(1-x)\ell)}{Df}. \tag{19}$$

With $b=\lambda/s$ and

$$K(q)=\begin{pmatrix}-5q/72&0&2q^2(1-q)\\-q/2&q(q-1)&0\\1-q&0&0\end{pmatrix},$$

set $R=K(s)R^{\mathrm{ord}}K(b)^t$ and $U=K(s)U^{\mathrm{ord}}K(b)^t$. The table also defines the nine rational functions $\eta_{ij}(s,x)$, each a Laurent polynomial in s with pole order at most three and $\deg_s(s^3\eta_{ij})\le 6$.

Here and below Res_s means the coefficient of s^{-1} in the formal Laurent expansion at $s=0$ over $\mathbb{Q}(x)$; it does not mean an analytic contour choice. Let $f_i(q)=\theta^iF_p(q)$ for $0\le i\le 2$, $E_{ij}=f_i(s)f_j(b)$, and write $\langle r,E\rangle=\sum_{i,j=0}^{2}r_{ij}E_{ij}$. For reference the native jet matrix is

$$B(q)=\begin{pmatrix}0&1&0\\0&0&1\\\frac{5q}{72(1-q)}&\frac{23q}{36(1-q)}&\frac{3q}{2(1-q)}\end{pmatrix},\qquad \omega=(5/72,23/36,3/2).$$

The coefficient recurrence gives

$$\theta f_i(q) = \sum_j B(q)_{ij} f_j(q) - \mathbf{1}_{i=2} \frac{A_p q^p}{1-q}.$$

It follows by the product rule that

$$\partial_s E_{ij} = \frac{1}{s}\left(\sum_u B(s)_{iu} E_{uj} - \sum_v B(b)_{jv} E_{iv}\right) + \epsilon^s_{ij}, \tag{20}$$

$$\partial_x E_{ij} = \frac{3}{x}\sum_v B(b)_{jv} E_{iv} + \epsilon^x_{ij}, \tag{21}$$

$$\epsilon^s_{ij} = -\frac{A_p}{s}\left(\mathbf{1}_{i=2}\frac{s^p}{1-s} f_j(b) - \mathbf{1}_{j=2}\frac{b^p}{1-b} f_i(s)\right), \tag{22}$$

$$\epsilon^x_{ij} = -\frac{3A_p}{x}\mathbf{1}_{j=2}\frac{b^p}{1-b} f_i(s). \tag{23}$$

Define $\mathcal{C}(r)_{ij} = (\sum_u r_{uj} B(s)_{ui} - \sum_v r_{iv} B(b)_{vj})/s$ and $\mathcal{C}_2(r)_{ij} = \sum_v r_{iv} B(b)_{vj}$. The following nine rational identities hold entry by entry:

$$\frac{\omega_i V_j}{s-1} + \frac{V_{i+3}\omega_j}{s-\lambda} = -R_{ij} + 2(1-x)\partial_x R_{ij} + \frac{6(1-x)}{x}\mathcal{C}_2(R)_{ij} + \partial_s(U-\eta)_{ij} + \mathcal{C}(U-\eta)_{ij}. \tag{24}$$

They are finite polynomial calculations: substitute (16), (19) and the table, differentiate fractions by the quotient rule, clear $x, s, e_x, h, s-1, s-\lambda, f$ and $1-\lambda$, and collect coefficients. These factors are nonzero rational functions. Thus the identity verifies the actual witness and is not a source-equation premise. The exact cleared polynomials, degree bounds, coefficient ranges and zero arrays are supplied in Section 1.9.

Define

$$T^{\mathrm{res}}_p = \mathrm{Res}_s \langle R, E\rangle, \tag{25}$$

$$\Delta_p = \mathrm{Res}_s\left(-2(1-x)\langle R, \epsilon^x\rangle + \langle \eta - U, \epsilon^s\rangle\right), \tag{26}$$

$$S_p = \Delta_p + A_p \sum_{i=0}^{2} V_{i+3} f_i(\lambda), \tag{27}$$

$$T_p = T^{\mathrm{nat}}_p + T^{\mathrm{res}}_p. \tag{28}$$

The residue of an s derivative is zero, and the residue commutes with ∂_x. Contracting (24) with E and using (23) therefore expresses its left residue as $2\beta(T^{\mathrm{res}}_p) + \Delta_p$. To identify that same residue with the angular six state, expand the two unit-pole fractions:

$$\mathrm{Res}_s \frac{s^u b^v}{s-1} = -\mathbf{1}_{u<v}\lambda^v, \qquad \mathrm{Res}_s \frac{s^u b^v}{s-\lambda} = -\mathbf{1}_{u<v}\lambda^u.$$

Here $u, v \geq 0$ are integer exponents and $b = \lambda/s$ is the reciprocal variable. If $u(n) = \frac{5}{72} + \frac{23}{36}n + \frac{3}{2}n^2$, then

$$u(n)c_n = (n+1)^3 c_{n+1} - n^3 c_n.$$

Hence the first residue triangle has coefficients $-n^{j+3}c_n^2$, and the second has coefficients $n^j b(n) c_n^2 - A_p n^j c_n$. In particular, the upper index $p-1$ is included in the second triangle. The six resulting components are exactly $Y - A_p(0,0,0,F_p,\theta F_p,\theta^2 F_p)$, evaluated at λ. For completeness, the finite prefix and suffix sums which give these coefficients are

$$\sum_{a=0}^{n-1} u(a)c_a = n^3 c_n, \qquad \sum_{b=n+1}^{p-1} u(b)c_b = A_p - (n+1)^3 c_{n+1}.$$

The first triangle is $-n^j c_n$ times the prefix; the second is $-n^j c_n$ times the suffix. Substituting $(n+1)^3 c_{n+1} = b(n)c_n$ gives exactly the two stated coefficients. Together with (17), this proves the actual complete source identity

$$2xG'_p + G_p = 6750\beta(T_p) + 3375 S_p + J_p. \tag{29}$$

1.5 The whole conductor, its cutoff and all source errors

The full conductor used here is the actual rational function

$$B_p = \lambda^p \operatorname{Res}_s \frac{\sum_i (6(1-x)sR_{i2}/x - U_{i2})f_i(s)}{s^{p+1}(1-b)}. \tag{30}$$

Its reduction to explicit polynomial coefficients can be stated without the tensor notation. Put $L_0 = P_{20}$, $R_0 = Z$, $\kappa = (\ell - 2(1-x)\ell')h + 9(1-x)\ell$, and

$$\begin{aligned} g_0 &= 2x(1-x)L_0\lambda' + 2xR_0(-2(1-x)\ell hm' - m\kappa),\\ g_1 &= 12(1-x)(-L_0 m + \ell h R_0) - 4x(1-x)L_0 m' + 2xR_0\kappa,\\ g_2 &= 12(1-x)L_0. \end{aligned}$$

Let $I_0 = g_0$, $I_1 = g_1 - g_0$, $I_2 = g_2 - g_1$, $I_3 = -g_2$ and $I_k = 0$ for $k \geq 4$. Define $K_0 = I_0$, $K_1 = 2mI_0 + \lambda I_1$ and

$$K_{k+2} = 2mK_{k+1} - \lambda K_k + \lambda^{k+2} I_{k+2}.$$

Equivalently,

$$\frac{(1-s)(g_0 + g_1 s + g_2 s^2)}{s^2 - 2ms + \lambda} = \sum_{k\geq 0} \frac{K_k}{\lambda^{k+1}} s^k.$$

The ordinary-state conversion and the zero entries of column zero leave only the ordinary component $2s^2(1-s)F_p(s)$ in (30). More explicitly, $K(b)_{2a} = (1-b)\mathbf{1}_{a=0}$, $R^{\mathrm{ord}}_{00} = R^{\mathrm{ord}}_{10} = U^{\mathrm{ord}}_{00} = U^{\mathrm{ord}}_{10} = 0$, and hence

$$\frac{\sum_i (6(1-x)sR_{i2}/x - U_{i2})f_i(s)}{1-b} = \left(\frac{6(1-x)s}{x} R^{\mathrm{ord}}_{20} - U^{\mathrm{ord}}_{20}\right) 2s^2(1-s)F_p(s).$$

Substitution of the displayed ordinary rows gives

$$\frac{6(1-x)s}{x} R^{\mathrm{ord}}_{20} - U^{\mathrm{ord}}_{20} = \frac{g_0 + g_1 s + g_2 s^2}{x^5 h e_x^2 f}.$$

Thus the coefficient selected by the residue for index n is precisely $[s^{p-2-n}](1-s)(g_0+g_1 s+g_2 s^2)/f = K_{p-2-n}/\lambda^{p-1-n}$. The kernel identity therefore proves exactly

$$B_p = \frac{\sum_{n=0}^{p-2} 2c_n \lambda^{n+1} K_{p-2-n}}{x^5 h e_x^2}. \tag{31}$$

Define Q_0, Q_1 by the explicit polynomials in the witness table and $Q_{k+2} = 2mQ_{k+1} - \lambda Q_k$. The two initial expanded numerators are

$$\begin{aligned} K_2 &= (4m^2 - \lambda)g_0 + 2m\lambda(g_1 - g_0) + \lambda^2(g_2 - g_1),\\ K_3 &= (8m^3 - 4m\lambda)g_0 + (4m^2\lambda - \lambda^2)(g_1 - g_0)\\ &\quad + 2m\lambda^2(g_2 - g_1) - \lambda^3 g_2. \end{aligned}$$

The displayed g_i and kernel tables give

$$[x^k](K_2 - x^5 h e_x^2 Q_0) = 0 \ (0 \leq k \leq 19), \qquad [x^k](K_3 - x^5 h e_x^2 Q_1) = 0 \ (0 \leq k \leq 22).$$

The degrees of the separate summands are at most 19 and 22, respectively, so these coefficient ranges are exhaustive. Therefore $K_2 = x^5 h e_x^2 Q_0$ and $K_3 = x^5 h e_x^2 Q_1$; the recurrence proves $K_{k+2} = x^5 h e_x^2 Q_k$ for every $k \geq 0$. Thus

$$B_{p,t}(x) = \sum_{n=0}^{t} 2c_n \lambda^{n+1} Q_{p-4-n}(x) \quad (t+4 \leq p)$$

is a polynomial with degree at most $3p-1$. Its coefficients are p-integral: Q_0, Q_1, m, λ have coefficient denominators involving only 2 and 5, and the recurrence preserves integrality.

Write

$$H_{n,k} = 2c_n a^{n+1} x^{3(n+1)-5} K_k.$$

Cancelling the actual common x^5 factor in (31) gives the full formula, with both top terms retained,

$$B_p = B_{p,p-4} + \frac{H_{p-3,1} + H_{p-2,0}}{h e_x^2}. \tag{32}$$

For $p \geq 19$, the two top polynomials have coefficient precision three. The inverse series of he_x^2 is integral at the origin, since its constant term is $62500 = 2^2 5^6$. The difference $B_{p,p-4} - B_{p,e}$ is a sum of terms with $n > e$, all with $c_n = O(p^3)$. Therefore

$$[x^k](B_p - B_{p,e}) = O(p^3), \qquad [x^k] A_p (B_p - B_{p,e}) = O(p^6) \tag{33}$$

for every $k \geq 0$. Negative origin coefficients vanish because the regularized expression (32) is a power series.

For clarity the other three finite errors in the source are defined here. Let $\eta^*_{ij} = s^3 \eta_{ij}$, a polynomial of degree at most six in s, and put

$$\begin{aligned} E_p^{\mathrm{ang}} &= -\operatorname{Res}_s \frac{s^{p-1}}{1-s} \sum_j \eta_{2j} f_j(b), \\ E_p^{\mathrm{con}} &= \operatorname{Res}_s \frac{s^{p-1}}{1-s} \sum_j U_{2j} f_j(b), \\ Z_p(s,x) &= \sum_i \eta^*_{i2}(s,x) f_i(s), \\ E_p^{\mathrm{high}} &= \sum_{a \geq p+3} [s^a] Z_p(s,x)\, \lambda^{a-3}. \end{aligned}$$

The last sum is finite. Expanding (23) gives the exact identity

$$S_p = A_p (B_p + E_p^{\mathrm{ang}} + E_p^{\mathrm{con}} + E_p^{\mathrm{high}}). \tag{34}$$

In obtaining it, the last-column residue is $-Z_p(\lambda, x)/\lambda^3 + E_p^{\mathrm{high}}$. The evaluations $\eta_{i2}(\lambda, x) = V_{i+3}(x)$, verified from the table, cancel its first term against the endpoint correction in (28).

Here is the precision argument for every retained error, both at $x = 0$ and at $x = \alpha = 50/99$. Expansion of the first angular kernel gives

$$\operatorname{Res}_s \frac{s^{p-1}}{1-s} \eta_{2j} b^n = \sum_k [s^k] \eta^*_{2j}\, \lambda^n \mathbf{1}_{p-1+k<n+3}.$$

It is zero when $n + 4 \leq p$. For the conductor kernel put

$$\begin{aligned} n_0 &= 2(1-x) L_0 \lambda' + 4(1-x) \ell h R_0 m' + 2 R_0 m \kappa, \\ n_1 &= -4(1-x) L_0 m' - 2 R_0 \kappa, \end{aligned}$$

and, with $u_k = [s^k](s^2 - 2ms + \lambda)^{-1}$, put $q(a,b) = \mathbf{1}_{a<b} u_{b-a-1}$. The conductor error is the sum $\sum_{n<p} c_n \mathcal{K}_{p,n}$, where

$$\begin{aligned} \mathcal{K}_{p,n} = \frac{2}{D} \big(&\lambda^{n+2} n_0 q(p-1, n+2) + \lambda^{n+2} n_1 q(p, n+2) \\ &- \lambda^{n+3} n_0 q(p-1, n+3) - \lambda^{n+3} n_1 q(p, n+3) \big). \end{aligned}$$

This too is zero for $n+4 \le p$. The u_k satisfy $u_0 = \lambda^{-1}$, $u_1 = 2m\lambda^{-2}$ and $\lambda u_{k+2} = 2mu_{k+1} - u_k$. The reduction to this four-coefficient expression keeps the complete ordinary numerator:

$$U_{02}^{\mathrm{ord}} = (n_0 + n_1 s)/(Df), \qquad U_{2j} = (1-s)U_{02}^{\mathrm{ord}} K(b)_{j2}, \qquad \sum_j K(b)_{j2} f_j(b) = 2b^2(1-b)F_p(b).$$

These identities cancel $1-s$ in the residue, leaving $2s^{p-1}(n_0 + n_1 s)b^{n+2}(1-b)/(Df)$ for native coefficient c_n. Its two affine numerator terms and the two terms of $1-b$ give the four coefficients in $\mathcal{K}_{p,n}$, with none dropped. Finally $\deg_s \eta^* \le 6$ implies that a term $[s^a]Z_p$ with $a \ge p+3$ involves only $n \ge p-3$ in f_i.

All kernel coefficients in these formulas are integral Laurent series at the two indicated x points. At the origin, powers of x and $\lambda = ax^3$ merely shift Laurent indices; inverse collision and escape series have unit constants 25 and -50. At α, x, λ and h have the nonzero unit values

$$\alpha = 50/99, \qquad \lambda(\alpha) = 8/1331, \qquad h(\alpha) = 225/11;$$

inverse powers of $e_x = 99(x-\alpha)$ only shift Laurent indices. Here are explicit factorizations of the numerical witness denominators. Put $B = 7^2\,11^2\,13\,17$, and let c_{ij} be the numerical factor of D_{ij} in the angular table. The displayed integer values multiply out to

$$(d_0, d_1, d_2, d_3, d_4, d_5)/B = (2^3 3^3 5^6,\ 2^2 3^3 5^6,\ 3^5 5^6,\ 2^3 3\,5^9,\ 3\,5^{10},\ 3\,5^{10}),$$

$$(c_{ij}/B)_{0\le i,j\le 2} = \begin{pmatrix} 2^5 3^5 5^8 & 2^4 3^5 5^5 & 2^3 3^9 5^5 \\ 2^4 3^3 5^9 & 2^3 3^3 5^6 & 2^2 3^7 5^6 \\ 2^3 3^3 5^9 & 2^2 3^3 5^6 & 2\,3^7 5^6 \end{pmatrix}. \tag{35}$$

Thus their only prime divisors are $2, 3, 5, 7, 11, 13, 17$, and none causes a loss for $p \ge 19$. Since every nonzero selected native coefficient has $n \ge p-3$, it has valuation three, and multiplication by A_p supplies three further digits. This proves

$$[x^k](S_p - A_p B_p) = O(p^6), \qquad [(x-\alpha)^k](S_p - A_p B_p) = O(p^6) \tag{36}$$

for every signed Laurent index k.

The square forcing has the completely regular origin expression

$$J_p = A_p^2 \frac{a^p x^{3p-5}\, 20250(1-x)(-P_2/d_2 + x^3 P_5/d_5)}{(1-\lambda)e_x^2}. \tag{37}$$

Its denominator has constant term $2500 = 2^2 5^4$ at the origin. At α, the regular factor $1-\lambda$ has value $1323/1331$, whose inverse is $1331/1323$; the only additional primes are 3 and 7. Thus every origin coefficient and every Laurent coefficient at α of J_p has precision six, coming from $v_p(A_p^2) = 6$.

1.6 Restoration of both primitive endpoints

The numerator of T_p over e_x^2 is an actual polynomial divisible by x and of degree at most $3p+7$. Here is the cancellation argument, including the terms which prevent an unjustified origin substitution. The native part is $\sum_{n<p,i} w_i(n) c_n^2 a^n x^{3n} P_i/(d_i x^{r_i} e_x^2)$, where

$$(w_i(n))_{i=0}^5 = (-n^3, -n^4, -n^5, b(n), nb(n), n^2 b(n)).$$

For $n \ge 2$ the numerator term contains x^{3n-r_i}, hence x^2; its degree is at most $3n+10$. The two indices $n = 0, 1$ combine to $L(x)/(xe_x^2)$, where

$$L(x) = \frac{P_3 - P_2/25}{884458575000000} + x^3 \sum_{i\ne 2} \frac{9}{40000} \frac{w_i(1)}{d_i} P_i.$$

Both the constant and linear coefficients of this displayed polynomial are zero. Thus L/x is a polynomial divisible by x of degree at most 13; the low terms have been combined before division by x. The cancellations are already visible in the first two entries of the primitive tables:

$$-509162500000000000 - \frac{-12729062500000000000}{25} = 0,$$
$$2871676500000000000 - \frac{71791912500000000000}{25} = 0.$$

The other part of L has an explicit factor x^3.

For the residue part define the ordinary polynomials

$$O_0 = (1-z)\theta^2 F_p - \tfrac{z}{2}\theta F_p - \tfrac{5z}{72}F_p, \quad O_1 = z(z-1)\theta F_p, \quad O_2 = 2z^2(1-z)F_p.$$

Each has degree at most $p+2$ and has zero constant and linear coefficients, using $c_0 = 1, c_1 = 5/72$. Write $P = h\overline{P}$, $\overline{A} = -m\overline{P} + \ell Q$, and $I_0 = 1, I_1 = 2m, I_{k+2} = 2mI_{k+1} - \lambda I_k$. Expansion of the quadratic inverse proves $[s^k]f^{-1} = I_k/\lambda^{k+1}$. Consequently the numerator of T_p^{res} over e_x^2 is the sum over $i, j = 0, 1, 2$, $0 \le a, b \le p+2$ and $\epsilon = 0, 1$ of

$$\mathbf{1}_{a\ge 2,\, a+\epsilon<b}\, 2[z^a]O_i[z^b]O_j\, a_0^{a+\epsilon} x^{3(a+\epsilon)-4} M_{\epsilon,ij} I_{b-a-\epsilon-1}, \quad a_0 = 729/15625,$$

where $M_0 = \overline{A}$ and $M_1 = \overline{P}$. This keeps all nine states and both numerator contributions. The selection $a + \epsilon < b$ follows from the coefficient of s^{-1} in $s^{a+\epsilon-b}f^{-1}$, not from an imposed vanishing rule. Each term is divisible by x^2. Since $\deg M_\epsilon \le 8$ and $\deg I_k \le 3k$, its degree is at most $3(a+\epsilon) - 4 + 8 + 3(b-a-\epsilon-1) = 3b+1 \le 3p+7$. This proves the asserted numerator properties for T_p, and in particular

$$[x^0]T_p = 0. \tag{38}$$

Euclidean division of its actual numerator by e_x^2 gives uniquely

$$T_p = P_p(x) + \frac{a_1}{e_x} + \frac{a_2}{e_x^2}, \qquad \deg P_p \le 3p+5.$$

If the remainder is $r_0 + r_1 x$, then $a_1 = r_1/99$ and $a_2 = r_0 + 50a_1$. These are the genuine escape coefficients. To bound them, first retain the full conductor poles. With $r = 8/1331$, their coefficients in e_x^{-2} and e_x^{-1} are respectively

$$b_2 = \frac{51450}{11} r^p c_{p-3},$$
$$b_1 = \frac{7}{33} r^p \big(242 c_{p-2} + (1323p - 681)c_{p-3}\big).$$

These follow by evaluating the numerator in (31) and its first derivative at α: all canceled terms have an e_x^2 factor, and the two retained top terms give precisely the displayed values. Thus $b_1, b_2 = O(p^3)$ and B_p has no lower Laurent powers. Equations (36) and $v_p(A_p) = 3$ show that every negative escape coefficient of S_p is $O(p^6)$; the same holds for J_p.

Apply the coefficient of $(x-\alpha)^n$ to (29) for $n = -3, -2$. The left side is a polynomial and has zero negative coefficients. Since the negative coefficients of T_p are $a_2/99^2$ at -2 and $a_1/99$ at -1, the resulting equations imply

$$\frac{-6750 \cdot 98 a_2}{99^3} = O(p^6), \qquad \frac{6750(-49a_1 + 3a_2/2)}{99^2} = O(p^6).$$

All divided numerical factors here have prime divisors among $2, 3, 5, 7, 11$. Therefore $a_1, a_2 = O(p^6)$ for $p \ge 19$. At the origin, inverse powers of e_x have integral coefficients, so these two tails and their beta images have coefficient precision six.

By (36) and (33), $[x^k](S_p - A_p B_{p,e}) = O(p^6)$ for every integer k. As $\deg B_{p,e} \le 3p-1$, $[x^k]S_p = O(p^6)$ for $k \ge 3p$. The polynomial G_p has degree at most $3(p-1)$, and the square forcing has coefficient precision six. Equation (29) now gives

$$(k+1)[x^{k+1}]P_p - (k+\tfrac{1}{2})[x^k]P_p = O(p^6) \quad (k \ge 3p).$$

Start from the genuinely zero coefficient $[x^{3p+6}]P_p$ and descend for $k = 3p+5, \dots, 3p$. The divided factors $2k+1$ have residues $11, 9, 7, 5, 3, 1$ modulo p, all units for $p \ge 19$. Thus all six divisions preserve precision six and, after reattaching both tails,

$$[x^{3p}]T_p = O(p^6). \tag{39}$$

This is the extended coefficient at $3p$, rather than the last coefficient of G_p.

1.7 The exact conductor moment and the precision-five adapter

It remains to attach the polynomial conductor to the genuine moments in (7). Define

$$\begin{aligned} u(r) &= (4r^2-1)/(3r), & x(r) &= 1-u(r)^2,\\ q_1(r) &= (16r^2-1)(1-r^2)^2/(500r^6), & q_2(r) &= 4(16r^2-1)^2(1-r^2)/125,\\ v_1(r) &= -(16r^2-11)/(3(1-r^2)(16r^2-1)), & v_2(r) &= 4(11r^2-1)/(3(1-r^2)(16r^2-1)). \end{aligned}$$

Rational substitution gives $q_1 q_2 = \lambda(x(r))$, $q_1 + q_2 = 2m(x(r))$, and, for $k = 0, 1$,

$$2\lambda(x(r))Q_k(x(r))u'(r) = v_1(r)q_2(r)^{k+4} + v_2(r)q_1(r)^{k+4}.$$

The two roots satisfy the same quadratic recurrence as Q_k, so induction proves that formula for every $k \ge 0$. Multiplying by the native coefficients and using $n + (p-4-n+4) = p$ gives

$$B_{p,t}(x(r))u'(r) = \sum_{n=0}^{t} c_n\big(v_1(r)q_2(r)^p q_1(r)^n + v_2(r)q_1(r)^p q_2(r)^n\big).$$

The reciprocal substitution $r \mapsto 1/(4r)$ exchanges q_1, q_2 and sends $v_1(r)\,dr$ to $-v_2(r)\,dr$. Multiply the density identity by p and set $r = s/2$, retaining $dr = ds/2$. The first density becomes

$$\frac{2p}{3\,125^{p+n}} \sum_{a=0}^{3(p+n)-1} [X^a]N_{p,n}(X)\, s^{d_{n,a}-1},$$

and the reflected one has exponent $-d_{n,a} - 1$. Formal monomial integration from $s = 1$ to $s = 2$ joins the two primitives:

$$\frac{2^d - 1}{d} + \frac{2^{-d} - 1}{-d} = \frac{2^d - 2^{-d}}{d}.$$

Here $d = d_{n,a} \ne 0$; no logarithmic slot is discarded. On the other hand $u(s/2) = \frac{2}{3}(s - s^{-1})$ takes the values $0, 1$ at $s = 1, 2$. For any polynomial P of degree at most $3p-1$, $\Gamma_{3p-1}([x^k]P) = \int_0^1 P(1-u^2)\,du$, where the integral denotes finite monomial antiderivative evaluation. To verify it, integration by parts gives $(2k+3)I_{k+1} = (2k+2)I_k$, $I_0 = 1$ for $I_k = \int_0^1 (1-u^2)^k du$; this is the factorial recurrence for γ_k. The change of variable in the finite Laurent primitive is therefore exact, and proves, for $t + 4 \le p$,

$$p\Gamma_{3p-1}([x^k]B_{p,t}) = \sum_{n=0}^{t} c_n m_{p,n}. \tag{40}$$

In particular $t = e$ is allowed for $p \ge 19$, since $\lfloor p/6 \rfloor \ge 3$.

Applying the finite Stokes identity (11) to (29), using (12) and (38), gives the exact equation

$$W_p = 6750(3p)\gamma_{3p-1}[x^{3p}]T_p + 3375\Gamma_{3p-1}([x^k]S_p) + \Gamma_{3p-1}([x^k]J_p). \tag{41}$$

The upper coefficient in (39) has precision six; $p\gamma_{3p-1} \in R_p$ by (13). The retained $3p$ factor thus restores the digit lost by γ_{3p-1}, and the entire upper endpoint in (41) is $O(p^6)$. The square forcing and source comparison have coefficient precision six, so their Gamma errors are $O(p^5)$, paying exactly the possible one-digit loss in (13). The conductor full/short comparison has precision three before its multiplication by A_p, precision six after that multiplication, and precision five after Gamma. Finally (40) and $A_p = p^3 c_p$ give the exact cancellation

$$A_p \Gamma_{3p-1}(B_{p,e}) = p^2 c_p C_p.$$

This cancellation uses the identity $p\Gamma(B_{p,e}) = C_p$ before any precision estimate. Replacing C_p by M_p uses (9) and the multiplier $p^2 c_p$, of valuation two, hence produces an $O(p^5)$ error. We have proved

$$\boxed{W_p \equiv 3375 p^2 c_p M_p \pmod{p^5} \quad (p \geq 19).} \tag{42}$$

Since M_p is integral, (10) may be multiplied by M_p without losing precision, and then by p^2, gaining two digits. The native moment assertion $M_p \equiv 1728^p/45000 \pmod{p^3}$ consequently implies the target literal congruence, because

$$3375 \cdot \frac{120}{1728^p} \cdot \frac{1728^p}{45000} = 9.$$

Conversely $K_p = 3375p^2(120/1728^p)$ is nonzero and $v_p(K_p) = 2$. If $K_p(M_p - 1728^p/45000) = O(p^5)$, either the difference is zero or the ordinary nonzero valuation formula subtracts exactly two digits and gives $O(p^3)$ for that difference. Thus the literal precision-five target and the complete native precision-three target are equivalent for $p \geq 19$.

1.8 The four smaller primes

For every n, the integer $a_n = (6n)!/(n!)^6 = \prod_{j=2}^{6} \binom{jn}{n}$ is a multinomial coefficient. Put

$$D_p = 10^{6(p-1)}, \qquad L_p = \sum_{n=0}^{p-1} a_n w(n) 10^{6(p-1-n)} - 9p^2 D_p.$$

Then $W_p - 9p^2 = L_p/D_p$, and D_p is a p-adic unit for every prime $p \geq 7$. The four exact integer certificates in the table below say $L_p = p^5 Q_p$. They establish the congruence for $p = 7, 11, 13, 17$ by integer arithmetic, independently of the uniform branch. There are no other primes in $7 \leq p < 19$.

1.9 Exact polynomial and integer witnesses

The notation $\mathrm{pol}(a_0, \ldots, a_d) = \sum_{j=0}^{d} a_j x^j$ lists coefficients in increasing order. In a broken list, read from left to right, then from top to bottom. This fixes every coefficient below. All displayed tables are exact integer data; no numerical approximation or unstated row selection enters the definitions.

The primitive polynomials. The denominator scalars, in the same order as $P_0, \ldots, P_5$, are

$$(d_0, d_1, d_2) = (4422292875000, 2211146437500, 4975079484375),$$
$$(d_3, d_4, d_5) = (61420734375000, 38387958984375, 38387958984375).$$

$$P_0(x) = \mathrm{pol}\begin{pmatrix} 2545812500000000, -10591244125000000, \\ 43747191704947500, -103207329500259600, \\ 40956081646390479, 2169455656771053, \\ -3723606976500000, 1503132522131250, \\ -221773650806250 \end{pmatrix}.$$

$$P_1(x) = \operatorname{pol}\begin{pmatrix} 10437831250000000, 213163861968750000, \\ -75558073465239750 0, 662287068835761600, \\ -271088081060128209, 77231074057061337, \\ -31176189825210000, 12585094275029625, \\ -1856817188119125 \end{pmatrix}.$$

$$P_2(x) = 2\operatorname{pol}\begin{pmatrix} -6364531250000000000, 35895956250000000000, \\ -73688625527343750000, 68640605627343750000, \\ -27903747231140625000, -3295566944859026250, \\ 8112395820448285200, -3319630016857857423, \\ 772142970292880589, -292627723225212000, \\ 118126926507824550, -17428562927383950 \end{pmatrix}.$$

$$P_3(x) = \operatorname{pol}\begin{pmatrix} -509162500000000000, 2871676500000000000, \\ -5892141846937500000, 5480192337716250000, \\ -2194872802872525000, -415424800467337500, \\ 806217628510566000, -330459542986715235, \\ 75389755821710031, -28123937355843360, \\ 11352971770851474, -1675028621928906 \end{pmatrix}.$$

$$P_4(x) = \operatorname{pol}\begin{pmatrix} -26094578125000000000, 14717342062500000000, \\ -3019796401436718750 0, 2816192347461281250 0, \\ -11315100845004496875, -1740150277592146875, \\ 3703643019960507000, -1517452083064009170, \\ 349201112876671032, -132746063056189920, \\ 53586462219006078, -7906199343787782 \end{pmatrix}.$$

$$P_5(x) = \operatorname{pol}\begin{pmatrix} -42642359375000000000, 24101474632812500000, \\ -4921910481865781250 0, 4491724993665525000 0, \\ -18100041531561838125, -2774200475601016875, \\ 6078803129564364000, -2490691894429221540, \\ 568012052156440284, -212018524827053040, \\ 85586889801508911, -12627573905140659 \end{pmatrix}.$$

The two canceled conductor kernels.

$$Q_0 = -\frac{81}{2000000}\operatorname{pol}\begin{pmatrix} 10240000, -65536000, \\ 173637632, -251265024, \\ 218546910, -117490743, \\ 38244069, -6908733, \\ 531441 \end{pmatrix},$$

$$Q_1 = -\frac{2187x}{1000000000}\operatorname{pol}\begin{pmatrix} 819200000, -6258688000, \\ 20666122240, -39081738240, \\ 47067955200, -37830511674, \\ 20594814975, -7511426460, \\ 1759069710, -239148450, \\ 14348907 \end{pmatrix}.$$

The complete angular primitive. Write $\eta_{ij} = \sum_k s^k T_{ij,k}(x)/D_{ij}(s,x)$, with the denominator and every nonzero numerator coefficient given below. Unlisted coefficients $T_{ij,k}$ are zero.

$$D_{00} = 3980063587500000 s^2 x^2 (9x-25) e_x^3.$$

$$T_{00,0}(x) = 2089055775807 x^6 \,\mathrm{pol}\begin{pmatrix} 40000000, -408800000, \\ 1745406000, -4033722240, \\ 5471483256, -4426575480, \\ 2077757163, -517623534, \\ 52612659 \end{pmatrix}.$$

$$T_{00,1}(x) = 4652022375 x^3 \,\mathrm{pol}\begin{pmatrix} -385000000000, 3934700000000, \\ -16803600750000, 38853547560000, \\ -52736243535000, 42687316418400, \\ -20036318346495, 4984554609630, \\ -505017222039 \end{pmatrix}.$$

$$T_{00,2}(x) = 689188500 x^2 \,\mathrm{pol}\begin{pmatrix} -828125000000, 6114843750000, \\ -7577578125000, -3871424531250, \\ 16777469401875, -13561459364250, \\ 5350669950429, -2475383169402, \\ 1002048125877, -149823315279 \end{pmatrix}.$$

$$T_{00,3}(x) = 390625 \,\mathrm{pol}\begin{pmatrix} 198583000000000000, -1675047605000000000, \\ 5730691619152500000, -10237049104156875000, \\ 10283611063410877500, -5861658531540055050, \\ 1828355476093736535, -305399011826168316, \\ 51659078495565825, -19253645046037863, \\ 4697621777691504, -468603118021599 \end{pmatrix}.$$

$$T_{00,4}(x) = 160673583984375 \,\mathrm{pol}\begin{pmatrix} -320000000, 2699200000, \\ -9232860000, 16485438600, \\ -16543077444, 9400815126, \\ -2884169673, 413815392, \\ -17537553 \end{pmatrix}.$$

$$D_{01} = 15920254350000 s x^2 (9x-25) e_x^3.$$

$$T_{01,0}(x) = 12058041996 x^5 \,\mathrm{pol}\begin{pmatrix} -2000000, 14550000, \\ -40309200, 52344144, \\ -31979772, 8181567, \\ -649539 \end{pmatrix}.$$

$$T_{01,1}(x) = 16540524 x^2 \,\mathrm{pol}\begin{pmatrix} -931250000000, 5908593750000, \\ -10056661875000, 4342565925000, \\ 5253654289500, -6344393673420, \\ 3269737007550, -1881451419705, \\ 781615787868, -119272897953 \end{pmatrix}.$$

$$T_{01,2}(x) = 125\,\mathrm{pol}\begin{pmatrix} 20354757500000000000, -171692379512500000000, \\ 586469576151834375000, -106181759677365312500, \\ 111666979802765168750, -69660007700839179637 5, \\ 25432369142810254465 5, -56152882923640590591, \\ 11779181176836775860, -3891351261079484523, \\ 950586991155562779, -94918457116638639 \end{pmatrix}.$$

$$T_{01,3}(x) = 90234375\,\mathrm{pol}\begin{pmatrix} -18689440000000, 157645426400000, \\ -538818955805000, 968088704388300, \\ -992448729606978, 588686508699057, \\ -194853790786671, 32318495019999, \\ -1977072654051 \end{pmatrix}.$$

$$D_{02} = 644770301175000sx^5(9x-25)e_x^3.$$

$$T_{02,0}(x) = 976701401676x^8\,\mathrm{pol}\begin{pmatrix} 2000000, -14550000, \\ 40309200, -52344144, \\ 31979772, -8181567, \\ 649539 \end{pmatrix}.$$

$$T_{02,1}(x) = 1339782444x^5\,\mathrm{pol}\begin{pmatrix} -1025000000000, 7456875000000, \\ -19255140000000, 22915825425000, \\ -11418743684250, 674773440705, \\ 1705623259437, -1359338331186, \\ 569021318874, -87337013940 \end{pmatrix}.$$

$$T_{02,2}(x) = 225\,\mathrm{pol}\begin{pmatrix} -5091625000000000000000, 42947856875000000000000, \\ -1470338948203125000000000, 26405062266515625000000000, \\ -2730466175012296875000000, 17858607783280340625000000, \\ -9651056557540321687500000, 6215484595273381260000000, \\ -3878692467585162019650 0, 1677657616371852442879 5, \\ -5266112411955166155954, 1844193944303395421355, \\ -713416998460391103402, 174421803221135835516, \\ -17428463216991827946 \end{pmatrix}.$$

$$T_{02,3}(x) = 781250\,\mathrm{pol}\begin{pmatrix} -4964575000000000000, 41876190125000000000, \\ -1432672904788125000000, 255793952621851875000, \\ -25584940391232048750 0, 1419416738803224112 50, \\ -3691861345683313357 5, -1798552068553414314, \\ 4503579848578495746, -1501411129930172178, \\ 226514469805312617, -11144935452182598 \end{pmatrix}.$$

$$T_{02,4}(x) = 8033679199218750\,\mathrm{pol}\begin{pmatrix} 320000000, -2699200000, \\ 9232860000, -16485438600, \\ 16543077444, -9400815126, \\ 2884169673, -413815392, \\ 17537553 \end{pmatrix}.$$

$$D_{10} = 1105573218750000s^2x^2(9x-25)e_x^3.$$

$$T_{10,0}(x) = 464234616846x^6 \operatorname{pol}\begin{pmatrix} 410000000, -4031700000, \\ 16642584000, -37432235160, \\ 49839306804, -39947435820, \\ 18699814467, -4658611806, \\ 473513931 \end{pmatrix}.$$

$$T_{10,1}(x) = 5168913750x^3 \operatorname{pol}\begin{pmatrix} -789250000000, 7761022500000, \\ -32051550600000, 72093491073000, \\ -95965771369500, 76826595967992, \\ -35847973042101, 8871430470246, \\ -892211317173 \end{pmatrix}.$$

$$T_{10,2}(x) = 382882500x^2 \operatorname{pol}\begin{pmatrix} 334375000000, 3400171875000, \\ -46061125312500, 96164958712500, \\ -86151263181750, 34498751040270, \\ 266443850250, -5644354259115, \\ 2344847363604, -357818693859 \end{pmatrix}.$$

$$T_{10,3}(x) = 78125 \operatorname{pol}\begin{pmatrix} -111881000000000000, 943716235000000000, \\ -3227608071187500000, 5763632000135625000, \\ -5776101894760312500, 3207492714616431750, \\ -757188371364321105, -247221347845970610, \\ 328158590540022414, -158060365144874289, \\ 39022754467917609, -3930176518120413 \end{pmatrix}.$$

$$T_{10,4}(x) = 13027587890625 \operatorname{pol}\begin{pmatrix} 320000000, -2699200000, \\ 9232860000, -16485438600, \\ 16543077444, -9400815126, \\ 2884169673, -413815392, \\ 17537553 \end{pmatrix}.$$

$$D_{11} = 4422292875000sx^2(9x - 25)e_x^3.$$

$$T_{11,0}(x) = 148864716x^4 \operatorname{pol}\begin{pmatrix} -962500000, -369000000, \\ -3646080000, 20631323700, \\ -45989954982, 43621667991, \\ -19939902516, 3256139007 \end{pmatrix}.$$

$$T_{11,1}(x) = 204204x \operatorname{pol}\begin{pmatrix} 15039062500000, -236784375000000, \\ 1139349375000000, -3327892872187500, \\ 4881600075600000, -3334853438282250, \\ 885690427748085, 164923069227291, \\ -250124632060743, 106234529241969, \\ -16483966965972 \end{pmatrix}.$$

$$T_{11,2}(x) = 125 \operatorname{pol}\begin{pmatrix} -2293560500000000000, 19346182817500000000, \\ -6688152100317187500, 10866468680882718750, \\ -7032862856879330625, -9201312413071077375, \\ 4008852961325106147, -3055901004162563501, \\ 17087428762029865377, -7235740389567548268, \\ 1779727318564677795, -178705403848066392 \end{pmatrix}.$$

$$T_{11,3}(x) = 90234375\,\mathrm{pol}\begin{pmatrix} 1515360000000, -12782061600000, \\ 43686478355000, -78512972937300, \\ 80565045187662, -47876594069223, \\ 15895723039029, -2650284988641, \\ 163794899169 \end{pmatrix}.$$

$$D_{12} = 179102861437500sx^5e_x^3.$$

$$T_{12,0}(x) = 108522377964x^8\,\mathrm{pol}\begin{pmatrix} 7600000, 56524200, \\ -167640228, 229893714, \\ -163559169, 43158258 \end{pmatrix}.$$

$$T_{12,1}(x) = 148864716x^5\,\mathrm{pol}\begin{pmatrix} 840500000000, -7059457500000, \\ 18708303600000, -22328188751250, \\ 11224262909835, -1139984995554, \\ -874042825608, 638889867144, \\ -173253486087 \end{pmatrix}.$$

$$T_{12,2}(x) = 25\,\mathrm{pol}\begin{pmatrix} 4175132500000000000000, -3371419493750000000000, \\ 1084122573548437500000000, -176666899901920312500000, \\ 15392003141550853125000000, -6955519593824012250000000, \\ 1555144511597463780000000, -585605197337971729275000, \\ 62912356717036892722650, -37040300405097319171980, \\ 15881014092504961677150, -7575808570486078122540, \\ 2635549271687041031970, -365520953427485835870 \end{pmatrix}.$$

$$T_{12,3}(x) = 390625\,\mathrm{pol}\begin{pmatrix} -44752400000000000, 361375630000000000, \\ -1160948001675000000, 1885276632704850000, \\ -1622900556359775000, 696763303470962100, \\ -97112212292558862, -31103534184361002, \\ 16900446327271146, -3319980960654459, \\ 163675825029981 \end{pmatrix}.$$

$$T_{12,4}(x) = 651379394531250\,\mathrm{pol}\begin{pmatrix} 12800000, -103360000, \\ 332104800, -539859816, \\ 467373564, -207778122, \\ 40566663, -1948617 \end{pmatrix}.$$

$$D_{20} = 552786609375000s^3x^2(9x - 25)e_x^3.$$

$$T_{20,0}(x) = 4178111551614x^6\,\mathrm{pol}\begin{pmatrix} -40000000, 408800000, \\ -1745406000, 4033722240, \\ -5471483256, 4426575480, \\ -2077757163, 517623534, \\ -52612659 \end{pmatrix}.$$

$$T_{20,1}(x) = 74432358x^3\,\mathrm{pol}\begin{pmatrix} 48125000000000, -491837500000000, \\ 2100450093750000, -485437016250000, \\ 65690692382250000, -5238217535652000, \\ 2278114862813955, -315938556594702, \\ -185349808663965, 116630742830679, \\ -29055761834022, 2953306387647 \end{pmatrix}.$$

$$T_{20,2}(x) = 344594250x^2 \operatorname{pol}\begin{pmatrix} 3312500000000, -35215312500000, \\ 178088231250000, -554538673125000, \\ 1118001193267500, -1442890772292600, \\ 1149974211613860, -536449467584745, \\ 132686236730430, -13344341615253 \end{pmatrix}.$$

$$T_{20,3}(x) = 127627500x \operatorname{pol}\begin{pmatrix} 3007812500000, -71883750000000, \\ 354657065625000, -634769031375000, \\ 714356707612500, -445785744007800, \\ 99160963333551, 8733690588033, \\ -12234044980674, 5121191869866, \\ -786033125460 \end{pmatrix}.$$

$$T_{20,4}(x) = 156250 \operatorname{pol}\begin{pmatrix} 29260000000000000, -246808100000000000, \\ 843999906750000000, -1504103932485000000, \\ 1506672133659000000, -883538015953608000, \\ 372285753643597860, -217333666519981956, \\ 155840299590015801, -72732895581612564, \\ 17892353680278075, -1796822479352736 \end{pmatrix}.$$

$$T_{20,5}(x) = 4342529296875 \operatorname{pol}\begin{pmatrix} -320000000, 2699200000, \\ -9232860000, 16485438600, \\ -16543077444, 9400815126, \\ -2884169673, 413815392, \\ -17537553 \end{pmatrix}.$$

$$D_{21} = 2211146437500s^2x^2e_x^3.$$

$$T_{21,0}(x) = 24116083992x^5 \operatorname{pol}\begin{pmatrix} -80000, 553200, \\ -1413216, 1585008, \\ -708588, 72171 \end{pmatrix}.$$

$$T_{21,1}(x) = 16540524x^2 \operatorname{pol}\begin{pmatrix} -74500000000, 515167500000, \\ -1316057400000, 1491490260000, \\ -646498943640, -384441066, \\ 94804927848, -69954641712, \\ 19145162025 \end{pmatrix}.$$

$$T_{21,2}(x) = 1837836x^2 \operatorname{pol}\begin{pmatrix} 1966625000000, -13448484375000, \\ 25043458387500, -25106800317750, \\ 10988012584755, 97873953696, \\ -874042825608, 638889867144, \\ -173253486087 \end{pmatrix}.$$

$$T_{21,3}(x) = 125 \operatorname{pol}\begin{pmatrix} -47986400000000000, 387490180000000000, \\ -1212946034107500000, 2448953425888200000, \\ -3705759035043912000, 3391382078065901880, \\ -1761098008216537242, 726810070558483674, \\ -351051815408393250, 122370795739918107, \\ -16999661465332821 \end{pmatrix}.$$

$$T_{21,4}(x) = 90234375\,\mathrm{pol}\begin{pmatrix} 20204800000, -163153760000, \\ 522798219800, -870530371956, \\ 820151053974, -432674449227, \\ 115853173533, -12047649372 \end{pmatrix}.$$

$$D_{22} = 89551430718750 s^2 x^5 e_x^3.$$

$$T_{22,0}(x) = 1953402803352 x^8\,\mathrm{pol}\begin{pmatrix} 80000, -553200, \\ 1413216, -1585008, \\ 708588, -72171 \end{pmatrix}.$$

$$T_{22,1}(x) = 1339782444 x^5\,\mathrm{pol}\begin{pmatrix} -82000000000, 567030000000, \\ -1448546400000, 1639119240000, \\ -715087832040, 13779041868, \\ 83116362006, -59805712935, \\ 15965019081 \end{pmatrix}.$$

$$T_{22,2}(x) = 5513508 x^2\,\mathrm{pol}\begin{pmatrix} 414062500000000, -2863242187500000, \\ 7314496875000000, -8249041687500000, \\ 3413796181875000, 348993907256250, \\ -856239541719750, 483324173903925, \\ -75859411627053, -19635717143106, \\ 14281739766333, -3853929884409 \end{pmatrix}.$$

$$T_{22,3}(x) = 25\,\mathrm{pol}\begin{pmatrix} 3411388750000000000000, -2754696415625000000000, \\ 8860501554351562500000, -1445619271456335937500000, \\ 1265791775968587656250000, -586331831665251646875000, \\ 151864507532332393312500, -68927557256135859802500, \\ 6151706542609587446490, -3251374548858802630368, \\ 1302215853198789261540, -607875890038334667639, \\ 211196321671618523052, -29258346604599637767 \end{pmatrix}.$$

$$T_{22,4}(x) = 390625\,\mathrm{pol}\begin{pmatrix} 23408000000000000, -189019600000000000, \\ 607152869400000000, -985661311219200000, \\ 848529205898220000, -366342580170394200, \\ 55178816399589204, 13336939045585134, \\ -8831009826152307, 2331609597698328, \\ -237496331196927 \end{pmatrix}.$$

$$T_{22,5}(x) = 217126464843750\,\mathrm{pol}\begin{pmatrix} -12800000, 103360000, \\ -332104800, 539859816, \\ -467373564, 207778122, \\ -40566663, 1948617 \end{pmatrix}.$$

Small-prime integer certificates. For these four rows only, write $\mathrm{int}(b_0, \ldots, b_t) = \sum_{j=0}^{t} b_j 10^{12j}$. The following signed integer values are the exact quotients $Q_p = L_p/p^5$.

$$Q_7 = -\,\mathrm{int}\begin{pmatrix} 46751667200, 385364382356, \\ 25673894153 \end{pmatrix}.$$

$$Q_{11} = -\operatorname{int}\begin{pmatrix}306050246400, 875359720448,\\ 189497429545, 925649907122,\\ 6702852754\end{pmatrix}.$$

$$Q_{13} = -\operatorname{int}\begin{pmatrix}697102720000, 393853893417,\\ 273211013438, 331972876309,\\ 441464918395, 4070912026\end{pmatrix}.$$

$$Q_{17} = -\operatorname{int}\begin{pmatrix}50757210000, 386170036604,\\ 481872002877, 773239567485,\\ 267773894931, 300758361178,\\ 938154922592, 1825184605\end{pmatrix}.$$

The six actual source fractions. The derivative recipe in (16) equals

$$V_i(x) = \frac{N_i(x)}{b_i x^{t_i} e_x^3}, \qquad (t_i) = (2, 2, 5, 2, 2, 2),$$

where

$$(b_0, b_1, b_2) = (884458575000, 442229287500, 1990031793750),$$
$$(b_3, b_4, b_5) = (61420734375000, 153551835937500, 38387958984375).$$

$$N_0(x) = \operatorname{pol}\begin{pmatrix}-203665000000000000, 1620026581250000000,\\ -4776671745037500000, 6390265799696625000,\\ -4568259094506607500, 967095462947044050,\\ -452996231267528739, 1504649505484526625,\\ -1024896781084880250, 358366114625568750,\\ -49911735537576000\end{pmatrix}.$$

$$N_1(x) = \operatorname{pol}\begin{pmatrix}-835026500000000000, 6742838987500000000,\\ -20412363692634375000, 42274489174403062500,\\ -72991314248116398750, 72681314187914929575,\\ -40033847500084452231, 17980716578328028350,\\ -8776295385209831250, 3059269893497952675,\\ -424991536633320525\end{pmatrix}.$$

$$N_2(x) = \operatorname{pol}\begin{pmatrix}2036650000000000000000, -1644594875000000000000000,\\ 5294374830937500000000, -8661423963740625000000000,\\ 7632856729553343750000, -3576464533193841562500000,\\ 915531305016847443750, -370450247425654071375000,\\ 32848530276068812572 75, -17481757726371667043 88,\\ 712568763360020254725, -337708827799074815355,\\ 117331289817565846140, -16254637002555354315\end{pmatrix}.$$

$$N_3(x) = \operatorname{pol}\begin{pmatrix}203665000000000000000, -164459487500000000000000,\\ 52943748309375000000000, -866078700145312500000000,\\ 76277077312156875000000, -358190852299997625000000,\\ 96813878517759690000000, -474160356450500328000000,\\ 42353397040332314088 0, -22116703899020475014 1,\\ 88343270723508495705, -41475305233234945818,\\ 14368114735064950374, -1985653292967065304\end{pmatrix}.$$

$$N_4(x)=\operatorname{pol}\begin{pmatrix}41751325000000000000,-337141949375000000000,\\ 1085346840342187500000 0,-17748778590210937500000 0,\\ 1562368635300020625000 00,-72792768566568622500000,\\ 1803289599373120582500 0,-710443680443468774775 0,\\ 6706460023297782552765,-372390385707084404962 8,\\ 1569822791226846121485,-757044868395948363864,\\ 263435458016134499112,-36543294297940253472\end{pmatrix}.$$

$$N_5(x)=\operatorname{pol}\begin{pmatrix}1705694375000000000000,-13773482078125000000000,\\ 4437355640566406250000 0,-72733992740694140625000,\\ 6418738241599999218750 0,-30095841001386345468750,\\ 7639960648920801665625,-311887813200449682262 5,\\ 2892773628120033493245,-156671694557589025040 4,\\ 642442623371264688075,-305192775314146313547,\\ 106064566372879052661,-14697320943081344436\end{pmatrix}.$$

Cleared coefficient cancellations for both row identities. Set $e=99x-50$, $h=25-9x$, $f=s^2-2ms+\lambda$ and form

$$\mathcal{O}_i=x^5e^3(1-\lambda)\Big(w_i+6w_i^{+}-3375V_i-6750(1-x)\partial_xU_i \\ -20250(1-x)x^{-1}\sum_j U_jL_{ji}+3375U_i\Big).$$

Every separate cleared summand has degree at most 19. The coefficient cancellations are

$$([x^k]\mathcal{O}_i)_{i=0}^{5}=(0,0,0,0,0,0)\qquad(0\le k\le 19).$$

For the tensor identity put $T=U-\eta$ and

$$\mathcal{E}_{ij}=x^5h^2e^3s^4(s-1)(s-\lambda)f^2\Big(\omega_iV_j/(s-1)+V_{i+3}\omega_j/(s-\lambda)+R_{ij} \\ -2(1-x)\partial_xR_{ij}-6(1-x)x^{-1}\mathcal{C}_2(R)_{ij}-\partial_sT_{ij}-\mathcal{C}(T)_{ij}\Big).$$

Every separate cleared summand has degree at most 24 in x and 12 in s. The complete coefficient cancellations are

$$([x^as^b]\mathcal{E}_{ij})_{0\le i,j\le 2}=0_{3\times 3}\qquad(0\le a\le 24,\ 0\le b\le 12).$$

These arrays use only the following coefficient rules, which specify every intermediate operation: for $P=\sum p_{ab}x^as^b$, $[x^as^b]\partial_xP=(a+1)p_{a+1,b}$, $[x^as^b]\partial_sP=(b+1)p_{a,b+1}$, and $[x^as^b](PQ)=\sum_{u=0}^{a}\sum_{v=0}^{b}p_{uv}q_{a-u,b-v}$. For a fraction $P/\prod F_j^{r_j}$ use

$$\partial(P/\prod F_j^{r_j})=\frac{\partial P}{\prod F_j^{r_j}}-\sum_j\frac{r_jP\,\partial F_j}{F_j\prod F_j^{r_j}}.$$

The complete numerator tables, the displayed matrices and these convolution rules give the zero arrays without an unevaluated source constant. Their degree bounds exhaust the coefficient ranges; hence the polynomial equalities and both rational row identities follow.

2 The full signed moment and its field decision

Throughout this section, $p\ge 19$ is prime, $d=\lfloor p/6\rfloor$, $N=(p-1)/2$ and $e=p-1-d$. A bar denotes reduction from $\mathbb{Z}_{(p)}=\{a/b\in\mathbb{Q}:p\nmid b\}$ to $\mathbb{F}_p$. Every division by p displayed below takes

place in $\mathbb{Q}$, before this reduction. Put

$$c_n = \frac{(6n)!}{(3n)!(n!)^3\,1728^n}, \qquad F_e(z) = \sum_{n=0}^{e} c_n z^n, \qquad H_e(z) = \sum_{n=1}^{e} \frac{c_n}{n} z^n.$$

In particular, $zH_e'(z) = F_e(z) - 1$. For $0 \le r < p$, write $H_r(z) = \sum_{n=1}^{r} c_n z^n/n$ for the truncated native Euler polynomial, reducing its coefficients when it occurs in a field formula. Without an argument, H_r denotes the scalar partial harmonic sum $\sum_{j=1}^{r} 1/j$ in $\mathbb{F}_p$. For a nonzero integer m write

$$w_p(m) = \frac{p(2^m - 2^{-m})}{m}.$$

The actual full moment is

$$N_{p,n}(X) = (4X-1)^{2p+n-1}(4-X)^{p+2n-1}(11-4X),$$

$$m_{p,n} = \frac{2}{3\,125^{p+n}} \sum_{a=0}^{3(p+n)-1} [X^a]N_{p,n}(X)\, w_p(2a-6n+1), \tag{43}$$

$$M_p = \sum_{n=0}^{p-1} c_n m_{p,n}, \qquad C_p = \sum_{n=0}^{e} c_n m_{p,n}. \tag{44}$$

Both $s = 1/2$ and $s = 2$ occur in w_p. The degree of $N_{p,n}$ is $3(p+n)-1$, so the coefficient range in (43) is complete.

2.1 Integrality, the cutoff, and formal endpoint integration

If $n < p$ and $0 \le a < 3(p+n)$, the integer $2a - 6n + 1$ is odd and nonzero, and its absolute value is less than $6p < p^2$. Its valuation is therefore at most one. Thus $p/(2a-6n+1)$ is p-integral. The other factors in each summand of (43) are integral: 2 is a unit, including under negative integer powers, and $3\,125^{p+n}$ is a unit. Consequently $m_{p,n} \in \mathbb{Z}_{(p)}$. Legendre's factorial valuation formula, whose factorial arguments here are below p^2, gives

$$v_p(c_n) = \lfloor 6n/p \rfloor - \lfloor 3n/p \rfloor \ge 0.$$

For $e < n < p$, these two floors are 5 and 2. Each omitted term is therefore in $p^3\mathbb{Z}_{(p)}$, and

$$M_p \equiv C_p \pmod{p^3}. \tag{45}$$

No argument here removes an endpoint or a Laurent coefficient of a moment.

For a finite Laurent polynomial $L(s) = \sum_j L_j s^j$, define the linear endpoint functional

$$\mathcal{I}(L) = \sum_{j \ne -1} L_j \frac{2^{j+1} - 2^{-(j+1)}}{j+1}.$$

Its coefficient at s^{-1} is assigned zero. All densities used as periods below have zero s^{-1} coefficient; for them this is the ordinary finite monomial primitive with the two stated endpoints. For every Laurent polynomial A, $\mathcal{I}(A') = A(2) - A(1/2)$. Set

$$V(x) = \frac{(4x-1)^2(4-x)}{125}, \qquad Q(x) = \frac{(4x-1)(4-x)^2}{125x^3} = V(1/x),$$

$$v(s) = V(s^2), \qquad q(s) = Q(s^2), \qquad S(s) = v(s)/s,$$

$$B(s) = s^{-p}v(s)^{p-1}v'(s), \qquad D(s) = B(s) - v(s)^p s^{-p-1},$$

$$\chi(x) = \frac{F_e(V(x)) - x^N F_e(Q(x))}{p}, \qquad A(x) = \chi(x) + H_e(V(x)).$$

The symbols B, D here are finite Laurent densities, not differential forms requiring a branch choice. In particular $(S^p)' = pD$. Expanding the numerator of (43) gives the exact density identity

$$\sum_{n=0}^{e} c_n \frac{2p\, N_{p,n}(s^2)}{3\,125^{p+n} s^{6n}} = \frac{p}{9}\, s^{-1} v^{p-1} v' F_e(q).$$

Its endpoint functional is C_p. Since $p = 2N+1$ and $p\chi = F_e(V) - x^N F_e(Q)$, differentiating $S^p H_e(v)$ yields

$$\frac{p}{9} s^{-1} v^{p-1} v' F_e(q) = \frac{p}{9} B - \frac{p^2}{9}\{\chi(s^2)B + H_e(v)D\} + \frac{p}{9}(S^p H_e(v))'.$$

At both endpoints $v = 0$ and $H_e(v) = 0$. Thus, with

$$b_p = \frac{p}{9}\mathcal{I}(B), \qquad \eta_p = p\mathcal{I}\{\chi(s^2)B + H_e(v)D\},$$

we have the exact identity $C_p = b_p - p\eta_p/9$.

2.2 Pole subtraction and every resonance

For a finite Laurent polynomial L and $a \neq 0$, let $\Delta_a L = (L(x) - L(a))/(x-a)$, including its negative powers. Define

$$\begin{aligned} P(x) &= 2\Delta_{1/4} A + \Delta_4 A - \frac{H_e(V(x))}{2x}, \\ r_k &= [x^{kp-1}]P(x) \quad (k \in \mathbb{Z}), \\ \mathcal{A}(x) &= \sum_{p \nmid j} \frac{[x^{j-1}]P(x)}{j} x^j. \end{aligned}$$

The canonical primitive has its coefficients at every x^{kp} set to zero. Coefficientwise differentiation gives $\mathcal{A}' = P - \sum_k r_k x^{kp-1}$. The support of P is contained in $[N - 3e, 3e - 1]$; divided differences reduce the upper bound by one and preserve the lower negative bound. Hence $-2p < j + 1 < 3p$ on this support. Its only resonance slots are

$$k = -1, 0, 1, 2.$$

Write

$$\begin{aligned} a_p &= (1 - 4^{-N})/p, & a'_p &= (1 - 4^N)/p, \\ A_0(s) &= \frac{4}{125}(4s^2 - 1)(4 - s^2), & B_0(s) &= -\frac{1}{125}(4s^2 - 1)^2, \\ E(s) &= 2s^{1-p} v(s)^{p-1}\{2a_p A_0(s) + a'_p B_0(s)\}, & & \\ \ell_p &= p\mathcal{I}(E), & K_k &= p\mathcal{I}\{2v(s)^p s^{(2k-1)p-1}\}, \\ T_p &= p\mathcal{I}\{D(s)\mathcal{A}(s^2)\}. & & \end{aligned}$$

These definitions retain both actual pole endpoints. To check them, $V = (x - 1/4)\,4(4x-1)(4-x)/125 = (x-4)(-(4x-1)^2/125)$, $A(1/4) = a_p$, and $A(4) = a'_p$. Substitution into the divided differences gives

$$S^p\, 2sP(s^2) = \chi(s^2)B + H_e(v)D - E(s).$$

Integration of $(S^p \mathcal{A}(s^2))'$ now gives

$$\eta_p = \ell_p + \sum_{k=-1}^{2} r_k K_k - pT_p.$$

Both product endpoints vanish because $S(1/2) = S(2) = 0$. Likewise $\mathcal{I}(D) = 0$, and the definition of K_0 gives $K_0 = 18b_p$. Therefore the full exact short-moment identity is

$$C_p = (1 - 2pr_0)b_p - \frac{p}{9}\ell_p - \frac{p}{9}(r_{-1}K_{-1} + r_1K_1 + r_2K_2) + \frac{p^2}{9}T_p. \tag{46}$$

This identity precedes every valuation approximation.

The reciprocal argument in Section 2.7 proves that χ is p-integral. For $a = 1/4, 4$, the divided difference of a monomial is

$$\Delta_a x^j = \sum_{i=0}^{j-1} a^{j-1-i}x^i \quad (j > 0), \qquad \Delta_a x^{-j} = -\sum_{i=0}^{j-1} a^{i-j}x^{-i-1} \quad (j > 0).$$

Both a and a^{-1} are units, so these formulas prove coefficientwise integrality of the full divided differences, including their negative powers, and hence of P. For $\mathcal{A}$, division is only by integers prime to p. The canonical density has no logarithmic monomial: it is s^{-p-1} times a Laurent polynomial in s^2, and p is odd. Its nonzero primitive denominators satisfy $|j + 1| < 11p < p^2$. The resonance densities similarly have no logarithmic monomial and $|j + 1| \le 9p < p^2$ for $-1 \le k \le 2$. The factor p in their definitions makes all these periods integral.

2.3 The base and pole precision calculation

Let $q_a = (a^{p-1} - 1)/p$ for $a = 2, 3, 5$ and $u_a = a^{p-1} = 1 + pq_a$, all formed in the integers or rationals. Set

$$\alpha_p = \sum_{r=1}^{p-1} \frac{16^r}{r^2}, \qquad \beta_p = 15q_3 + 15q_5 - 64q_2.$$

These are p-integral. Here is a finite coefficient calculation of the base and pole terms, including the carry terms whose cancellation is needed. For $P \in \mathbb{Z}[X]$ of degree at most $3p$, put

$$J_p(P) = 125^{-p} \sum_{n=0}^{3p} [X^n]P\, w_p(2n - 5p).$$

The denominator is nonzero because p is odd, and its absolute value is at most $5p < p^2$. Thus $J_p(P)$ is integral. Reflection $s \mapsto 1/s$ of the density gives

$$b_p = \frac{1}{9}J_p((4 - X)^{2p}(4X - 1)^p), \qquad E_p = 16J_p((4 - X)^{2p-1}(4X - 1)^p),$$

where E_p is the first reflected endpoint period. The second endpoint period differs from E_p by $9b_p$. As $a_p = q_2/u_2$ and $a_p' = -q_2$,

$$\ell_p = q_2\{(1 + u_2^{-1})E_p - 9b_p\}. \tag{47}$$

Put $f = 4 - X$, $g = 4X - 1$, $F = 4^p - X^p$, $G = 4^pX^p - 1$, and define the integral carry polynomials

$$h = \frac{f^p - F}{p}, \qquad k = \frac{g^p - G}{p}, \qquad L = \sum_{j=0}^{p-1} 4^{p-1-j}X^j, \qquad t = h/f.$$

The quotient t is integral: $h(4) = 0$, so monic division by $X - 4$ has zero remainder. The exact carry identities are

$$f^{2p}g^p = F^2G + p(2FhG + F^2k) + p^2(h^2G + 2Fhk) + p^3h^2k,$$
$$f^{2p-1}g^p = LFG + p(2LhG + LFk) + p^2(thG + 2Lhk) + p^3thk.$$

They follow by multiplying $f^p = F + ph$, $g^p = G + pk$, $f^{p-1} = L + pt$ and using $tF = Lh$.

For $1 \le r < p$, the reduced carry coefficients are

$$\overline{h_r} = -\frac{4}{4^r r}, \qquad \overline{k_r} = \frac{4^r}{r}, \qquad \overline{L_r} = 4^{-r}.$$

Indeed $\binom{p}{r}/p \equiv (-1)^{r-1}/r$ follows by canceling p in $\binom{p}{r} = p\binom{p-1}{r-1}/r$ before reduction. All indices $r + cp$, $1 \le r < p$, in the primitive give, after division of $w_p(2r + (2c-5)p)$ by p, the field weights

$$W_0 = \frac{z/32 - 32/z}{2r}, \quad W_1 = \frac{z/8 - 8/z}{2r}, \quad W_2 = \frac{z/2 - 2/z}{2r}, \qquad z = 4^r.$$

The complete first carry of the base vanishes after its additional division by p, since its summand is exactly

$$-\frac{4}{zr}(-8W_0 + 34W_1 - 8W_2) + \frac{z}{r}(16W_0 - 8W_1 + W_2) = -\frac{225}{r^2}.$$

Summing gives zero in the field: $\sum_{r=1}^{p-1} r^{-2} = 0$ for $p > 3$, as multiplication by an element whose square is not one permutes the sum. Consequently its contribution, including the outside factor p, is zero modulo p^3.

For the second base carry, reflection $r \mapsto p - r$ gives $\sum h_r h_{p-r} = 0$ and $\sum h_r k_{p-r} = \overline{\alpha}_p$. Explicitly these sums reduce to $-4\sum r^{-2} = 0$ and $16\sum 16^{-r} r^{-2}$, respectively. The latter uses the substitution $r \mapsto p - r$ in $16\sum 16^{-r}/r^2 = \sum 16^r/r^2$. Thus the coefficients at X^p, X^{2p} of $h^2G + 2Fhk$ are $8\overline{\alpha}_p, -2\overline{\alpha}_p$; its constant and X^{3p} coefficients are zero. The only primitive weights surviving reduction are those at indices divisible by p. At $n = p, 2p$ they are $21/8, 3/2$. Hence

$$J_p(h^2G + 2Fhk) \equiv \frac{18}{125}\alpha_p \pmod{p}.$$

The cubic carry is integral and is multiplied by p^3.

The four uncarried base coefficients are $-4^{2p}, 4^{3p} + 2 \cdot 4^p, -(2 \cdot 4^{2p} + 1), 4^p$ at indices $0, p, 2p, 3p$. Canceling p in their primitive weights gives

$$J_p(F^2G) = 125^{-p} S(2^p), \qquad S(l) = \frac{2l^9 - 20l^5 + 40l^3 - 30l + 8/l}{15}.$$

We have proved

$$b_p \equiv \frac{S(2^p)}{9\,125^p} + p^2 \frac{2}{125}\alpha_p \pmod{p^3}. \tag{48}$$

For the endpoint numerator, every interior uncarried coefficient cancels at its first field digit because $-4W_0 + 17W_1 - 4W_2 = 0$ pointwise. Indeed $FG = -4^p + (4^{2p}+1)X^p - 4^p X^{2p}$; multiplication by L gives coefficients 4^{p-1-r} times $-4^p, 4^{2p}+1, -4^p$ in the three bands $r, r+p, r+2p$, $0 \le r < p$. At $r = 0$ the three resonant coefficients are $-4^{2p-1}, 4^{p-1}(4^{2p}+1), -4^{2p-1}$. Their primitive, with $l = 2^p$, is $125^{-p}u_2^2$ times

$$-l^2(l^5 - l^{-5})/5 + (l^4+1)(l^3 - l^{-3})/3 - l^2(l - l^{-1}) = \tfrac{2}{15}(l^7 - 5l^3 + 5l - l^{-3}).$$

The nonresonant coefficients are multiples of p after integration; the pointwise field cancellation makes their total a multiple of p^2. Thus the full expression gives

$$J_p(LFG) \equiv u_2^2\, 125^{-p} T(2^p) \pmod{p^2}, \qquad T(l) = \frac{2}{15}(l^7 - 5l^3 + 5l - l^{-3}).$$

The middle convolution coefficients of Lh, Lk are respectively $0, \ell(16)/4$, where $\ell(z) = \sum_{r=1}^{p-1} z^r/r$ in $\mathbb{F}_p$. Their explicit sums are $-\sum_{r=1}^{p-1} 1/r = 0$ and $-4\sum_{r=1}^{p-1} 16^{-r}/r = \ell(16)/4$; substitute $r \mapsto p - r$ in the second sum. Both carry polynomials have zero constant, and their degrees are below $2p$, so these are all their coefficients at multiples of p. Multiplication by G and F in $2LhG + LFk$ consequently gives the next two coefficients and no others. Thus the first endpoint carry has coefficients $\ell(16), -\ell(16)/4$ at $p, 2p$, and its primitive is $9\ell(16)/(4 \cdot 125)$ modulo p. Expanding

$(1-z)^p$ and canceling p gives $\ell(16) = 15\bar{q}_3 + 15\bar{q}_5 - 64\bar{q}_2 = \bar{\beta}_p$. The second and cubic endpoint carries are integral, hence do not enter precision two. It follows that

$$E_p \equiv 16u_2^2\, 125^{-p} T(2^p) + p\frac{180}{625}\beta_p \pmod{p^2}.$$

To perform the remaining Taylor calculation, write $2^p = 2(1+pq_2)$, $125^p = 125(1+pq_5)^3$. Direct expansion gives

$$\begin{aligned} S(2(1+t)) &= \frac{216}{5} + \frac{2304}{5}t + \frac{10476}{5}t^2 + t^3\frac{R_S(t)}{15(1+t)}, \\ \frac{16}{125}(1+t)^2 T(2(1+t)) &= \frac{1044}{625} + \frac{10476}{625}t + t^2\frac{4R_T(t)}{1875(1+t)}, \\ (1+t)^{-3} &= 1 - 3t + 6t^2 - t^3\frac{10+15t+6t^2}{(1+t)^3}. \end{aligned}$$

Here the complete integer remainder polynomials are

$$\begin{aligned} R_S(t) &= 79932 + 205760t + 254208t^2 + 214400t^3 \\ &\quad + 122880t^4 + 46080t^5 + 10240t^6 + 1024t^7, \\ R_T(t) &= 33903 + 116800t + 210320t^2 + 256128t^3 \\ &\quad + 214720t^4 + 122880t^5 + 46080t^6 + 10240t^7 + 1024t^8. \end{aligned}$$

Multiplication by the displayed denominators verifies every coefficient. Their denominators at $t = pq_a$ are units, so these identities specify the full discarded precision. They yield

$$\begin{aligned} b_p &\equiv \frac{24}{625}\left[1 + p\left(\frac{32}{3}q_2 - 3q_5\right) + p^2\left(\frac{97}{2}q_2^2 - 32q_2q_5 + 6q_5^2 + \frac{5}{12}\alpha_p\right)\right] \pmod{p^3}, \qquad (49) \\ E_p &\equiv \frac{1044 + p(10476q_2 - 3132q_5 + 180\beta_p)}{625} \pmod{p^2}. \end{aligned}$$

Finally $1 + u_2^{-1} \equiv 2 - pq_2 \pmod{p^2}$ in (47). In units of $1/625$, the constant of $(1+u_2^{-1})E_p - 9b_p$ is $2\cdot 1044 - 9\cdot 24 = 1872 = 36\cdot 52$. Its coefficient of p is

$$2(10476q_2 - 3132q_5 + 180\beta_p) - 1044q_2 - 216(32q_2/3 - 3q_5) = 36(-151q_2 + 150q_3 - 6q_5).$$

Multiplication by the remaining q_2 gives

$$\ell_p \equiv \frac{36}{625}q_2\{52 + p(-151q_2 + 150q_3 - 6q_5)\} \pmod{p^2}. \qquad (50)$$

2.4 Resonance denominators and the three-digit defect

For a p-integral Laurent density with no logarithmic monomial and primitive denominators of absolute value below p^2, reduction of $p\mathcal{I}$ keeps only exponents j with $p \mid j+1$. For $j+1 = pt$ one first cancels p in $\mathbb{Q}$ and obtains $(2^t - 2^{-t})/t$ in $\mathbb{F}_p$. Applied to K_k, the reduced density is the complete Laurent polynomial

$$2V(s^2)s^{2k-2} = \frac{-32s^{2k+4} + 144s^{2k+2} - 66s^{2k} + 8s^{2k-2}}{125}.$$

Its nonzero odd primitive denominators have absolute value at most 9, and direct endpoint evaluation gives

$$K_{-1} \equiv \frac{54}{125}, \qquad K_1 \equiv \frac{6102}{4375}, \qquad K_2 \equiv \frac{28809}{8750} \pmod{p}. \qquad (51)$$

The denominator primes are only $2, 3, 5, 7$. To make these finite endpoint evaluations explicit, put $\psi_m = (2^m - 2^{-m})/m$. The weights used are

$$\psi_{-3} = \psi_3 = 21/8, \quad \psi_{-1} = \psi_1 = 3/2, \quad \psi_5 = 1023/160, \quad \psi_7 = 16383/896, \quad \psi_9 = 262143/4608.$$

In particular the numerators of the three primitive values, before division by 125, are

$$\begin{aligned} -32\psi_3 + 144\psi_1 - 66\psi_{-1} + 8\psi_{-3} &= 54, \\ -32\psi_7 + 144\psi_5 - 66\psi_3 + 8\psi_1 &= 6102/35, \\ -32\psi_9 + 144\psi_7 - 66\psi_5 + 8\psi_3 &= 28809/70. \end{aligned}$$

Set

$$\begin{aligned} r_* &= -2q_2 - \tfrac{3}{2}q_3 - \tfrac{3}{2}q_5, & \rho_p &= (r_0 - r_*)/p, \\ s_k &= r_k/p \quad (k = -1, 1, 2), & \mathcal{U}_p &= \tfrac{625}{216}T_p, \\ \mathcal{R}_p &= 2\rho_p + \tfrac{5}{4}s_{-1} + \tfrac{113}{28}s_1 + \tfrac{1067}{112}s_2, \\ \mathcal{P}_p &= \tfrac{304}{3}q_2^2 - 11q_2q_3 - 11q_2q_5 - 9q_3q_5 - 3q_3^2 - 3q_5^2 + \tfrac{5}{12}\alpha_p, \\ \mathcal{D}_p &= \mathcal{P}_p - \mathcal{R}_p + \mathcal{U}_p. \end{aligned}$$

The central and noncentral calculations at the end of this section establish $\bar{r}_0 = \bar{r}_*$ and $\bar{r}_{-1} = \bar{r}_1 = \bar{r}_2 = 0$. Since the residues themselves are integral, this proves the integrality of ρ_p, s_{-1}, s_1, s_2 by rational division; they are never obtained by dividing a field residue by zero. T_p is integral by the bounds above, so all three digits and $\mathcal{D}_p$ are integral.

Let $b = 24/625$, $b_1 = 32q_2/3 - 3q_5$, $b_2 = 97q_2^2/2 - 32q_2q_5 + 6q_5^2 + 5\alpha_p/12$, $l_1 = 36 \cdot 52q_2/625$ and $l_2 = 36q_2(-151q_2 + 150q_3 - 6q_5)/625$. The rational expression obtained by substituting the proved precisions into (46) and dividing by b is

$$Z = \frac{(1 - 2p(r_* + p\rho_p))b(1 + pb_1 + p^2b_2) - \frac{p}{9}(l_1 + pl_2) - \frac{p^2}{9}\left(\frac{54}{125}s_{-1} + \frac{6102}{4375}s_1 + \frac{28809}{8750}s_2\right) + p^2b\mathcal{U}_p}{b}.$$

Each substitution error is in $p^3\mathbb{Z}_{(p)}$: respectively it has precision 3, $2 + 1$, or $1 + 2$. The central multiplier is integral, and no substitution is made in $\mathcal{U}_p$. An exact polynomial expansion gives

$$Z = 1 + p(6q_2 + 3q_3) + p^2\{15q_2^2 + 18q_2q_3 + 3q_3^2 + \mathcal{D}_p\} + p^3\{-2r_*b_2 - 2\rho_pb_1 - 2p\rho_pb_2\}.$$

To display the digit collection, $l_1/(9b) = 26q_2/3$ and $l_2/(9b) = q_2(-151q_2 + 150q_3 - 6q_5)/6$. Thus the linear digit is $b_1 - 2r_* - 26q_2/3 = 6q_2 + 3q_3$. The arithmetic part of the quadratic digit is

$$b_2 - 2r_*b_1 - l_2/(9b) = \frac{349}{3}q_2^2 + 7q_2q_3 - 11q_2q_5 - 9q_3q_5 - 3q_5^2 + \frac{5}{12}\alpha_p.$$

Subtracting $15q_2^2 + 18q_2q_3 + 3q_3^2$ gives exactly $\mathcal{P}_p$. The other three period coefficients are $(625/216)(54/125) = 5/4$, $(625/216)(6102/4375) = 113/28$ and $(625/216)(28809/8750) = 1067/112$, exactly the coefficients of $\mathcal{R}_p - 2\rho_p$. Meanwhile, the full degree-nine product has the exact Taylor expression

$$u_2^6u_3^3 = 1 + p(6q_2 + 3q_3) + p^2(15q_2^2 + 18q_2q_3 + 3q_3^2) + p^3R_u,$$

where $R_u = \sum_{\substack{0 \le i \le 6,\, 0 \le j \le 3 \\ i+j \ge 3}} \binom{6}{i}\binom{3}{j}p^{i+j-3}q_2^iq_3^j$ is integral. Consequently (45), and the fact that b and b^{-1} are integral units, give

$$\frac{M_p}{b} - u_2^6u_3^3 \equiv p^2\mathcal{D}_p \pmod{p^3}. \tag{52}$$

2.5 The actual endpoint pair and HJet cancellation

For a Laurent polynomial L over $\mathbb{F}_p$, let $L^-(a) = \sum_{j<0} L_ja^j$ and $L^{\ge p}(a) = \sum_{j \ge p} L_ja^j$. Define the field primitive

$$\Pi\left(\sum_j L_jx^j\right) = \sum_{p \nmid m} \frac{L_{m-1}}{m}x^m.$$

The polynomial $J(z)$ is the reduction of the actual parameter derivative of F_e, with its coefficients through N; coefficients above N vanish in the field. Set

$$H_J(z)=\sum_{n=1}^{N}\frac{J_n}{n}z^n,\quad P_J(x)=H_J(V(x)),\quad C_0=[x^0]P_J,\quad C_1=[x^p]P_J,$$

and use the actual source to define, rather than fit at the endpoints,

$$B_p(x)=\Pi\left(\bar{P}(x)-\tfrac{1}{6}P_J'(x)\right),\quad \mathcal{B}_p=\frac{29}{6}B_p(1/4)-\frac{23}{6}B_p(4)+2\{-2B_p^-(1/4)-B_p^-(4)\}.$$

The source-selected B_p has support in $[-p+2,N]$ and has no x^{kp} coefficient. In particular its high part is empty, whereas its entire negative half remains in $\mathcal{B}_p$.

Here is the finite endpoint projection used on the original canonical primitive $\bar{\mathcal{A}}$. Its value on any Laurent polynomial in the relevant support band is

$$\begin{aligned}\Lambda(L)=&\frac{1044}{625}L(1/4)-\frac{828}{625}L(4)+\frac{432}{625}\{-2L^-(1/4)-L^-(4)\}\\&+\frac{6102}{4375}\{8L^{\geq p}(1/4)+L^{\geq p}(4)/4\}.\end{aligned}$$

It is obtained by reducing the canonical density with the same denominator rule used for (51). The four elementary canceled kernels and their signed primitives are

$125\,L(s)$	$\mathcal{I}(L)$
$-32s^4+136s^2-32$	$522/625$
$-32s^4+16s^2-2$	$-828/625$
$-32s^4+144s^2-66+8s^{-2}$	$432/625$
$-32s^6+144s^4-66s^2+8$	$6102/4375$

The first kernel occurs twice. For the divisions by $x-a$, splitting a Laurent monomial into its geometric quotient and its endpoint value separates precisely the negative and high bands in Λ. Thus $\bar{T}_p=\Lambda(\bar{\mathcal{A}})$. For the first three kernels the numerator sums before division by 125 are, using the weights just listed,

$$\begin{aligned}-32\psi_5+136\psi_3-32\psi_1&=522/5,\\-32\psi_5+16\psi_3-2\psi_1&=-828/5,\\-32\psi_5+144\psi_3-66\psi_1+8\psi_{-1}&=432/5.\end{aligned}$$

The fourth kernel is the already evaluated first positive kernel.

As $\deg P_J<2p$, coefficientwise integration gives

$$\bar{\mathcal{A}}=B_p+\tfrac{1}{6}(P_J-C_0-C_1x^p).$$

Furthermore $H_J(0)=0$, so $P_J(1/4)=P_J(4)=0$. The complete high endpoint pair vanishes: $8P_J^{\geq p}(1/4)+P_J^{\geq p}(4)/4=0$. To see its source, put $E_J(z)=\sum_{n=1}^{N}J_nz^n/n^2$. Then $zE_J'(z)=H_J(z)$, so $P_J\,d\log V=d(E_J(V))$. All denominators here are units because $N<p$. An exact rational differential has zero Cartier operator, coefficientwise: its coefficient at x^{kp-1} is kp times the coefficient of its primitive at x^{kp}, which is zero in the field. The geometric pole formula in the same band selects the displayed high pair; the endpoint and negative terms are zero. Its low pair then vanishes as well, since the full endpoint values are zero. It follows that $\Lambda(P_J)=0$, $\Lambda(1)=216/625$, and $\Lambda(x^p)=-3051/4375$. Consequently

$$\bar{\mathcal{U}}_p=\mathcal{B}_p-C_0/6+\frac{113}{336}C_1.$$

Indeed multiplication of $\Lambda(B_p)$ by $625/216$ gives $1044/216=29/6$, $828/216=23/6$ and $432/216=2$; its high term is zero by the proved remainder support. The constant correction is

$-(625/216)(216/625)/6 = -1/6$, and the degree-$p$ correction is $(625/216)(3051/4375)/6 = 113/336$. The full positive-residue extraction gives $\bar{s}_1 = C_1/12$ and $\bar{s}_2 = 0$. Indeed its exact coefficient is $s_k = (k-1/2)[x^{kp}](H_e(V) - H_d(V))/p + k[x^{kp}]\sum_{n=1}^{e} c_n V^n/n^2$. The second summand is zero modulo p above $3d$, and the divided first summand is $P_J/6$ modulo p in that same upper band. Since $\deg P_J < 2p$, these formulas give both assertions, retaining the full cutoff e. We obtain

$$\bar{\mathcal{R}}_p - \bar{\mathcal{U}}_p = 2\bar{\rho}_p + \tfrac{5}{4}\bar{s}_{-1} - \mathcal{B}_p + C_0/6. \tag{53}$$

For clarity, the rational polynomials in the final negative calculation are specified before reduction. Put

$$a_n(t) = \prod_{j=0}^{n-1} \frac{(6j+1+6t)(6j+5+6t)}{36(j+1)(j+1+t)}, \qquad Y_k(T) = \sum_{n=0}^{e} a_n(-kp/6)((1-T)/2)^n,$$

for $k = 1, 5$, and $u(T) = (Y_1(T) - Y_5(T))/(4p)$. The low node Y_1 terminates at degree d; both nodes and the rationally divided u have integral coefficients. For an even polynomial $A(T)$ of degree at most $2e$, define its actual q fold by

$$\mathrm{Fold}(A)(z) = \sum_{n=0}^{e} [T^{2n}]A(T)\,(1-z)^n.$$

Then $\mathrm{Fold}(A)(1-T^2) = A(T)$, without choosing a residue lift. Write $G_1 = \mathrm{Fold}(Y_1 u)$ and $G_2 = \mathrm{Fold}(u^2)$; their degree bounds are N and e. To specify the native parameter without an implicit normalization, set

$$A_n(t) = \prod_{j=0}^{n-1} (6j+1+6t)(2j+1+2t)(6j+5+6t),$$
$$D_n(t) = \prod_{j=0}^{n-1} 72(j+1)(j+1+t)(j+1+2t), \qquad C_n(t) = A_n(t)/D_n(t), \qquad t_0 = -p/6.$$

Indeed the ratio of consecutive factorial coefficients is

$$\frac{c_{n+1}}{c_n} = \frac{\prod_{i=1}^{6}(6n+i)}{1728(n+1)^3\prod_{i=1}^{3}(3n+i)} = \frac{(6n+1)(2n+1)(6n+5)}{72(n+1)^3}.$$

The three even factors cancel the three denominator factors, and $6n+3 = 3(2n+1)$; both products have constant coefficient 1 at $n = 0$. Thus $C_n(0) = c_n$. Define the actual full polynomials, with the original upper cutoff retained in both derivatives,

$$F^{(0)}(z) = \sum_{n=0}^{e} C_n(t_0)z^n, \quad J_e(z) = \sum_{n=0}^{e} C_n'(t_0)z^n, \quad K_e(z) = \sum_{n=0}^{e} C_n''(t_0)z^n.$$

The derivative in K_e is the ordinary second derivative, namely

$$C_n'' = \frac{A_n''}{D_n} - \frac{2A_n'D_n'}{D_n^2} - \frac{A_nD_n''}{D_n^2} + \frac{2A_n(D_n')^2}{D_n^3}.$$

All evaluations in this display are at the same node t_0. The denominator factors are units: modulo p, they are $72(j+1)^3$, with $j+1 < p$. The numerator has a zero factor for $n > d$, so $\deg F^{(0)} \le d$. Finally set

$$L_2 = \mathrm{Fold}(K_e(1-T^2) - 144u(T)^2).$$

The full Taylor calculation, with all coefficients through e, gives

$$F_e = F^{(0)} + 2pG_1 + p^2(6G_2 + 5L_2/72) + p^3E, \qquad E \in \mathbb{Z}_{(p)}[z]. \tag{54}$$

This uses the rational quotient Taylor formula with unit denominators at the node. The second derivative is the full quotient derivative, including the derivative of its denominator. Coefficientwise the Taylor formula is

$$C_n(0) = C_n(t_0) + \frac{p}{6}C_n'(t_0) + \frac{p^2}{72}C_n''(t_0) + p^3E_n, \qquad E_n \in \mathbb{Z}_{(p)}.$$

For clarity about the quotient remainder, expand the two finite numerator and denominator products in $h = t - t_0$. The coefficients of both expansions are integral and the denominator constant is a unit. The recursion $b_0 = D_n(t_0)^{-1}$, $b_r = -D_n(t_0)^{-1}\sum_{i=1}^{r} D_{n,i}b_{r-i}$ gives integral inverse coefficients through $r = 2$, with an integral rational remainder at $h = p/6$. Multiplication by the numerator expansion proves the displayed cubic precision without differentiating a chosen lift. The terminating square and the two-node divided expansion give, before field reduction,

$$\begin{aligned} F^{(0)}(1-T^2) &= Y_1(T)^2, \\ J_e(1-T^2) &= 12Y_1u + pA_1, \qquad A_1 \in \mathbb{Z}_{(p)}[T], \\ \frac{A_1}{6} + \frac{K_e(1-T^2)}{72} &= 6u^2 + \frac{5}{72}\{K_e(1-T^2) - 144u^2\} + pE_1, \qquad E_1 \in \mathbb{Z}_{(p)}[T]. \end{aligned}$$

Substituting these three equalities into the full Taylor formula and applying the even fold yields (54). The integral remainders are explicit: the two-node identity below gives $E_1(T) = E_5(1-T^2)/4$. If $E_{\mathrm{nat}}(z) = \sum_{n=0}^{e} E_n z^n$ is the coefficientwise Taylor remainder above, then $E(z) = E_{\mathrm{nat}}(z) + E_5(z)/4$. Thus the fold introduces no additional division by p or precision loss. The coefficient of the logarithmic subtraction has degree at most d after reduction, as explained next. Therefore it has the integral splitting $L_2 = L_2^{\le d} + pL_2^{>d}/p$; the upper divided polynomial is integral. This is the degree assertion needed to remove its contribution below; it does not assert that the full second derivative has degree at most d.

Here is the polynomial degree argument, including its characteristic-p kernel. Write $q = 1 - T^2$, $Y = \bar{Y}_1 = P_d(T)$, $R = 6\bar{u}$, $K = \overline{K_e}(q)$ and $L = K - 4R^2$. Define

$$\mathfrak{F}_c(G) = q(YG' - cY'G), \qquad \mathfrak{G}(G) = \mathfrak{F}_1(\mathfrak{F}_2(G)).$$

The full rank source is $q(YR' - Y'R) = TY^2 - T^p$; its T^p row is retained. For $Q_2 = K - 2R^2$, the differentiated full native equation gives

$$\big[\mathfrak{G}(Q_2) - 2qY^3(4TR' + 2R + 2Y) - 4(T^{2p} - 1)\big]' = 0.$$

The degree is at most $2p$. Its coefficient at T^{2p} is zero: if $y = [T^d]Y$, $r = [T^e]R$, the rank source gives $yr(1 + 2d) = -1$, the actual two-node top coefficient gives $[T^{2e}]K = 4r^2$, and direct leading-coefficient differentiation gives

$$[T^{2d+m+2}]\mathfrak{G}(G) = y^2[T^m]G\,(m+1)(m-2d).$$

At $m = 2e$, this is 4 for Q_2, the same as the retained $4T^{2p}$ source. After cancellation the defect has degree below $2p$. The kernel of differentiation in that band is $a + bT^p$. At both $T = 1, -1$ the defect is zero; since p is odd, these endpoints give $a + b = a - b = 0$. Thus the defect is identically zero. The identity

$$\mathfrak{G}(R^2) = 4qTY^3R' + 2qY^3R - 2T^2Y^4 + 2T^{2p}$$

follows by differentiating the complete rank Wronskian. Subtracting twice this identity, including its $2T^{2p}$ term, proves $\mathfrak{G}(L) = 4(Y^4 - 1)$.

Let P_n denote the ordinary Legendre polynomials, with $P_n(1) = 1$, and put

$$B = -\sum_{n=0}^{d-1} \frac{P_{n+1}P_n}{n+1}.$$

The product identity

$$q(P_{n+1}P_n)' = (n+1)(P_n^2 - P_{n+1}^2)$$

telescopes to $qB' = Y^2 - 1$, with $\deg B \le 2d-1$. The actual proper polynomial n satisfies $q(Yn' - Y'n) = 1 - Y^2$, $\deg n < d$, and $n(1) = B(1)$. It is the same finite-field proper polynomial in (102); that identity is used only in this field degree argument. Its endpoint is $n(1) = -H_d = B(1)$. Set $P = \mathfrak{F}_2(L) - 4(BY - n)$. Expanding the derivative with the last two equations gives $YP' - Y'P = 0$. The explicit polynomial $H = (Y + qn')P - qnP'$ obeys $P = YH$ and $H' = 0$; these equalities follow by multiplying the Wronskian equation and canceling the nonzero polynomial Y. The initial bounds $\deg L \le 2e$, $\deg P \le d + 2e + 1$ give $\deg H < 2p$, so $H = a + bT^p$. The parity identities $Y(-T) = (-1)^d Y(T)$, $B(-T) = -B(T)$ and $n(-T) = -(-1)^d n(T)$, together with $n(1) = B(1)$, give $P(1) = P(-1) = 0$. Since $Y(1)$ and $Y(-1)$ are nonzero, $a = b = 0$. Hence

$$q(YL' - 2Y'L) = 4(BY - n), \qquad \deg(BY - n) \le 3d - 1.$$

If $m = \deg L > 2d$, comparison at degree $d + m + 1$ gives $p \mid m - 2d$. The initial bound gives $0 < m - 2d < 2p$, so $m - 2d = p$. But L is even, while $2d + p$ is odd. This contradiction proves $\deg L \le 2d$, and its even fold therefore has degree at most d.

For $G \in \mathbb{Q}[z]$, define the original negative coefficient

$$I_j(G) = [x^{-jp-1}]\{2\Delta_{1/4}(x^N G(Q(x))) + \Delta_4(x^N G(Q(x)))\} \quad (j = 0, 1),$$

and $\mathcal{E}_e(G)(z) = \sum_{n=1}^{e} G_n z^n/n$. These are integral for integral coefficient polynomials. If $3\deg G \le N + jp$, its negative coefficient is zero, by the Laurent support bound for the divided differences. In particular both I_j annihilate $F^{(0)}$ and $L_2^{\le d}$, while $I_1(G_1) = 0$. Directly from the full source P,

$$r_{-1} = -I_1(F_e)/p, \qquad r_0 = -\tfrac{1}{2}\mathcal{E}_e(F_e)(4/125) - I_0(F_e)/p.$$

Insert (54) into these two exact rational equations. The displayed cutoff bounds and the integral splitting of L_2 retain all remaining terms in an integral error. With

$$H = \mathcal{E}_e(G_1)(4/125), \qquad \Delta_U = -2I_0(G_1) - \tfrac{1}{2}\mathcal{E}_e(F^{(0)})(4/125) - r_*,$$

the result is

$$\begin{aligned} s_{-1} &= -6I_1(G_2) + pR_-, \\ \rho_p &= \Delta_U/p - 6I_0(G_2) - H + pR_0, \qquad R_-, R_0 \in \mathbb{Z}_{(p)}. \end{aligned}$$

Explicitly write $L_2 = L_2^{\le d} + pL_2^{\mathrm{up}}$ with integral L_2^{up}, put $C_2 = 6G_2 + 5L_2/72$, and use the integral cubic remainder E in (54). Then the entire tails in these equations are

$$\begin{aligned} R_- &= -\tfrac{5}{72}I_1(L_2^{\mathrm{up}}) - I_1(E), \\ R_0 &= -\tfrac{5}{72}I_0(L_2^{\mathrm{up}}) - I_0(E) - \tfrac{1}{2}\mathcal{E}_e(C_2)(4/125) - \tfrac{p}{2}\mathcal{E}_e(E)(4/125). \end{aligned}$$

Each is integral because the divided differences use only the unit endpoint values $1/4, 4$, and Euler division uses $1, \dots, e < p$. The second equation proves Δ_U/p integral, since every other term is already integral. Reducing it is now legitimate. The actual first-jet polynomial satisfies $J = 12\bar{G}_1$. Euler division is by $1, \dots, e < p$, so $H_J = 12\overline{\mathcal{E}_e(G_1)}$ coefficientwise and

$$C_0 = H_J(V(0)) = 12\bar{H} \qquad (V(0) = 4/125).$$

Substitution into (53) cancels the same actual endpoint: $-2\bar{H} + C_0/6 = 0$. We obtain the actual scalar

$$\bar{\mathcal{R}}_p - \bar{\mathcal{U}}_p = \mathcal{S}_p := 2\overline{\Delta_U/p} - 12\overline{I_0(G_2)} - \frac{15}{2}\overline{I_1(G_2)} - \mathcal{B}_p. \tag{55}$$

No endpoint value, scalar value, or Hasse-nonvanishing assertion has been used as an extra hypothesis in this calculation.

2.6 Equivalence with the original moment value

Since $\mathcal{D}_p$ is integral, $p^2\mathcal{D}_p \equiv 0 \pmod{p^3}$ holds exactly when $\bar{\mathcal{D}}_p = 0$. If $\mathcal{D}_p = 0$ this is immediate; otherwise $v_p(p^2\mathcal{D}_p) = 2 + v_p(\mathcal{D}_p)$, so cancellation loses exactly two digits. Thus (52) and (55) give

$$M_p/b \equiv u_2^6 u_3^3 \pmod{p^3} \quad \Longleftrightarrow \quad \mathcal{S}_p = \bar{\mathcal{P}}_p.$$

Finally $u_2^6 u_3^3 = 1728^{p-1}$ and $b\,1728^{p-1} = 1728^p/45000$. Both multiplication and division by $b = 24/625$ preserve precision, because it is a p-integral unit. The desired decision is therefore exactly

$$M_p \equiv \frac{1728^p}{45000} \pmod{p^3} \quad \Longleftrightarrow \quad \mathcal{S}_p = \bar{\mathcal{P}}_p. \tag{56}$$

The scalar on the right is the same source-selected scalar that is evaluated in the finite field argument; the full moment on the left is (44), including its p terms and both signed endpoints. In that argument write $S = \mathcal{S}_p$ and $P = \bar{\mathcal{P}}_p$. Indeed $\bar{\alpha}_p = \sum_{r=1}^{p-1} 16^r/r^2 = R(16)$, so its displayed P is exactly this reduction, with $q = \bar{q}_2$, $r = \bar{q}_3$ and $k = \bar{q}_5$. The rational digit $\rho_p = (r_0 - r_*)/p$ is distinct from the whole-source field residue denoted ρ in the balanced calculation.

2.7 Native reciprocal source and integral pullbacks

For $n < p$, the valuation already calculated is zero for $n \le d$ and at least one for $n > d$: if $a = \lfloor 3n/p \rfloor$, then $\lfloor 6n/p \rfloor \ge 2a$, and for $n > d$ the latter floor is at least one. Thus $\bar{F}_e = F := \sum_{n=0}^{d} \bar{c}_n z^n$ and $\bar{H}_e = H := \sum_{n=1}^{d} \bar{c}_n z^n/n$. Canceling successive factorials gives the exact rational recurrence

$$72(n+1)^3 c_{n+1} = (2n+1)(6n+1)(6n+5)c_n.$$

Its reduction proves that F satisfies

$$\mathcal{L}(F) = 72z^2(1-z)F''' + (216z - 324z^2)F'' + (72 - 226z)F' - 5F = 0.$$

Indeed, after multiplication by z, the coefficient at z^{n+1} is the difference of the two sides of the recurrence. At the cutoff $n = d$ the right side vanishes because $p = 6d+1$ or $6d+5$; every higher coefficient is zero. The coefficient $\bar{c}_d$ is nonzero, because $6d < p$, so $\deg F = d$.

Here is the full reciprocal verification, with a finite uniqueness argument that does not assume a Hasse value to be nonzero. Set $D = (4-x)(4x-1)$ and $C = (4x-1)^2(4-x)$ and define

$$\begin{aligned} A_3 &= 8(x+1)D^2, & A_2 &= 12D^2 + 24(x+1)DD', \\ A_1 &= 16DD' + 8(x+1)(DD'' + (D')^2) - 10C, & A_0 &= -5C', \\ \mathcal{C}(G) &= A_3G''' + A_2G'' + A_1G' + A_0G. \end{aligned}$$

The chain-rule identities $G' = V'F'(V)$, $G'' = (V')^2F''(V) + V''F'(V)$, $G''' = (V')^3F'''(V) + 3V'V''F''(V) + V'''F'(V)$ give

$$\mathcal{C}(F(V)) = 3(4x-1)(11-4x)\mathcal{L}(F)(V) = 0.$$

All factors here are polynomial factors; no root of F has been divided out. Put $\vartheta = x\,d/dx$ and $\mathcal{E} = x^3\mathcal{C}$. Its action on x^v is $x^v I(v,x)$, where the complete coefficient list is

$$\begin{aligned} I(v,x) = {} & (128v^3 - 384v^2 + 256v) \\ & + (-960v^3 + 1440v^2 - 480v)x + (1480v^3 - 40v)x^2 \\ & + (1480v^3 + 2220v^2 + 1070v + 165)x^3 \\ & + (-960v^3 - 2880v^2 - 2640v - 720)x^4 \\ & + (128v^3 + 576v^2 + 736v + 240)x^5. \end{aligned}$$

Replacing v by $-1/2-v$ reverses these six coefficients and changes their sign. Since $N=-1/2$ in $\mathbb{F}_p$, linearity on monomials therefore gives

$$\mathcal{E}(x^N G(1/x)) = -x^{N+5}(\mathcal{E}G)(1/x) \quad (\deg G \le N).$$

Both $G=F(V)$ and its reciprocal have degree at most $3d \le N < p$. They satisfy $\mathcal{C}=0$ and take the same value at $x=4$: $G(4)=1$, and $4^N G(1/4) = 4^N = 1$.

To justify uniqueness at that endpoint, write $x=4+y$ and use $\theta = y\,d/dy$. Multiplication of the translated equation by y converts it into $E_3\theta^3+E_2\theta^2+E_1\theta+E_0$, with

$$\begin{aligned} E_3 &= 8(y+5)(4y+15)^2, \\ U &= 12y(4y+15)^2 + 24(y+5)(4y+15)(8y+15), \\ E_2 &= U-3E_3, \qquad E_1 = A_1(y+4)-U+2E_3, \qquad E_0 = yA_0(y+4). \end{aligned}$$

Their constant coefficients are respectively $9000, 0, 0, 0$. For a solution with zero constant, suppose its coefficients below degree j have already vanished. The coefficient equation at degree j is then $9000j^3G_j=0$. Since $1 \le j < p$ and $9000 = 2^3\cdot 3^2\cdot 5^3$ is a unit, $G_j=0$. Induction proves uniqueness for every polynomial of degree below p. Apply this to the difference of the two reciprocal solutions. It proves

$$F(V(x)) = x^N F(Q(x)),$$

coefficientwise in the complete Laurent ring. Consequently the rational numerator defining χ is in $p\mathbb{Z}_{(p)}[x,x^{-1}]$, and its division by p is integral. This establishes the integrality used above.

2.8 The terminating square and the two-node Taylor source

The following verifies the polynomial attachments used in (54). The rank coefficient products satisfy

$$36(n+1)(n+1+t)a_{n+1}(t) = (6n+1+6t)(6n+5+6t)a_n(t).$$

At $t=-p/6$ they terminate at degree d, and at $t=-5p/6$ at degree e. Their denominators are rationally nonzero and are units modulo p; the latter assertion follows from $n+1<p$. In the coordinate T their equation is

$$\mathcal{R}_t(Y) = 36qY'' - 72(1+t)TY' - (1+6t)(5+6t)Y = 0, \qquad q = 1-T^2.$$

The coefficient products and $Y(1)=1$ verify this equation including the terminating row. Since it has coefficients depending only on T^2 except for the factor T multiplying Y', its polynomial solution of terminating degree m obeys $Y(-T)=(-1)^m Y(T)$: its top coefficient recursion downward determines coefficients of its parity, and the other parity would have a different terminating degree. More explicitly the coefficient at T^j is

$$36(j+2)(j+1)Y_{j+2} - \{36j(j-1)+72(1+t)j+(1+6t)(5+6t)\}Y_j = 0.$$

The coefficient in braces equals $36(j-m)(j+m+1+2t)$ at a terminating node. The second root $-m-1-2t$ is not a nonnegative integer below m for either actual node. This proves the stated parity by descending induction. Since d and e have the same parity, u has that parity, and $Y_1 u$ and u^2 are even.

For an arbitrary polynomial G define

$$\begin{aligned} \mathcal{J}_t(G) = {} & 36q^2G''' - 216(1+t)TqG'' \\ & + \{-72(1+t)q + 144(1+t)(1+2t)T^2 - 4(1+6t)(5+6t)q\}G' \\ & + 4(1+6t)(5+6t)(1+2t)TG. \end{aligned}$$

The product derivatives $(Y^2)' = 2YY'$, $(Y^2)'' = 2(Y')^2 + 2YY''$, $(Y^2)''' = 6Y'Y'' + 2YY'''$ give the exact identity

$$\mathcal{J}_t(Y^2) = 2qY(\mathcal{R}_tY)' + (6qY' - 4(1+2t)TY)\mathcal{R}_tY.$$

Hence Y_k^2 solves this third-order equation. As it is even, write it uniquely as $B_k(q)$. Substitution of $G' = -2TB_k'$, $G'' = -2B_k' + 4T^2B_k''$ and $G''' = 12TB_k'' - 8T^3B_k'''$ gives $\mathcal{J}_t(B_k(q)) = -4T\mathcal{L}_t(B_k)(q)$, where

$$\begin{aligned}\mathcal{L}_t(B) = 72z^2(1-z)B''' &+ \{216(1+t)z(1-z) - 108z^2\}B'' \\ &+ \{72(1+t)(1+2t) - (226+432t+216t^2)z\}B' \\ &- (1+6t)(5+6t)(1+2t)B.\end{aligned}$$

Equating coefficients in $z\mathcal{L}_t(B) = 0$ gives exactly

$$72(n+1)(n+1+t)(n+1+2t)B_{n+1} = (6n+1+6t)(2n+1+2t)(6n+5+6t)B_n.$$

At either actual node every factor on the left is nonzero in $\mathbb{Q}$. With $B_0 = Y_k(1)^2 = 1$, induction identifies all coefficients as $C_n(-kp/6)$. Thus the two terminating squares are precisely the native parameter polynomials, without a hypergeometric transformation assumption.

The two actual parameter values differ by $-2p/3$. Apply the integral quotient Taylor recursion already displayed at t_0 through the full cutoff e to obtain

$$Y_5^2 = Y_1^2 - \frac{2p}{3}J_e(q) + \frac{2p^2}{9}K_e(q) + p^3E_5(q), \qquad E_5 \in \mathbb{Z}_{(p)}[z].$$

The rational difference $Y_1 - Y_5 = 4pu$ is integral after division because the two coefficient products reduce to the same products at $t = 0$. Expanding its square, and then canceling p in $\mathbb{Q}$, gives

$$J_e(q) - 12Y_1u = p\{K_e(q)/3 - 24u^2 + (3p/2)E_5(q)\}.$$

This proves both the integral first carry and its complete second-order source. In particular $\bar{J}_e(q) = 12\bar{Y}_1\bar{u}$, so folding and the injectivity of $G(z) \mapsto G(1-T^2)$ give $J = 12\bar{G}_1$. The integral quotient Taylor formula at $t = 0$, together with this last identity, gives the claimed rational full-q Taylor expansion with an integral cubic remainder. Its first-jet tail above N also vanishes: differentiate the parameter recurrence and reduce; at $n = N$ the factor $2n+1$ is zero, while $\bar{C}_n = 0$ for $n > d$, and at each subsequent row the left coefficient $72(n+1)^3$ is a unit. Induction therefore forces all first-jet coefficients for $N < n < p$ to be zero.

The identification $\bar{Y}_1 = P_d$ follows directly from the rank equation. For either $p = 6d+1$ or $p = 6d+5$ one has $d(d+1) = -5/36$ in the field. Hence $\bar{Y}_1$ and P_d satisfy the same equation $qY'' - 2TY' + d(d+1)Y = 0$, have degree at most d, and equal 1 at $T = 1$. Its solution space of degree at most d has dimension one: subtract a multiple of P_d to cancel the leading coefficient. A nonzero remainder of degree $m < d$ would have top coefficient multiplied by $(d-m)(d+m+1)$, a unit because $0 < d-m < p$ and $0 < d+m+1 < p$. It must therefore be zero. The value at 1 fixes the multiple to be one. The actual terminating-square identity consequently gives $F(1-T^2) = P_d(T)^2$.

The full second source and its top coefficient. For the native numerator and denominator factors $U_n(t)$ and $L_n(t)$ used above, differentiate $L_nC_{n+1} = U_nC_n$ twice, before reducing:

$$L_nK_{n+1} + 2L_n'J_{n+1} + L_n''C_{n+1} = U_nK_n + 2U_n'J_n + U_n''C_n.$$

At $t = 0$, their values and first two derivatives are

	value	first derivative	second derivative
L_n	$72(n+1)^3$	$216(n+1)^2$	$288(n+1)$
U_n	$(2n+1)(6n+1)(6n+5)$	$216n^2+216n+46$	$432n+216.$

All denominator factors at t_0 are units in $\mathbb{Z}_{(p)}$, so reduction of this rational identity is legitimate. Thus

$$72(n+1)^3\bar{K}_{n+1} - U_n(0)\bar{K}_n = -432(n+1)^2\bar{J}_{n+1} + 2(216n^2+216n+46)\bar{J}_n$$

$$- 288(n+1)\bar{c}_{n+1} + (432n+216)\bar{c}_n.$$

The terminal row is included: at $n = e$, $U_e(0) = 0$, whereas the native and first-jet coefficients at $e, e+1$ vanish. Therefore $\bar{K}_{e+1} = 0$. The same recurrence inductively gives $\bar{K}_n = 0$ for $e < n < p$. Collecting coefficients gives

$$\begin{aligned}\mathcal{L}_0(\bar{K}_e) + 2\mathcal{A}(J) + \mathcal{A}_2(F) &= 0,\\ \mathcal{A}(G) &= 216z(1-z)G'' + (216-432z)G' - 46G,\\ \mathcal{A}_2(G) &= (288-432z)G' - 216G.\end{aligned}$$

Under $z = q = 1 - T^2$, multiplication by $-4T$ turns these operators into $\mathcal{J}_0$, and

$$\begin{aligned}\mathcal{J}_1(G) &= -216TqG'' + (432T^2 - 216q)G' + 184TG,\\ \mathcal{J}_2(G) &= (576T^2 - 288q)G' + 864TG.\end{aligned}$$

Since $J(q) = 2YR$, the complete second equation is

$$\mathcal{J}_0(K) + 2\mathcal{J}_1(2YR) + \mathcal{J}_2(Y^2) = 0.$$

Here and below $R = 6\bar{u}$ and $K = \bar{K}_e(q)$; the factor 6 is essential. Put $\mathcal{R}(G) = 36qG'' - 72TG' - 5G$ and $S = 36(2TY' + Y)$. For clarity, these actual rank sources follow from the two rational nodes as well. Subtracting their operators gives

$$\mathcal{R}_{-5p/6} - \mathcal{R}_{-p/6} = 48pT\, d/dT + 24p(1-p).$$

Since $Y_1 - Y_5 = 4pu$, this proves the rational identity $\mathcal{R}_{-p/6}(6u) = 36\{2TY_5' + (1-p)Y_5\}$. Reduction gives $\mathcal{R}(Y) = 0$ and $\mathcal{R}(R) = S$. Now $W = q(YR' - Y'R)$ satisfies $W' = Y(qR'' - 2TR') - R(qY'' - 2TY') = (TY^2)'$. Its degree is at most $d + e + 1 = p$, so $W - TY^2 = a + bT^p$. The node endpoint values $Y_k(1) = 1$ give $R(1) = 0$, while their equal reflection parity gives $R(-1) = 0$. At $T = 1, -1$ we obtain $a + b = -1$ and $a - b = 1$, hence $a = 0, b = -1$. This proves the entire source $q(YR' - Y'R) = TY^2 - T^p$. Expanding products gives the following identity, which retains both defects until the actual equations are substituted:

$$\begin{aligned}&- 2\mathcal{J}_1(2YR) - \mathcal{J}_2(Y^2) - 2\{2qRS' + (6qR' - 4TR)S\}\\ &\quad = 36\{2qY(4TR' + 2R + 2Y)' + (6qY' - 4TY)(4TR' + 2R + 2Y)\}\\ &\quad + 16TR\mathcal{R}(Y) + 16TY(\mathcal{R}(R) - S).\end{aligned}$$

Together with $\mathcal{J}_0(R^2) = 2qRS' + (6qR' - 4TR)S$, this proves the displayed quadratic source for $Q_2 = K - 2R^2$. The complete Green differentiation identity is

$$\begin{aligned}36\mathfrak{G}(G)' = Y^2\mathcal{J}_0(G) &- 4qYG'\mathcal{R}(Y) - 2qYG\mathcal{R}(Y)'\\ &+ (2qY' + 4TY)G\mathcal{R}(Y).\end{aligned}$$

After substituting the rank equations and the full Wronskian $q(YR' - Y'R) = TY^2 - T^p$, its right side equals $36\{2qY^3(4TR' + 2R + 2Y) + 4(T^{2p} - 1)\}'$. This proves the differentiated Green source used in the degree argument; the Frobenius term survives the integration even though its derivative vanishes.

The full top normalization also follows over $\mathbb{Q}$, rather than from field division by p. In $A_e(t_0 + h)$ the factor indexed by d has a zero at $h = 0$, and the factor indexed by N contains $2h + 2p/3$. Hence

$$A_e(t_0 + h) = h(2h + 2p/3)B(h), \qquad B \in \mathbb{Z}_{(p)}[h].$$

Write $B(h)/D_e(t_0+h) = g_0 + g_1h + \cdots$; every g_i is integral. Then $J_e = (2p/3)g_0$, $K_e = 4g_0 + (4p/3)g_1$, and

$$J_e/p - K_e/6 = -2pg_1/9.$$

Thus J_e/p is integral and reduces to $K_e/6$. Since composition with $1-T^2$ takes its top coefficient to $(-1)^e$ times that coefficient, and $\deg(Y_1 u) = d+e < 2e$, the top coefficient of the first carry $(J_e(q) - 12Y_1 u)/p$ is $K_{2e}/6$ after reduction. The two-node identity already proved says its field value is $K/3 - 24\bar{u}^2 = K/3 - 2R^2/3$. Equating degree $2e$ coefficients gives $K_{2e}/6 = K_{2e}/3 - 2R_e^2/3$, hence $K_{2e} = 4R_e^2$ and $(Q_2)_{2e} = 2R_e^2$. If $y = Y_d$ and $r = R_e$, the coefficient of T^p in the full Wronskian is $-yr(e-d) = -1$, so $yr(1+2d) = -1$ in $\mathbb{F}_p$. The degree-$2p$ coefficient of the Green polynomial is $y^2(Q_2)_{2e}(2e+1)(2e-2d) = 4$. This supplies precisely the top row used above to remove the final Frobenius ambiguity.

The native quotient differential. The central computation uses the actual native polynomials $F(z) = \bar{F}_e(z)$ and $J(z) = \bar{J}_e(z)$, with degrees d and at most N. Their equations are $\mathcal{L}_0(F) = 0$ and $\mathcal{L}_0(J) + \mathcal{A}(F) = 0$. To obtain the first integral without assuming a quotient value, define

$$D_F = 72z(1-z)(2FF'' - F'^2) + 72(2-3z)FF' - 10F^2.$$

Product differentiation gives $zD_F' + D_F = 2F\mathcal{L}_0(F) = 0$. Its degree is at most $2d+2 < p-1$, and each coefficient is multiplied by a unit $j+1$, so $D_F = 0$. Put $B = z(FJ' - F'J) + F^2$ and

$$\begin{aligned}\mathcal{B}_F(G) = {} & 72z(1-z)FG'' + \{(72-180z)F - 72z(1-z)F'\}G' \\ & + \{-46F + 72(1-z)F' + 72z(1-z)F''\}G.\end{aligned}$$

The exact product identity is $\mathcal{B}_F(B) = F^2(\mathcal{L}_0(J) + \mathcal{A}(F)) - FJ\mathcal{L}_0(F) = 0$. For $h = (1-z)^N$, the identities $2(1-z)h' = h$ and $2(1-z)h'' = 3h'$ hold because $2N = -1$ in the field. A second expansion gives

$$\mathcal{B}_F(Fh) - D_F h = 36F(zF' + F)\{2(1-z)h' - h\} + 36zF^2\{2(1-z)h'' - 3h'\} = 0.$$

The difference $B - Fh$ has degree at most $d + N + 1 < p$ and constant zero. After multiplication of the equation by z, its first nonzero coefficient in degree $j > 0$ would be $72F(0)j^2$ times that coefficient. This is a unit, a contradiction. Therefore

$$z(FJ' - F'J) = F\{(1-z)^N - F\}. \tag{57}$$

In the rational function field $F(V)$ is nonzero (it is a nonconstant polynomial substitution of $F(0) = 1$). No assertion that its value at $x = 0$ is nonzero is made. Equation (57) therefore gives the complete Laurent differential

$$d\{J(V)/F(V)\} = \{(1-V)^N/F(V) - 1\}\, d\log V.$$

The conic cross residue. All residues here mean the coefficient of $x^{-1}dx$ in the formal Laurent expansion. Let $H(z) = \sum_{n=1}^{d} \bar{c}_n z^n/n$ and $H_0 = H(4/125)$. Then $dH(Q) = (F(Q) - 1)d\log Q$. By reciprocity, $x^{-N}J(V) = \{J(V)/F(V)\}F(Q)$. Thus, with $I_u = \operatorname{Res}_{x=0} x^{-N}J(V)d\log Q$ and $C = \operatorname{Res}_{x=0} H(Q)d\{J(V)/F(V)\}$, the zero residue of the derivative of their product gives

$$I_u = -C + \operatorname{Res}_{x=0}\{J(V)/F(V)\}d\log Q.$$

The quotient differential just proved writes C as the difference of the cross residue $\operatorname{Res} H(Q)(1-V)^N d\log V/F(V)$ and $\operatorname{Res} H(Q)d\log V$. The second density is a Laurent polynomial: $Q(1/4) = Q(4) = 0$ and $H(0) = 0$ cancel both finite poles. At infinity $H(Q) = H_0 + O(x^{-1})$ and $d\log V = (3/x + O(x^{-2}))dx$; its Laurent coefficient is consequently $3H_0$.

For the cross term, adjoin a root $a^2 = 5$ to the residue field and use

$$t = \frac{1+z^2}{a(1-z^2)}, \quad w = \frac{2z}{1-z^2}, \quad x = w^2, \quad u = t/w = \frac{1+z^2}{2az}, \quad v = 1/u = \frac{2az}{1+z^2}.$$

Here $w^2 = 5t^2 - 1$, the x-cover has order 2 at zero, and v is an unramified parameter with linear coefficient $2a \neq 0$. Write $T_c(u) = 4u^3 - 3u$, $q_c(u) = (1-u^2)(4u^2-1)^2$, $A_c(u) = (1-u^2)(4u^2-1)$ and $L(u) = u(3-11u^2)/(5u^2-1)$. Direct substitution gives

$$V(w^2) = q_c(t), \quad Q(w^2) = q_c(u), \quad T_c(t)/w = L(u), \quad d\log V = 6L(u)du/A_c(u).$$

The identities $1-q_c(t) = T_c(t)^2$ and $F(q_c(t)) = w^{p-1}F(q_c(u))$ now turn the pulled-back cross density into $6L(u)^p\omega(u)du$, where

$$\omega(u) = \frac{H(q_c(u))}{A_c(u)F(q_c(u))} = \frac{1}{3}r(T_c(u))T_c'(u), \qquad r(T) = \frac{H(1-T^2)}{(1-T^2)Y(T)^2}.$$

The last equality uses the actual native square $F(1-T^2) = Y(T)^2$; it retains its full denominator. The numerator of r has degree at most $2d$, and its denominator has degree $2d+2$ and nonzero leading coefficient. Hence $g(s) = -s^{-2}r(1/s)$ is regular at $s = 0$. At infinity put

$$s = 1/T_c(1/v) = \frac{v^3}{4-3v^2}, \qquad L(1/v) = v^{-1}U(v), \quad U(v) = \frac{3v^2-11}{5-v^2}.$$

The cross density is $2v^{-p}U(v)^p g(s(v))s'(v)dv$. Since s has order 3, $g(s)s'$ has order at least 2. Write $g(s) = h'(s) + s^{p-1}k(s)$ with $h(s) = \sum_{n=0}^{p-2} g_n s^{n+1}/(n+1)$. The derivative $h(s(v))'$ has coefficient zero in degree $p-1$; the other term has order at least $3(p-1)+2 > p-1$. Moreover $U(v)^p$ has only powers divisible by p, so only its constant can contribute to degree $p-1$. That coefficient is zero. The cross residue on the conic is zero.

For completeness the residue substitution rule used here is $\operatorname{Res} f(x(z))x'(z)dz = m \operatorname{Res} f(x)dx$ when $x = z^m u(z)$, $u(0) \neq 0$. For $f = x^{-1}$ it follows from $x'/x = m/z + u'/u$. For every other monomial, expand the unit and its inverse as formal series with integer polynomial coefficients in the unit coefficients and $u(0)^{-1}$. Over $\mathbb{Q}$ the coefficient vanishes by differentiating $x^{j+1}/(j+1)$; thus that same integer polynomial is identically zero and remains zero in characteristic p. Each residue uses only finitely many unit coefficients, so linearity extends the identity to the Laurent series here. The two chart factors are respectively 2 and 1, both units. The original cross residue is therefore zero. The field extension is injective, so the vanishing descends to $\mathbb{F}_p$. We have proved $C = -3H_0$ and

$$I_u = 3H_0 + \operatorname{Res}_{x=0}\{J(V)/F(V)\}d\log Q. \tag{58}$$

The complete cubic midpoint quotient and its coordinate correction. Use $k = \lfloor p/3 \rfloor$ and distinguish the following cubic coordinate from the conic coordinate above:

$$A(t) = 64t(1-t)^3, \quad h(t) = 1+8t, \quad D(t) = 1-20t-8t^2, \quad z(t) = A(t)/h(t)^3.$$

Here is the full native cubic dependence, proved for these actual polynomials. Let

$$P_i(t) = \sum_{j=0}^{d} [z^j]F^{(i)}(z)\, A(t)^j h(t)^{N-3j} \quad (0 \le i \le 3), \qquad b = A'h - 24A = 64(1-t)^2 D.$$

In this definition $[z^j]F^{(i)}$ means the coefficient of degree j of the ith derivative of F in its own polynomial variable. The inequality $3d \le N$ ensures every exponent is nonnegative. Termwise differentiation, using $8N = -4$, gives $h^4P_i' + 4h^3P_i = bP_{i+1}$ for $i = 0, 1, 2$. Differentiating the first equation and substituting the second gives

$$h^8P_0'' = 48h^6P_0 + (h^4b' - 40h^3b)P_1 + b^2P_2.$$

Differentiate this identity and substitute all three first equations:

$$h^{12}P_0''' = -960h^9P_0 + (h^8b'' - 76h^7b' + 1808h^6b)P_1$$

$$+ (3h^4bb' - 108h^3b^2)P_2 + b^3P_3.$$

Define the full square operator

$$\mathcal{S}(P) = 9t^2(1-t)^2P''' + 27t(1-t)(1-2t)P'' + (9 - 62t + 62t^2)P' - 4(1-2t)P.$$

Substitution of the three displayed derivative formulas yields

$$\begin{aligned} 8h^{12}\mathcal{S}(P_0) = b\{&72A^2(h^3 - A)P_3 + (216Ah^6 - 324A^2h^3)P_2 \\ &+ (72h^9 - 226Ah^6)P_1 - 5h^9P_0\}. \end{aligned}$$

For a direct check, the third-, second-, first- and zeroth-derivative terms of $8h^{12}\mathcal{S}$ are respectively replaced by the formulas for $h^{12}P_0'''$, h^8P_0'' multiplied by h^4, $h^4P_0' = bP_1 - 4h^3P_0$ multiplied by h^8, and $h^{12}P_0$. Collecting P_3, P_2, P_1, P_0 gives exactly the four printed coefficients. In the rational function field $P_i = h^N F^{(i)}(A/h^3)$, so the braces are $h^{N+9}\mathcal{L}_0(F)(A/h^3) = 0$. Since $8h^{12}$ is a nonzero polynomial, $\mathcal{S}(P_0) = 0$.

Now let $G(t) = \sum_{j=0}^{k}(-1)^j\binom{k}{j}\binom{k+j}{j}t^j$. The coefficient recurrence proves its Legendre equation below. Since $k(k+1) = -2/9$, it also satisfies $\mathcal{G}(G) = 9t(1-t)G'' + 9(1-2t)G' - 2G = 0$. The full symmetric-square product identity is

$$\mathcal{S}(G^2) = 2t(1-t)G\mathcal{G}(G)' + (6t(1-t)G' + 2(1-2t)G)\mathcal{G}(G) = 0.$$

Both P_0 and G^2 have constant 1 and degree below p: $\deg P_0 \le N + d = 2k$ in both prime classes, and $\deg G^2 = 2k < p$. With $\theta = t\,d/dt$, multiplication by t rewrites $\mathcal{S}$ as

$$9(1-t)^2\theta^3 - 27t(1-t)\theta^2 + (-17t + 26t^2)\theta + (-4t + 8t^2).$$

The first nonzero coefficient of their difference, in degree $j > 0$, would therefore be multiplied by $9j^3$, a unit for $j < p$. The difference is zero, proving the actual identity $h^N F(A/h^3) = G^2$. Thus $G(0) = 1$, $G(1) = (-1)^k$, $\deg G = k < p$, and

$$t(1-t)G'' + (1-2t)G' + k(k+1)G = 0.$$

Its proper polynomial is explicitly $g(t) = \sum_{j=0}^{k-1}(H_k - H_j)G_jt^j$, where $H_j = \sum_{i=1}^{j} i^{-1}$ in the field. The coefficient recurrence $(j+1)^2G_{j+1} = (j-k)(j+k+1)G_j$ and $H_{j+1} - H_j = 1/(j+1)$ give

$$t(1-t)g'' + (1-2t)g' + k(k+1)g = -G', \qquad \deg g < k.$$

Differentiating $2t(1-t)(Gg' - G'g) + G^2 - 1$ gives zero by these two equations. Its degree is below p and its constant is zero, so

$$2t(1-t)(Gg' - G'g) = 1 - G^2. \tag{59}$$

Reflection of the two differential equations, followed by comparison of leading coefficients, gives $G(1-t) = (-1)^kG(t)$ and $g(1-t) = -(-1)^kg(t)$. Indeed a homogeneous polynomial of degree $m < k$ would have nonzero top coefficient multiplied by $(m-k)(m+k+1)$, and both factors are units, so it must be zero. These reflections prove $g(1)G(1) = -H_k$.

Set $W = h^{3N+1}J(z)$, formed as the polynomial $\sum_{j=0}^{N} J_jA^jh^{3N+1-3j}$; it has degree at most $2p-1$ and $W(0) = W(1) = 0$. To transport the native first integral, put $B = A'h - 24A$ and note the exact polynomial identities

$$h^3 - A = D^2, \quad t(1-t)B = AD, \quad z' = B/h^4.$$

Differentiating $G^2 = h^NF(z)$ and $W = h^{3N+1}J(z)$, using $8N = -4$ and $8(3N+1) = -4$ in the field, gives

$$2h^4GG' + 4h^3G^2 = Bh^NF'(z), \qquad h^4W' + 4h^3W = hBh^{3N}J'(z).$$

Multiply the first by W, the second by G^2, and subtract. The $4h^3WG^2$ terms cancel. Substitution of $z(FJ' - F'J) = F((1-z)^N - F)$, followed by $1 - z = D^2/h^3$ and $G^2 = h^N F$, gives

$$t(1-t)(W'G - 2WG') = D^pG - Dh^{p-1}G^3. \tag{60}$$

All cancellations take place in the rational function field, where the nonzero polynomials h, G are invertible; they require no nonzero midpoint value.

Let $\ell(t) = \sum_{j=1}^{p-1} t^j/j$ and define its full polynomial primitive

$$M(t) = (4 + 8t^p)\ell(t) - 3\ell(-8t).$$

It has degree at most $2p-1$, $M(0) = 0$, and

$$M' = (4 + 8t^p)((1-t)^{p-1} - t^{p-1}) + 24(h^{p-1} - t^{p-1}).$$

Multiply by $th(1-t)$ and use $(1-t)^p = 1 - t^p$, $h^p = 1 + 8t^p$ and $D^p = 1 - 20t^p - 8t^{2p}$; the numerator becomes $h(D^p - Dh^{p-1})$. Canceling h yields $t(1-t)M' = D^p - Dh^{p-1}$. Consequently $E = W - 2D^pgG - MG^2$ satisfies $E'G - 2EG' = 0$. Equation (59) shows $\gcd(G, G') = 1$: its rearrangement expresses 1 as a polynomial combination of G, G'. Reduction modulo G first gives $G \mid E$ since 2 is a unit. Write $E = GH$; its equation reduces to $GH' - G'H = 0$, so the same coprimality gives $G \mid H$. Thus $E = G^2K$ and $K' = 0$. The bound $\deg E \le 2p + 2k - 1$ gives $\deg K < 2p$, hence $K = a + bt^p$. At 0 we find $a = -2H_k$. At 1, using $D(1) = -27$ and $g(1)G(1) = -H_k$, we find $a + b = -54H_k - M(1)$. Therefore

$$M_{\mathrm{reg}} = M - 2H_k + (-52H_k - M(1))t^p, \qquad \frac{J(z)}{F(z)} = \frac{2D^pg + M_{\mathrm{reg}}G}{h^pG}.$$

This is the full actual quotient, including its degree-p row.

The entropy identity and inversion of the finite logarithm give

$$\ell(1/2) = q_2, \quad \ell(-4) = 8q_2 - 5q_5, \quad \ell(-8) = 24q_2 - 18q_3.$$

For example, the exact integer relation $q_{ab} = q_a + q_b + pq_aq_b$ gives $q_8 = 3q_2$ and $q_9 = 2q_3$ after reduction. Applying $\ell(a) = -aq_a - (1-a)q_{1-a}$ gives the last two evaluations; $\ell(a^{-1}) = -a^{-1}\ell(a)$ and $\ell(2) = -2q_2$ give the first. Thus $M(1) = -72q_2 + 54q_3$ and $M(1/2) = -16q_2 + 15q_5$. The harmonic identities in the shared field derivation are $H_k = -3q_3/2$ and $H_d = -2q_2 - 3q_3/2$. Since $(1/2)^p = 1/2$,

$$\beta_{\mathrm{CM}} := M_{\mathrm{reg}}(1/2)/5 = \{M(1/2) - 28H_k - M(1)/2\}/5 = -2H_d + 3q_5.$$

If k is even, reflection gives $g(1/2) = 0$ and $G'(1/2) = 0$. The Wronskian gives $G(1/2)^2 = 1$, so the quotient is regular and its constant is β_{CM}. If k is odd, reflection gives $G(1/2) = g'(1/2) = G''(1/2) = 0$, while the Wronskian gives $G'(1/2)g(1/2) = -2$. In particular $G'(1/2) \neq 0$. The numerator and denominator of the quotient have, at the midpoint, denominator derivative $5G'$, denominator second derivative zero, and numerator derivative $M_{\mathrm{reg}}G'$. Its Laurent expansion is therefore

$$\frac{J(z(1/2+\delta))}{F(z(1/2+\delta))} = \frac{\kappa}{\delta} + \beta_{\mathrm{CM}} + O(\delta), \qquad \kappa = \frac{44}{5G'(1/2)^2} \neq 0.$$

Indeed $D(1/2) = -11$, the numerator constant is $-22g(1/2)$, and division by $5G'(1/2)$ gives the displayed pole coefficient. The finite part of $N(\delta)/D(\delta)$ is $N'(0)/D'(0) - N(0)D''(0)/(2D'(0)^2)$; the second term is zero and the first is $M_{\mathrm{reg}}(1/2)/5$.

The original x-coordinate is related to this midpoint by the unique formal series $\delta(x)$ with $\delta(0) = 0$ and $z(1/2 + \delta(x)) = V(x)$. Its existence follows coefficient by coefficient from the nonzero linear coefficient $-176/625$ of the centered cubic; all later coefficients are obtained by dividing by that same unit. The exact centered numerator and denominator are

$$4 - 16\delta + 64\delta^3 - 64\delta^4, \qquad (5 + 8\delta)^3 = 125 + 600\delta + 960\delta^2 + 512\delta^3.$$

Their quotient begins $4/125 - (176/625)\delta + (3456/3125)\delta^2$. Since $V(x) = 4/125 - (33/125)x + (72/125)x^2 + \cdots$, comparison gives

$$\delta(x) = \frac{15}{16}x + \frac{45}{32}x^2 + O(x^3).$$

In the odd case write $\delta = xf$ with $f_0 = 15/16, f_1 = 45/32$. The exact logarithmic differential is

$$d\log Q = \left(-\frac{3}{x} + \frac{1}{x-1/4} + \frac{2}{x-4}\right)dx, \qquad D_0 = -9/2.$$

Since $(f^{-1})_0 = f_0^{-1}$ and $(f^{-1})_1 = -f_1/f_0^2$, its residue against the full quotient is

$$\kappa\left(\frac{3f_1}{f_0^2} + \frac{D_0}{f_0}\right) - 3\beta_{\mathrm{CM}} = \kappa(24/5 - 24/5) - 3\beta_{\mathrm{CM}} = 6H_d - 9q_5.$$

The even case has the same residue, directly from its regular constant. Thus the nonzero odd pole is retained and its coordinate correction is explicitly canceled.

Finally, inversion $x \mapsto 1/x$ changes the original density $x^N J(Q)d\log V$ into $x^{-N}J(V)d\log Q$ and reverses its residue. The first integral Taylor source gives

$$\bar{r}_0 = I_u/6 - H_0/2.$$

Combining this with (58) and the just computed midpoint residue yields $\bar{r}_0 = H_d - 3q_5/2 = -2q_2 - 3q_3/2 - 3q_5/2 = \bar{r}_*$. This proves the required field equality before the actual rational difference $r_0 - r_*$ is divided by p.

The source remainder, all noncentral slots, and the entire endpoint functional. Write $F^{(0)} = \sum_{n=0}^{d} C_n(t_0)z^n$ and

$$\delta_F(x) = \{F^{(0)}(V) - x^N F^{(0)}(Q)\}/p.$$

Both numerator terms are polynomials of degree at most N because $3d \le N$; their field reductions agree by reciprocity. Thus $\delta_F \in \mathbb{Z}_{(p)}[x]$ and $\deg \delta_F \le N$. Taylor expansion of the actual full F_e gives, coefficientwise,

$$\bar{\chi} = \bar{\delta}_F + \tfrac{1}{6}\{J(V) - x^N J(Q)\}.$$

Since $H_J(0) = J(0) = 0$, the exact derivative $P_J' = J(V)d\log V/dx$ equals the weighted divided-difference source of $J(V)$. Hence the actual remainder source is the reduction of

$$\mathrm{WD}(\delta_F) - \tfrac{1}{6}\,\mathrm{WD}(x^N J_e(Q)) + \mathrm{WD}(H_d(V)) - H_d(V)/(2x), \quad \mathrm{WD} = 2\Delta_{1/4} + \Delta_4.$$

Use J_e through N in this representative; its higher coefficients reduce to zero. A divided difference lowers the upper exponent by one and leaves a negative lower exponent unchanged. The displayed representative has support in $[-2N, N-1] = [-p+1, N-1]$. Its canonical primitive therefore has support in $[-p+2, N]$ and coefficients zero at every multiple of p, proving the stated support of B_p. Together with $\bar{\mathcal{A}} = B_p + (P_J - C_0 - C_1x^p)/6$ and $\deg P_J \le 3N < 2p$, this proves $-p < j < 2p$ for every exponent of the actual field primitive.

For the negative slot, the exact formula $r_{-1} = -I_1(F_e)/p$ and the full Taylor expansion give $\bar{r}_{-1} = -I_1(J)/6 = 0$, since $3\deg J \le 3N \le N + p$. For a positive slot the exact source, before coefficient extraction, is

$$P = \frac{1}{p}(H_e(V))' + \left(\sum_{n=1}^{e} c_n V^n/n^2\right)' - \frac{H_e(V)}{2x} - \frac{1}{p}\,\mathrm{WD}(x^N F_e(Q)).$$

The last term has upper exponent at most $N - 1 < kp - 1$ for $k > 0$. Taking the coefficient of x^{kp-1} in the other three terms gives

$$r_k = (k - 1/2)[x^{kp}]H_e(V) + kp[x^{kp}]\sum_{n=1}^{e} c_n V^n/n^2 \quad (k = 1, 2).$$

Reduction gives zero, because $\bar{H}_e = H_d$ and $3d < p$. These arguments prove integrality of each $s_k = r_k/p$ by rational division. For the next digit, terms with $n \le d$ cancel in $(H_e - H_d)/p$. For $n > d$, $C_n(t_0) = 0$, and integral Taylor gives $\overline{c_n/p} = J_n/6$. Thus, in degrees above $3d$, the reduced divided pullback is $P_J/6$. The Euler primitive with denominators n^2 reduces to a polynomial of degree at most $3d$. This proves the complete formulas $\bar{s}_1 = C_1/12$ and $\bar{s}_2 = 0$.

Here is the full monomial computation behind Λ. For a field Laurent density put $\mathscr{C}(\sum a_j x^j) = \sum a_{p(k+1)-1} x^k$. For $a \neq 0$ the geometric identities are

$$\frac{x^n}{x-a} = \frac{a^n}{x-a} + \sum_{i=0}^{n-1} a^i x^{n-1-i} \quad (n \ge 0), \qquad \frac{x^{-m}}{x-a} = \frac{a^{-m}}{x-a} - \sum_{i=1}^{m} a^{-(m+1-i)} x^{-i} \quad (m > 0).$$

Also $\mathscr{C}(1/(x-a)) = 1/(x-a)$: multiply the denominator to the pth power to write $(x-a)^{p-1}/(x^p-a)$, whose only selected numerator power is x^{p-1} with coefficient one. In the band $-p < j < 2p$, the positive geometric sum selects only exponent $p-1$, when $j \ge p$; its coefficient is $a^{j-p} = a^j/a$. The negative geometric sum selects only exponent -1, with coefficient $-a^j$. Consequently

$$\mathscr{C}\left(\frac{x^j}{x-a}\right) = \frac{a^j}{x-a} - \mathbf{1}_{j<0}\frac{a^j}{x} + \mathbf{1}_{j\ge p}\frac{a^j}{a}.$$

The logarithmic term L/x selects just the $0, p$ coefficients of L; both are zero for the canonical primitive. Summing every monomial, at $a = 1/4$ and $a = 4$ with weights $2, 1$, gives

$$\mathscr{C}\left[L\left(\frac{2}{x-1/4} + \frac{1}{x-4} - \frac{1}{2x}\right)\right] = \frac{2L(1/4)}{x-1/4} + \frac{L(4)}{x-4} + \frac{-2L^-(1/4) - L^-(4)}{x} + 8L^{\ge p}(1/4) + L^{\ge p}(4)/4.$$

This proves every negative and high-band term, without deleting any coefficient.

The canonical density is $S(s)^p\, 2sL(s^2)\{2/(s^2-1/4) + 1/(s^2-4) - 1/(2s^2)\}$. Cartier obeys $\mathscr{C}(f^p g) = f\mathscr{C}(g)$ coefficientwise. For odd p, quadratic substitution gives $\mathscr{C}_s(2sf(s^2)) = 2s(\mathscr{C}_x f)(s^2)$: the selected condition is $p \mid 2j+2$, equivalently $p \mid j+1$. Multiplying by $S = v/s$ therefore gives the four canceled kernels $2V(s^2)/(s^2-1/4)$, $2V(s^2)/(s^2-4)$, $2V(s^2)/s^2$ and $2V(s^2)$ printed above. Their signed primitives are obtained by substituting $\mathcal{I}(s^{2j}) = (2^{2j+1} - 2^{-2j-1})/(2j+1)$ into their four coefficients. All primitive denominators are among $1, 3, 5, 7$ and all coefficient denominators divide powers of $2, 5, 7$, hence are units. Together with the earlier rational denominator reduction this proves $\bar{T}_p = \Lambda(\bar{\mathcal{A}})$. The exact differential $P_J d\log V = d(E_J(V))$ has zero Cartier operator; the displayed full formula with no logarithmic term gives its whole high pair equal to zero. This closes the full HJet endpoint cancellation used in (53).

3 Finite logarithms and the corrected consumer

3.1 The full finite consumer

Fix a prime $p \ge 19$, a field K of characteristic p, and $u \in K$ with $u^2 = 1/5$. Integer and rational constants whose denominators are prime to p are interpreted in K. Write

$$Q_a = \frac{a^{p-1} - 1}{p} \in \mathbb{Z} \quad (p \nmid a), \qquad q_a = \overline{Q_a} \in \mathbb{F}_p \subset K, \qquad n = \frac{p-1}{2}.$$

The division defining Q_a takes place in the integers. It is exact by Fermat's theorem; no division by p is made in K. In this subsection only q_2, q_3, q_5 and their indicated integer analogues occur.

Define the full finite logarithm and dilogarithm by

$$\ell(x) = \sum_{j=1}^{p-1} \frac{x^j}{j}, \qquad R(x) = \sum_{j=1}^{p-1} \frac{x^j}{j^2}, \tag{61}$$

$$E(x) = \frac{\ell(x) + \ell(-x)}{2}, \qquad O(x) = \frac{\ell(x) - \ell(-x)}{2}.$$

Set

$$K_e(t) = \frac{R(t) + R(-t) - R(1-t) - R(1+t)}{2}, \qquad C = E(2u) - E(u). \tag{62}$$

The loop uses the following finite triangular sums, for $0 \le N < p$:

$$\begin{aligned} L_N(x) &= \sum_{j=1}^{N} \frac{x^j}{j}, \qquad (63)\\ D_N(x,y) &= \sum_{j=1}^{N} \sum_{k=1}^{j} \frac{x^k y^{j-k}}{kj},\\ S(x,y) &= D_n(x^2, y^2),\\ A(t) &= S(2t,1) - S(t,1/2), \qquad B(t) = S(2t,2) - S(t,1),\\ \chi(t) &= S(t,t) + S(2t,2t) - S(2t,t) - S(t,2t). \end{aligned}$$

Write $\mathcal{L} = A(u) - B(u) + \chi(u)$. All indices inverted in (61) and (63) lie between 1 and $p-1$. The fixed denominators below have prime factors only $2, 3, 5$, so every displayed division is legitimate.

Finite functional equations. For $x \neq 0$, reversing the full index interval by $j \mapsto p - j$ gives

$$\ell(x^{-1}) = -x^{-p}\ell(x), \qquad R(x^{-1}) = x^{-p}R(x). \tag{64}$$

Indeed $(p-j) = -j$ in K, and $(x^{-1})^{p-j} = x^{-p}x^j$. The two quadratic distribution identities are

$$\begin{aligned} \ell(x^2) &= (1+x^p)\ell(x) + (1-x^p)\ell(-x), \qquad (65)\\ R(x^2) &= 2(1+x^p)R(x) + 2(1-x^p)R(-x). \end{aligned}$$

For completeness, put $P_h(x) = \sum_{j=1}^{p-1} x^j/j^h$ for $h = 1, 2$, and

$$V_h = \sum_{j=1}^{n} \frac{x^{2j}}{(2j)^h}, \qquad T_h = \sum_{j=1}^{n} \frac{x^{p-2j}}{(p-2j)^h}.$$

Then $P_h(x) = V_h + T_h$, $P_h(-x) = V_h - T_h$. Splitting the sum for $P_h(x^2)$ into indices j and $p-j$, $1 \le j \le n$, gives $P_h(x^2) = 2^h(V_h + x^p T_h)$: its upper term uses $2(p-j) = p + (p-2j)$ and $(p-j) = -j$. Substitution proves (65), including $x = 0$.

Reflection holds as polynomial identities, and consequently at every element of the extension field K:

$$\ell(1-x) = \ell(x), \qquad x^p R(x) + (1-x)^p R(1-x) = -\frac{\ell(x)^2}{2}. \tag{66}$$

Here is a polynomial proof, which is needed to justify extension-field evaluation. Regard the sums $\ell(X), R(X)$ as polynomials over $\mathbb{F}_p$. Inversion at 1 gives $\ell(1) = 0$; distribution at 1 gives $R(1) = 4R(1)$, hence $R(1) = 0$. Both polynomials also vanish at 0. Termwise differentiation and the finite geometric sum give

$$XR'(X) = \ell(X), \qquad \ell'(X) = \sum_{j=0}^{p-2} X^j = (1-X)^{p-1} - X^{p-1}.$$

Thus the derivative of $\ell(X) - \ell(1-X)$ is zero, its degree is less than p, and its value at 0 is zero. It is the zero polynomial. Now put

$$H(X) = X^p R(X) + (1-X)^p R(1-X) + \ell(X)^2/2.$$

Using the just-proved logarithm reflection,

$$H'(X) = \big(X^{p-1} - (1-X)^{p-1} + \ell'(X)\big)\ell(X) = 0.$$

Also $\deg H < 2p$ and $H(0) = H(1) = 0$. A polynomial of degree less than $2p$ with zero derivative in characteristic p has the form $a + bX^p$: the coefficient of X^{j-1} in its derivative is j times the coefficient of X^j, so only indices $0, p$ survive. The two endpoint values force $a = b = 0$. This proves (66) over $\mathbb{F}_p[X]$, and evaluation proves it over K.

The seven rational sign rows. These values use the same triangular sums, including every $k = j$ term. Define homogeneous sums

$$H_r(a,b) = \sum_{j=1}^{p-1} \frac{a^j b^{p-j}}{j^r} \quad (r = 1, 2).$$

They are polynomials, including at $b = 0$; in particular $H_r(a, 0) = 0$, because $p - j \geq 1$. For $b \neq 0$, $H_2(a,b) = b^p R(a/b)$. The full triangular sum satisfies

$$D_{p-1}(x,y) = H_2(x-y, 1-y) - H_2(-y, 1-y) + H_2(y-x, y). \tag{67}$$

Here is a proof without division by $1-y$. Regarding x as an indeterminate, termwise differentiation gives

$$(x-y)D'_{p-1}(x,y) = \ell(x) - \ell(y), \qquad a\,\partial_a H_2(a,b) = H_1(a,b),$$

and the geometric sum gives

$$a(a-b)\partial_a H_1(a,b) = ba^p - b^p a.$$

Use the last equation at $(a,b) = (x-y, 1-y), (y-x, y)$. Multiplying by $x(x-1)(x-y)$, adding the two equations and using $(x-y)^p = x^p - y^p$ shows that

$$\partial_x\{H_1(x-y, 1-y) + H_1(y-x, y) - \ell(x)\} = 0.$$

The expression has degree $< p$ and value $-\ell(y)$ at $x = y$. Thus $H_1(x-y, 1-y) + H_1(y-x, y) = \ell(x) - \ell(y)$. The first two derivative identities now show that the difference between the sides of (67) has zero derivative. Its degree is $< p$, so it is constant. At $x = 0$ it is zero: the first two homogeneous terms cancel and $H_2(y, y) = y^p R(1) = 0$. This proves the identity for every y.

Since p is odd, $H_2(-a,-b) = -H_2(a,b)$. Add (67) at the four sign choices of x, y. The terms with second argument $\pm y$ cancel in pairs, and the remaining expression is

$$\begin{aligned} S(x,y) = {} & H_2(x-y, 1-y) + H_2(-x-y, 1-y) - 2H_2(-y, 1-y) \\ & + H_2(x+y, 1+y) + H_2(-x+y, 1+y) - 2H_2(y, 1+y). \end{aligned} \tag{68}$$

The identification with the original S follows directly from its definition: summing the four signs in D_{p-1} keeps precisely the even $j = 2a$, even $k = 2b$, and the four signs cancel the denominator $4ab$. Thus the answer is $\sum_{a=1}^{n}\sum_{b=1}^{a} x^{2b}y^{2(a-b)}/(ab)$. For $y = 1$, the entire first line of (68) is zero by $H_2(a, 0) = 0$. No zero denominator is introduced. At $y = 1, 2, 1/2$, scaling the nonzero second arguments gives

$$\begin{aligned} S(x,1) &= 2\{R((1+x)/2) + R((1-x)/2) - 2R(1/2)\}, \\ S(x,2) &= -R(2-x) - R(2+x) + 2R(2) \\ &\quad + 3\{R((2+x)/3) + R((2-x)/3) - 2R(2/3)\}, \\ S(x,1/2) &= \{R(2x-1) + R(-2x-1)\}/2 \\ &\quad + \tfrac{3}{2}\{R((1+2x)/3) + R((1-2x)/3) - 2R(1/3)\}. \end{aligned} \tag{69}$$

Put $q = q_2$, $r = q_3$, $r_- = R(-2)$, and $a = 2q - 3r$. The rational values used in these three equations are

$$\begin{aligned} R(0) = R(1) = R(-1) &= 0, \\ R(2) &= -q^2, \\ R(1/2) &= -q^2/2, \\ R(4) &= -6q^2 - 2r_-, \\ R(1/4) &= -3q^2/2 - r_-/2, \\ R(-1/2) &= -r_-/2, \end{aligned}$$

$$
\begin{aligned}
R(3) &= 2r_-/3 - a^2/6,\\
R(1/3) &= 2r_-/9 - a^2/18,\\
R(2/3) &= -r_-/9 - a^2/18,\\
R(3/2) &= -r_-/6 - a^2/12,\\
R(3/4) &= r_-/6 - 13q^2/6 + 2qr - 3r^2/8,\\
R(4/3) &= \tfrac{4}{3}R(3/4),\\
R(-1/3) &= -R(-3)/3.
\end{aligned}
$$

To derive them, reflection at -1, with $\ell(-1) = -2q$, gives $2R(2) = -2q^2$. Quadratic distribution at 2 gives $R(4) = 6R(2) - 2R(-2)$. Inversion gives the values at $1/2, 1/4, -1/2$. Reflection at -2, with $\ell(-2) = 2q - 3r = a$, gives $-2r_- + 3R(3) = -a^2/2$, hence the stated $R(3)$; inversion gives $R(1/3)$. Reflection at $1/3$, using $\ell(1/3) = r - 2q/3 = -a/3$, gives

$$R(1/3) + 2R(2/3) = -a^2/6.$$

Inversion at $2/3$ gives $R(3/2)$. Finally $\ell(1/4) = 2q - 3r/4$ follows from integer entropy at 4 and inversion. Reflection there gives

$$R(1/4) + 3R(3/4) = -2(2q - 3r/4)^2,$$

which yields the stated value of $R(3/4)$. Inversion at $3/4$ and at -3 gives the last two entries.

Before expanding the squares, the seven substitutions in (69) are respectively

$$
\begin{gathered}
-4R(1/2), \quad 2(R(3/4) + R(1/4) - 2R(1/2)),\\
-R(3) + 2R(2) + 3(R(1/3) - 2R(2/3)),\\
R(-2)/2 + \tfrac{3}{2}(R(2/3) - 2R(1/3)),\\
-R(4) + 2R(2) + 3(R(4/3) - 2R(2/3)),\\
2(R(3/2) + R(-1/2) - 2R(1/2)), \quad -3R(1/3).
\end{gathered}
$$

In the last entry the two $R(-3)$ terms cancel by $R(-1/3) = -R(-3)/3$. Substitution and expansion give

(x, y)	$S(x, y)$
$(1, 1)$	$2q^2$
$(1/2, 1)$	$-\frac{2}{3}r_- - \frac{16}{3}q^2 + 4qr - \frac{3}{4}r^2$
$(1, 2)$	$\frac{2}{3}r_- - \frac{2}{3}q^2 - 4qr + 3r^2$
$(1/2, 1/2)$	$-\frac{1}{3}r_- + \frac{(2q-3r)^2}{12}$
$(2, 2)$	$\frac{10}{3}r_- - \frac{10}{3}q^2 + 4qr + \frac{3}{2}r^2$
$(2, 1)$	$-\frac{4}{3}r_- + \frac{4}{3}q^2 + 2qr - \frac{3}{2}r^2$
$(1, 1/2)$	$-\frac{2}{3}r_- + \frac{(2q-3r)^2}{6}$

(70)

The integer binomial identity also fixes the rational logarithm values. For an integer a, the numerator $1 - a^p - (1-a)^p$ is divisible by p, and

$$\ell(\overline{a}) = \overline{\frac{1 - a^p - (1-a)^p}{p}}. \tag{71}$$

To see this, expand $(1-a)^p$ and use, for $1 \le j < p$,

$$\overline{\frac{\binom{p}{j}}{p}} = \frac{(-1)^{j-1}}{j}.$$

This equality follows from $j\binom{p}{j}/p = \binom{p-1}{j-1}$ and $\binom{p-1}{j-1} \equiv (-1)^{j-1} \pmod p$. When a and $1-a$ are prime to p, the quotient in (71) equals $-aQ_a - (1-a)Q_{1-a}$ in the integers. Since $p-1$ is even and $Q_4 = 2Q_2 + pQ_2^2$, this yields

$$\ell(2) = -2q_2, \qquad\qquad \ell(-2) = 2q_2 - 3q_3, \tag{72}$$

$$\ell(-4) = 8q_2 - 5q_5, \qquad \ell(1/5) = q_5 - \frac{8}{5}q_2.$$

The last equality uses reflection at 5 and inversion at 5; prime-field constants satisfy $a^p = a$.

The fifth entropy bridge. This bridge determines the fifth endpoint from an integer expression. In the ring $\mathbb{Z}[s]/(s^2-5)$, let $\mathrm{Re}(a+bs) = a$ and define

$$T_p(a) = \mathrm{Re}\big((5+as)^p\big) = \sum_{j=0}^{n} \binom{p}{2j} 5^{p-j} a^{2j}, \tag{73}$$

$$B_p(a) = -\sum_{j=1}^{n} \frac{\binom{p}{2j}}{p}\, 5^{p-j} a^{2j}.$$

Every quotient in $B_p(a)$ is an integer. Expanding the binomial gives the exact identity

$$T_p(a) = 5^p - pB_p(a). \tag{74}$$

Equivalently, $B_p(a)$ is the real part of the integral polynomial

$$-\sum_{j=1}^{p-1} \frac{\binom{p}{j}}{p}\, 5^{p-j}(as)^j.$$

Apply the two homomorphisms $s \mapsto 5u$ and $s \mapsto -5u$ to this polynomial. They exist because $(5u)^2 = 5$. The divided-binomial congruence above shows that their respective values are $5^p\ell(-au)$ and $5^p\ell(au)$. Adding them and dividing by 2 gives

$$\overline{B_p(a)} = 5E(au), \qquad C = \frac{\overline{B_p(2) - B_p(1)}}{5}. \tag{75}$$

In particular C belongs to the prime field and the formula is unchanged for either root of $u^2 = 1/5$.

The relation between $E(u)$ and $E(2u)$ comes from an exact cubic trace, rather than a prescribed endpoint value. Put $z = 5 + s$. Then $z^3 = 40(5+2s)$ and $z\overline{z} = 20$, where conjugation sends s to $-s$. For $v = A + Bs$, one has $\mathrm{Re}(v^3) = 4A^3 - 3(v\overline{v})A$. Applying this to $v = z^p$ proves

$$40^p T_p(2) = 4T_p(1)^3 - 3\,20^p T_p(1). \tag{76}$$

Use $2^p = 2(1+pQ_2)$, $5^p = 5(1+pQ_5)$ and (74) in (76), and cancel the integer factor 4. The resulting exact equation is

$$\begin{aligned} &2(1+pQ_2)^3\, 5(1+pQ_5)\big(5(1+pQ_5) - pB_p(2)\big) \\ &\quad = \big(5(1+pQ_5) - pB_p(1)\big)^3 - 3(1+pQ_2)^2\, 5(1+pQ_5)\big(5(1+pQ_5) - pB_p(1)\big). \end{aligned}$$

Modulo p^2, discard terms containing p^2 and compare the coefficients of the remaining p term. This gives

$$p\,5\big(2B_p(2) - 12B_p(1) - 60Q_2 + 25Q_5\big) \equiv 0 \pmod{p^2}.$$

Cancel p in the integer divisibility statement and then reduce modulo p; since 5 is a unit, $2\overline{B_p(2)} - 12\overline{B_p(1)} - 60q_2 + 25q_5 = 0$. By (75),

$$E(2u) = 6E(u) + 6q_2 - \frac{5}{2}q_5, \tag{77}$$

$$E(u) = \frac{q_5}{2} - \frac{6}{5}q_2 + \frac{C}{5}, \qquad E(2u) = E(u) + C.$$

The distribution identity in (65) says $E(x) + x^p O(x) = \ell(x^2)/2$. For $x = u, 2u$, the right sides agree, since $u^2 = 1/5$ and $(2u)^2 = 4/5 = 1 - 1/5$. Using (72) gives

$$E(u) + u^p O(u) = E(2u) + 2u^p O(2u) = \frac{q_5}{2} - \frac{4}{5}q_2.$$

Also $(u^p)^2 = 1/5$ and $u^p \neq 0$. Hence

$$O(u) = u^p(2q_2 - C), \qquad O(2u) = u^p(q_2 - 3C). \tag{78}$$

The finite loop primitive. The elementary triangular-sum identity underlying the loop is

$$\begin{aligned} &D_N(as,1) - D_N(s,a^{-1}) - D_N(as,a) + D_N(s,1) \\ &\quad + D_N(s,s) + D_N(as,as) - D_N(as,s) - D_N(s,as) \\ &= (L_N(1) - L_N(a))L_N(s) + (L_N(1) - L_N(a^{-1}))L_N(as), \qquad a \neq 0. \end{aligned} \tag{79}$$

Here its needed instances have $N = n < p$, so its denominators are nonzero. To prove it, put

$$r_{m,j}(a) = \frac{1 + a^m - a^j - a^m a^{-j}}{mj}.$$

Reversing the order of summation in the first four sums on the left gives $\sum_{m=1}^{N}\sum_{j=m}^{N} s^m r_{m,j}(a)$. The last four sums give $\sum_{m=1}^{N}\sum_{j=1}^{m} s^m r_{m,j}(a)$. The repeated diagonal is zero because $r_{m,m}(a) = 0$. The inner sum over the joined interval is

$$\sum_{j=1}^{N} r_{m,j}(a) = \frac{(1 + a^m)L_N(1) - L_N(a) - a^m L_N(a^{-1})}{m}.$$

Summing over m now gives (79). Even-index pairing in the full logarithm gives $L_n(t^2) = 2E(t)$. By logarithm reflection and inversion and (72),

$$L_n(1) = -2q_2, \qquad L_n(4) = -3q_3, \qquad L_n(1/4) = 2q_2 - \frac{3}{2}q_3. \tag{80}$$

For example, these are respectively $\ell(1)+\ell(-1)$, $\ell(2)+\ell(-2)$ and $\ell(1/2)+\ell(-1/2)$; use $\ell(-1) = \ell(2)$ and $\ell(\pm 1/2) = -(\pm 2)^{-1}\ell(\pm 2)$. Taking $a = 4$, $s = t^2$ in (79) therefore yields

$$A(t) - B(t) + \chi(t) = -(4q_2 - 6q_3)E(t) - (8q_2 - 3q_3)E(2t).$$

At $t = u$, substitution from (77) proves the full loop identity

$$\mathcal{L} = -6q_2q_5 + \frac{9}{2}q_3q_5 + \frac{72}{5}q_2^2 - \frac{54}{5}q_2q_3 - \frac{52}{5}q_2C + \frac{24}{5}q_3C. \tag{81}$$

The full weight-two endpoint. Reflection at t and $-t$, together with quadratic distribution, gives the cleared identity

$$(1 - t^{2p})K_e(t) = \frac{R(t^2)}{4} + t^p E(t)O(t) + \frac{E(t)^2 + O(t)^2}{2}. \tag{82}$$

To verify the combination explicitly, write $v = t^p$. The two reflection equations are

$$vR(t) + (1 - v)R(1 - t) = -\ell(t)^2/2, \quad -vR(-t) + (1 + v)R(1 + t) = -\ell(-t)^2/2.$$

Multiply them by $-2(1+v)$ and $-2(1-v)$, respectively, and add $R(t^2) = 2(1+v)R(t)+2(1-v)R(-t)$. This gives

$$2(1 - v^2)\big(R(t) + R(-t) - R(1 - t) - R(1 + t)\big) = R(t^2) + (1 + v)\ell(t)^2 + (1 - v)\ell(-t)^2.$$

Substitution of $\ell(t) = E(t) + O(t)$ and $\ell(-t) = E(t) - O(t)$ proves (82). No division by $1 - t^{2p}$ was used.

For the two fifth endpoints, this factor is $4/5$ and $1/5$, respectively, and is a unit. Write $e = E(u)$, $q = q_2$, $k = q_5$. By (78) and $(u^p)^2 = 1/5$, solving (82) gives

$$\begin{aligned} K_e(u) &= \frac{5}{16}R(1/5) + \frac{1}{4}e(2q - C) + \frac{5}{8}e^2 + \frac{1}{8}(2q - C)^2, \\ K_e(2u) &= \frac{5}{4}R(4/5) + 2(e + C)(q - 3C) + \frac{5}{2}(e + C)^2 + \frac{1}{2}(q - 3C)^2. \end{aligned} \tag{83}$$

Reflection at $1/5$ and inversion at 5 give

$$\begin{aligned}\frac{1}{5}R(1/5)+\frac{4}{5}R(4/5)&=-\frac{1}{2}(k-8q/5)^2, & R(1/5)&=\frac{1}{5}R(5),\\ \frac{5}{4}R(4/5)-\frac{5}{16}R(1/5)&=-\frac{1}{8}R(5)-\frac{25}{32}(k-8q/5)^2. & &\end{aligned}\tag{84}$$

Subtracting the two lines of (83) first gives

$$\begin{aligned}K_e(2u)-K_e(u)=&\frac{5}{4}R(4/5)-\frac{5}{16}R(1/5)+\frac{15}{8}e^2\\ &+\frac{3}{2}qe-\frac{3}{4}Ce-\frac{1}{2}qC+\frac{7}{8}C^2.\end{aligned}$$

Use (84) and $e=k/2-6q/5+C/5$ to obtain

$$K_e(2u)-K_e(u)=-\frac{1}{8}R(5)-\frac{5}{16}k^2+qk-\frac{11}{10}q^2-\frac{1}{5}qC+\frac{4}{5}C^2.\tag{85}$$

The elimination of $R(5)$ uses only the rational weight-two rows of the same full sums. Inversion at -1 gives $R(-1)=0$; reflection at 2 then gives $R(2)=-q^2$. Distribution at 2, reflection at -4, and distribution at 4, respectively, give

$$\begin{aligned}R(4)&=-6q^2-2R(-2),\\ -4R(-4)+5R(5)&=-\frac{1}{2}(8q-5k)^2,\\ R(16)&=10R(4)-6R(-4).\end{aligned}\tag{86}$$

The middle row uses $\ell(-4)=8q-5k$ from (72). Eliminating $R(4),R(-4)$ gives

$$R(16)+20R(-2)+\frac{15}{2}R(5)=-108q^2+60qk-\frac{75}{4}k^2.\tag{87}$$

In particular, dividing this equation by 60 gives $R(-2)/3+R(16)/60=-9q^2/5+qk-5k^2/16-R(5)/8$. Substituting this in (85) proves

$$K_e(2u)-K_e(u)=\frac{7}{10}q_2^2-\frac{1}{5}q_2C+\frac{4}{5}C^2+\frac{1}{3}R(-2)+\frac{1}{60}R(16).\tag{88}$$

Exact cancellation of the corrected consumer. Define the expression produced by the native source contraction as

$$\begin{aligned}F=&\frac{77}{75}\mathcal{L}+\frac{22}{5}\big(K_e(2u)-K_e(u)\big)-\frac{2233}{125}q_2^2+\frac{1386}{125}q_2q_3\\ &+\frac{154}{25}q_2q_5-\frac{231}{50}q_3q_5+\frac{4334}{375}q_2C-\frac{616}{125}q_3C-\frac{88}{25}C^2\\ &-\frac{22}{15}R(-2)-\frac{11}{150}R(16).\end{aligned}\tag{89}$$

Substituting (81) and (88) into the first two terms gives, in the order

$$(q_2^2,q_2q_3,q_2q_5,q_3q_5,q_2C,q_3C,C^2,R(-2),R(16)),$$

the coefficient vector

$$\left(\frac{2233}{125},-\frac{1386}{125},-\frac{154}{25},\frac{231}{50},-\frac{4334}{375},\frac{616}{125},\frac{88}{25},\frac{22}{15},\frac{11}{150}\right).$$

Every entry cancels its corresponding remaining term in (89). In particular

$$\frac{77}{75}\frac{72}{5}+\frac{22}{5}\frac{7}{10}=\frac{1848}{125}+\frac{385}{125}=\frac{2233}{125},$$

which fixes the quadratic coefficient exactly. Therefore

$$F=0\quad\text{for every prime } p\geq 19 \text{ and every } u\in K \text{ with } u^2=1/5.\tag{90}$$

All identities were proved in the full finite alphabet, and (75) supplies the same prime-field entropy for either root. Thus neither a choice of root sign nor a square-class restriction on 5 occurs in this finite consumer theorem.

4 The whole native field calculation

Throughout this section $p \geq 19$, $N = (p-1)/2$, $d = \lfloor p/6 \rfloor$, and K is any field of characteristic p. We use the finite sums of Section 3.1. Put $q = q_2$, $r = q_3$, $k = q_5$, and $h = \sum_{i=1}^{d} 1/i = -2q - 3r/2$. The last identity follows by splitting the interval $1, \ldots, p-1$ as follows. Its full harmonic sum is zero by $j \mapsto p - j$. Pairing even and odd indices gives $h_N = \ell(-1) = -2q$. Put $m = \lfloor (p-1)/3 \rfloor$. In $\mathbb{Z}[\zeta]/(\zeta^2 + \zeta + 1)$ the integral trace $T = (1-\zeta)^p + (1-\zeta^2)^p$ satisfies $T \equiv 3 \pmod{p}$ and $T^2 = 3^{p+1}$: the ratio of the two bases is $-\zeta$, and $-\zeta^p - \zeta^{-p} + 2 = 3$ for $p \neq 3$. Write $T = 3 + pB$. Reducing $6pB + p^2B^2 = 9pQ_3$ modulo p^2 gives $B = 3q_3/2$ modulo p. The divided binomial entropy (71), evaluated in this integral quadratic ring and then traced, gives $\ell(\zeta) + \ell(\zeta^2) = (3 - T)/p = -3q_3/2$. On the other hand $\zeta^j + \zeta^{2j}$ is 2 when $3 \mid j$ and -1 otherwise; hence this sum is h_m. In h_m, the even indices contribute $h_d/2$. Reflection $j \mapsto p - j$ sends its odd indices to the even indices whose halves run from $m+1$ through N. Thus $h_m = h_d/2 - (h_N - h_m)/2$, or $h_d = h_m + h_N$. This proves the identity in both classes $p = 6d + 1, 6d + 5$. We may work in an algebraic closure of $\mathbb{F}_p$ and choose $u^2 = 1/5$, $w^2 = 3/11$. These roots exist because the closure is algebraically closed. All conclusions below are unchanged for either choice of either root.

4.1 Finite polynomial and residue conventions

The variable x is indeterminate. A reciprocal always means $F^\vee(x) = F(x^{-1})$ with its full signed support. Set

$$\begin{aligned} a_* &= \frac{1}{4}, \quad b_* = 4, \quad s = \frac{x+1}{5}, \\ A_* &= (4-x)(4x-1), \quad L = 33 - 12x, \\ V &= \frac{(4x-1)^2(4-x)}{125}, \\ \widetilde{V} &= s\left(\frac{4x-11}{5}\right)^2 = 1 - V. \end{aligned}$$

For $F = \sum_{j \in \mathbb{Z}} F_j x^j \in K[x, x^{-1}]$ define

$$\Omega(F) = \frac{1}{15} \sum_{j<0} F_j \left(a_*^{-j} - b_*^{-j}\right). \tag{91}$$

This is the residue at zero of F/A_*, expanded in $K((x))$. Indeed $A_* = -4(x - a_*)(x - b_*)$; expanding each simple pole at zero gives (91). It follows, without a support restriction, that

$$\Omega(A_* F) = [x^{-1}]F, \qquad \Omega(A_* F') = 0, \qquad \Omega(F^\vee) = \Omega(F) + \frac{F(a_*) - F(b_*)}{15}. \tag{92}$$

The second identity uses $[x^{-1}]F' = 0$ even in characteristic p. For a polynomial B with $\deg B \leq m$ the same expansion gives

$$\Omega(x^{-m} B) = \frac{a_*^m B(b_*) - b_*^m B(a_*)}{15}. \tag{93}$$

For $m = jp$, retain the distinct factors a_*^j, b_*^j: $a_*^{jp} = a_*^j$, $b_*^{jp} = b_*^j$. They are not all 1.

Here are explicit recipes for the finite sources used below:

$$\begin{aligned} D(x) &= \frac{1}{6} \sum_{j=1}^{N} (1 - 4^j) \frac{s^j}{j}, \\ D_s &= D + D^\vee, \\ a(x) &= q - h + \sum_{j=1}^{N} \frac{\widetilde{V}^j}{2j}, \end{aligned} \tag{94}$$

$$
\begin{aligned}
e &= a^{\vee} - (q-h),\\
W(x) &= \frac{2\epsilon}{15}\left(\sum_{j=1}^{N}\frac{(4^j-1)s^j}{2j} - C\right), \qquad \epsilon = 5^N,\\
A_+(x) &= -\frac{4}{15}(q+C) - \frac{2W(x)}{\epsilon},\\
B_\chi(x) &= \frac{17C}{15} + \frac{17W(x)}{2\epsilon} + \frac{2}{15}\left(4\ell(x/4) - \frac{\ell(4x)}{4}\right),\\
C_-(x) &= \frac{4}{15}(q-C) - \frac{2W(x)}{\epsilon} + \frac{2}{15}\big(\ell(4x) - \ell(x/4)\big),\\
\mathcal{P}(x) &= x^p A_+(x) + B_\chi(x) + x^{-p}C_-(x).
\end{aligned}
$$

The variable in $\ell(cx)$ in these recipes is a polynomial variable; thus each logarithm has all $p-1$ terms. The entropy identity (75) gives $\sum_{j=1}^{N}(4^j-1)(1/5)^j/(2j) = C$. Consequently $W(0) = 0$. Since $4^N = 1$, $\deg D \le N-1$. Coefficient differentiation and a geometric sum give

$$
\begin{aligned}
A_*D' &= \frac{5}{2}(1-s^N), & A_*a' &= \frac{3}{2}\big(4x - 11 - s^N(4x^p-11)\big), \qquad (95)\\
A_*\mathcal{P}' &= \frac{1}{2}s^N(17 - 4x^p - 4x^{-p}) - \frac{A_*}{2x}, & \mathcal{P}(a_*) &= q, \quad \mathcal{P}(b_*) = -q.
\end{aligned}
$$

These are identities of Laurent polynomials, not just of values at prime-field points. For example the $\mathcal{P}$ derivative uses $D(x^p) = 0$, keeps all three phases, and is $\mathcal{R} - 1/(2x)$, where $\mathcal{R} = \frac{1}{2}x^{-p}A_*^{p-1}s^N$. All constants and summation denominators in (94) are units.

Define the three-band kernel

$$
\begin{aligned}
K_0 &= \frac{35\ell(x/4) - 5\ell(4x)}{6}, & K_1 &= \frac{92\ell(x/4) - (29/4)\ell(4x)}{6},\\
K_2 &= \frac{368\ell(x/4) - (29/16)\ell(4x)}{6}, & \mathcal{K} &= K_0 + x^{-p}K_1 + x^{-2p}K_2. \qquad (96)
\end{aligned}
$$

Its coefficient at zero is zero. No band of $\mathcal{K}$ is projected away when it is multiplied by another source.

The complete cubic logarithm source. Put $T(t) = 4t^3 - 3t$, $Q(t) = 1 - t^2$, $B(t) = 4t^2 - 1$, and

$$
U_W = q(1-4t^{2p}) + (1+4t^{2p})E(t) + (2-4t^{2p})E(2t) + 4t^pO(t) - 2t^pO(2t).
$$

Termwise differentiation of the full finite sums gives the four cleared rows

$$
\begin{array}{c|c}
QE'(t) & t - t^p\\
B\{E(2t)\}' & -4(t-t^p)\\
QO'(t) & 1 - t^{p-1}\\
B\{O(2t)\}' & -2(1-t^{p-1}).
\end{array}
$$

Using $T' = 3B$, $1 - T^2 = QB^2$ and $T^p = 4t^{3p} - 3t^p$, the derivative of $q + E(T)$, multiplied by QB, is $3(T - T^p)$. The derivative of U_W, with the same multiplier, is the following fully collected numerator:

$$
\begin{aligned}
QBU_W' &= B(1+4t^{2p})(t-t^p) - 4Q(2-4t^{2p})(t-t^p)\\
&\quad + 4t^pB(1-t^{p-1}) + 4t^pQ(1-t^{p-1})\\
&= 3(4t^3 - 3t - 4t^{3p} + 3t^p).
\end{aligned}
$$

Cancellation of the nonzero polynomials Q, B proves equality of derivatives. Their difference is even and has degree less than $3p$. Its derivative being zero, only degrees $0, p, 2p$ can survive; evenness

removes degree p since p is odd. At $t = 0$ both sides equal q. At $t = 1$, the endpoint rows are $E(1) = -q$, $O(1) = q$, $E(2) = -3r/2$, $O(2) = -2q+3r/2$. Thus $U_W(1) = -3q-5q+3r+4q+4q-3r = 0 = q + E(1)$. The degrees 0 and $2p$ vanish at these two points, proving

$$q + E(T) = U_W. \tag{97}$$

The quadratic distribution identity gives $\ell(t^2) = 2E(t) + 2t^pO(t)$ and $\ell(4t^2) = 2E(2t) + 4t^pO(2t)$. Substitution in (97) therefore yields

$$E(T) = (4t^{2p} - 3)\{E(t) - E(2t)\} + 2\ell(t^2) - \ell(4t^2)/2 - 4qt^{2p}.$$

Now $3D(5t^2 - 1) = E(t) - E(2t)$ by its finite definition, and $a(5t^2 - 1) = q - h + E(T)$. Since $(5t^2 - 1)^p = 5t^{2p} - 1$, the displayed formula descends to

$$a = \frac{3}{5}(4x^p - 11)D + 2\ell(s) - \ell(4s)/2 - 4qx^p/5 + q/5 - h.$$

Descent is injective: substitution by the nonconstant polynomial $5t^2 - 1$ multiplies the degree of any nonzero polynomial by two, and its leading coefficient is a unit. For $F_{\text{aff}} = 2\ell(s) - \ell(4s)/2$ and $T_{\log} = 4\ell(x/4) - \ell(4x)/4$, their cleared derivatives are $A_*F'_{\text{aff}} = 6(x - x^p)$ and $A_*T'_{\log} = 15(x - x^p)$, by the finite geometric derivative. Their degrees are at most $p - 1$, so the difference $F_{\text{aff}} - 2T_{\log}/5$ is constant. Its value at 0 is $3\ell(1/5)/2$; integer entropy and inversion give $\ell(1/5) = k - 8q/5$. Consequently

$$a = x^p(12D/5 - 4q/5) - 33D/5 + \frac{2}{5}(4\ell(x/4) - \ell(4x)/4) + 3k/2 - 11q/5 - h. \tag{98}$$

This determines its full positive and reciprocal bands and its origin; it has not been inferred from a truncated tail.

For a polynomial B of degree less than p, put

$$H_B(c, y) = \sum_{i=1}^{p-1} B_i \sum_{j=1}^{i} \frac{c^{i-j}y^j}{j}. \tag{99}$$

Then $(y-c)\partial_y H_B(c, y) = B(y)-B(c)$ and $H_B(c, 0) = 0$. This remains valid for $c = 0$ and $c = y$ after evaluation. Write $\mathsf{H}(c, y) = H_{-3D}(c, y)$. The same primitive has the explicit triangular expression

$$\begin{aligned} 2\mathsf{H}(c, y) = {} & D_N(4(1 + y)/5, 4(1 + c)/5) - D_N(4/5, 4(1 + c)/5) \\ & - D_N((1 + y)/5, (1 + c)/5) + D_N(1/5, (1 + c)/5). \end{aligned} \tag{100}$$

For a proof, multiply the y derivative of each double sum by $y - c$. For its degree-i terms the inner geometric numerator is $(1+y)^i-(1+c)^i$. The four differences consequently give $2\{-3D(y)+3D(c)\}$. Both sides of (100) have degree at most $N < p$ in y and value zero at $y = 0$. Their difference has zero derivative and constant zero, proving the identity. This proof never divides by $y - c$ or by c. At the six used endpoints, write $D_5 = S(2u, 2u) - S(u, u)$. The exact rows are

$$\begin{array}{c|l} 2\mathsf{H}(1/4, 1/4) & S(1, 1) - S(1/2, 1/2) - A(u) \\ 2\mathsf{H}(4, 1/4) & S(1, 2) - S(1/2, 1) - B(u) \\ 2\mathsf{H}(1/4, 4) & S(2, 1) - S(1, 1/2) - A(u) \\ 2\mathsf{H}(4, 4) & S(2, 2) - S(1, 1) - B(u) \\ 2\mathsf{H}(0, 1/4) & \mathcal{A}_u - D_5 \\ 2\mathsf{H}(0, 4) & \mathcal{B}_u - D_5 \end{array} \tag{101}$$

where $\mathcal{A}_t = S(1, 2t)-S(1/2, t)$ and $\mathcal{B}_t = S(2, 2t)-S(1, t)$. These follow by the four literal argument substitutions in (100), using $u^2 = 1/5$.

4.2 The actual metric and its branch factor

Set

$$b_i = \frac{(6i)!}{(3i)!(i!)^3\,1728^i}, \qquad H_0(z) = \sum_{i=1}^{d} \frac{\overline{b}_i}{i} z^i, \qquad H(x) = H_0(V(x)).$$

For $0 \le j \le d$ let

$$Y_j(t) = \sum_{i=0}^{j} \frac{(-j)_i (j+1)_i}{(i!)^2} \left(\frac{1-t}{2}\right)^i, \quad h_j = \sum_{i=1}^{j} \frac{1}{i}, \quad n_j(t) = -\sum_{i=0}^{j} (h_j - h_i) \frac{(-j)_i (j+1)_i}{(i!)^2} \left(\frac{1-t}{2}\right)^i.$$

Thus $Y = Y_d$ and $n = n_d$ are the actual Legendre and proper polynomials; $n(1) = -h$, $\deg n < d$, and $Y(1) = 1$. Their adjacent recurrences, obtained by clearing the factorial coefficients, are

$$(j+2)Y_{j+2} = (2j+3)tY_{j+1} - (j+1)Y_j, \qquad (j+2)n_{j+2} = (2j+3)tn_{j+1} - (j+1)n_j,$$

with $Y_0 = 1, Y_1 = t, n_0 = 0, n_1 = -1$. The proper Wronskian follows directly from the displayed coefficients. Write $z = (1-t)/2$ and $a_i = (-d)_i (d+1)_i/(i!)^2$. For $0 \le i < d$,

$$(i+1)^2 a_{i+1} = \{i(i+1) - d(d+1)\} a_i.$$

For $\mathscr{L} = z(1-z)\partial_z^2 + (1-2z)\partial_z + d(d+1)$, this recurrence gives $\mathscr{L}Y = 0$. Since $n_i = -(h_d - h_i)a_i$, its coefficient at z^i is

$$[z^i]\mathscr{L}n = (i+1)^2 n_{i+1} + \{d(d+1) - i(i+1)\} n_i = -\frac{d(d+1) - i(i+1)}{i+1} a_i = [z^i]\partial_z Y.$$

At $i = d$ both sides are zero. Thus in the t coordinate $((1-t^2)n')' + d(d+1)n = -2Y'$. Differentiating $(1-t^2)(Yn' - Y'n) + Y^2$ gives zero; its degree is at most $2d < p$ and its value at $t = 1$ is 1. This proves the first identity below:

$$(1-t^2)(Yn' - Y'n) = 1 - Y^2, \qquad Y(t)^2 = \sum_{i=0}^{d} \overline{b}_i (1-t^2)^i. \tag{102}$$

The square identity is the terminating third-order coefficient identity of Section 2.8, after setting $z = 1 - t^2$ and reducing the actual factorial coefficients. The two terminating parameter choices agree because $p = 6d + 1$ or $6d + 5$; both give $-d, d+1$ after the quadratic square transformation.

Put $B = Yn$ and

$$U = 2H_0(1-t^2) + 4tB + 4h, \qquad J = \frac{U}{1-t^2}, \qquad M(t) = \sum_{i \ge 0} \frac{J_i}{i+1} t^{i+1}.$$

These are polynomial definitions. The adjacent recurrences and their initial values give $Y_j(-t) = (-1)^j Y_j(t)$ and $n_j(-t) = (-1)^{j+1} n_j(t)$ by induction. Thus $B = Y_d n_d$ is odd. Therefore U is even and $U(1) = U(-1) = 0$, proving the stated division. Here $\deg J \le 2d - 2$ and $\deg M \le 2d - 1 < p$. All denominators $i + 1$ in M are consequently units.

If an even polynomial $F(t) = \sum F_{2i} t^{2i}$ is given, define $\operatorname{desc} F(x) = \sum F_{2i} s^i$. For an even polynomial F, also define $\operatorname{inv}_2 F(z) = \sum F_{2i}(1-z)^i$. Let $T(t) = 4t^3 - 3t$, and set

$$C_B = \operatorname{desc}\left(\frac{t^p B(T(t))}{5}\right), \quad C_M = \operatorname{desc}\left(\frac{t^p M(T(t))}{10}\right), \quad P_M(z) = \operatorname{inv}_2(M(t)/t)(z). \tag{103}$$

Oddness makes both descended inputs even. Under $x = 5t^2 - 1$, $V(x) = 1 - T(t)^2$ and $A_*(x) = 25(1-t^2)(4t^2-1)$. Applying the product and chain rules gives

$$2A_* C_M' = 3s^N(H + 2h) + 6(4x - 11)C_B, \qquad C_M = \frac{s^{N+1}(4x-11)P_M(V)}{50}. \tag{104}$$

The polynomial pullback by $5t^2-1$ is injective by its leading term. Thus (104) descends from its pulled-back equality. It uses $H+2h$, whereas H alone occurs in the Laurent collector.

The polynomial $x^{N-1}D + x^{N-1}D^\vee$ is divisible by s^{N+1}. Define β_D by the exact quotient

$$x^{N-1}(D+D^\vee) = s^{N+1}\beta_D, \qquad \deg\beta_D \le N-3. \tag{105}$$

To see the divisibility, put $R_D = x^{N-1}D_s$. Both terms are polynomials of degree at most $2N-2$, since $\deg D \le N-1$. The derivative in (95) and its reciprocal give

$$xA_*R_D' - (N-1)A_*R_D = \frac{5}{2}s^N(1-x^N).$$

At $x=-1$, $D(-1)=0$, hence $R_D(-1)=0$. Write $R_D(-1+y) = \sum r_i y^i$. In the coefficient of y^k for $0 \le k < N$, the right side vanishes and the only term involving r_{k+1} on the left is $25(k+1)r_{k+1}$. Starting with $r_0=0$, induction gives $r_1 = \cdots = r_N = 0$; 25 and $1,\dots,N$ are units. Thus $(x+1)^{N+1}$ divides R_D. The quotient by s^{N+1} differs only by the unit 5^{N+1}, and its degree is at most $2N-2-(N+1) = N-3$. Only polynomial division is involved.

Define

$$\begin{aligned}
\mathcal{O} &= \sum_{i=0}^{N-1} \frac{x^{-3i}s^i(4-11x)^{2i+1}}{5^{2i}(2i+1)}, & O_s &= x^{N-1}\mathcal{O},\\
\varphi &= 4x^{-p}-11, & j &= \frac{O_s + s^N\varphi(e+a/3)}{s^N},\\
\delta &= \frac{2}{5}D_sj + \frac{2}{3}De, & N_j &= \frac{2}{5}LD_sj - 2hs^Nj,\\
Z_{52} &= (a-a^\vee)^2 + 6a^\vee(\mathcal{P}-D), & R_{52} &= -\varphi a^\vee - (4x^p-11)a,\\
B_0 &= -2(1-\tfrac{2}{3}x^{-p})\mathcal{K} + \frac{5}{12}x^pR_{52}, & b &= -\frac{2}{3}j - \frac{10}{9}e.
\end{aligned} \tag{106}$$

The denominators $2i+1$ run through $1,3,\dots,p-2$. Although j is written in $K(x)$, D_sj, s^Nj, δ and N_j are finite Laurent polynomials: use (105). Furthermore the complete product

$$\Pi = C_M(B_0+b) = C_MB_0 - \frac{2}{3}s\frac{(4x-11)P_M(V)}{50}(O_s + s^N\varphi(e+a/3)) - \frac{10}{9}C_Me \tag{107}$$

is a finite Laurent polynomial. This is the actual half-scale metric C_M; replacing it by $2C_M$ would change the answer. The only cancellations in (107) use the nonzero polynomial s, whose coefficient of x is $1/5$. They do not require a Hasse value to be nonzero.

4.3 The actual balanced quotient and its five homogeneous terms

Throughout this subsection $p \ge 19$ is prime, $N=(p-1)/2$, $d=\lfloor p/6\rfloor$, and reduction means the coefficient map $\mathbb{Z}_{(p)} \longrightarrow \mathbb{F}_p$, followed, when necessary, by the injective map to a characteristic-p field K. All rational quotients by p below are taken in characteristic zero. Write $F^\iota(x) = F(x^{-1})$, $D = d/dx$, and

$$s = \frac{x+1}{5}, \qquad A = (4-x)(4x-1), \qquad V = \frac{(4x-1)^2(4-x)}{125}, \qquad T_c(t) = 4t^3-3t.$$

For an even polynomial $F(t) = \sum a_{2i}t^{2i}$, put $\operatorname{ct}_2 F(z) = \sum a_{2i}z^i$, $\operatorname{qi} F(z) = \operatorname{ct}_2 F(1-z)$, and $\operatorname{ds} F(x) = \operatorname{ct}_2 F(s)$. Thus $\operatorname{ds} F(5t^2-1) = F(t)$. This substitution is injective: the leading term of a nonzero polynomial of degree e becomes a nonzero term of degree $2e$, because $5 \ne 0$.

Here is the rational family underlying the balanced quotient; it is essential to specify it before reducing. Define two polynomials in (η,t) by

$$a_0 = 1, \qquad a_1 = (1-\eta)t, \qquad a_{j+2} = \frac{(2j+3-\eta)t\,a_{j+1} - (j+1-\eta)a_j}{j+2},$$

$$b_0 = 0, \qquad b_1 = -1, \qquad b_{j+2} = \frac{(2j+3-\eta)t\, b_{j+1} - (j+1-\eta)b_j}{j+2}. \tag{108}$$

Set

$$\eta_* = \frac{p}{3}, \qquad m_* = \prod_{j=1}^{d}\left(1 - \frac{p}{3j}\right), \qquad Y_{\mathbb{Q}}(t) = \frac{a_d(\eta_*, t)}{m_*}, \qquad J_{\mathbb{Q}}(t) = b_d(\eta_*, t).$$

Every $j \le d < p$ is a unit in $\mathbb{Z}_{(p)}$; $m_* \equiv 1 \pmod p$, so m_*^{-1} is also integral. The parity of a_j is j, and that of b_j is $j-1$. The recurrence gives $a_d(\eta, 1) = \prod_{j=1}^{d}(1-\eta/j)$ and

$$(1-t^2)(a_d b_d' - a_d' b_d) - \eta t\, a_d b_d = \prod_{j=1}^{d}(1-\eta/j) - a_d^2.$$

For completeness, these are obtained by subtracting the two three-term recurrences after multiplication by the preceding opposite solution: the resulting adjacent determinant telescopes. Equivalently, the two ladder equations are

$$\begin{aligned}(1-t^2)a_j' &= (j-\eta)a_{j-1} - jta_j,\\ (1-t^2)b_j' &= (j-\eta)(b_{j-1} - tb_j) - a_j,\end{aligned}$$

with the adjacent determinant supplying the preceding displayed metric identity. More explicitly, if $m_j = \prod_{i=1}^{j}(1-\eta/i)$, then

$$(j-\eta)(a_j b_{j-1} - a_{j-1}b_j) = m_j.$$

For $j = 1$ this is $1-\eta = m_1$. The recurrence gives $a_{j+1}b_j - a_j b_{j+1} = (j-\eta)(a_j b_{j-1} - a_{j-1}b_j)/(j+1)$; combine this with $m_{j+1} = (j+1-\eta)m_j/(j+1)$ to prove the determinant formula. Multiplying the two ladder equations by a_j, b_j and subtracting leaves this determinant minus $a_j^2 + \eta t a_j b_j$. Substitution of the recurrence proves each ladder by induction, and their values at $t = 1$ give the endpoint product. After division by m_*, the metric is

$$(1-t^2)(Y_{\mathbb{Q}} J_{\mathbb{Q}}' - Y_{\mathbb{Q}}' J_{\mathbb{Q}}) - \eta_* t Y_{\mathbb{Q}} J_{\mathbb{Q}} = 1 - m_* Y_{\mathbb{Q}}^2. \tag{109}$$

Reduction gives the already specified $Y = P_d$ and proper solution n, and hence

$$(1-t^2)(Yn' - Y'n) = 1 - Y^2. \tag{$*$}$$

In particular these rational definitions give precisely the normalized rank-two node: differentiating the ladder yields $(1-t^2)a_d'' - (2-\eta)ta_d' + d(d+1-\eta)a_d = 0$; its value at 1 is m_*. This differential equation and endpoint determine the polynomial solution, since the top coefficient of any nonzero difference of degree $e < d$ is $(d-e)(d+e+1-\eta_*)$ times its leading coefficient, which is nonzero over $\mathbb{Q}$.

Put

$$e_{\mathbb{Q}}(z) = \sum_{i=1}^{N} \frac{z^i}{2i}, \qquad e_N(z) = \overline{e_{\mathbb{Q}}}(z) = \sum_{i=1}^{N} \frac{z^i}{2i} \in K[z]. \tag{110}$$

Every $2i$, $1 \le i \le N < p$, is a unit before reduction. This half-log polynomial is distinct from the native Laurent carrier $e = a^\vee - (q-h)$, whose notation retains that meaning. Define the following actual rational Laurent polynomials:

$$\begin{aligned} W_{\mathbb{Q}}(x) &= e_{\mathbb{Q}}(4s) - e_{\mathbb{Q}}(s), & c_{\mathbb{Q}} &= e_{\mathbb{Q}}(1/4) - e_{\mathbb{Q}}(4),\\ \phi_{\mathbb{Q}}(x) &= 17 - 4(x^p + x^{-p}), & S_{\mathbb{Q}} &= \frac{\phi_{\mathbb{Q}}}{15}(W_{\mathbb{Q}} + W_{\mathbb{Q}}^{\iota}) + \frac{17 - 4(4^{-p} + 4^p)}{15} c_{\mathbb{Q}},\\ F_{\mathbb{Q}} &= \mathrm{qi}(Y_{\mathbb{Q}}^2)(V), & M_{\mathbb{Q}} &= x^N F_{\mathbb{Q}}^{\iota},\\ C_{B,\mathbb{Q}} &= \mathrm{ds}\left(\frac{t^p}{5}(Y_{\mathbb{Q}} J_{\mathbb{Q}})(T_c(t))\right), & \zeta_{\mathbb{Q}} &= \frac{x^{-p}A^p}{3\,5^{2(p-1)}}, \end{aligned}$$

$$G_{\mathbb{Q}} = \zeta_{\mathbb{Q}} C_{B,\mathbb{Q}}, \qquad G^*_{\mathbb{Q}} = x^N G^{\iota}_{\mathbb{Q}},$$
$$E_{\mathbb{Q}} = \frac{G_{\mathbb{Q}} + G^*_{\mathbb{Q}} + M_{\mathbb{Q}} S_{\mathbb{Q}}}{p}, \qquad \delta_{\mathbb{Q}} = \frac{F_{\mathbb{Q}} - M_{\mathbb{Q}}}{p}. \tag{111}$$

The arguments of both quadratic operations are even, by the parity just proved and the oddness of T_c and p. Their reductions will be denoted f, G, G^*, E, δ, S. In particular $f(5t^2-1) = Y(T_c(t))^2$. Here is the reflection seed, with its forward dependencies explicit. Use the actual original half-cutoff source of (184):

$$F_N(z) = \sum_{n=0}^{N} \frac{(6n)!}{(n!)^3 (3n)!\, 1728^n} z^n, \qquad \chi_{N,\mathbb{Q}} = \frac{F_N(V) - x^N F_N(V)^{\iota}}{p}, \qquad \chi_N = \overline{\chi_{N,\mathbb{Q}}}.$$

The valuation and native reciprocal calculation in Section 2.7, repeated in (184), make $\chi_{N,\mathbb{Q}}$ integral and give $\overline{F_N(V)} = f = x^N f^{\iota}$. In characteristic zero, involutivity of ι gives directly

$$\begin{aligned} p\{\chi_{N,\mathbb{Q}} + x^N \chi^{\iota}_{N,\mathbb{Q}}\} &= F_N(V) - x^N F_N(V)^{\iota} \\ &\quad + x^N \{F_N(V)^{\iota} - x^{-N} F_N(V)\} = 0. \end{aligned}$$

Cancel p in $\mathbb{Q}[x, x^{-1}]$ and then reduce the integral identity to obtain $\chi_N + x^N \chi^{\iota}_N = 0$. The independent original-source attachment (SSA75a), writing its finite carrier as P, proves $\chi_N = fP + \zeta C_B$; the finite comparison calculation (176) proves $P + P^{\iota} = S$. Also $A/x = 17 - 4x - 4x^{-1}$ and $5^{p-1} = 1$ give

$$\zeta = \frac{17 - 4x^p - 4x^{-p}}{3} = \zeta^{\iota}.$$

Thus every term in the involution expands as

$$\begin{aligned} 0 = \chi_N + x^N \chi^{\iota}_N &= fP + \zeta C_B + x^N f^{\iota} P^{\iota} + x^N \zeta^{\iota} C^{\iota}_B \\ &= f(P + P^{\iota}) + \zeta(C_B + x^N C^{\iota}_B) = fS + G + G^*. \end{aligned}$$

Consequently

$$f = x^N f^{\iota}, \qquad G + G^* + fS = 0. \tag{112}$$

These are forward references to independent arguments: the original chi Wronskian and degree bound, its two unit endpoints and full moment source coefficient, and the finite P sums prove the cited attachments without the balanced quotient, its integrality or its homogeneous normalization. There is no mathematical dependency cycle. They imply that both numerators divided by p in (111) lie in $p\mathbb{Z}_{(p)}[x, x^{-1}]$. This proves the integrality of $\delta_{\mathbb{Q}}, E_{\mathbb{Q}}$; reducing their numerators to zero is used only to establish divisibility, and is never used to compute their divided residues.

The full first source of E can also be obtained before reducing. Define

$$L_{\mathrm{cl}} = 2x(x+1)A, \quad \Gamma = -12x^3 + 9x^2 + 4x + 8, \quad U_{\mathbb{Q}} = 3x(x+1)\zeta_{\mathbb{Q}} s^N,$$
$$T_{B,\mathbb{Q}} = \mathrm{qi}(tY_{\mathbb{Q}} J_{\mathbb{Q}})(V), \quad B_{\mathbb{Q}} = \frac{x^{-p} s^N (1 - x^{-N}) A^{p-1}}{2\, 5^{2(p-1)}}, \quad \sigma_{\mathbb{Q}} = \frac{S'_{\mathbb{Q}} - B_{\mathbb{Q}}}{p}.$$

Since $L_{\mathrm{cl}} B_{\mathbb{Q}} = U_{\mathbb{Q}}(1 - x^{-N})$, the rational comparison derivative has an integral first carry: direct differentiation of the two $e_{\mathbb{Q}}$ sums gives $\overline{S'_{\mathbb{Q}}} = \overline{B_{\mathbb{Q}}}$. Here $2(1-z)e'_{\mathbb{Q}} = 1 - z^N$; inserting $z = s, 4s$ and the reflected arguments proves this coefficient identity, including the term of degree N. Let

$$\kappa_* = \frac{1 - m_*}{p} = \sum_{j=1}^{d} \frac{1}{3j} \prod_{i=1}^{j-1} \left(1 - \frac{p}{3i}\right).$$

The product telescoping proves this equality exactly; it proves integrality and $\bar{\kappa}_* = H_d/3 = h/3$, while $\bar{m}_* = 1$. The rational metric (109), the cubic identities

$$1 - T_c^2 = (1 - t^2)(4t^2 - 1)^2, \quad T'_c = 3(4t^2 - 1), \quad V(5t^2 - 1) = 1 - T_c(t)^2$$

give the two cleared, rational proper sources

$$L_{\mathrm{cl}}(F_{\mathbb{Q}}G'_{\mathbb{Q}} - G_{\mathbb{Q}}F'_{\mathbb{Q}}) - p\Gamma F_{\mathbb{Q}}G_{\mathbb{Q}} = U_{\mathbb{Q}}F_{\mathbb{Q}}(1 - m_*F_{\mathbb{Q}} + \eta_*T_{B,\mathbb{Q}}),$$
$$L_{\mathrm{cl}}(M_{\mathbb{Q}}(G^*_{\mathbb{Q}})' - G^*_{\mathbb{Q}}M'_{\mathbb{Q}}) - p(\Gamma - L_{\mathrm{cl}}/(2x))M_{\mathbb{Q}}G^*_{\mathbb{Q}} = -U_{\mathbb{Q}}M_{\mathbb{Q}}(1 - m_*x^{-N}M_{\mathbb{Q}} + \eta_*T^{\iota}_{B,\mathbb{Q}}). \quad (113)$$

The second is the first under $x \mapsto x^{-1}$: $L^{\iota}_{\mathrm{cl}} = x^{-5}L_{\mathrm{cl}}$, $x^3\Gamma^{\iota} = -\Gamma + (x+1)A$, $\zeta^{\iota}_{\mathbb{Q}} = \zeta_{\mathbb{Q}}$, and $x^{2N+3}U^{\iota}_{\mathbb{Q}} = U_{\mathbb{Q}}$. These identities also show where the shift derivative $pL_{\mathrm{cl}}/(2x)$ in the reflected source enters.

To see the complete carry calculation, put

$$A^{\mathrm{pr}}_{\mathbb{Q}} = \frac{M_{\mathbb{Q}}(G_{\mathbb{Q}} + G^*_{\mathbb{Q}})' - (G_{\mathbb{Q}} + G^*_{\mathbb{Q}})M'_{\mathbb{Q}} + M^2_{\mathbb{Q}}B_{\mathbb{Q}}}{p}.$$

Differentiating (111) first gives

$$M_{\mathbb{Q}}E'_{\mathbb{Q}} - E_{\mathbb{Q}}M'_{\mathbb{Q}} = A^{\mathrm{pr}}_{\mathbb{Q}} + M^2_{\mathbb{Q}}\sigma_{\mathbb{Q}}.$$

This proves that $A^{\mathrm{pr}}_{\mathbb{Q}}$ is integral. Substitute $F_{\mathbb{Q}} - M_{\mathbb{Q}} = p\delta_{\mathbb{Q}}$, $1 - m_* = p\kappa_*$, and $\eta_* = p/3$ in (113), add the sources, and use $L_{\mathrm{cl}}B_{\mathbb{Q}} = U_{\mathbb{Q}}(1 - x^{-N})$. Cancelling p over $\mathbb{Q}$ gives the exact identity

$$\begin{aligned} L_{\mathrm{cl}}A^{\mathrm{pr}}_{\mathbb{Q}} &= \Gamma(F_{\mathbb{Q}}G_{\mathbb{Q}} + M_{\mathbb{Q}}G^*_{\mathbb{Q}}) - \frac{L_{\mathrm{cl}}}{2x}M_{\mathbb{Q}}G^*_{\mathbb{Q}} - L_{\mathrm{cl}}\delta_{\mathbb{Q}}G'_{\mathbb{Q}} + L_{\mathrm{cl}}G_{\mathbb{Q}}\delta'_{\mathbb{Q}} \\ &\quad + U_{\mathbb{Q}}\delta_{\mathbb{Q}}(1 - m_*(M_{\mathbb{Q}} + F_{\mathbb{Q}})) + \kappa_*U_{\mathbb{Q}}M^2_{\mathbb{Q}}(1 - x^{-N}) \\ &\quad + \frac{U_{\mathbb{Q}}}{3}(F_{\mathbb{Q}}T_{B,\mathbb{Q}} - M_{\mathbb{Q}}T^{\iota}_{B,\mathbb{Q}}). \end{aligned} \quad (114)$$

In this calculation, $F^2_{\mathbb{Q}} - M^2_{\mathbb{Q}} = p\delta_{\mathbb{Q}}(F_{\mathbb{Q}} + M_{\mathbb{Q}})$ and $F_{\mathbb{Q}} - M_{\mathbb{Q}} = p\delta_{\mathbb{Q}}$ give the term $pU_{\mathbb{Q}}\delta_{\mathbb{Q}}(1 - m_*(F_{\mathbb{Q}} + M_{\mathbb{Q}}))$; the unpaired phase term is $(1 - m_*)U_{\mathbb{Q}}M^2_{\mathbb{Q}}(1 - x^{-N})$. Changing the local multiplier from $F_{\mathbb{Q}}$ to $M_{\mathbb{Q}}$ gives $-pL_{\mathrm{cl}}\delta_{\mathbb{Q}}G'_{\mathbb{Q}} + pL_{\mathrm{cl}}G_{\mathbb{Q}}\delta'_{\mathbb{Q}}$. This specifies all terms surviving the division.

Reduce (114), multiply by f, and use $L_{\mathrm{cl}}(fG' - Gf') = Uf(1 - f)$ and the seed (112). The result is

$$\begin{aligned} fL_{\mathrm{cl}}(fE' - Ef') &= f^3L_{\mathrm{cl}}\sigma - \Gamma f^3S - \frac{L_{\mathrm{cl}}}{2x}f^2G^* + fL_{\mathrm{cl}}G\delta' - L_{\mathrm{cl}}G\delta f' - f^2U\delta \\ &\quad + \frac{h}{3}f^3U(1 - x^{-N}) + \frac{1}{3}f^2U(T_B - T^{\iota}_B). \end{aligned} \quad (115)$$

Here $U = 3x(x+1)\zeta s^N$, $G = \zeta C_B$, $G^* = x^NG^{\iota}$, and $\zeta = (17 - 4x^p - 4x^{-p})/3$; in particular $\zeta' = 0$.

The particular numerator requires every proper and finite-log term. Write $L_V(t) = q_2 + \sum_{i=1}^{N} t^{2i}/(2i)$, and

$$J_L(t) = \frac{R((1+t)/2) - R((1-t)/2)}{2}, \quad R_d(t) = Y(t)(L_V(t) - h) - t^pn(t),$$

$$K_d(t) = n(t)R_d(t) + \frac{t^p}{2}n(t)^2 + Y(t)^2J_L(t).$$

Define

$$T_{\mathrm{loc}} = \zeta \ \mathrm{ds}\left(\frac{t^p}{5}K_d(T_c(t))\right), \qquad T_{\mathrm{pr}} = T_{\mathrm{loc}} + x^NT^{\iota}_{\mathrm{loc}}, \qquad T_0 = UV - UV(-1).$$

The exact UV recipe is in (178) below. Differentiation of these finite sums gives

$$(1 - t^2)L'_V = t - t^p, \qquad (1 - t^2)J'_L = L_V.$$

Together with $(*)$, expansion of K_d gives

$$(1 - t^2)(Y^2K'_d - (Y^2)'K_d) = YR_d + hY^4 + tY^3n.$$

One can check its three summands separately: apply $(*)$ to nR_d and $t^p n^2/2$, noting $(t^p)' = 0$, and apply the second finite-log derivative to $Y^2 J_L$. Cubic descent and reflection therefore give

$$L_{\mathrm{cl}}(fT'_{\mathrm{pr}} - T_{\mathrm{pr}}f') = U\{3(fP + G - \delta) + hf^2(1 - x^{-N}) + f(T_B - T_B^{\iota})\}.$$

The other source identities needed here are

$$2L_{\mathrm{cl}}P' = 2U - L_{\mathrm{cl}}/x, \qquad 2L_{\mathrm{cl}}T'_0 = 2L_{\mathrm{cl}}\sigma - 2\Gamma S + L_{\mathrm{cl}}(S - P)/x. \tag{116}$$

These are the full previously established P and UV sources; their rational first carries, including the resonant terms, are also recorded below.

Form the cleared Laurent numerator

$$Q_6 = 6f^2T_0 - 3f^2P^2 - 3G^2 - 6fGP + 6G\delta + 2fT_{\mathrm{pr}}. \tag{117}$$

The product rule proves that its source is six times (115). More explicitly, use $3f^3$ times the T_0-equation, $-3(f^3P + f^2G)$ times the P-equation, $6(\delta - G - fP)$ times the G-equation, $2f$ times the T_{pr}-equation, and $3L_{\mathrm{cl}}f^2/x$ times $G+G^*+fS = 0$. Every derivative term becomes $L_{\mathrm{cl}}(fQ'_6 - 2Q_6f')$, and the remaining terms are

$$6f^3L_{\mathrm{cl}}\sigma - 6\Gamma f^3S - 3L_{\mathrm{cl}}f^2G^*/x + 6fL_{\mathrm{cl}}G\delta' - 6L_{\mathrm{cl}}G\delta f' - 6f^2U\delta + 2hf^3U(1 - x^{-N}) + 2f^2U(T_B - T_B^{\iota}).$$

Consequently, for $M = Q_6 - 6fE$, cancellation of the nonzero polynomial L_{cl}, rather than any value of f, yields

$$fM' - 2Mf' = 0. \tag{118}$$

Here is the division argument at all zeros of f. Pull back by $x = 5t^2 - 1$, put $y = Y(T_c(t))$, $m = n(T_c(t))$, and $a = (1 - t^2)(4t^2 - 1)/3$. The cubic identities give

$$a(ym' - y'm) = 1 - y^2, \qquad y \neq 0$$

(indeed $y(1) = 1$). If $yB' - ky'B = 0$, for a nonzero $k \in \{1, 2, 3, 4\}$, define the polynomial

$$C = B(y + am') - \frac{a}{k}mB'.$$

Then

$$yC = B\{y^2 + a(ym' - y'm)\} = B, \qquad yC' - (k - 1)y'C = 0.$$

The first equality follows by replacing yB'/k with $y'B$; differentiate $B = yC$ and cancel the globally nonzero y for the second. The pullback of (118), after one cancellation of y, has $k = 4$. Apply this construction for $k = 4, 3, 2, 1$ to obtain $M(5t^2 - 1) = y^4C$, $C' = 0$. Because $M(5t^2 - 1)$ and y^4 are even, C is even. Descend C through ds, and use injectivity of the conic substitution; the chain rule, with $10t \neq 0$ as a polynomial, gives

$$M = f^2R_h, \qquad R'_h = 0.$$

For a Laurent M, first multiply by x^{pr} for sufficiently large r; its derivative is zero, and it clears every negative exponent. Apply the polynomial argument and then multiply the quotient by x^{-pr}. This proves the same assertion with a Laurent R_h. Every division here is a polynomial Bezout construction or cancellation in a domain. It covers zeros of Y, and in particular the Hasse-zero branch.

The support bounds are needed to retain exactly five terms. The coefficient recipes above and below give

$$\deg f = 3d, \quad \operatorname{ord}_0 f = N - 3d, \quad \deg f^2 = 6d < p,$$

and the following upper bounds on a carrier and its inverse:

	$\max\operatorname{supp} F$	$\max\operatorname{supp} F^{\iota}$
f	$3d$	$3d-N$
G	$p+N+3d-1$	p
P	$3N$	p
δ	N	0
T_0	$3N+1$	$3N+1$
T_{pr}	$3p+3d-3$	$3p+3d-N-3$
E	$4N+1$	$2N+3d+1$

For example $\deg C_B \le N+3d-1$, since $\deg(Yn)\le 2d-1$, and the cubic composition followed by multiplication by t^p and contraction divides this degree bound by two. The same argument from $\deg K_d \le p+2d-2$ gives $\deg \mathrm{ds}(t^pK_d(T_c)/5)\le 2p+3d-3$. The δ and E bounds are obtained over $\mathbb{Q}$ from their whole divided numerators, so division introduces no new exponent. For each summand of (117), add the corresponding bounds in the table. With $2N+1=p$, they give

$$\max\operatorname{supp} M \le 3p+6d-3, \qquad \max\operatorname{supp} M^{\iota} \le 3p+6d-2N-3.$$

The leading term of f^2R_h, and then its reflected leading term, therefore give $|j|\le 3p-3$ for every exponent of R_h. Since $R_h'=0$, $j[R_h]_j=0$, so $p\mid j$ whenever $[R_h]_j\neq 0$. Exactly the exponents $-2p,-p,0,p,2p$ remain. No extreme term has been dropped. The rational definition also gives $E-x^NE^{\iota}=-S\delta$. Under $G^*=-fS-G$, $P^{\iota}=S-P$, and the reflected defect $-\delta$, expansion of (117) gives $x^{2N}Q_6^{\iota}=Q_6+6fS\delta$. Thus $M=x^{2N}M^{\iota}$, and cancellation of f^2 gives

$$R_h = c_0+c_1(x^p+x^{-p})+c_2(x^{2p}+x^{-2p}). \tag{119}$$

To fix these three coefficients, retain the original $Z=-P^2/2+G_J$ and define $\beta=[x^{2p}]Z$, $\mu_{\mathrm{bal}}=UV(1/4)-UV(-1)-q_2^2/2$. Here G_J is the full finite J_L pair, equivalently

$$G_J=\frac{1}{3}\left(\zeta\ \mathrm{ds}(t^pJ_L(T_c)/5)+\left(\zeta\ \mathrm{ds}(t^pJ_L(T_c)/5)\right)^{\iota}\right).$$

It is independent of the chosen fifth root, because every coefficient has already descended to $\mathbb{F}_p$. Put $n_2=s^p\,\mathrm{qi}(n^2)(V)$, $T_p=(4s-3)^p$, and $K_{\mathrm{sq}}=\zeta n_2T_p/5$. The exact proper expansions, by conic injectivity, are

$$\begin{aligned}
C_B^2 &= fn_2/25,\\
2fT_{\mathrm{loc}} &= 2G\Sigma+fK_{\mathrm{sq}}+2f^2\zeta\ \mathrm{ds}(t^pJ_L(T_c)/5),\\
2f(x^NT_{\mathrm{loc}}^{\iota}) &= 2G^*\Sigma^*+fx^NK_{\mathrm{sq}}^{\iota}+2f^2\left(\zeta\ \mathrm{ds}(t^pJ_L(T_c)/5)\right)^{\iota},\\
3\delta &= 3fP+3G-\Sigma+\Sigma^*, \qquad \Sigma=\mathrm{qi}(YR_d)(V), \quad \Sigma^*=x^N\Sigma^{\iota}.
\end{aligned}$$

In particular the $T_c^p=t^p(4t^2-3)^p$ factor has been retained. Substitution in (117), using $G+G^*+fS=0$, gives

$$Q_6=6f^2(T_0+Z)+fL_{\mathrm{pr}}, \qquad L_{\mathrm{pr}}=\frac{3\zeta^2n_2}{25}+K_{\mathrm{sq}}+x^NK_{\mathrm{sq}}^{\iota}-2S\Sigma^*. \tag{120}$$

For instance the δ-term supplies $6G\delta=6fGP+6G^2-2G\Sigma+2G\Sigma^*$; the local proper source cancels $-2G\Sigma$, and its reflected source combines $2(G+G^*)\Sigma^*=-2fS\Sigma^*$. The remaining $3G^2=3f\zeta^2n_2/25$ is precisely the square term.

The apparent third positive band of the proper square cancels:

$$5T_p=4x^p-11, \qquad 3\zeta+5T_p=6-4x^{-p}, \qquad \frac{3\zeta^2n_2}{25}+K_{\mathrm{sq}}=\frac{\zeta n_2}{25}(6-4x^{-p}).$$

Since $\deg n \le d-1$, $\deg n_2 \le p+3d-3$. The last expression has upper Laurent exponent at most $2p+3d-3$. The reflected square has upper exponent at most $p+N$. Also S has upper exponent at most $3N$, while $\Sigma^* = x^N\Sigma^\iota$ has upper exponent at most N. Thus $fS\Sigma^*$ has upper exponent at most $3d+4N$, which is strictly below $2p+6d$. The top coefficient of Z is at most $2p$. These bounds and the table show that $6fE$, $6f^2T_0$, and fL_{pr} have zero coefficient at $2p+6d$. Thus at that exponent

$$[Q_6 - 6fE]_{2p+6d} = 6\,\mathrm{lc}(f)^2\beta = \mathrm{lc}(f)^2 c_2.$$

The second equality uses $6d < p$: only the $2p$ term in (119) can meet the top $6d$ term of f^2 at this exponent. As $\mathrm{lc}(f) \neq 0$, $c_2 = 6\beta$. At $x = 1/4$, the rational endpoint recipes give $E = G = T_{\mathrm{pr}} = 0$, $f = 1$, and $P = q_2$. Consequently $R_h(1/4) = 6\mu_{\mathrm{bal}}$.

The branch $x = -1$ needs care. Set $b = -2q_2^2 - 3q_2q_3/2$. The full comparison phase in (111) gives

$$S_{\mathbb{Q}}(-1) = \frac{17 - 4(4^{-p} + 4^p)}{15}\big(e_{\mathbb{Q}}(1/4) - e_{\mathbb{Q}}(4)\big).$$

Its quotient by p reduces to b: the phase quotient is $-30q_2$, and the endpoint difference is $q_2 + 3q_3/4$. The rational comparison is inversion invariant, so differentiating $S_{\mathbb{Q}}(x) = S_{\mathbb{Q}}(x^{-1})$ at -1 gives $S'_{\mathbb{Q}}(-1) = -S'_{\mathbb{Q}}(-1) = 0$. The rational proper carriers and their first jets vanish there. Evaluating and differentiating $pE_{\mathbb{Q}} = G_{\mathbb{Q}} + G^*_{\mathbb{Q}} + M_{\mathbb{Q}}S_{\mathbb{Q}}$, and cancelling the nonzero rational p, therefore gives $E(-1) = f(-1)b$, $E'(-1) = f'(-1)b$. The conic carriers G, T_{loc} have their value and first two derivatives zero at -1, since they contain t^p, corresponding to order $(p+1)/2$ in $x+1$. Also $T_0(-1) = P(-1) = 0$. Either $f(-1) \neq 0$ or $f'(-1) \neq 0$: if d is even,

$$Y(0) = (-1)^{d/2}2^{-d}\binom{d}{d/2} \neq 0;$$

if d is odd,

$$Y(0) = 0, \quad Y'(0) = (-1)^{(d-1)/2} d\, 2^{-(d-1)}\binom{d-1}{(d-1)/2} \neq 0, \quad f'(-1) = 9Y'(0)^2/5.$$

All factorials here have arguments $< p$. In the first case evaluate $M = f^2R_h$. In the second differentiate it twice: every G-term and proper term vanishes by the preceding jets, while $f(-1) = 0$ kills their unsupplied second jets. What remains is

$$2f'(-1)^2R_h(-1) = 2f'(-1)^2(-6b).$$

Thus in both cases $R_h(-1) = -6b$. Since $4^p = 4$ in the prime field, the two scalar equations are

$$c_0 - 2c_1 + 2c_2 = -6b, \qquad c_0 + \frac{17}{4}c_1 + \frac{257}{16}c_2 = 6\mu_{\mathrm{bal}}.$$

Their difference is $\frac{25}{4}c_1 + \frac{225}{16}c_2 = 6(\mu_{\mathrm{bal}} + b)$. Solving, with $c_2 = 6\beta$, gives

$$\begin{aligned} c_1 &= \frac{24}{25}(\mu_{\mathrm{bal}} + b) - \frac{9}{4}(6\beta), \\ c_0 &= \frac{48}{25}(\mu_{\mathrm{bal}} + b) - 6b - \frac{13}{2}(6\beta). \end{aligned} \tag{121}$$

Equations (120) and $Q_6 - 6fE = f^2R_h$, cancelling the globally nonzero f, now give the actual normalization

$$E = f\left(T_0 + Z - \frac{R_h}{6}\right) + \frac{L_{\mathrm{pr}}}{6}. \tag{122}$$

4.4 Trace transport and the actual divided consumer

Here are the complete trace conventions. Put

$$r(z)=\frac{1}{(1-4z)(1-z/4)}=\sum_{j\ge 0}\frac{4}{15}(4^{j+1}-4^{-(j+1)})z^j.$$

For $F=\sum_j F_j x^j$, in either $\mathbb{Q}[x,x^{-1}]$ or $K[x,x^{-1}]$, define

$$\tau_m(F)=-\left(-\frac14\right)^m\sum_j F_j[z^{\,j-(2m-1)}]r(z)^m,\qquad [z^k]r^m=0\quad(k<0).\tag{123}$$

Thus τ_m is minus the coefficient of x^{-1} in the expansion of F/A^m at infinity, with precisely the sign in (123). The origin functional is

$$o_m(F)=\left(-\frac14\right)^m\sum_j F_j[z^{-j-1}]r(z)^m=[x^{-1}]_0(F/A^m).$$

Every coefficient of r^m is p-integral, since its only fixed denominators are powers of $2,3,5$. Both are finite sums for a Laurent polynomial; consequently they commute with reduction and with field extension. Directly from their coefficient formulas,

$$\tau_{m+r}(A^rF)=\tau_m(F),\qquad \tau_m(F)=-o_m(x^{2m-2}F^\iota),\qquad \tau_m(F')=m\,\tau_{m+1}(A'F).\tag{‡}$$

The last equality is the coefficient of the derivative $(F/A^m)'$, whose x^{-1} coefficient is zero.

Write

$$R_w=x(4x-11),\quad d_x=3x^{-1}(4-11x),\quad \Delta_w=AR_w'-R_wA'=24x^2-32x+44.$$

The rational trace rows actually used are

$$\begin{aligned}\kappa_{\mathbb{Q}}(F)&=\tau_p(3\,5^{p-1}R_w^pF),\\ \lambda_{\mathbb{Q}}(F)&=\tau_{p+1}(3\,5^{p-1}R_w^{p-1}\Delta_wF),\\ \xi_{\mathbb{Q}}(E_{\mathbb{Q}})&=\tau_{p+1}(3\,5^{p-1}R_w^pd_xx^{-N}E_{\mathbb{Q}}),\\ t_{\mathbb{Q}}(F)&=\tau_1(d_xF).\end{aligned}$$

Let $H_{\mathbb{Q},0}(z)=\sum_{i\ge1}[z^i]\,\mathrm{qi}(Y_{\mathbb{Q}}^2)\,z^i/i$, $H_{\mathbb{Q},c}=H_{\mathbb{Q},0}(V)^\iota$, $\mathcal{L}_{\mathbb{Q}}=e_{\mathbb{Q}}(1-V)$, and $\Pi_{\mathbb{Q}}=x^{-N}\delta_{\mathbb{Q}}\mathcal{L}_{\mathbb{Q}}$. All sums in $H_{\mathbb{Q},0}$ have $i\le d<p$. The common, balanced, and period rows are

$$C_{\mathbb{Q}}=\kappa_{\mathbb{Q}}(H_{\mathbb{Q},c}\sigma_{\mathbb{Q}})+\lambda_{\mathbb{Q}}(H_{\mathbb{Q},c}S_{\mathbb{Q}}),\quad B_{\mathbb{Q}}^{\mathrm{tr}}=\xi_{\mathbb{Q}}(E_{\mathbb{Q}}),\quad P_{\mathbb{Q}}^{\mathrm{tr}}=t_{\mathbb{Q}}(\Pi_{\mathbb{Q}}).$$

They are p-integral because their complete inputs are integral, and their weights in (123) are integral. The H in this trace is $H_c=H_0(V)^\iota$. The unreflected $H_0(V)$ has constant-normalized primitive $H_0(0)=0$; the first Green carrier used earlier has $H_0(V)+2h$. These two primitives have not been identified with one another.

In the field put

$$\kappa(F)=\tau_p(3R_w^pF),\qquad \mathcal{W}(F)=\tau_{p+1}(3R_w^pd_xF),\qquad \mathcal{W}_{\mathrm{low}}(F)=\mathcal{W}(x^{-N}fF).$$

The exact rational Euler source reduces to

$$AH_c'=d_x(x^{-N}f-1).$$

Multiplication by $3R_w^pF$ and the first identity (‡) give

$$\mathcal{W}_{\mathrm{low}}(F)=\mathcal{W}(F)+\kappa(H_c'F).$$

Because $(R_w^p)' = 0$, $\kappa(H_c' F) + \kappa(H_c F') = 0$. Hence $\mathcal{W}_{\mathrm{low}}(F) = \mathcal{W}(F)$ whenever $F' = 0$. In particular $\mathcal{W}_{\mathrm{low}}(R_h) = \mathcal{W}(R_h)$, and subtracting the branch constant in T_0 does not affect its trace.

The two nonconstant homogeneous weights can be calculated without assigning either homogeneous coefficient. Since $3R_w^p = -3A^p + 18x^p - 12$, and

$$\tau_1(d_x x^j) = \begin{cases} -(2\,4^j + 4^{-j}), & j > 0, \\ 0, & j \le 0, \end{cases}$$

the first weight is $\mathcal{W}(x^p + x^{-p}) = -3(-33/4) = 99/4$. The terms at pole order $p+1$ involving $d_x x^p, d_x x^{2p}$ vanish, since their exponents are below $2p+1$. For the next phase the only remaining high weight is

$$\tau_{p+1}(d_x x^{3p}) = \frac{33}{16}.$$

Indeed $r(z)^{p+1} = r(z^p) r(z)$, and the required coefficients have degrees $p-2, p-1 < p$, so they equal the coefficients of r. They are $r_{p-2} = 0, r_{p-1} = 1$; multiplying the coefficients $12, -33$ of $d_x x^{3p}$ by $-(-1/4)^{p+1} = -1/16$ gives $33/16$. The cleared lower pole gives $\tau_1(d_x x^{2p}) = -513/16$. Therefore

$$\mathcal{W}(x^{2p} + x^{-2p}) = -3(-513/16) + 18(33/16) = 2133/16.$$

Nonpositive exponents, including the constant, have zero weight by (123). Inserting (121) yields

$$\begin{aligned} \frac{1}{9}\mathcal{W}(R_h) &= \frac{11}{4} c_1 + \frac{237}{16} c_2 \\ &= \frac{66}{25}(\mu_{\mathrm{bal}} + b) + \left(-\frac{99}{16} + \frac{237}{16} \right) 6\beta = \frac{66}{25}(\mu_{\mathrm{bal}} + b) + \frac{207}{4}\beta. \end{aligned} \tag{124}$$

To display the unchanged low source, use the full B_γ and P already defined from the native chart, and put

$$\begin{aligned} &\Omega_{\mathrm{HF}} = \Delta_w S - A R_w (B_\gamma + Z'), \\ &D_{\mathrm{low}} = 3R_w^{p-1} H_c \Omega_{\mathrm{HF}} + 3R_w^p d_x x^{-N} L_{\mathrm{pr}}/6 - A^p (d_x \Pi), \qquad \Pi = x^{-N} \delta\, e_N (1 - V). \end{aligned} \tag{125}$$

No proper square, reflected Σ, or period term has been removed. For reference, the derivative used here is

$$Z' = R_{\mathrm{src}} \left(-P + \frac{L_V^{\mathrm{desc}} - x^{-N} (L_V^{\mathrm{desc}})^\iota}{3} \right) + \frac{P}{2x}, \qquad L_{\mathrm{cl}} P' = U - L_{\mathrm{cl}}/(2x), \quad L_{\mathrm{cl}} R_{\mathrm{src}} = U.$$

The full common source after reduction is

$$\bar{C}_{\mathbb{Q}} = \tau_{p+1}(3R_w^{p-1} \Delta_w H_c S) - \tau_p(3R_w^{p-1} H_c B_\gamma) - \kappa(H_c' U V).$$

This follows by applying integration by parts to $\kappa(H_c \sigma)$, using the full UV source (116); $R_w' = 4x - 11 + 4x$ enters through $\Delta_w = A R_w' - R_w A'$. Add $\kappa(H_c' UV) + \kappa(H_c' Z)$. The UV terms cancel, and the identity $\kappa(H_c' Z) = -\kappa(H_c Z')$ makes what remains $\tau_{p+1}(3R_w^{p-1} H_c \Omega_{\mathrm{HF}})$. From (122) the balanced row is

$$\bar{B}_{\mathbb{Q}}^{\mathrm{tr}} = \mathcal{W}(UV) + \kappa(H_c' UV) + \mathcal{W}(Z) + \kappa(H_c' Z) - \mathcal{W}(R_h)/6 + \tau_{p+1}(3R_w^p d_x x^{-N} L_{\mathrm{pr}}/6).$$

Finally $\bar{P}_{\mathbb{Q}}^{\mathrm{tr}} = \tau_1(d_x \Pi) = \tau_{p+1}(A^p d_x \Pi)$. Collecting these three equations and (124), with the displayed cancellations, gives

$$\frac{2}{3}(\bar{C}_{\mathbb{Q}} + \bar{B}_{\mathbb{Q}}^{\mathrm{tr}} - \bar{P}_{\mathbb{Q}}^{\mathrm{tr}}) = \frac{2}{3}\mathcal{W}(UV + Z) - \frac{66}{25}(\mu_{\mathrm{bal}} + b) - \frac{207}{4}\beta + \frac{2}{3}\tau_{p+1}(D_{\mathrm{low}}). \tag{126}$$

The two complete finite traces. We compute the actual carriers in (178) and (120). Work in characteristic $p \geq 19$, put $N = (p-1)/2$, $q = q_2$, $r = q_3$, $k = q_5$, and write $a = 1/4$, $b_* = 4$, $\epsilon = 5^N$, $s = (x+1)/5$. Here μ_{bal} is the balanced branch scalar defined above, and $b = -2q^2 - 3qr/2$ remains the branch carry in (126); it is distinct from the endpoint b_*. The pole polynomial is $A_* = (4-x)(4x-1)$. All divisions below have denominators with prime factors among $2, 3, 5$, or indices between 1 and $p-1$. In particular $\epsilon \neq 0$. Use the field half-logarithm e_N from (110), and put

$$\Delta e(z) = e_N(4z) - e_N(z).$$

The full $\ell, R, E, O, D_N, S, A(t), B(t), \chi(t)$ retain the definitions in (61)–(63). Thus, for clarity,

$$D_m(v,z) = \sum_{i=1}^{m} \sum_{j=1}^{i} \frac{v^j z^{i-j}}{ij} \quad (0 \leq m < p), \qquad S(v,z) = D_N(v^2, z^2).$$

Set

$$\begin{aligned} \mathcal{A}_t &= S(1,2t) - S(1/2,t), & \mathcal{B}_t &= S(2,2t) - S(1,t), \\ D_5 &= S(2u,2u) - S(u,u), & T_5 &= A(u) - 2\mathcal{A}_u + \mathcal{B}_u + D_5, \\ \mathcal{G}(z) &= D_N(4z,4) - D_N(z,1) + 2D_N(4z,1) - 2D_N(z,1/4), & D_0 &= D_N(4/5,0) - D_N(1/5,0). \end{aligned}$$

The root satisfies $u^2 = 1/5$; both choices are allowed. In later formulas denoted $G(z)$, the finite function is this same $\mathcal{G}(z)$; the proper carrier $G = \zeta C_B$ is distinct. The finite P below is $\mathcal{P}$ from (94), not the target scalar in (305). Every triangular sum includes $j = i$.

Endpoint weights and coefficient windows. For a finite Laurent polynomial $F = \sum F_j x^j$, put

$$Q(F) = \sum_{j>0} (-2b_*^j - a^j) F_j.$$

The partial fractions

$$d_x/A_* = \frac{2}{x-4} + \frac{1}{x-a} - \frac{3}{x}, \qquad d_x = 3x^{-1}(4-11x),$$

expanded at infinity, prove $Q(F) = \tau_1(d_x F)$. Consequently

$$Q(B) = -2B(b_*) - B(a) + 3B(0), \qquad Q(B^\iota) = 0 \quad (B \in K[x]).$$

These formulas apply coefficientwise to the full signed support. Write $r_*(z) = 1/((1-4z)(1-z/4))$, so $r_*(0) = 1$, $[z]r_* = 17/4$, and $r_*^{p+1} = r_*(z) r_*(z^p)$. If $\max \operatorname{supp} F < 3p$, substitution in (123) gives

$$\tau_{p+1}(d_x F) = -\frac{1}{4} Q(x^{-2p} F), \qquad \tau_{p+1}(d_x x^p F) = -\frac{1}{4} Q(x^{-p} F) - \frac{17}{16} Q(x^{-2p} F).$$

Indeed the first coefficient window is below p, and the second is below $2p$; the latter adds only the coefficient $[z]r_* = 17/4$ in the shifted window. The convention $[z^j]r_* = 0$ for $j < 0$ covers every negative exponent. Since

$$3R_w^p = -3A_*^p + 18x^p - 12,$$

clearing A_*^p in the first term gives

$$\mathcal{W}(F) = -3Q(F) - \frac{9}{2} Q(x^{-p} F) - \frac{129}{8} Q(x^{-2p} F), \quad \max \operatorname{supp} F < 3p. \tag{127}$$

Here $18(-17/16) - 12(-1/4) = -129/8$. The last term vanishes when the upper exponent is $< 2p$.

The actual polynomial and its unit primitives are

$$W(x) = \frac{2\epsilon}{15} \{\Delta e(s) - \Delta e(1/5)\}, \qquad C = \Delta e(1/5) = E(2u) - E(u),$$

$$I_z(y) = \sum_{i=1}^{N-1} W_i \sum_{j=1}^{i} \frac{z^{i-j} y^j}{j}, \qquad W_i = [x^i]W. \tag{128}$$

The top coefficient of W is zero because $4^N = 1$; also $W(0) = 0$. Thus $\deg W \le N-1$, $\deg I_z < p$, $I_z(0) = 0$, and $(y-z)I_z'(y) = W(y) - W(z)$. The same coefficient computation as in (100) gives the complete finite expression

$$I_z(y) = \frac{\epsilon}{15}\{D_N(4s_y, 4s_z) - D_N(s_y, s_z) - D_N(4/5, 4s_z) + D_N(1/5, s_z)\}, \quad s_v = (1+v)/5. \tag{129}$$

To verify it, multiply its derivative by $y - z$. The inner geometric numerator is $(1+y)^i - (1+z)^i$, giving $W(y) - W(z)$; both sides have degree $< p$ and origin zero. Their difference therefore has zero derivative and constant zero. This argument also covers $z = 0, -1$ and $z = y$.

The Frobenius factor

$$K_0 = \frac{17 - 4(x^p + x^{-p})}{2\epsilon} = \frac{x^{-p} A_*^p}{2\epsilon}$$

gives, without a support bound,

$$\mathcal{W}(K_0 F) = \frac{3}{2\epsilon} Q((4x^p - 11)F). \tag{130}$$

This follows by clearing A_*^p and using $R_w^p x^{-p} = 4x^p - 11$. For $B \in K[x]$, $\deg B < p$, the two reflected endpoint rows are

$$Q(x^p(B + B^\iota)) = -\frac{33}{4}\{B(a) + B(b_*)\}, \qquad Q(B + B^\iota) = -2B(b_*) - B(a) + 3B(0).$$

Indeed $a^p = a$, $b_*^p = b_*$, and reflection exchanges the endpoints. Hence

$$\mathcal{W}(K_0(B + B^\iota)) = -\frac{33}{2\epsilon}\{2B(a) + B(b_*) + 3B(0)\}, \quad \mathcal{W}(B + B^\iota) = 3\{B(a) + 2B(b_*) - 3B(0)\}. \tag{131}$$

The original UV carrier and its branch correction. The unit primitives I_z belong to W; the native harmonic polynomial H_0 in Section 4.2 retains its distinct meaning. Write $w_z = W(z)$, $I_m = I_{-1}$, and set

$$\begin{aligned}
I_{\mathrm{sym}} &= I_0 + I_0^\iota - I_a - I_{b_*} - I_a^\iota - I_{b_*}^\iota - (I_m + I_m^\iota)/2, \\
J_a &= \frac{2}{\epsilon}(x^{-p} - a)\ell(4x), & J_b &= \frac{2}{\epsilon}(x^{-p} - b_*)\ell(x/4), \\
J_2 &= \frac{2}{15}\{(x^{-p} - a)R(4x) - (x^{-p} - b_*)R(x/4)\}, & Z_E &= w_a + w_{b_*} + \frac{4\epsilon C}{15}, \\
U_{\mathrm{can}} &= K_0 I_{\mathrm{sym}} - Z_E(J_a + J_b) - J_2/2, & U &= U_{\mathrm{can}} - u_0.
\end{aligned}$$

The actual branch constant is $u_0 = U_{\mathrm{can}}(-1)$. Explicitly,

$$u_0 = \frac{25}{2\epsilon}\{2I_0(-1) - 2I_a(-1) - 2I_{b_*}(-1) - I_m(-1)\} + \frac{5Z_E}{\epsilon}\ell(-4) + \frac{1}{6}R(-4), \tag{132}$$

where every $I_z(-1)$ is the sum (129). This is the branch normalization of the source recipe in Section 4.5. For the actual counterterm in (178), retain

$$\begin{aligned}
\mathcal{N} &= (x^{-p} - 4)\ell(x/4) + (x^{-p} - a)\ell(4x), \\
K_1 &= 2(q-k)K_0 + 15q/\epsilon + 2\mathcal{N}/\epsilon, \\
W_1 &= -2\epsilon q\{e_N(4s) - e_N(4/5)\}/15, & T &= K_1 W + K_0 W_1, \\
J_{\mathrm{pr}} &= \epsilon J_a/15, & G_{\mathrm{cl}} &= K_0(I_a + I_{b_*} - I_0) + w_a J_a + w_{b_*} J_b.
\end{aligned}$$

Here $\operatorname{End} F = F + F^\iota - F(a) - F(b_*)$, and $\operatorname{Can}_0 T = T - [x^0]T - ([x^p]T)x^p$; its $-p$ slot is zero. The recipe says

$$UV = U - \operatorname{End}(\operatorname{Can}_0 T) + q \operatorname{End} J_{\mathrm{pr}} + \operatorname{End} G_{\mathrm{cl}}.$$

By (64), $J_a^\iota = J_b$; both endpoint values of each J are zero. Also $K_0(a) = K_0(b_*) = 0$. Thus the I_0, I_a, I_{b_*} pairs cancel, and

$$-Z_E + \epsilon q/15 + w_a + w_{b_*} = \epsilon(q - 4C)/15.$$

With $t_p = [x^p]T$, the resulting full identity is

$$\begin{aligned} UV &= UV_0 + t_p\Phi, & \Phi &= x^p + x^{-p} - 17/4, \\ UV_0 &= -\operatorname{End} T - K_0(I_m + I_m^\iota)/2 + \epsilon(q - 4C)(J_a + J_b)/15 - J_2/2 - u_0. \end{aligned} \tag{133}$$

No origin constant is lost: $\operatorname{End}(T_0) = 0$, while $\operatorname{End}(t_p x^p) = t_p\Phi$. The endpoint is $UV(a) = -u_0$, since all displayed nonconstant terms in (133) vanish at a. Therefore, writing $\mu_{\mathrm{bal},0} = -u_0 - UV_0(-1) - q^2/2$,

$$\mu_{\mathrm{bal}} = \mu_{\mathrm{bal},0} + \frac{25}{4}t_p, \qquad \mathcal{W}(UV) = \mathcal{W}(UV_0) + \frac{99}{4}t_p.$$

Here $\Phi(-1) = -25/4$, $\Phi(a) = 0$, and (124) gives its weight $99/4$. The actual p slot cancels because $(2/3)(99/4) - (66/25)(25/4) = 0$.

The full weighted product row. Put $W_s = W + W^\iota$. For any $z \in K$, direct finite convolution gives

$$\begin{aligned} Q(W_s\ell(z/x)) &= -2I_{b_*}(z) - I_a(z) + 3I_0(z), \\ Q(W_s\ell(zx)) &= -2(w_a + w_{b_*})\ell(b_*z) - (w_a + w_{b_*})\ell(az) + 2I_a(z) + I_{b_*}(z). \end{aligned} \tag{134}$$

For the first formula, only $1 \le j < i < p$ contributes; at endpoint v, the positive part is $\sum_i W_i \sum_{j<i} z^j v^{i-j}/j = I_v(z) - I_0(z)$. For the second, the positive part of $W^\iota\ell(zx)$ at v is

$$\sum_i W_i \sum_{j>i}^{p-1} \frac{z^j v^{j-i}}{j} = W(v^{-1})\ell(zv) - I_{v^{-1}}(z).$$

Adding $W(v)\ell(zv)$ proves the stated row. The omitted $j = i$ term is exactly the origin term $I_0(z)$, which has been retained in the first formula.

Reindexing $j \mapsto p - j$ in the complete finite logarithm gives

$$\mathcal{N} = -a\ell(4/x) - 4\ell(a/x) - 4\ell(ax) - a\ell(4x).$$

For the negative logs the second window of (127) is zero. For the positive logs it is the strict tail computed below in (141). Applying these two windows gives

$$\begin{aligned} \mathcal{E}_W := \mathcal{W}(\mathcal{N}W_s) &= \frac{15}{2}Q(W_s\ell(a/x)) - \frac{15}{4}Q(W_s\ell(4/x)) \\ &\quad + 12Q(W_s\ell(ax)) + \frac{3}{4}Q(W_s\ell(4x)) \\ &= -\frac{3}{4}\ell(16)(w_a + w_{b_*}) + \frac{33}{2}I_a(a) + \frac{21}{4}I_a(b_*) \\ &\quad - 3I_{b_*}(a) + \frac{33}{4}I_{b_*}(b_*) + \frac{45}{2}I_0(a) - \frac{45}{4}I_0(b_*). \end{aligned}$$

To see the shifted coefficients explicitly, the positive-log part is $12Q(W_s\ell(ax)) + (3/4)Q(W_s\ell(4x))$. Its second window adds $(9/2)\{4Q(x^{-p}W_s\ell(ax)) + aQ(x^{-p}W_s\ell(4x))\}$. The strict-product formula changes this to $-(9/2)Q(W_s\ell(4/x)) - (9/2)Q(W_s\ell(a/x))$. Combining with $3aQ(W_s\ell(4/x)) + 12Q(W_s\ell(a/x))$ gives $-15/4, 15/2$, as displayed. All products have upper exponent $< 2p$. The

displayed log pairs give $\mathcal{N}^\iota = \mathcal{N}$. Also $K_0^\iota = K_0$, $W_1(0) = 0$, and $\deg W_1 \le N < p$. Thus the symmetric part of T is

$$2(q-k)K_0W_s + 15qW_s/\epsilon + 2\mathcal{N}W_s/\epsilon + K_0(W_1 + W_1^\iota).$$

Using (131) on these remaining terms gives

$$\mathcal{W}(\operatorname{End} T) = -\frac{33(q-k)}{\epsilon}(2w_a + w_{b_*}) + \frac{45q}{\epsilon}(w_a + 2w_{b_*}) + \frac{2}{\epsilon}\mathcal{E}_W - \frac{33}{2\epsilon}(2W_1(a) + W_1(b_*)). \quad (135)$$

The endpoint subtraction is constant, and its weight is zero. Furthermore (127) and (64) give

$$\mathcal{W}(J_a + J_b) = -\frac{3}{2\epsilon}\ell(16), \qquad \mathcal{W}(J_2) = -\frac{1}{10}R(16).$$

For example the $x^{-p}\ell(cx)$ terms are nonpositive and have zero weight; the positive logs contribute $3\{2\ell(4c) + \ell(ac)\}$. Thus (133) gives

$$\frac{2}{3}\mathcal{W}(UV) - \frac{66}{25}\mu_{\mathrm{bal}} = -\frac{2}{3}\mathcal{W}(\operatorname{End} T) + \frac{11}{2\epsilon}(2I_m(a) + I_m(b_*)) - \frac{q-4C}{15}\ell(16) + \frac{1}{30}R(16) - \frac{66}{25}\mu_{\mathrm{bal},0}. \quad (136)$$

The actual endpoint and branch values. The half-log endpoint rows are $e_N(1) = -q$, $e_N(1/4) = q - 3r/4$, $e_N(4) = -3r/2$. They give the complete table

	a	b_*
$15W/\epsilon$	$-4q + 3r/2 - 2C$	$2q - 3r - 2C$
$15W_1/(2\epsilon)$	$q\{q + e_N(4/5)\}$	$q\{3r/2 + e_N(4/5)\}$.

At the branch, $\mathcal{N}(a) = \mathcal{N}(b_*) = 0$, $\mathcal{N}(-1) = -5\ell(-4)/2$, and

$$K_0(-1) = 25/(2\epsilon), \quad K_1(-1) = 0, \quad T(-1) = 5qe_N(4/5)/3, \quad T(a) + T(b_*) = 15q(w_a + w_{b_*})/\epsilon.$$

Indeed $K_1(-1) = \{25(q-k) + 15q - 5\ell(-4)\}/\epsilon = 0$, using $\ell(-4) = 8q - 5k$. Also

$$I_m(-1) = -\epsilon D_0/15, \quad (J_a + J_b)(-1) = -5\ell(-4)/\epsilon, \quad J_2(-1) = -R(-4)/3.$$

Substitution in the definition of $\mu_{\mathrm{bal},0}$ first gives

$$\mu_{\mathrm{bal},0} = \frac{10}{3}qe_N(4/5) - \frac{15q}{\epsilon}(w_a + w_{b_*}) + \frac{25}{2\epsilon}I_m(-1) + \frac{q-4C}{3}\ell(-4) - \frac{1}{6}R(-4) - \frac{1}{2}q^2,$$

and therefore

$$\mu_{\mathrm{bal},0} = \frac{10}{3}qe_N(4/5) + \frac{3}{2}q^2 + \frac{3}{2}qr + 4qC + \frac{q-4C}{3}\ell(-4) - \frac{5}{6}D_0 - \frac{1}{6}R(-4). \quad (137)$$

For the other I_m row, (129) at $s_m = 0$ gives

$$2I_m(a) + I_m(b_*) = \frac{\epsilon}{15}\{4R(-2) - 3D_0\}.$$

In detail the nonconstant sum is $D_N(1,0) - 2D_N(1/4,0) + D_N(4,0)$. Its first term is $\sum_{j=1}^N 1/j^2 = 0$, since $R(1) = 2\sum_{j=1}^N 1/j^2 = 0$. Splitting $R(-2)$ into even indices $2j$ and reflected odd indices $p-2j$ gives $4R(-2) = D_N(4,0) - 2D_N(1/4,0)$.

Put

$$H_{\mathrm{ord}} = \frac{-90I_0(a) + 45I_0(b_*) - 66I_a(a) - 21I_a(b_*) + 12I_{b_*}(a) - 33I_{b_*}(b_*)}{2\epsilon}.$$

For completeness, the arithmetic collection in (136), including the carry b, is shown below. Columns I, II, III are respectively $-2\mathcal{W}(\mathrm{End}\,T)/3 - 2H_{\mathrm{ord}}/3$, $11(2I_m(a) + I_m(b_*))/(2\epsilon) - (q-4C)\ell(16)/15 + R(16)/30$, and $-66(\mu_{\mathrm{bal},0} + b)/25$. Use $\ell(16) = -64q + 15r + 15k$, from (65) and (72).

	I	II	III	sum
q^2	$8/3$	$64/15$	$-143/25$	$91/75$
qr	$78/5$	-1	0	$73/5$
r^2	$-3/2$	0	0	$-3/2$
qk	$34/5$	-1	$22/5$	$51/5$
rk	$-3/2$	0	0	$-3/2$
qC	$304/15$	$-256/15$	$88/5$	$104/5$
kC	$24/5$	4	$-88/5$	$-44/5$
$qe_N(4/5)$	$22/5$	0	$-44/5$	$-22/5$
$R(-2)$	0	$22/15$	0	$22/15$
$R(-4)$	0	0	$11/25$	$11/25$
$R(16)$	0	$1/30$	0	$1/30$
D_0	0	$-11/10$	$11/5$	$11/10$

It follows that

$$\begin{aligned}\frac{2}{3}\mathcal{W}(UV) - \frac{66}{25}(\mu_{\mathrm{bal}} + b) = \frac{2}{3}H_{\mathrm{ord}} &+ \frac{22}{15}R(-2) + \frac{1}{30}R(16) + \frac{11}{25}R(-4) + \frac{11}{10}D_0 \\ &+ \frac{91}{75}q^2 + \frac{73}{5}qr - \frac{3}{2}r^2 + \frac{51}{5}qk - \frac{3}{2}rk \\ &+ \left(\frac{104}{5}q - \frac{44}{5}k\right)C - \frac{22}{5}qe_N(4/5). \end{aligned} \tag{138}$$

Here $I_z(y) = (2\epsilon/15)\mathsf{H}(z, y)$ by (100). Substitution of all six rows of (101) gives

$$H_{\mathrm{ord}} = \frac{3}{2}T_5 + \frac{7}{10}\mathcal{G}(1/5) + R_{\mathrm{ord}},$$

because the variable part is $-3\mathcal{A}_u + 3\mathcal{B}_u/2 + 3D_5/2 + 29A(u)/10 + 7B(u)/10$ and $\mathcal{G}(1/5) = 2A(u) + B(u)$. The rational part, with every endpoint retained, is

$$\begin{aligned} R_{\mathrm{ord}} = &-\frac{11}{10}S(1,1) + \frac{11}{5}S(1/2,1/2) - \frac{7}{10}S(2,1) + \frac{7}{10}S(1,1/2) \\ &+ \frac{2}{5}S(1,2) - \frac{2}{5}S(1/2,1) - \frac{11}{10}S(2,2) \\ = &-\frac{17}{5}R(-2) + \frac{18}{5}q^2 - \frac{63}{5}qr + \frac{18}{5}r^2. \end{aligned}$$

The last equality substitutes precisely the seven values (70). In (138) the remaining weight-two part is

$$-\frac{4}{5}R(-2) + \frac{1}{30}R(16) + \frac{11}{25}R(-4) = \frac{17}{15}R(4) - \frac{1}{25}R(16) + \frac{12}{5}q^2,$$

using (86). The quadratic coefficients are $91/75 + 12/5 + 12/5 = 451/75$, $73/5 - 42/5 = 31/5$, and $-3/2 + 12/5 = 9/10$. We have proved the complete first value

$$\begin{aligned}\mathcal{U} = &\frac{7}{15}\mathcal{G}(1/5) + \frac{17}{15}R(4) - \frac{1}{25}R(16) + \frac{11}{10}D_0 \\ &+ \frac{451}{75}q^2 + \frac{31}{5}qr + \frac{9}{10}r^2 + \frac{51}{5}qk - \frac{3}{2}rk \\ &+ \left(\frac{104}{5}q - \frac{44}{5}k\right)C - \frac{22}{5}qe_N(4/5), \end{aligned}$$

$$\frac{2}{3}\mathcal{W}(UV) - \frac{66}{25}(\mu_{\mathrm{bal}} + b) = T_5 + \mathcal{U}. \tag{139}$$

This $\mathcal{U}$ is exactly (301).

All five phases of the P-square. To avoid confusing phase polynomials with A_* or the triangular $A(t), B(t)$, write

$$\begin{aligned}
&u_A = -4(q+C)/15, && v_B = 17C/15, && w_C = 4(q-C)/15,\\
&A_P = u_A - 2W/\epsilon, && B_P^{(0)} = v_B + 17W/(2\epsilon), && C_P^{(0)} = w_C - 2W/\epsilon,\\
&L_a = \ell(x/4), && L_b = \ell(4x), && L_t = 4L_a - L_b/4, \quad L_d = L_b - L_a,\\
&B_P = B_P^{(0)} + 2L_t/15, && C_P = C_P^{(0)} + 2L_d/15.
\end{aligned}$$

These are the original phase polynomials in (94): $P = x^p A_P + B_P + x^{-p} C_P$. Their degrees are $\deg A_P \le N-1$, $\deg B_P, \deg C_P \le p-1$, and their origins are u_A, v_B, w_C. Hence $\max \operatorname{supp} P^2 < 3p$. For $\deg F < 2p$, define its strict tail and its endpoint pair by

$$T_p(F) = \sum_{j=1}^{p-1} F_{p+j} x^j, \qquad t(F) = T_p(F)(a) + 2T_p(F)(b_*).$$

The p coefficient is excluded from this tail and retained separately. The full phase expansion is

$$P^2 = x^{2p} A_P^2 + 2x^p A_P B_P + (B_P^2 + 2A_P C_P) + 2x^{-p} B_P C_P + x^{-2p} C_P^2.$$

Write $A_a = A_P(a)$, $A_b = A_P(b_*)$, and similarly for B, C. The three windows of (127) are, term by term,

$$\begin{aligned}
Q(P^2) = &-32A_b^2 - A_a^2/16 - 16A_bB_b - A_aB_a/2\\
&- 2B_b^2 - 4A_bC_b - B_a^2 - 2A_aC_a + 3(v_B^2 + 2u_Aw_C) - 2t(B_PC_P),\\
Q(x^{-p}P^2) = &-8A_b^2 - A_a^2/4 - 4A_bB_b - 2A_aB_a + 6u_Av_B - t(B_P^2) - 2t(A_PC_P),\\
Q(x^{-2p}P^2) = &-2A_b^2 - A_a^2 + 3u_A^2 - 2t(A_PB_P).
\end{aligned}$$

Each shifted negative phase contributes its strict tail, and each unshifted polynomial contributes its complete endpoint value and origin correction. The degree bounds make all further shifted phases zero. In particular the entire remaining tail is

$$6t(B_PC_P) + \frac{9}{2}t(B_P^2) + 9t(A_PC_P) + \frac{129}{4}t(A_PB_P). \tag{140}$$

Strict logarithm tails, including the separate diagonal. For $\deg B < p$, $B(0) = 0$, and $c \neq 0$, reversing $j = p - h$ in $B(x)\ell(cx)$ gives

$$T_p(B\ell(cx))(z) = -c^p\{I_B(z, c^{-1}) - I_B(0, c^{-1})\}, \qquad [x^p](B\ell(cx)) = -c^p I_B(0, c^{-1}), \tag{141}$$

where $I_B(v, y)$ is (99). Indeed $i + j = p + n$ gives $n = i - h$: the strict tail has $1 \le h < i$, whereas the p coefficient has $h = i$. Also $c^{p-h}/(p-h) = -c^p(c^{-1})^h/h$. Thus every diagonal and every denominator is accounted for. For $B = W$, $I_B(v, y) = I_v(y)$. For $B = \ell(dx)$, the same formula is

$$T_p(\ell(dx)\ell(cx))(z) = -c^p\{D_{p-1}(d/c, dz) - R(d/c)\}.$$

From (67), inversion, and reflection one obtains

$$D_{p-1}(t, 1) = R(1-t), \quad D_{p-1}(1, t) = 2\{R(t) - R(1-t)\}, \quad D_{p-1}(t, t) = R(t) - R(1-t)$$

at the rational values used here. The first follows by setting $y = 1$ in the homogeneous formula; the second and third follow by substituting $x = 1$ and $x = y$, respectively, then using $tR((t-1)/t) = R(t) - R(1-t)$, itself reflection at $1/t$ followed by inversion. The values $t = 1$ are obtained directly from $R(0) = R(1) = 0$, without division by $1 - t$. Write $\lambda = \ell(16)$, $R_{16} = R(16)$. Reflection at $16, 1/16$ gives

$$R(1-16) = \frac{16R_{16} + \lambda^2/2}{15}, \qquad R(1-1/16) = -\frac{R_{16} + \lambda^2/2}{240}.$$

The strict-product formula now yields the full table

	a	b_*
$T_p(L_b^2)$	0	$(8R_{16}+4\lambda^2)/15$
$T_p(L_a^2)$	$-(32R_{16}+\lambda^2)/960$	0
$T_p(L_bL_a)$	$-(2R_{16}+\lambda^2)/120$	$(32R_{16}+\lambda^2)/120.$

The mixed-tail cancellation can be checked before evaluation. Put

$$F_0 = 6B_P^{(0)}C_P^{(0)} + \frac{9}{2}(B_P^{(0)})^2 + 9A_PC_P^{(0)} + \frac{129}{4}A_PB_P^{(0)},$$
$$F_1 = \frac{2}{15}(6v_B + 9u_A)L_d + \frac{2}{15}(6w_C + 9v_B + 129u_A/4)L_t.$$

Both have degree $\le p$. Expansion of the polynomial underlying (140) gives exactly

$$F_0 + F_1 + \frac{22}{5\epsilon}WL_d - \frac{13}{600}L_b^2 + \frac{64}{75}L_a^2 + \frac{22}{75}L_bL_a.$$

For the WL_t coefficient the cancellation is

$$6\left(-\frac{2}{\epsilon}\right)\frac{2}{15} + \frac{9}{2}\,2\frac{17}{2\epsilon}\frac{2}{15} + \frac{129}{4}\left(-\frac{2}{\epsilon}\right)\frac{2}{15} = 0;$$

the WL_d coefficient is $6(17/(2\epsilon))(2/15) + 9(-2/\epsilon)(2/15) = 22/(5\epsilon)$. The quadratic-log coefficients result from $6(2/15)^2L_tL_d + (9/2)(2/15)^2L_t^2$, giving $-13/600, 64/75, 22/75$ in the displayed order. Since $T_p(F_0) = T_p(F_1) = 0$, only the four displayed tails remain. Define the explicit primitive row

$$H_P = -4\{I_a(a) + 2I_{b_*}(a) - 3I_0(a)\} + \frac{1}{4}\{I_a(b_*) + 2I_{b_*}(b_*) - 3I_0(b_*)\}.$$

The strict-product formula gives $t(WL_d) = H_P$. The log table gives

$$t(L_b^2) = \frac{16R_{16}+8\lambda^2}{15}, \quad t(L_a^2) = -\frac{32R_{16}+\lambda^2}{960}, \quad t(L_bL_a) = \frac{62R_{16}+\lambda^2}{120}.$$

Their coefficients $-13/600, 64/75, 22/75$ give the total $R_{16}/10 - \lambda^2/100$. Thus (140) is $22H_P/(5\epsilon) + R_{16}/10 - \lambda^2/100$.

The original phase endpoints, from the finite W table and inversion, are

	a	b_*
A_P	$(4q-3r)/15$	$-2(4q-3r)/15$
B_P	$-34q/15 + 17r/20 - \lambda/30$	$17q/15 - 17r/10 - \lambda/30$
C_P	$4q/5 - r/5 + \lambda/120$	$2r/5 + 2\lambda/15.$

In particular $P(a) = q$, $P(b_*) = -q$. Substituting these six endpoints and the three origins in the three windows, their endpoint part is

$$\frac{599}{25}q^2 - \frac{339}{25}qr + \frac{99}{50}r^2 + \frac{6}{25}q\lambda + \frac{33}{100}r\lambda - \frac{203}{25}C^2 + \frac{32}{25}qC + \frac{\lambda^2}{100}.$$

The coefficients of the three displayed endpoint polynomials are

	$-3Q_0$	$-(9/2)Q_1$	$-(129/8)Q_2$	sum
q^2	$577/25$	-6	$172/25$	$599/25$
qr	$-66/5$	$378/25$	$-387/25$	$-339/25$
r^2	$207/50$	$-1593/200$	$1161/200$	$99/50$
$q\lambda$	0	$6/25$	0	$6/25$
$r\lambda$	$51/100$	$-9/50$	0	$33/100$
C^2	$-321/25$	$204/25$	$-86/25$	$-203/25$
qC	0	$204/25$	$-172/25$	$32/25$
λ^2	$1/100$	0	0	$1/100.$

Here Q_j denotes only the endpoint part of window j. Its $\lambda^2/100$ cancels the strict-tail term. Consequently

$$P_{\mathrm{fin}} := \mathcal{W}(P^2) = \frac{22}{5\epsilon}H_P + \frac{1}{10}R_{16} + \frac{599}{25}q^2 - \frac{339}{25}qr + \frac{99}{50}r^2 + \frac{6}{25}q\lambda + \frac{33}{100}r\lambda - \frac{203}{25}C^2 + \frac{32}{25}qC. \quad (142)$$

The coefficient at $2p$ is also read from all five phases:

$$\beta_P = [x^{2p}](-P^2/2) = -u_A^2/2 - [x^p](A_P B_P).$$

Only $(-2W/\epsilon)(2L_t/15)$ contributes to the latter coefficient. Its diagonal, by (141), is $[x^p](WL_t) = I_0(a) - I_0(b_*)$. Thus, with $\Delta I_0 = I_0(a) - I_0(b_*)$,

$$\beta_P = -u_A^2/2 + 4\Delta I_0/(15\epsilon), \qquad -\frac{1}{3}\mathcal{W}(P^2) - \frac{207}{4}\beta_P = -P_{\mathrm{fin}}/3 + 207u_A^2/8 - 69\Delta I_0/(5\epsilon). \quad (143)$$

The complete G_J row and its strict high band. Use the actual finite polynomial $\mathscr{J}$ in (294):

$$\mathscr{J}(x) = \sum_{j=0}^{N-1} \frac{q + H_j/2}{(2j+1)5^{2j+1}} s^{N+j+1}(4x-11)^{2j+1}, \qquad H_j = \sum_{i=1}^{j} 1/i.$$

Its degree is $\le 2p-3$, and both endpoint values are zero. It is attached to the full primitive by $\mathscr{J}(5t^2-1) = t^p J_L(T_c(t))$, as proved in (294)–(295). The original proper pair in (120) is

$$G_J = \frac{2\epsilon}{45}K_0(\mathscr{J} + \mathscr{J}^\iota) = \frac{17 - 4(x^p + x^{-p})}{45}(\mathscr{J} + \mathscr{J}^\iota).$$

Put

$$J_{\mathrm{lo}} = \sum_{j=0}^{p-1} \mathscr{J}_j x^j, \quad J_{\mathrm{hi}} = \sum_{j=0}^{p-1} \mathscr{J}_{p+j} x^j, \quad J_> = J_{\mathrm{hi}} - \mathscr{J}_p.$$

These are complete bands, and $\mathscr{J} = J_{\mathrm{lo}} + x^p J_{\mathrm{hi}}$. At $v = a, b_*$, the endpoint zero gives $J_{\mathrm{lo}}(v) = -v\{\mathscr{J}_p + J_>(v)\}$. By (130),

$$\mathcal{W}(G_J) = Q\left(\frac{(4x^p - 11)(\mathscr{J} + \mathscr{J}^\iota)}{15}\right).$$

The polynomial half contributes $-33\,\mathscr{J}_0/15$. For the reflected shifted half,

$$Q(x^p \mathscr{J}^\iota) = -8J_{\mathrm{lo}}(a) - J_{\mathrm{lo}}(b_*)/4, \qquad Q(\mathscr{J}^\iota) = 0.$$

The p coefficient cancels in the first equality because the origin correction is $3\,\mathscr{J}_p$ and $8a + b_*/4 = 3$. Substitution of the two low endpoint values yields

$$\mathcal{W}(G_J) = -11\,\mathscr{J}_0/5 + 4\,\mathscr{J}_p/5 + 8J_>(a)/15 + 4J_>(b_*)/15, \qquad \beta_J = [x^{2p}]G_J = -4\,\mathscr{J}_p/45. \quad (144)$$

All other terms at $2p$ vanish by $\deg \mathscr{J} < 2p$. It follows, retaining this Frobenius coefficient, that

$$\frac{2}{3}\mathcal{W}(G_J) - \frac{207}{4}\beta_J = -\frac{22}{15}\mathscr{J}_0 + \frac{77}{15}\mathscr{J}_p + \frac{8}{45}\{2J_>(a) + J_>(b_*)\}. \quad (145)$$

Here is the source calculation that removes the remaining strict high coefficients. Let $\widetilde{V} = s((4x-11)/5)^2 = 1 - V$, $L_V^{\mathrm{desc}} = q + e_N(\widetilde{V})$, and $R_{\mathrm{src}} = x^{-p}A_*^{p-1}s^N/2$. The polynomial $B_+ = L_V^{\mathrm{desc}} - 3x^p A_P$ has degree $< p$. Indeed (97) and $A_P(5t^2-1) = -4\{q + E(2t) - E(t)\}/15$ give

$$B_+(5t^2-1) = q/5 + 3E(t) + 6\{E(2t) - E(t)\}/5 + 4t^p O(t) - 2t^p O(2t).$$

The right side has degree at most $2p-1$, while a nonzero polynomial composed with $5t^2-1$ has twice its degree. Hence $\deg B_+ < p$. Put $D_+ = B_+ - 3B_P$ and $F_+ = x^p D_+ - 3C_P = x^p(L_V^{\mathrm{desc}} - 3P)$.

Then $\deg F_+ \le 2p-1$. The finite geometric derivative and the source for P give the complete cleared equation

$$A_* F_+' = \frac{3}{2}\{x^p(4x-11) - 6x^p s^N + 4s^N\} + \frac{3}{2}A_* x^{p-1}.$$

The right side has degree $\le p+N$; since $\deg A_* = 2$, $\deg F_+' \le p+N-2$. For $p+N \le j \le 2p-1$, the derivative coefficient is $j(F_+)_j = 0$, and $j \equiv j-p \in \{1,\dots,p-1\}$ is a unit. Thus $\deg F_+ \le p+N-1$, and comparison of the $p+j$ coefficients gives $\deg D_+ \le N-1$. Therefore the bracket

$$-P + \{L_V^{\mathrm{desc}} - x^{-N}(L_V^{\mathrm{desc}})^\iota\}/3 = \{D_+ - 3x^{-p}C_P - x^{-N}(L_V^{\mathrm{desc}})^\iota\}/3$$

has upper exponent $\le N-1$. The derivative for the original $Z = -P^2/2 + G_J$, displayed after (125), now has upper exponent $\le 2p-4$: R_{src} has upper exponent $p+N-2$, the bracket has upper exponent $N-1$, and $P/(2x)$ has upper exponent $p+N-2 \le 2p-4$. Initially Z has upper exponent $\le 3p-3$. For $2p < j \le 3p-3$, j is a unit modulo p, so $jZ_j = 0$ forces $Z_j = 0$. The same argument removes Z_{2p-2}, Z_{2p-1}. It retains Z_{2p}.

$$\max \operatorname{supp} Z \le 2p, \qquad Z_{2p-2} = Z_{2p-1} = 0. \tag{146}$$

At exponent $2p+j$, $j>0$, the five square phases and the proper pair therefore give

$$-\frac{1}{2}[x^j]A_P^2 - [x^{p+j}](A_P B_P) - \frac{4}{45}\mathscr{J}_{p+j} = 0.$$

All other phases are zero by their stated degree bounds. The constant coefficient of $J_>$ is zero. Thus, coefficientwise,

$$J_> = -\frac{45}{8}(A_P^2 - u_A^2) - \frac{45}{4}T_p(A_P B_P).$$

The strict-product identity and $\deg W^2 < p$ give

$$T_p(A_P B_P) = -\frac{4}{15\epsilon}J_\Delta, \qquad J_\Delta(z) = I_z(a) - I_z(b_*) - I_0(a) + I_0(b_*).$$

Indeed $T_p(WL_t) = J_\Delta$; the remaining products have degree $\le p$. Hence

$$J_>(z) = -\frac{45}{8}\{A_P(z)^2 - u_A^2\} + \frac{3}{\epsilon}J_\Delta(z).$$

The complete origin and p coefficient of the same polynomial are (295); consequently its first two terms in (145) are (296). We obtain

$$\frac{2}{3}\mathcal{W}(G_J) - \frac{207}{4}\beta_J = \frac{11}{15}U_J(u) - \frac{22}{15}V_J(u) - 6\left(\frac{4q-3r}{15}\right)^2 + 3u_A^2 + \frac{8}{15\epsilon}\{2J_\Delta(a) + J_\Delta(b_*)\}. \tag{147}$$

For an explicit finite definition of the two fifth values, put $\mathscr{D} = D_{p-1}$ and

$$\begin{aligned}
\mathscr{T}_c(t) &= \{\mathscr{D}(ct,-c) - \mathscr{D}(ct,c)\}/2 + \mathscr{D}(ct,c/2) - \mathscr{D}(ct,-c/2),\\
F_b(t) &= 2\mathscr{T}_1(t) + 2\mathscr{T}_{-1}(t) - 2\mathscr{T}_2(t) - 2\mathscr{T}_{-2}(t),\\
F_d(t) &= 2\mathscr{T}_1(t) - 2\mathscr{T}_{-1}(t) - \mathscr{T}_2(t) + \mathscr{T}_{-2}(t),\\
F_e(t) &= \{\mathscr{T}_1(t) + \mathscr{T}_{-1}(t)\}/2 + \mathscr{T}_2(t) + \mathscr{T}_{-2}(t),\\
M_E(t) &= (8q-6r)E(t) + (4q-3r)E(2t),\\
M_O(t) &= -3rO(t) + 3rO(2t)/2, \qquad K_J = R(-2) - q^2,\\
U_J(t) &= t^p F_b(t) + F_d(t) + M_E(t) - K_J,\\
V_J(t) &= t^p\{F_e(t) + M_O(t)\}.
\end{aligned}$$

Here E, O are the finite even and odd logarithms in (61), not the balanced carrier E. Their attachment to $\mathscr{J}$, including its origin and high coefficient, is the finite cubic calculation (293)–(296).

The normalized Z value. By linearity and the actual coefficient split $\beta = \beta_P + \beta_J$, combine (143) and (147). The primitive terms remain their finite sums (128), and the arithmetic coefficients collect as follows:

	$-P_{\mathrm{fin}}/3$	$207u_A^2/8 + 3u_A^2$	$-6(4q-3r)^2/225$	sum
q^2	$-599/75$	$154/75$	$-32/75$	$-159/25$
qr	$113/25$	0	$16/25$	$129/25$
r^2	$-33/50$	0	$-6/25$	$-9/10$
qC	$-32/75$	$308/75$	0	$92/25$
C^2	$203/75$	$154/75$	0	$119/25.$

Thus the second complete finite value is

$$\begin{aligned}\mathcal{Z} = &-\frac{22}{15\epsilon}H_P - \frac{1}{30}R(16) - \frac{159}{25}q^2 + \frac{129}{25}qr - \frac{9}{10}r^2\\ &-\frac{2}{25}q\ell(16) - \frac{11}{100}r\ell(16)\\ &+\frac{119}{25}C^2 + \frac{92}{25}qC + \frac{11}{15}U_J(u) - \frac{22}{15}V_J(u)\\ &+\frac{8}{15\epsilon}\{2J_\Delta(a) + J_\Delta(b_*)\} - \frac{69}{5\epsilon}\{I_0(a) - I_0(b_*)\},\\ \frac{2}{3}\mathcal{W}(Z) - \frac{207}{4}\beta = &\,\mathcal{Z}. \end{aligned} \tag{148}$$

Every term on the right is now a specified finite sum. The two root choices give the same value because the left side is the original descended carrier; no sign choice has entered any endpoint or coefficient window. In the rest of the manuscript, including the final collection after (301), $U_{\mathrm{rem}} = \mathcal{U}$ and $Z_{\mathrm{norm}} = \mathcal{Z}$ mean exactly (139) and (148). We have proved

$$\frac{2}{3}\mathcal{W}(UV) - \frac{66}{25}(\mu_{\mathrm{bal}} + b) = T_5 + U_{\mathrm{rem}}, \qquad \frac{2}{3}\mathcal{W}(Z) - \frac{207}{4}\beta = Z_{\mathrm{norm}}. \tag{149}$$

In the moment consumer below, $a(t) = \mathcal{A}_e(t) = -\{R(1-t) + R(1+t)\}/2$ is a finite function; the endpoint $a = 1/4$ above remains a separate symbol.

We next record explicitly the rational V_{bal} that consumes T_5. Set

$$r_0 = -2Q_2 - \frac{3}{2}(Q_3 + Q_5), \quad c_{\log} = -N[x^N]e_{\mathbb{Q}}(1-V),$$

where $Q_a = (a^{p-1} - 1)/p$ is the integer Fermat quotient. Then

$$V_{\mathrm{bal},\mathbb{Q}} = -t_{\mathbb{Q}}(\mathcal{L}_{\mathbb{Q}}) - \mathcal{W}_{\mathbb{Q}}(S_{\mathbb{Q}}) - 3r_0 + 2J_{\mathbb{Q}}(1)c_{\log}, \qquad \mathcal{W}_{\mathbb{Q}}(F) = \tau_{p+1}(3\,5^{p-1}R_w^p d_x F). \tag{150}$$

The three bare logarithm values are actual evaluations at $1/4, 4, 0$: using $V(1/4) = V(4) = 0$ and $1 - V(0) = 121/125$,

$$t_{\mathbb{Q}}(\mathcal{L}_{\mathbb{Q}}) = -\frac{3}{2}H_N + 3e_{\mathbb{Q}}(121/125).$$

The complete rational carries in (150) are as follows:

$$\begin{aligned} H_N &= -2Q_2 + 2p\,k_H,\\ e_{\mathbb{Q}}(121/125) &= \frac{11}{5}C_{\mathbb{Q}}^{(5)} - \frac{16}{5}Q_2 + \frac{3}{2}Q_5 + p\,k_5, & C_{\mathbb{Q}}^{(5)} &= e_{\mathbb{Q}}(4/5) - e_{\mathbb{Q}}(1/5),\\ \mathcal{W}_{\mathbb{Q}}(S_{\mathbb{Q}}) &= F_{0,\mathbb{Q}} + p\,k_w, & F_{0,\mathbb{Q}} &= \frac{33}{5}(Q_2 - C_{\mathbb{Q}}^{(5)}) + p\,k_0,\\ -J_{\mathbb{Q}}(1) &= h_{\mathbb{Q}} + p\,k_e, & h_{\mathbb{Q}} &= -2Q_2 - \frac{3}{2}Q_3,\\ c_{\log} &= -\frac{3}{2} + p\,k_c. \end{aligned}$$

These equations are definitions of the rational carries, with their integrality verified by the coefficient recipes below. Substitution in (150) cancels every zeroth digit:

$$-3Q_2 + \frac{48}{5}Q_2 - \frac{33}{5}Q_2 + 6Q_2 = 6Q_2, \quad -\frac{33}{5}C_{\mathbb{Q}}^{(5)} + \frac{33}{5}C_{\mathbb{Q}}^{(5)} = 0,$$

and this $6Q_2$, together with $9Q_3/2$, cancels $3h_{\mathbb{Q}}$; the Q_5 terms are $-9Q_5/2 + 9Q_5/2 = 0$. Thus the exact divided quantity is

$$\frac{V_{\mathrm{bal},\mathbb{Q}}}{p} = 3k_H - 3k_5 - k_0 - k_w + 3k_e - 2h_{\mathbb{Q}}k_c - 2p\,k_e k_c \in \mathbb{Z}_{(p)}. \tag{151}$$

In particular the last term has survived the rational division; it vanishes only on subsequent reduction.

To make the resulting finite alphabet transparent, write $q = q_2, r = q_3, k = q_5, j = q_{11}$, $C = E(2u) - E(u)$, $C_w = E(2w) - E(w)$, with $u^2 = 1/5, w^2 = 3/11$, and $h = -2q - 3r/2$. The full carry reductions are

$$\begin{aligned}
\bar{k}_H &= q^2/2, \\
\bar{k}_e &= q^2/3 - qr + 3r^2/8, \\
\bar{k}_c &= 3q/5 + 11(C_w - C)/5, \\
\bar{k}_0 &= \frac{11}{5}r_1 + \Phi, \\
r_1 &= R(-2)/2 - 2q^2 + 3qr/2, \\
\Phi &= \frac{33}{5}k(q - C) + \frac{4}{5}(13q - 11j/4)(9r/4 + 3C) \\
&\quad + \frac{62}{5}q(q + 3r/4), \\
\bar{k}_5 &= \mathcal{P}_5 + a(-11u/5) - \frac{11}{5}(a(2u) - a(u)), \\
\mathcal{P}_5 &= -169q^2/50 + 24qk/5 - 15k^2/8 + 103qC/25 - 11kC/5 - 37C^2/50.
\end{aligned} \tag{152}$$

For the remaining row $\bar{k}_w = T_w$, no trace is assigned a value without its full triangular evaluation. Define

$$\begin{aligned}
H_{\mathrm{mov}}(y, v) &= \frac{1}{2}\{D_N(4(1+v)/5, 4(1+y)/5) - D_N(4/5, 4(1+y)/5) \\
&\qquad - D_N((1+v)/5, (1+y)/5) + D_N(1/5, (1+y)/5)\}, \\
D_e &= q + 3r/4, \\
M_-(a, b, z) &= -2H_{\mathrm{mov}}(b, z) - H_{\mathrm{mov}}(a, z) + 3H_{\mathrm{mov}}(0, z), \\
M_+(z) &= D_e\{2\ell(4/z) + \ell(1/(4z))\} + 2H_{\mathrm{mov}}(1/4, z^{-1}) + H_{\mathrm{mov}}(4, z^{-1}), \\
T_w &= -2M_-(1/4, 4, 1/4) + M_-(1/4, 4, 4) \\
&\quad - \frac{4}{5}\{M_+(1/4)/4 + 4M_+(4) - (11/4)M_+(11/4)\}.
\end{aligned}$$

Only the nonzero rational arguments displayed here are inverted. The finite polynomial identity

$$(x - y)\left(\sum_{n=0}^{N} w_n \sum_{a=1}^{n} \frac{y^{n-a}x^a}{a}\right)' = W_{\mathbb{Q}}(x) - W_{\mathbb{Q}}(y)$$

for the coefficients of the moving polynomial proves the displayed four-D_N expression: expand each $(1 + x)^n$, then set the outer exponent to n and the inner integrated exponent to a. All degrees are at most $N < p$, and its value at $x = 0$ is zero; the derivative identity and origin therefore identify the entire polynomial, including the diagonal. Combining the three M_+ rows uses

$$2\ell(16/11) + \ell(1/11) = 3j - 138q/11, \qquad \ell(16) = 15(r + k) - 64q.$$

These follow from the integer entropy identity at a/b, with $(a,b) = (16,11),(1,11),(16,1)$, and $Q_{ab} \equiv Q_a + Q_b \pmod p$. Collecting the four H_{mov} endpoint rows gives

$$\begin{aligned} -T_w &= \mathcal{R} - \frac{3}{2}T_5 + \frac{2}{5}\mathcal{G}(1/5) - \frac{11}{10}\mathcal{G}(3/11) \\ &\quad - \frac{D_e}{5}(33j - 138q - \ell(16)), \\ \mathcal{R} &= \{-4(S(1,2) - S(1/2,1)) + 22(S(1,1) - S(1/2,1/2)) \\ &\qquad + 11(S(2,2) - S(1,1)) + 7(S(2,1) - S(1,1/2))\}/10 \\ &= 17R(-2)/5 - 18q^2/5 + 63qr/5 - 18r^2/5. \end{aligned} \tag{153}$$

The first equality is obtained by substituting the four-D_N formula above into T_w; the fifth-root terms are exactly $A(u) - 2A_u + B_u + D_5$, and the two unspecialized square arguments are exactly $\mathcal{G}(1/5), \mathcal{G}(3/11)$. For the last equality the seven full rational rows, with $D = R(-2)$, are

$S(a,b)$	D	q^2	qr	r^2
$S(1,1)$	0	2	0	0
$S(1/2,1)$	$-2/3$	$-16/3$	4	$-3/4$
$S(1,2)$	$2/3$	$-2/3$	-4	3
$S(1/2,1/2)$	$-1/3$	$1/3$	-1	$3/4$
$S(2,2)$	$10/3$	$-10/3$	4	$3/2$
$S(2,1)$	$-4/3$	$4/3$	2	$-3/2$
$S(1,1/2)$	$-2/3$	$2/3$	-2	$3/2$

Each is the full sign trace $D_N(a^2, b^2)$; the reflection, inversion and distribution arguments of the finite-consumer subsection give these rows with the same diagonal convention. Multiplying the rows by the four coefficients $-4, 22, 11, 7$ and their indicated differences gives the displayed $\mathcal{R}$, so its rational part can be checked column by column.

After (152), the first three endpoint terms in (151) are

$$\frac{69}{10}q^2 - \frac{63}{10}qr + \frac{9}{8}r^2 - \frac{11}{10}R(-2) - 3\bar{k}_5 - \Phi - T_w + (4q + 3r)\bar{k}_c.$$

Insert (153), $\ell(16) = 15(r+k) - 64q$, and the two fifth and conic carry digits. The Q_{11} coefficient is

$$-\frac{11}{5}j(-9r/4 - 3C + 3(q + 3r/4)) = -\frac{33}{5}(q - C)j.$$

The remaining ordinary product coefficients, in the order $q^2, qr, r^2, qk, rk, k^2, qC, rC, kC, C^2$, are

$$\left(\frac{456}{25}, -\frac{21}{2}, -\frac{9}{40}, -18, \frac{9}{4}, \frac{45}{8}, -\frac{1309}{25}, -\frac{33}{5}, \frac{66}{5}, \frac{111}{50}\right).$$

Denote that explicit quadratic polynomial by $\mathcal{P}_{\mathrm{bal}}$, and put

$$\mathcal{C}_{\mathrm{con}} = \mathcal{G}(3/11) + 4hC_w + 6(q - C)j.$$

The full result of the substitution is

$$\overline{V_{\mathrm{bal},\mathbb{Q}}/p} = \mathcal{P}_{\mathrm{bal}} + \frac{23}{10}R(-2) + \frac{33}{5}(a(2u) - a(u)) + \frac{2}{5}\mathcal{G}(1/5) - \frac{3}{2}T_5 - 3a(-11u/5) - \frac{11}{10}\mathcal{C}_{\mathrm{con}}.$$

Define the explicit moment consumer

$$\mathcal{V} = \frac{2}{3}\mathcal{P}_{\mathrm{bal}} + \frac{23}{15}R(-2) + \frac{22}{5}(a(2u) - a(u)) + \frac{4}{15}\mathcal{G}(1/5) - 2a(-11u/5) - \frac{11}{15}\mathcal{C}_{\mathrm{con}}. \tag{154}$$

Thus, since $(2/3)(-3T_5/2) + T_5 = 0$,

$$\frac{2}{3}\overline{V_{\mathrm{bal},\mathbb{Q}}/p} + T_5 = \mathcal{V}. \tag{155}$$

In particular the exact whole rational source, before reduction, is

$$\mathcal{C}_{\mathbb{Q}} = \frac{2}{3p}\{V_{\mathrm{bal},\mathbb{Q}} + p(C_{\mathbb{Q}} + B^{\mathrm{tr}}_{\mathbb{Q}} - P^{\mathrm{tr}}_{\mathbb{Q}})\}.$$

All four divided inputs have been proved integral. Reduction of (126), then use of (149) and (155), gives

$$\overline{\mathcal{C}_{\mathbb{Q}}} = \mathcal{V} + U_{\mathrm{rem}} + Z_{\mathrm{norm}} + \frac{2}{3}\tau_{p+1}(D_{\mathrm{low}}). \tag{156}$$

This identity keeps the entire low density at its original pole.

To connect it to the original residual scalar, let $\mathrm{op}(F) = [x^{-1}]_0(F/A)$. The full proper source is

$$\mathcal{Q}(t) = Y(t)n(t)\left(\sum_{\substack{1\le i<p\\ i\ \mathrm{odd}}} \frac{t^i}{i} + t^p \sum_{i=1}^{N} \frac{t^{2i}}{2i}\right),$$

$$Q_{\mathrm{pr}} = [z^{N+1}]\left\{\frac{\mathrm{qi}\,\mathcal{Q}(V(z)) - \mathrm{qi}\,\mathcal{Q}(0)}{V(z)}\,\frac{132z^3 - 81z^2 + 12z}{125}\right\}.$$

Its last odd index is $p-2$, not p.

The original low-density and coupled origin. Here $p \ge 19$, $N = (p-1)/2$, $d = \lfloor p/6 \rfloor$, and $h = h_d$. All formulas below are first over $\mathbb{F}_p$, then mapped coefficientwise into the field K used for the native chart. Write

$$A = (4-x)(4x-1) = A_*, \quad s = (x+1)/5, \quad B = 4x-11, \quad C_x = 4-11x, \quad L_{\mathrm{n}} = 33-12x = -3B,$$
$$R_w = xB, \quad d_x = 3x^{-1}C_x, \quad \alpha = 4x^p - 11, \quad \varphi = 4x^{-p} - 11,$$
$$\zeta = (17 - 4x^p - 4x^{-p})/3, \quad B^{\vee}_{\mathrm{n}} = x^N C^{\iota}_B.$$

The symbol $B^{\vee}_{\mathrm{n}}$ denotes this whole Laurent polynomial. Reflection is always $F^{\iota}(x) = F(x^{-1})$. We keep the native

$$a = L^{\mathrm{desc}}_V - h, \qquad e = a^{\iota} - (q-h) = (L^{\mathrm{desc}}_V)^{\iota} - q, \qquad D_s = D + D^{\iota}.$$

Use the half-log polynomial e_N defined above in the balanced period,

$$\delta_{\mathrm{bal}} = \overline{(F_{\mathbb{Q}} - x^N F^{\iota}_{\mathbb{Q}})/p}, \qquad \Pi = x^{-N}\delta_{\mathrm{bal}} e_N(1-V).$$

Thus the period's e_N is distinct from the native e. All fixed divisions below use only $2, 3, 5$. Finite logarithmic indices are at most $p-1$; the new primitive's indices will be at most $p-2$. These denominators are units.

Put $H = H_0(V)$, so $H_c = H^{\iota}$. The Euler definition $VH_0'(V) = f - 1$, together with $V'/V = L_{\mathrm{n}}/A$, gives

$$AH' = (f-1)L_{\mathrm{n}}. \tag{157}$$

This uses the zero-constant primitive H_0. The metric identity (104) continues to use $H + 2h$.

Define the original proper-square carriers by

$$P_2 = \mathrm{qi}(n^2)(V), \qquad n_2 = s^p P_2, \qquad T_p = \alpha/5, \qquad K_{\mathrm{sq}} = \zeta n_2 T_p/5,$$
$$J_{\mathrm{sq}} = \frac{x^{-N}}{6}\left(\frac{3\zeta^2 n_2}{25} + K_{\mathrm{sq}} + x^N K^{\iota}_{\mathrm{sq}}\right), \qquad F_{\Sigma} = -R^p_w d_x S\Sigma^{\iota}, \qquad F_{\mathrm{HF}} = 3R^{p-1}_w H_c \Omega_{\mathrm{HF}}.$$

Here $\Sigma = \mathrm{qi}(YR_d)(V)$, with the same original $R_d = Y(L_V - h) - t^p n$ used in the proper calculation. Substituting the full L_{pr} of (120) into (125) gives the whole-source split

$$D_{\mathrm{low}} = F_{\mathrm{HF}} + 3R^p_w d_x J_{\mathrm{sq}} + F_{\Sigma} - A^p d_x \Pi. \tag{158}$$

Indeed, its reflected summand is

$$3R^p_w d_x x^{-N}(-2Sx^N\Sigma^{\iota})/6 = -R^p_w d_x S\Sigma^{\iota}.$$

No reflection or proper term has been suppressed.

The proper-square contribution. The proper polynomial has parity $n(-t) = (-1)^{d+1}n(t)$, degree at most $d-1$, and $n(1) = -h$. Consequently

$$\deg P_2 \le 3d-3 < N, \qquad P_2(1/4) = P_2(4) = h^2.$$

The endpoint statements follow from $V(1/4) = V(4) = 0$ and $\mathrm{qi}(n^2)(0) = n(1)^2$. Use $s^p = (x^p+1)/5$ and $3\zeta + 5T_p = 6 - 4x^{-p}$. Expansion of all three square terms gives

$$\begin{aligned} 2250 J_{\mathrm{sq}} = {} & \left(-24x^{2p-N} + 94x^{p-N} + 26x^{-N} - 76x^{-p-N} + 16x^{-2p-N}\right) P_2 \\ & + \left(44x^{p} - 159 - 91x^{-p} + 96x^{-2p} - 16x^{-3p}\right) P_2^{\iota}. \end{aligned} \tag{159}$$

For clarity, the two coefficient multiplications, with $U = x^p$, are

$$(17 - 4U - 4U^{-1})(U+1)(6 - 4U^{-1}) = -24U^2 + 94U + 26 - 76U^{-1} + 16U^{-2},$$

$$(17 - 4U - 4U^{-1})(U^{-1}+1)(4U^{-1} - 11) = 44U - 159 - 91U^{-1} + 96U^{-2} - 16U^{-3}.$$

The factor $2250 = 6 \cdot 375$ comes from the original $1/6$, ζ, $n_2 = s^p P_2$, and the two original $1/25$ factors.

For a Laurent polynomial define the positive-coefficient functional

$$\mathcal{E}(F) = -\sum_{j>0} F_j (2\,4^j + 4^{-j}) = \tau_1(d_x F).$$

The second equality is the geometric-series calculation at (123). Since $4^N = 1$, the three positive phases in (159) have values

$$\mathcal{E}(x^{p-N}P_2) = -33h^2/4, \quad \mathcal{E}(x^{2p-N}P_2) = -513h^2/16, \quad \mathcal{E}(x^p P_2^{\iota}) = -33h^2/4.$$

Every coefficient in these three products has strictly positive exponent because $\deg P_2 < N$. The other seven phases have nonpositive exponents and contribute zero to $\mathcal{E}$. Hence

$$\mathcal{E}(J_{\mathrm{sq}}) = \frac{-24(-513/16) + (94+44)(-33/4)}{2250} h^2 = -\frac{41}{250} h^2,$$

$$\mathcal{E}(x^{-p} J_{\mathrm{sq}}) = \frac{-24(-33/4)}{2250} h^2 = \frac{11}{125} h^2, \qquad \mathcal{E}(x^{-2p} J_{\mathrm{sq}}) = 0.$$

The constant slot of $x^{-p}J_{\mathrm{sq}}$ is retained; its $\mathcal{E}$-weight is zero.

Here J_{sq} has no exponent above $2p-1$. The weight reduction for such a carrier is

$$\mathcal{W}(J_{\mathrm{sq}}) = -3\mathcal{E}(J_{\mathrm{sq}}) - \frac{9}{2}\mathcal{E}(x^{-p} J_{\mathrm{sq}}).$$

To verify this reduction directly, use $3R_w^p = -3A^p + 18x^p - 12$ in $\mathcal{W}(F) = \tau_{p+1}(3R_w^p d_x F)$. The unshifted last term has no exponent reaching $2p+1$, so its successor trace is zero. For the middle term use $r(z)^{p+1} = r(z^p)r(z)$ in (123). The coefficient of F_j in $\tau_{p+1}(d_x x^p F)$ is zero for $j \le p$. For $p < j \le 2p-1$, put $k = j - p$, $1 \le k < p$. It is

$$-\frac{1}{16}(12 r_{k-2} - 33 r_{k-1}) = \frac{1}{4}(2\,4^k + 4^{-k}), \qquad r_{-1} = 0.$$

Here $12r_{k-2} - 33r_{k-1} = -4(2\,4^k + 4^{-k})$, also at $k = 1$. Thus this middle trace is $-\mathcal{E}(x^{-p}F)/4$, giving the displayed reduction. It follows that

$$\frac{2}{3}\tau_{p+1}(3R_w^p d_x J_{\mathrm{sq}}) = \frac{2}{3}\left(\frac{123}{250} - \frac{99}{250}\right) h^2 = \frac{8}{125} h^2. \tag{160}$$

The complete proper and j product. Let

$$O(t)=\sum_{\substack{1\le i<p\\ i\ \text{odd}}}\frac{t^i}{i},\quad E(t)=\sum_{i=1}^{N}\frac{t^{2i}}{2i},\quad O_{\mathrm{pr}}=\mathrm{qi}(YnO)(V),\quad E_{\mathrm{pr}}=\mathrm{desc}\{E(T_c(t))\}.$$

Equivalently $E_{\mathrm{pr}}=L_V^{\mathrm{desc}}-q=a-(q-h)$. Write $Q_{\mathrm{pr}}^{\mathrm{pull}}=\mathrm{qi}\,\mathcal{Q}(V)$ for the whole proper polynomial. The conic pullback $x=5t^2-1$, for which $V=1-T_c(t)^2$, gives

$$Q_{\mathrm{pr}}^{\mathrm{pull}}=O_{\mathrm{pr}}+\alpha C_B E_{\mathrm{pr}}. \tag{161}$$

In fact the second pulled-back product is

$$T_c(t)^pY(T_c)n(T_c)E(T_c)=\alpha(5t^2-1)\frac{t^pY(T_c)n(T_c)}{5}E(T_c),$$

because $T_c(t)^p=t^p(4t^{2p}-3)$ and $\alpha(5t^2-1)=5(4t^{2p}-3)$. The conic pullback is injective by its nonzero leading coefficient. At $t=1$, $O(1)=q$ and $E(1)=-q$: these are $H_{p-1}=0$, $H_N=-2q$, split into odd and even indices. Thus $\mathrm{qi}\,\mathcal{Q}(0)=\mathcal{Q}(1)=0$.

Define explicitly the original reciprocal proper density

$$\mathcal{D}_{\mathscr{Q}}(z)=\left(\sum_{i\ge1}\mathscr{Q}_iV(z)^{i-1}\right)\frac{132z^3-81z^2+12z}{125},\qquad \mathscr{Q}(z)=\mathrm{qi}\,\mathcal{Q}(z),\qquad D_{\mathrm{pr}}=x^N\mathcal{D}_{\mathscr{Q}}(x^{-1}).$$

This is a polynomial recipe before reflection, including the subtracted constant. Coefficient inversion gives

$$[x^{-1}]D_{\mathrm{pr}}=Q_{\mathrm{pr}},\qquad AD_{\mathrm{pr}}=-3B\,x^N(Q_{\mathrm{pr}}^{\mathrm{pull}})^\iota.$$

For the second identity, if $F_{\mathrm{rev}}(z)=(132z^3-81z^2+12z)/125$, then $AF_{\mathrm{rev}}^\iota=-3B\,V^\iota$. Multiplying by $x^N(\mathscr{Q}(V^\iota)-\mathscr{Q}_0)/V^\iota$ proves the claim; the quotient is interpreted by the displayed finite sum. Also $\mathrm{op}(AD_{\mathrm{pr}})=Q_{\mathrm{pr}}$.

The entire original j-product becomes Laurent after multiplication by $B_{\mathrm{n}}^\vee$. Its explicit carrier is

$$\begin{aligned}J_{\mathrm{prod}}&=x^NO_{\mathrm{pr}}^\iota+\varphi B_{\mathrm{n}}^\vee(E_{\mathrm{pr}}^\iota+a/3)\\&=x^NO_{\mathrm{pr}}^\iota+\varphi B_{\mathrm{n}}^\vee(a^\iota-(q-h)+a/3)=B_{\mathrm{n}}^\vee j.\end{aligned}\tag{162}$$

To check the last equality for the actual j of (106), set $F_O(z)=\sum_{i=0}^{N-1}z^i/(2i+1)$, so $O(t)=tF_O(t^2)$. The same conic substitution gives

$$s^NO_{\mathrm{pr}}=C_BBF_O(\widetilde{V}),\qquad s^Nx^NO_{\mathrm{pr}}^\iota=B_{\mathrm{n}}^\vee x^{N-1}C_xF_O(\widetilde{V}^\iota)=B_{\mathrm{n}}^\vee O_s,$$

$$\widetilde{V}^\iota=\frac{s\,x^{-3}C_x^2}{25},\qquad C_xF_O(\widetilde{V}^\iota)=\sum_{i=0}^{N-1}\frac{x^{-3i}s^iC_x^{2i+1}}{5^{2i}(2i+1)}=\mathcal{O}.$$

These equalities keep the pivot $2N-1=p-2$. They prove $s^NJ_{\mathrm{prod}}=B_{\mathrm{n}}^\vee\{O_s+s^N\varphi(e+a/3)\}$. Since $s(0)=1/5$, s is a unit in the origin series, proving (162) for that same series, not an assigned residue.

Put

$$J_{\mathrm{sub}}=-2BJ_{\mathrm{prod}},\qquad J_{\mathrm{coupled}}=\frac{2}{3}[x^{-1}]_0\left\{(\tfrac{3}{5}fD_s-C_B)j\,\frac{-3B}{A}\right\}.$$

The independently proved identity (SSA83) in Section 6.6 is $\delta_{\mathrm{bal}}=fD-\frac{5}{3}C_B$, for this same rational square reflection quotient. Its anti-reflection is $x^N\delta_{\mathrm{bal}}^\iota=-\delta_{\mathrm{bal}}$: reflect $F_{\mathbb{Q}}-x^NF_{\mathbb{Q}}^\iota=p\delta_{\mathbb{Q}}$, multiply by x^N, and cancel the nonzero rational p before reduction. Adding the weighted reflection to (SSA83), and using $x^Nf^\iota=f$, gives

$$0=f(D+D^\iota)-\frac{5}{3}(C_B+x^NC_B^\iota),\qquad \frac{3}{5}fD_s=C_B+B_{\mathrm{n}}^\vee.$$

This turns the full series above into $J_{\mathrm{coupled}} = \mathrm{op}(J_{\mathrm{sub}})$. The original normalized chi attachment gives

$$\chi_N = fP + \zeta C_B, \quad \chi_N + x^N \chi_N^{\iota} = 0, \quad P + P^{\iota} = S.$$

Use $x^N f^{\iota} = f$ and $\zeta^{\iota} = \zeta$ in this weighted sum. Its two whole P terms give fS, so

$$fS + \zeta(C_B + B_{\mathrm{n}}^{\vee}) = 0, \qquad S^{\iota} = S.$$

Its independent proofs are the original chi normalization in Section SSA75a and the full symmetric $P + P^{\iota} = S = K_0(W + W^{\iota} + 4\epsilon C/15)$ calculation in Section 4.5. Combining this seed with the metric identity, and cancelling the globally nonzero f, gives $S = -3\zeta D_s/5$. Consequently, with $K_{\Sigma} = -2\varphi D_s/5$,

$$A^p K_{\Sigma} = 2C_x^p S.$$

Here $A^p = 3x^p\zeta$ and $C_x^p = x^p\varphi$. Since $R_w^{\iota} = x^{-2}C_x$, $d_x^{\iota} = 3B$, inversion of the full reflected term in (158) gives

$$-\frac{2}{3}x^{2p}F_{\Sigma}^{\iota} = A^p K_{\Sigma} B\Sigma, \qquad \frac{2}{3}\tau_{p+1}(F_{\Sigma}) = \mathrm{op}(K_{\Sigma}B\Sigma).$$

The trace equality is precisely $\tau_{p+1}(F) = -o_{p+1}(x^{2p}F^{\iota})$; the A^p clears to the original simple pole.

Now specify the whole coupled carrier by

$$J_{\mathrm{joint}} = K_{\Sigma}B\Sigma + \frac{2}{3}AD_{\mathrm{pr}} - J_{\mathrm{sub}}.$$

It follows, with the original proper scalar, that

$$\frac{2}{3}\tau_{p+1}(F_{\Sigma}) + \frac{2}{3}Q_{\mathrm{pr}} - J_{\mathrm{coupled}} = \mathrm{op}(J_{\mathrm{joint}}). \tag{163}$$

Reflection of (161) is

$$x^N(Q_{\mathrm{pr}}^{\mathrm{pull}})^{\iota} = x^N O_{\mathrm{pr}}^{\iota} + \varphi B_{\mathrm{n}}^{\vee} E_{\mathrm{pr}}^{\iota}.$$

Subtract it from (162), before taking a residue. The entire odd and even proper terms cancel, leaving

$$J_{\mathrm{joint}} = K_{\Sigma}B\Sigma + \frac{2}{3}B\varphi B_{\mathrm{n}}^{\vee} a.$$

The actual square product is $\Sigma = fa - \alpha C_B$: under the injective conic pullback, $YR_d = Y^2(L_V - h) - t^p Yn$ gives the first term fa, and the same equality $\alpha C_B \mapsto T_c^p Y(T_c)n(T_c)$ used in (161) gives the second. This identity and $\frac{3}{5}fD_s = C_B + B_{\mathrm{n}}^{\vee}$, now give

$$fJ_{\mathrm{joint}} = -\frac{2}{3}B\varphi(C_B\Sigma - \alpha C_B B_{\mathrm{n}}^{\vee}). \tag{164}$$

Indeed, substitute $K_{\Sigma} = -2\varphi D_s/5$ into the preceding line and replace fa by $\Sigma + \alpha C_B$.

Its primitive and all four cross phases. Let

$$\mathcal{NR}(t) = n(t)R_d(t), \qquad Q_{\mathcal{NR}}(t) = \frac{\mathcal{NR}(t)}{t^2 - 1}, \qquad E_{\mathcal{NR}}(t) = \sum_{i=1}^{p-2} \frac{[t^i](2t^2 Q_{\mathcal{NR}})}{i} t^i.$$

The independent hat comparison in Section 6.1 identifies this same R_d with the rank-divided polynomial and proves $\deg R_d \le p - 1 - d$, $R_d(1) = 0$, and $R_d(-t) = (-1)^d R_d(t)$. Thus $\mathcal{NR}$ is odd, has degree at most $p - 2$, and vanishes at 1 and -1. The two distinct linear factors therefore divide it; their product gives the displayed polynomial $Q_{\mathcal{NR}}$, odd and of degree at most $p - 4$. Its zero-constant Euler primitive satisfies

$$(1 - t^2)E'_{\mathcal{NR}} = -2t\mathcal{NR}, \qquad \deg E_{\mathcal{NR}} \le p - 2.$$

Indeed $tE'_{\mathcal{NR}} = 2t^2 Q_{\mathcal{NR}}$, and cancellation of the polynomial t gives the identity. Define the actual even descents

$$NR_{\mathrm{n}} = \mathrm{desc}\{t^p \mathcal{NR}(T_c(t))/5\}, \qquad E_{\mathrm{prim}} = \mathrm{desc}\{t^p E_{\mathcal{NR}}(T_c(t))/5\}.$$

Oddness makes both inputs even. The conic product identity gives $C_B\Sigma = fNR_{\mathrm{n}}$. Its two sides pull back respectively to $t^p Y(T_c)^2 n(T_c) R_d(T_c)/5$. The chain rule gives

$$AE'_{\mathrm{prim}} = -3BNR_{\mathrm{n}}. \tag{165}$$

In detail, $1 - T_c^2 = (1-t^2)(4t^2-1)^2$, $T_c' = 3(4t^2-1)$, and $dx/dt = 10t$. After multiplying by $10t$, the left side pulls back to $-30t^{p+1}(4t^2-3)\mathcal{NR}(T_c)$, as does the right side. Polynomial cancellation of t and conic injectivity prove (165); no endpoint denominator is inverted.

Here is an explicit cross carrier. Set

$$\begin{aligned} &r_Y = d \bmod 2, \quad r_n = 1 - r_Y, \quad M = (N - r_Y)/2, \\ &e_c = (N + 1 + 3r_n)/2 = M + 2 - r_Y, \quad \Delta = N - 3d, \quad \eta = (-1)^d 2^N, \\ &\\ &Y(t) = t^{r_Y} Z_Y(t^2), \quad n(t) = t^{r_n} Z_n(t^2), \\ &Y_0 = Z_Y(1-V), \quad N_0 = Z_n(1-V), \quad b_0 = B/5, \quad c_0 = C_x/5. \end{aligned}$$

Both M and e_c are integers: $N + d$ is even for $p = 6d+1$ and $p = 6d+5$, and Δ is respectively 0 or 2. The parity pullback gives

$$C_B = \frac{s^{N+1} b_0 Y_0 N_0}{5}, \qquad f = s^{r_Y}(b_0^{r_Y} Y_0)^2.$$

For example $T_c^2 = s b_0^2$, $r_Y + r_n = 1$, and $t^p T_c = t^{p+1}(4t^2 - 3)$ give the first formula. Let $P_Y = b_0^{r_Y} Y_0$. The original $f = x^N f^\iota$ and $s = x s^\iota$ give $P_Y^2 = (x^M P_Y^\iota)^2$, cancelling only the nonzero polynomial s^{r_Y}. Since $\eta^2 = 4^N = 1$, the two possible signs are $P_Y = \pm \eta x^M P_Y^\iota$. At $x = 4$, $P_Y(4) = 1$, $P_Y(1/4) = (-2)^{r_Y}$, and

$$\eta 4^M (-2)^{r_Y} = (-1)^{d + r_Y} 2^{N + 2M + r_Y} = 2^{p-1} = 1.$$

Characteristic different from 2 excludes the minus sign. Thus

$$b_0^{r_Y} Y_0 = \eta x^{M - r_Y} c_0^{r_Y} Y_0^\iota, \qquad b_0 c_0 Y_0 Y_0^\iota = \eta x^{r_Y - M} b_0^{r_Y + 1} c_0^{1 - r_Y} Y_0^2.$$

The second identity follows directly for $r_Y = 0$ and for $r_Y = 1$ from the first; it makes no division by Y_0, b_0, c_0. Define

$$D_x = 6\eta x^{-e_c} C_x s^{r_n} (b_0 c_0)^{r_n} N_0 N_0^\iota, \qquad C_{\mathrm{cross}} = \frac{\eta}{25} x^{-e_c} s^{p + r_n} (b_0 c_0)^{r_n} N_0 N_0^\iota.$$

Direct multiplication of the two C_B factors gives

$$C_B B_{\mathrm{n}}^\vee = \frac{x^{-2} s^{p+1} b_0 c_0 Y_0 Y_0^\iota N_0 N_0^\iota}{25} = f C_{\mathrm{cross}}, \qquad D_x s^p = 150 C_x C_{\mathrm{cross}}.$$

Use these identities in (164). Since $f(4) = 1$, it is globally nonzero in the Laurent polynomial domain, and cancellation gives the whole identity

$$J_{\mathrm{joint}} = -\frac{2}{3} B\varphi(NR_{\mathrm{n}} - \alpha C_{\mathrm{cross}}).$$

Keep all four original signed phases in

$$N_{\mathrm{cross}} = \frac{1}{250}\left(-44x^{p+N} + 93x^N + 93x^{-N-1} - 44x^{-p-N-1}\right) D_x. \tag{166}$$

Inverting the explicit D_x gives

$$C_x D_x^{\iota} = x^N B D_x.$$

The exponent identity behind this is $2e_c - 1 - 3r_n = N$, with $s^{\iota} = x^{-1}s$, $b_0^{\iota} = x^{-1}c_0$, $c_0^{\iota} = x^{-1}b_0$. If the parenthesis divided by 250 in (166) is G_4, then

$$x^N G_4^{\iota} = \frac{-44x^{-p} + 93 + 93x^p - 44x^{2p}}{250} = \frac{\varphi\alpha(x^p+1)}{250}.$$

Multiply by BD_x, use $x^p + 1 = 5s^p$ and $D_x s^p = 150 C_x C_{\mathrm{cross}}$, and cancel only the globally nonzero polynomial C_x. This proves

$$N_{\mathrm{cross}}^{\iota} = 3B\varphi\alpha C_{\mathrm{cross}}.$$

Together with (165) and $\varphi' = 0$, this yields the full primitive decomposition

$$J_{\mathrm{joint}} = \frac{2}{9} A(\varphi E_{\mathrm{prim}})' + \frac{2}{9} N_{\mathrm{cross}}^{\iota}. \tag{167}$$

To evaluate the second term, put $k_n = \deg Z_n$. The proper degree bound is $2k_n + r_n \le d - 1$. Since $\deg N_0 \le 3k_n$, the explicit D_x has bounds

$$\max \operatorname{supp} D_x \le -e_c + 1 + 3r_n + 3k_n \le -1 - \Delta/2, \qquad -\min \operatorname{supp} D_x \le e_c + 3k_n \le N - 1 - \Delta/2 < N.$$

Consequently $X_{\mathrm{cross}} = x^N D_x$ is an actual polynomial with strictly positive exponents. In (166) the two lower phases are strictly negative, so the full nonnegative projection is exactly

$$T_{\mathrm{cross}} := \operatorname{trunc}(N_{\mathrm{cross}}) = (93 - 44x^p) X_{\mathrm{cross}}/250.$$

This projection is a consequence of both support bounds, not a replacement of the original four-phase carrier.

At the two reciprocal endpoints, $N_0(1/4) = N_0(4) = -h$. Also sb_0c_0 is $-1/8$ at $1/4$ and -8 at 4. The powers in the explicit D_x satisfy

$$\eta 4^{e_c}(-1/8)^{r_n} = (-1)^{d+r_n} 2^{2N+1} = -2, \qquad \eta 4^{-e_c}(-8)^{r_n} = -1/2.$$

Thus

$$D_x(1/4) = 6(-2)(5/4)h^2 = -15h^2, \qquad D_x(4) = 6(-1/2)(-40)h^2 = 120h^2.$$

Since $4^N = (1/4)^N = 1$, these are also the values of X_{cross}. Since $c^p = c$ for $c = 1/4, 4$,

$$T_{\mathrm{cross}}(1/4) = \frac{(93-11)(-15)}{250}h^2 = -\frac{123}{25}h^2, \qquad T_{\mathrm{cross}}(4) = \frac{(93-176)120}{250}h^2 = -\frac{996}{25}h^2.$$

For every Laurent F, the origin geometric series gives

$$\operatorname{op}(F^{\iota}) = \frac{\operatorname{trunc}(F)(1/4) - \operatorname{trunc}(F)(4)}{15}.$$

Indeed a coefficient F_i, $i > 0$, contributes $(4^{-i} - 4^i)F_i/15$, and the constant contributes zero. It follows that

$$\operatorname{op}(N_{\mathrm{cross}}^{\iota}) = \frac{-123/25 + 996/25}{15}h^2 = \frac{291}{125}h^2.$$

The first term in (167) has origin $\frac{2}{9}[x^{-1}](\varphi E_{\mathrm{prim}})' = 0$, because a derivative has zero coefficient at x^{-1}, including the characteristic-p phases. Therefore

$$\operatorname{op}(J_{\mathrm{joint}}) = \frac{2}{9}\frac{291}{125}h^2 = \frac{194}{375}h^2. \tag{168}$$

The original period and HF successor pole. The already derived rational anti-reflection of the divided defect is

$$x^N \delta_{\mathrm{bal}}^{\iota} = -\delta_{\mathrm{bal}}.$$

The finite period log reduces to $e_N(1-V) = L_V^{\mathrm{desc}} - q$, so $\Pi^{\iota} = -\delta_{\mathrm{bal}} e$. Since $A^{\iota} = x^{-2}A$ and $d_x^{\iota} = -L_{\mathrm{n}}$,

$$\frac{2}{3} x^{2p} (A^p d_x \Pi)^{\iota} = \frac{2}{3} A^p L_{\mathrm{n}} \delta_{\mathrm{bal}} e.$$

Identity (SSA83), for δ_{bal} itself, gives both original period rows:

$$-\frac{2}{3} \tau_{p+1}(A^p d_x \Pi) = \mathrm{op}\left\{ L_{\mathrm{n}} \left(\frac{2}{3} f D e - \frac{10}{9} C_B e \right) \right\}. \tag{169}$$

The sign here includes the minus in (158), the minus in $\tau_m = -o_m(x^{2m-2}(\cdot)^{\iota})$, and the two reflection minuses just displayed. The defect is the original rational quotient, not a replacement field solution.

We next transport the whole HF density at its original pole order. Use the clearing polynomial $L_{\mathrm{cl}} = 2x(x+1)A$, distinct from L_{n}, and

$$\Gamma = -12x^3 + 9x^2 + 4x + 8, \quad R_{\mathrm{src}} = \tfrac{1}{2} x^{-p} A^{p-1} s^N, \quad P = \mathcal{P}, \quad L_V = L_V^{\mathrm{desc}}.$$

The full positive source and the Z' source in (125) are, with this notation,

$$L_{\mathrm{cl}} B_{\gamma} = -\Gamma S + \frac{L_{\mathrm{cl}}}{2x}(S - P), \qquad Z' = R_{\mathrm{src}} \left(-P + \frac{L_V - x^{-N} L_V^{\iota}}{3} \right) + \frac{P}{2x}.$$

Adding them cancels only the printed $P/(2x)$ pair:

$$L_{\mathrm{cl}}(B_{\gamma} + Z') = -\Gamma S + \frac{L_{\mathrm{cl}}}{2x} S + L_{\mathrm{cl}} R_{\mathrm{src}} \left(-P + \frac{L_V - x^{-N} L_V^{\iota}}{3} \right).$$

The polynomial identity

$$2x(x+1)\Delta_w + R_w \Gamma = 3x^2 A(4x-1)$$

is obtained by multiplying $\Delta_w = 24x^2 - 32x + 44$, $R_w = 4x^2 - 11x$, and the displayed Γ. Substitution into $\Omega_{\mathrm{HF}} = \Delta_w S - A R_w (B_{\gamma} + Z')$ gives

$$\begin{aligned} L_{\mathrm{cl}} \Omega_{\mathrm{HF}} = {} & \left(3x^2 A^2 (4x-1) - \tfrac{1}{2} L_{\mathrm{cl}} A R_w x^{-1} \right) S \\ & + L_{\mathrm{cl}} A R_w R_{\mathrm{src}} \left(P - \frac{L_V - x^{-N} L_V^{\iota}}{3} \right). \end{aligned} \tag{170}$$

Here $S = P + P^{\iota}$ is the independently derived symmetric comparison, so $S^{\iota} = S$.

Put

$$g = -11x^2 - 4x - 8, \quad \mathcal{D}_Z = 2x(x+1)C_x, \quad Z_{\mathrm{inv}} = P^{\iota} - \tfrac{1}{3}(L_V^{\iota} - x^N L_V).$$

Inverting (170), using $L_{\mathrm{cl}}^{\iota} = x^{-5} L_{\mathrm{cl}}$, and multiplying by x^7, gives

$$x^2 L_{\mathrm{cl}} \Omega_{\mathrm{HF}}^{\iota} = -A^2 g S + 2(x+1) A^2 C_x x^{-1} R_{\mathrm{src}}^{\iota} Z_{\mathrm{inv}}.$$

The first coefficient is

$$A^2 \{ 3(4-x) - (x+1) C_x \} = -A^2 g.$$

Define the actual origin successor numerator

$$F_{\mathrm{HF}}^{o} = -\tfrac{2}{3} x^{2p} F_{\mathrm{HF}}^{\iota} = -2x^2 C_x^{p-1} H \Omega_{\mathrm{HF}}^{\iota}, \qquad \Phi_H = -x^{-N} Z_{\mathrm{inv}} = -x^{-N} P^{\iota} + \frac{x^{-N}(a^{\iota} + h) - (a+h)}{3}.$$

Both the harmonic constant h and the full reciprocal a^{ι} remain in this bracket. Direct inversion of R_{src} gives

$$2x^{-2} C_x^p R_{\mathrm{src}}^{\iota} = A^{p-1} s^N x^{-N} \varphi, \qquad A^p K_{\Sigma} = 2 C_x^p S.$$

Multiply the preceding inverted equation by $-2C_x^pH$; its left side is $A\mathcal{D}_ZF^o_{\mathrm{HF}}$. The two displayed substitutions turn its right side into

$$A\mathcal{D}_ZF^o_{\mathrm{HF}} = A^{p+2}HK_\Sigma g + A\mathcal{D}_ZA^pHs^N\varphi\Phi_H.$$

Cancel the globally nonzero polynomial A. Finally the original branch factor (105) gives the explicit Laurent numerator

$$\Sigma_H = -\frac{1}{25}Ag\,C_x^{p-1}x^{-p-N}s^N\beta_D, \qquad \mathcal{D}_Z\Sigma_H = AK_\Sigma g.$$

For the last equality use $x+1=5s$, $C_x^px^{-p}=\varphi$, and $D_s = x^{1-N}s^{N+1}\beta_D$. Cancelling the globally nonzero $\mathcal{D}_Z$ therefore proves

$$F^o_{\mathrm{HF}} = A^pH(\Sigma_H + s^N\varphi\Phi_H), \qquad \frac{2}{3}\tau_{p+1}(F_{\mathrm{HF}}) = \mathrm{op}\{H(\Sigma_H + s^N\varphi\Phi_H)\}. \tag{171}$$

The trace equality keeps the denominator A^{p+1} until the shown A^p is cleared. Its coefficient formula commutes with the map into K.

Attachment to the same original W_{need}. Now use the native carriers already specified in (106):

$$\delta_{\mathrm{n}} = \tfrac{2}{5}D_sj + \tfrac{2}{3}De, \qquad b_{\mathrm{n}} = -\tfrac{2}{3}j - \tfrac{10}{9}e,$$

$$W_{\mathrm{need}} = \delta_{\mathrm{n}}L_{\mathrm{n}} - AH\delta_{\mathrm{n}}' + C_Bb_{\mathrm{n}}L_{\mathrm{n}} + H(\Sigma_H + s^N\varphi\Phi_H). \tag{172}$$

This is exactly the source in (SSA95), with the same $\Sigma_H, \Phi_H, H, D_s, j, e$. Its products are finite: (105) clears D_sj, and $C_B = s^{N+1}BY_0N_0/25$ clears C_Bj. No coefficient of W_{need} is chosen to force its origin.

Integration by parts at the origin gives $[x^{-1}]_0(H\delta_{\mathrm{n}})' = 0$. With (157), the first three terms of (172) consequently have origin

$$[x^{-1}]_0\frac{L_{\mathrm{n}}}{A}(f\delta_{\mathrm{n}} + C_Bb_{\mathrm{n}}) = J_{\mathrm{coupled}} + \mathrm{op}\left\{L_{\mathrm{n}}\left(\tfrac{2}{3}fDe - \tfrac{10}{9}C_Be\right)\right\}.$$

Indeed the j-coefficient is $\frac{2}{5}fD_s - \frac{2}{3}C_B = \frac{2}{3}(\frac{3}{5}fD_s - C_B)$, and $L_{\mathrm{n}} = -3B$. Using (169) and (171) proves

$$J_{\mathrm{coupled}} + \frac{2}{3}\tau_{p+1}(F_{\mathrm{HF}}) - \frac{2}{3}\tau_{p+1}(A^pd_x\Pi) = \mathrm{op}(W_{\mathrm{need}}). \tag{173}$$

Apply $\frac{2}{3}\tau_{p+1}$ to (158), add the original $\frac{2}{3}Q_{\mathrm{pr}}$, and use (163), (160), (168), and (173). The full low-density/proper contribution is therefore

$$\frac{2}{3}\tau_{p+1}(D_{\mathrm{low}}) + \frac{2}{3}Q_{\mathrm{pr}} - \mathrm{op}(W_{\mathrm{need}}) = \left(\frac{8}{125} + \frac{194}{375}\right)h^2 = \frac{218}{375}h^2. \tag{174}$$

Every term has reached the same origin functional on its original carrier.

The independent whole-source Cartier calculation in (179), before subtracting W_{need}, gives

$$\mathrm{op}(S_{\mathrm{origin}}) = T_5 + U_{\mathrm{rem}} + Z_{\mathrm{norm}} + \frac{2}{3}\tau_{p+1}(D_{\mathrm{low}}) + \frac{2}{3}Q_{\mathrm{pr}}.$$

Here the complete finite U_{rem} and Z_{norm} are those evaluated in (149); their independent derivations are not replaced by a definition through this equation. For the original $\rho = \mathrm{op}(S_{\mathrm{origin}} - W_{\mathrm{need}})$, (174) now proves

$$\rho = T_5 + U_{\mathrm{rem}} + Z_{\mathrm{norm}} + \frac{218}{375}h^2.$$

Finally the actual external divided scalar is

$$V_{\mathrm{div}} = \frac{2}{3}\overline{V_{\mathrm{bal},\mathbb{Q}}/p} - \frac{103}{1000}h^2.$$

Add (155); the exact scalar arithmetic is

$$\frac{218}{375}-\frac{103}{1000}=\frac{1744-309}{3000}=\frac{287}{600}.$$

We obtain the required original-source contraction

$$\boxed{V_{\mathrm{div}}+\rho=\mathcal{V}+U_{\mathrm{rem}}+Z_{\mathrm{norm}}+\frac{287}{600}h^2}. \tag{175}$$

All fixed denominators used in the normalization and trace reduction have prime factors $2,3,5$; 11 occurs only in the conic finite arguments, and is also a unit for $p\geq 19$. The construction and coefficient maps work for either fifth or conic root and descend injectively to the prime field; no root-existence or nonzero-Hasse condition has been imposed on that field.

4.5 Coefficient recipes for the balanced source

This appendix fixes every integration constant and every rational division used above. Put $\epsilon=5^N\in\mathbb{F}_p^\times$ and $\Delta e(z)=\sum_{i=1}^N(4^i-1)z^i/(2i)$. Define

$$W(x)=\frac{2\epsilon}{15}\{\Delta e(s)-\Delta e(1/5)\},\qquad K_0=\frac{17}{2\epsilon}-\frac{2}{\epsilon}(x^p+x^{-p}).$$

The coefficient of x^N in W is zero, because $4^N=2^{p-1}=1$; also $W(0)=0$. Thus $\deg W\leq N-1$. For $W=\sum_i w_ix^i$, define the four actual finite primitives

$$H_z(x)=\sum_{i=1}^{N-1}w_i\sum_{a=1}^{i}\frac{z^{i-a}x^a}{a}\quad(z=0,1/4,4,-1).$$

There is no unspecified primitive: these formulas have value zero at $x=0$, degree at most $N-1<p$, and

$$(x-z)H_z'=W(x)-W(z).$$

Indeed the derivative of the inner sum is $\sum_{a=1}^i z^{i-a}x^{a-1}=(x^i-z^i)/(x-z)$, as a polynomial identity even at $x=z$. Put

$$\begin{aligned}
H_{\mathrm{sym}}&=H_0+H_0^\iota-H_{1/4}-H_4-H_{1/4}^\iota-H_4^\iota-(H_{-1}+H_{-1}^\iota)/2,\\
J_a&=\frac{2}{\epsilon}(x^{-p}-1/4)\ell(4x), & J_b&=\frac{2}{\epsilon}(x^{-p}-4)\ell(x/4),\\
J_2&=\frac{2}{15}\{(x^{-p}-1/4)R(4x)-(x^{-p}-4)R(x/4)\}, & Z_E&=W(1/4)+W(4)+4\epsilon C/15,\\
U_{\mathrm{can}}&=K_0H_{\mathrm{sym}}-Z_E(J_a+J_b)-J_2/2.
\end{aligned}$$

Its actual branch constant is the full triangular expression

$$\begin{aligned}
u_0=\frac{5}{3}\{&S(2u,1)+S(2u,2)-S(u,1/2)-S(u,1)-S(2u,2u)+S(u,u)\\
&+(S(2u,0)-S(u,0))/2\}\\
&-\left(\frac{2}{3}q+\frac{1}{2}r\right)\ell(-4)+\frac{1}{6}R(-4).
\end{aligned}$$

Thus $U=U_{\mathrm{can}}-u_0$. The expression descends independently of the fifth root: replace $S(a,b)$ by $D_N(a^2,b^2)$, so all arguments in u_0 lie in $\mathbb{F}_p$. The branch normalization follows from the finite primitives themselves. In the full triangle defining D_N, termwise differentiation gives $(X-b)\partial_XD_N(X,b)=L_N(X)-L_N(b)$: the inner geometric sum has last index j, so its product by $X-b$ is X^j-b^j. Also

$$W=\frac{\epsilon}{15}\{L_N(4s)-L_N(s)-L_N(4/5)+L_N(1/5)\}.$$

For $s_z = (1+z)/5$, the polynomial

$$\frac{\epsilon}{15}\{D_N(4s,4s_z) - D_N(s,s_z) - D_N(4/5,4s_z) + D_N(1/5,s_z)\}$$

has origin zero and, after multiplication of its derivative by $x-z$, equals $W(x)-W(z)$. Its degree is $N<p$. It equals H_z: the difference has zero derivative after cancellation of the nonzero $x-z$, degree $<p$, and origin zero. Evaluating at $x=-1$, where $s=0$, gives

$$H_z(-1) = -\frac{\epsilon}{15}\{D_N(4/5,4(1+z)/5) - D_N(1/5,(1+z)/5)\}.$$

Thus $H_0(-1) = -\epsilon D_5/15$, $H_{1/4}(-1) = -\epsilon A(u)/15$, $H_4(-1) = -\epsilon B(u)/15$, and $H_{-1}(-1) = -\epsilon\{S(2u,0) - S(u,0)\}/15$. Inversion fixes -1, so $H_{\mathrm{sym}}(-1) = 2H_0(-1) - 2H_{1/4}(-1) - 2H_4(-1) - H_{-1}(-1)$. Also

$$K_0(-1) = \frac{25}{2\epsilon}, \qquad (J_a+J_b)(-1) = -\frac{5}{\epsilon}\ell(-4), \qquad J_2(-1) = -R(-4)/3.$$

For the last two equations use $\ell(-1/4) = \ell(-4)/4$ and $R(-1/4) = -R(-4)/4$. Since $Z_E = -\epsilon(2q/15 + r/10)$, substitution gives exactly the displayed u_0, hence $U(-1)=0$.

Here are the pole-cleared differentiations which verify its source. Put $a=1/4$, $\Pi = x(x+1)(x-a)(x-4)$, $\Pi_0 = (x+1)(x-a)(x-4)$, $w_z = W(z)$. From the four primitive equations and $(H_z^\iota)' = -x^{-2}(H_z')^\iota$, one obtains

$$\begin{aligned} 2\Pi H'_{\mathrm{sym}} = {} & 2\Pi_0(W - W^\iota) - 2x(x+1)(x-4)(W - w_a) \\ & - 2x(x+1)(x-a)(W - w_4) - 8(x+1)(x-a)(W^\iota - w_a) \\ & - 2a(x+1)(x-4)(W^\iota - w_4) - x(x-a)(x-4)(W - w_{-1}) \\ & + (x-a)(x-4)(W^\iota - w_{-1}). \end{aligned}$$

For example inversion changes the primitive at a to $x(x-4)(H_a^\iota)' = 4(W^\iota - w_a)$; this accounts for the coefficient 8 above. Define

$$\Gamma_2 = x(x-a)(x-4) + 2x(x+1)(2x-a-4) - 2\Pi_0, \quad \Gamma_a = x(x+1)(2x-a-4) - \Pi_0.$$

Use $2w_{-1} + 4\epsilon C/15 = 0$, obtained by evaluating $W(-1)$, and collect the preceding seven terms. The result is

$$4\Pi H'_{\mathrm{sym}} = 2\Pi_0 W^\iota - 2\Gamma_2(W + W^\iota + 4\epsilon C/15) + 4Z_E\Gamma_a + (4\epsilon C/15)\Pi_0.$$

The finite logarithm derivative gives, for $c \in \mathbb{F}_p^\times$,

$$(x-c)\{\ell(x/c)\}' = x^{p-1} - 1.$$

Inserting $c = a, 4$ in J_a, J_b, and using $K_0' = 0$, gives

$$\Pi(J_a+J_b)' = K_0\Gamma_a + \frac{2}{\epsilon}\Pi(x^{p-1} - x^{-p-1}), \qquad xJ_2' = \frac{2}{15}\{(x^{-p}-a)\ell(4x) - (x^{-p}-4)\ell(x/4)\}.$$

Retain the actual invisible part of P in its decomposition $P = K_0 W + J_{\log} + I_P$, where

$$J_{\log} = \frac{2}{15}\{(x^{-p}-a)\ell(4x) - (x^{-p}-4)\ell(x/4)\}, \quad I_P = \left(-\frac{8C}{15} - \frac{4(q-C)}{15}\right)x^p + \frac{17C}{15} + \frac{4(q-C)}{15}x^{-p}.$$

The full finite-log inversion (64) gives

$$\ell(4/x) = -4x^{-p}\ell(x/4), \qquad \ell(1/(4x)) = -\tfrac{1}{4}x^{-p}\ell(4x).$$

Here $4^p = 4$ and $(1/4)^p = 1/4$. Substitution, with $a = 1/4$, therefore yields

$$J_{\log}^\iota = \frac{2}{15}\{(x^p - a)(-4x^{-p})\ell(x/4) + (x^p - 4)\tfrac{1}{4}x^{-p}\ell(4x)\}$$

$$= \frac{2}{15}\{(x^{-p}-4)\ell(x/4) - (x^{-p}-a)\ell(4x)\} = -J_{\log}.$$

The coefficients of the two phases in $I_P + I_P^\iota$ are both $-8C/15$, and its constant coefficient is $34C/15$; hence

$$I_P + I_P^\iota = \frac{2C}{15}(17 - 4x^p - 4x^{-p}) = \frac{4\epsilon C}{15}K_0.$$

The fifth entropy row (77) gives $\Delta e(1/5) = E(2u) - E(u) = C$. Thus $W = 2\epsilon\{\Delta e(s) - C\}/15$ and

$$W + W^\iota + 4\epsilon C/15 = \frac{2\epsilon}{15}\{\Delta e(s) + \Delta e(s)^\iota\}.$$

Since $K_0 = (17 - 4x^p - 4x^{-p})/(2\epsilon)$, this proves

$$P + P^\iota = K_0(W + W^\iota + 4\epsilon C/15) = \frac{17 - 4x^p - 4x^{-p}}{15}\{\Delta e(s) + \Delta e(s)^\iota\} = S. \tag{176}$$

This is precisely the reduction of $S_{\mathbb{Q}}$: $\overline{W_{\mathbb{Q}}} = \Delta e(s)$, while its retained branch scalar vanishes because $17 - 4(4^{-p} + 4^p) = 17 - 4(1/4 + 4) = 0$ in K. Substituting these formulas in $4\Pi U'_{\mathrm{can}}$ gives

$$4\Pi U' = 2\Pi_0 P^\iota - 2\Gamma_2 S - 4\alpha\Pi(x^{p-1} - x^{-p-1}), \qquad \alpha = -\frac{2q+3r}{15}.$$

The remaining scalar coefficient vanishes because

$$-\frac{8q}{15} - \frac{8}{\epsilon}Z_E + 4\alpha = 0, \qquad Z_E = -\epsilon(2q/15 + r/10).$$

Thus the actual polynomial quotient B_γ, rather than an arbitrary source, is

$$4\Pi B_\gamma = 2\Pi_0 P^\iota - 2\Gamma_2 S, \qquad U' = B_\gamma - \alpha(x^{p-1} - x^{-p-1}).$$

Existence of this quotient follows from the displayed equation for U'; uniqueness follows because $\Pi \neq 0$ in the Laurent domain. As $A = -4(x-a)(x-4)$, $L_{\mathrm{cl}} = -8\Pi$, and $4\Gamma_2 = -\Gamma$, its equivalent balanced form is

$$L_{\mathrm{cl}}B_\gamma = -\Gamma S + \frac{L_{\mathrm{cl}}}{2x}(S - P).$$

To specify the actual rational counterterm, define

$$\mathrm{End}(F) = F + F^\iota - F(1/4) - F(4), \qquad \mathrm{Can}_{\mathbb{Q}}(F) = F - F_0 - F_p x^p.$$

For a Laurent density $F = \sum F_j x^j$, set

$$\mathcal{J}_{\mathbb{Q}}(F) = \sum_{j\neq 0,p} \frac{F_{j-1}}{j}x^j, \qquad \mathcal{J}_{\mathrm{ord}}(F) = \sum_{j\neq 0} \frac{F_{j-1}}{j}x^j.$$

These are characteristic-zero finite sums. Put

$$K_{\mathrm{act}} = \frac{x^{-p}A^p}{2\,5^{5N}}, \quad R_{\mathrm{act}} = \frac{x^{-p}(x+1)^N A^{p-1}}{2\,5^{5N}}, \quad k_{\mathrm{act}} = \frac{[x^{p-1}]R_{\mathrm{act}}}{p}, \quad Z_p = x^p + x^{-p} - 4^p - 4^{-p}.$$

The ordinary primitive is $\mathcal{J}_{\mathrm{ord}}(R_{\mathrm{act}}) = \mathcal{J}_{\mathbb{Q}}(R_{\mathrm{act}}) + k_{\mathrm{act}}x^p$. Centering its full symmetric pair and subtracting $(k_{\mathrm{act}} + 8C_{\mathbb{Q}}^{(5)}/15)Z_p$ therefore gives exactly

$$S_{\mathrm{act},\mathbb{Q}} = \mathrm{End}(\mathcal{J}_{\mathbb{Q}}(R_{\mathrm{act}})) - \frac{8}{15}C_{\mathbb{Q}}^{(5)}Z_p.$$

This equality cancels the actual k_{act} over $\mathbb{Q}$; no reduction or omission of its positive-p coefficient is used.

For the comparison product define

$$E_{4,c} = e_{\mathbb{Q}}(4s) - e_{\mathbb{Q}}(4/5), \qquad E_{1,c} = e_{\mathbb{Q}}(s) - e_{\mathbb{Q}}(1/5),$$
$$W_{\mathrm{bar}} = \frac{2\,5^N}{15}(E_{4,c} - E_{1,c}), \qquad W_{\mathrm{act}} = \frac{2\,5^N}{15}(4^{-N}E_{4,c} - E_{1,c}),$$
$$K_{\mathrm{bar}} = \frac{\phi_{\mathbb{Q}}}{2\,5^N}, \qquad W_1 = \frac{W_{\mathrm{act}} - W_{\mathrm{bar}}}{p} = -\frac{2\,5^N Q_2}{15\,4^N}E_{4,c},$$
$$K_1 = (K_{\mathrm{act}} - K_{\mathrm{bar}})/p, \qquad c_{\mathrm{pr}} = 5^{-2(p-1)}(4^{-N} - 1),$$
$$R_{\mathrm{pr}} = \frac{2}{15}x^{-p}(4-x)A^{p-1}, \qquad J_{\mathrm{pr},\mathbb{Q}} = \sum_{-p<j<p} \frac{[x^{j-1}]R_{\mathrm{pr}}}{j}x^j,$$

where the $j = 0$ summand is zero. Every other denominator in $J_{\mathrm{pr},\mathbb{Q}}$ is a p-unit. Let

$$F_{\log,\mathbb{Q}} = \frac{x^{-p}A^{p-1}W_{\mathrm{act}}A'}{2\,5^{5N}} - K_{\mathrm{act}}W_{\mathrm{act}}/x.$$

Product differentiation gives the exact rational primitive equation

$$\mathcal{J}_{\mathbb{Q}}(R_{\mathrm{act}}) = \mathrm{Can}_{\mathbb{Q}}(K_{\mathrm{act}}W_{\mathrm{act}}) + c_{\mathrm{pr}}J_{\mathrm{pr},\mathbb{Q}} - p\mathcal{J}_{\mathbb{Q}}(F_{\log,\mathbb{Q}}).$$

It can be checked coefficientwise: $D(K_{\mathrm{act}}W_{\mathrm{act}})$ is $R_{\mathrm{act}} - c_{\mathrm{pr}}R_{\mathrm{pr}} + pF_{\log,\mathbb{Q}}$; applying $\mathcal{J}_{\mathbb{Q}}$ to a derivative removes exactly its $0, p$ coefficients. The support of R_{pr} leaves exactly the stated unit integration band. The same centered product for $K_{\mathrm{bar}}W_{\mathrm{bar}}$, with the same $C_{\mathbb{Q}}^{(5)}Z_p$, is $S_{\mathbb{Q}}$ in (111). Indeed its $0, p$ product coefficients are zero because $W_{\mathrm{bar}}(0) = 0$, $\deg W_{\mathrm{bar}} \le N$, and its phase exponents are $0, \pm p$; its remaining endpoint constant is the one printed there. Consequently the complete quotient is

$$VS_{\mathbb{Q}} = \frac{S_{\mathrm{act},\mathbb{Q}} - S_{\mathbb{Q}}}{p} = \mathrm{End}\,\mathrm{Can}_{\mathbb{Q}}(K_1W_{\mathrm{bar}} + K_{\mathrm{bar}}W_1 + pK_1W_1) + \frac{c_{\mathrm{pr}}}{p}\,\mathrm{End}(J_{\mathrm{pr},\mathbb{Q}}) - \mathrm{End}(\mathcal{J}_{\mathbb{Q}}(F_{\log,\mathbb{Q}})). \tag{177}$$

All terms on the right are integral. For W_1 and $c_{\mathrm{pr}}/p = -5^{-2(p-1)}Q_2/4^N$, this is explicit above. For K_1, expand A^p. Every mixed multinomial coefficient $p!/(i_0!i_1!i_2!)$, $i_0 + i_1 + i_2 = p$, has valuation at least one unless a single $i_\nu = p$. The pure terms are $(-4)^p, 17^p x^p, (-4)^p x^{2p}$. Their difference from $5^{4N}\phi_{\mathbb{Q}}$ is divisible by p, because $5^{4N} \equiv 1$ and $a^p \equiv a \pmod p$. Thus the coefficient numerator of K_1 is divisible by p. The support of $F_{\log,\mathbb{Q}}$ is $-p \le j \le 2p-2$. Its integration indices lie in $-p < j < 2p$; the only nonunits there are $0, p$, both omitted. The possible lower endpoint $j = -p$ is absent because $W_{\mathrm{act}}(0) = 0$. No denominator of valuation -1 survives. The integrality proof of (177) establishes the divisibility of the whole original numerator, including all its Frobenius coefficients and endpoint constants.

Here is its field recipe. Set

$$\mathcal{N} = (x^{-p} - 4)\ell(x/4) + (x^{-p} - 1/4)\ell(4x), \quad K_1^{(0)} = 2(q-k)K_0 + \frac{15q}{\epsilon} + \frac{2}{\epsilon}\mathcal{N},$$

$$W_1^{(0)} = -2\epsilon q\,\{e_N(4s) - e_N(4/5)\}/15, \qquad T_{\mathrm{prod}} = K_1^{(0)}W + K_0W_1^{(0)}.$$

For the mixed multinomials above, divide first by p and use $\binom{p}{i}/p = (-1)^{i-1}/i \pmod p$; collecting the two linear factors of A gives exactly the two finite logarithms in $\mathcal{N}$. The pure coefficients and the denominator 5^{4N} give $2(q-k)K_0 + 15q/\epsilon$. This proves the stated residue of K_1. The reduction of the proper primitive and of the logarithmic primitive are

$$J_{\mathrm{pr}} = \frac{2}{15}(x^{-p} - 1/4)\ell(4x), \qquad G_{\mathrm{cl}} = K_0(H_{1/4} + H_4 - H_0) + w_{1/4}J_a + w_4J_b.$$

For the first, expand the geometric factor $(4-x)A^{p-1}$, integrate every coefficient in the open unit band, and use the finite geometric identity for ℓ'. Its values at $1/4, 4$ are zero. For the second, differentiate the three H_z equations and the two J equations above to obtain

$$G'_{\mathrm{cl}} = \bar{F}_{\log,\mathbb{Q}} - \frac{2}{\epsilon}(4w_{1/4} + w_4/4)x^{-1} + \frac{2}{\epsilon}(w_{1/4} + w_4)x^{p-1}.$$

All coefficients of G_{cl} at exponents divisible by p are zero: every H_z has degree $< p$ and zero origin, and each shifted finite logarithm has exponents in the open bands $(-p, 0)$, $(0, p)$. The nonresonant integration operator therefore annihilates the two displayed resonant densities and sends G'_{cl} to G_{cl} exactly. Let $\mathrm{Can}_0(F) = F - F_0 - F_p x^p$ in the field; its value here equals the three-hole prescription with the $-p$ coefficient also removed, since $T_{\mathrm{prod},-\mathrm{p}} = 0$. Reduction of (177) gives

$$\boxed{VS = \mathrm{End}(\mathrm{Can}_0\, T_{\mathrm{prod}}) - q\,\mathrm{End}(J_{\mathrm{pr}}) - \mathrm{End}(G_{\mathrm{cl}}), \qquad UV = U - VS.} \tag{178}$$

This fixes the actual UV, not just its derivative. The original counterterm derivative, before division, is

$$S'_{\mathrm{act},\mathbb{Q}} = B_{\mathbb{Q}} - p\left(k_{\mathrm{act}} + \frac{8}{15}C^{(5)}_{\mathbb{Q}}\right)(x^{p-1} - x^{-p-1}).$$

Thus $VS' = -\sigma - \alpha(x^{p-1} - x^{-p-1})$. The closed value of α also follows by inspecting the $p-1, -p-1$ coefficients of σ: a Laurent derivative has zero coefficients at those indices, and the displayed equation there gives $\sigma_{p-1} = -\alpha$, $\sigma_{-p-1} = \alpha$. Subtracting this derivative from U' proves

$$UV' = \sigma + B_\gamma, \qquad 2L_{\mathrm{cl}}UV' = 2L_{\mathrm{cl}}\sigma - 2\Gamma S + \frac{L_{\mathrm{cl}}}{x}(S - P).$$

The support of U is contained in $[-3N, 3N]$: $K_0 H_{\mathrm{sym}}$ has extremes $\pm(p+N-1) = \pm 3N$, and all J sums have smaller extremes. The product carry, proper primitive, and closed logarithmic primitive in (178) have extremes at most $3N+1$ in either direction. Therefore UV, T_0 have the bound used in the five-slot argument, with no erased $2p$ term.

The full source sent to the native origin. The common, balanced and period densities are combined before taking any Laurent coefficient. In the notation above their actual field sum is

$$D_{\mathrm{whole}} = 3R_w^{p-1}\Delta_w H_c S - A(3R_w^{p-1}H_c B_\gamma) - A(3R_w^p H'_c UV) + 3R_w^p d_x x^{-N} E - A^p d_x \Pi,$$

where $H_c = H_0(V)^\iota$ and $\Pi = x^{-N}\delta\, e_N(1-V)$. Each of these is a finite Laurent polynomial fixed by its earlier coefficient recipe. Consequently $\tau_{p+1}(D_{\mathrm{whole}}) = \bar{C}_{\mathbb{Q}} + \bar{B}^{\mathrm{tr}}_{\mathbb{Q}} - \bar{P}^{\mathrm{tr}}_{\mathbb{Q}}$. There is an explicit conversion to the native simple pole. For $F = \sum F_j x^j$, define its differential Cartier selection by

$$\mathfrak{C}(F) = \sum_j F_{pj+p-1}x^j, \qquad \mathfrak{S}(F) = \frac{8C_F + A'C'_F}{225}, \quad C_F = \mathfrak{C}(A^{p-1}F).$$

The discriminant and its derivative are

$$(A')^2 + 16A = 225, \qquad A'' = -8.$$

Multiplying the first by F, taking τ_2, and integrating the derivative of $A'F/A$ gives

$$225\tau_2(F) = 16\tau_1(F) + \tau_1(-8F + A'F') = \tau_1(8F + A'F').$$

The denominator $225 = 3^2 5^2$ is a unit. Since A^{2p} is a Frobenius polynomial, its reciprocal series has exponents divisible by p. In the coefficient $[x^{-1}]F/A^{2p}$, therefore only numerator exponents congruent to $p-1$ modulo p contribute. Taking those coefficients is exactly $\mathfrak{C}$; its coefficients lie in $\mathbb{F}_p$, so their p-th powers are unchanged. Clearing A^{p-1} proves

$$\tau_{p+1}(F) = \tau_2(\mathfrak{C}(A^{p-1}F)) = \tau_1(\mathfrak{S}(F)) = -\operatorname{op}(\mathfrak{S}(F)^\iota).$$

All positive and negative Laurent exponents are retained by this selection. The last equality uses the exact inversion convention in (123) and (‡), not an endpoint approximation.

For the proper source $\mathcal{Q}$, put $q(z) = \mathrm{qi}\,\mathcal{Q}(z)$,

$$\mathcal{D}_q(z) = \left(\sum_{i\geq 1} q_i V(z)^{i-1}\right) \frac{132z^3 - 81z^2 + 12z}{125}, \qquad D_{\mathrm{pr}}(x) = x^N \mathcal{D}_q(x^{-1}).$$

The sum is $q.\,\mathrm{div}\,z$ evaluated at V; it is a polynomial even when $V(z) = 0$. Its definition thus includes the subtracted constant q_0. Direct coefficient inversion gives $[x^{-1}]D_{\mathrm{pr}} = [z^{N+1}]\mathcal{D}_q = Q_{\mathrm{pr}}$. As the native origin satisfies $\mathrm{op}(AF) = [x^{-1}]F$, the SAME original whole source is the following explicit Laurent polynomial:

$$S_{\mathrm{origin}} = -\frac{2}{3}\mathfrak{S}(D_{\mathrm{whole}})^{\iota} + \frac{2}{3}AD_{\mathrm{pr}}, \qquad \rho = \mathrm{op}(S_{\mathrm{origin}} - W_{\mathrm{need}}). \tag{179}$$

Its origin is

$$\mathrm{op}(S_{\mathrm{origin}}) = \frac{2}{3}(\bar{C}_{\mathbb{Q}} + \bar{B}^{\mathrm{tr}}_{\mathbb{Q}} - \bar{P}^{\mathrm{tr}}_{\mathbb{Q}}) + \frac{2}{3}Q_{\mathrm{pr}}.$$

This proves the whole-source attachment used in (175). It neither supplies nor replaces the full mixed Green calculation of the original W_{need}.

The actual entropy and harmonic carries. The following calculations justify the carries in (151) before applying a field map. For $r < p$, put

$$P_r = \prod_{j=1}^{r}(1 - p/j), \quad H_r = \sum_{j=1}^{r} 1/j, \quad P_r = 1 - pH_r + p^2\mathcal{T}_r.$$

The remainder is integral and completely specified by $\mathcal{T}_0 = 0$, $\mathcal{T}_{r+1} = \mathcal{T}_r + H_r/(r+1) - p\mathcal{T}_r/(r+1)$. The identity follows by multiplying the preceding product by its last factor. Every denominator is a p-unit. For $1 \leq i \leq N$, the exact binomial coefficient equation is

$$-\binom{p}{2i}/p = P_{2i-1}/(2i).$$

Therefore, for a characteristic-zero argument t, define

$$\mathcal{E}(t) = \frac{2 - (1-t)^p - (1+t)^p}{2p}, \qquad \mathcal{O}(t) = \frac{(1+t)^p - (1-t)^p - 2t^p}{2p},$$

$$a_{\mathbb{Q}}(t) = \sum_{i=1}^{N} \frac{H_{2i-1} - p\mathcal{T}_{2i-1}}{2i}\, t^{2i}.$$

They have integral polynomial coefficients and the exact relation

$$e_{\mathbb{Q}}(t^2) = \mathcal{E}(t) + p\, a_{\mathbb{Q}}(t).$$

Its top index is $i = N$, with $2N = p - 1$, and is retained. The field reduction of $a_{\mathbb{Q}}$ is

$$a(t) = \sum_{i=1}^{N} \frac{H_{2i-1}}{2i} t^{2i} = -\frac{R(1-t) + R(1+t)}{2}.$$

For a direct proof of the last equality, consecutive coefficients in the first polynomial give $(1-t^2)a' = tE + O$. The last coefficient is $-H_{p-2} = -1$, since $H_{p-1} = 0$. Differentiation of the second polynomial, followed by finite reflection, gives the same equation. Both have degree $< p$, and value zero at 0, so their difference is zero. At $t = 1$, $\mathcal{E}(1) = -Q_2$. Thus $k_H = a_{\mathbb{Q}}(1)$ is integral and $\bar{k}_H = -R(2)/2 = q^2/2$.

The endpoint carry needs the full sixth harmonic, not just its zeroth residue. Put $m = \lfloor p/3 \rfloor$ and

$$C_n^{\mathrm{ref}} = \sum_{j=1}^{n} \frac{1}{j(p-j)}, \qquad C^{\mathrm{odd}} = \sum_{j=d+1}^{N} \frac{1}{j(p-2j)}.$$

All their denominators are units. The elementary identities

$$\frac{1}{p-j} = -\frac{1}{j} + \frac{p}{j(p-j)}, \qquad \frac{1}{j} = -\frac{2}{p-2j} + \frac{p}{j(p-2j)}$$

give, on the full reflected intervals,

$$H_{p-1-n} = H_{p-1} + H_n - pC_n^{\mathrm{ref}}, \quad H_{p-1} = pC_N^{\mathrm{ref}},$$

$$H_d = H_N + H_m + 2H_{p-1} - p(2C_m^{\mathrm{ref}} + C^{\mathrm{odd}}).$$

In the last identity $j \mapsto p - 2j$, $d < j \le N$, is a bijection onto the odd indices $1 \le j \le p - 1 - m$; this is checked in the two classes $p = 6d + 1, 6d + 5$. The even indices in that interval have sum $H_m/2$. This proves the exact suffix formula and keeps both remainders.

To obtain the first digit of H_m, let

$$A_3 = \sum_{\substack{1 \le j < p \\ 3 \mid j}} \frac{H_{j-1}}{j}, \quad S_3 = \sum_{\substack{1 \le j < p \\ 3 \mid j}} (-1)^j \binom{p}{j}/p.$$

The signed product expansion gives

$$S_3 = -H_m/3 + pC_3, \qquad C_3 = \sum_{3 \mid j} \frac{H_{j-1} - p\mathcal{T}_{j-1}}{j}.$$

In characteristic zero adjoin $\omega^2 + \omega + 1 = 0$. The cubic binomial filter and $p \equiv \pm 1 \pmod 3$ give

$$(1-\omega)^p + (1-\omega^2)^p = 3 + 3pS_3.$$

The quotient $B = (1-\omega)^p/(1-\omega^p)$ is the same for the two conjugates: using $1 - \omega^2 = -\omega^2(1-\omega)$ verifies this for the two residue classes. Therefore $B = 1 + pS_3$, and

$$B^2 = 3^{p-1} = 1 + pQ_3, \qquad S_3 = Q_3/2 + pU_3, \qquad U_3 = -S_3^2/2 \in \mathbb{Z}_{(p)}.$$

Combining the two equations for S_3 proves

$$H_m = -3Q_3/2 + pC_m, \qquad C_m = 3(C_3 - U_3).$$

There is no assumption of a cube root in the prime field. To reduce A_3, extend the field injectively to include a root ω, put $w = \omega^p$, and define $A(z) = \sum_{j=1}^{p-1} H_{j-1} z^j/j$. Termwise telescoping gives $(1-z)A' = \ell(z)$, and hence $A(z) = -R(1-z)$, by degree $< p$ and origin zero. The cubic filter gives

$$A(\omega) + A(\omega^2) = 3A_3, \quad R(\omega) + R(\omega^2) = H_m^{(2)}/3, \quad R(\omega^2) = w^2 R(\omega).$$

The equation for B above gives $\ell(\omega) = -(1-w)r/2$; reflection in each conjugate gives

$$(1-w)A(\omega) = wR(\omega) + (1-w)^2 r^2/8,$$

and the same equation with w^2, ω^2. Multiply these equations respectively by $1 - w^2, 1 - w$; use $(1-w)(1-w^2) = 3$, $w^2 + w + 1 = 0$. Their sum is

$$3(A(\omega) + A(\omega^2)) = 3wR(\omega) + 9r^2/8.$$

Since $R(\omega)+R(\omega^2)=-wR(\omega)$, substitution of the two filtered traces yields

$$A_3 = r^2/8 - H_m^{(2)}/9, \quad \bar{U}_3 = -r^2/8, \quad \bar{C}_m = 3r^2/4 - H_m^{(2)}/3.$$

These expressions are in the prime field, so injectivity returns the identity to the original field.

The full sum of inverse squares is zero: multiplication of its index by 2 permutes $\mathbb{F}_p^\times$, and $4-1\neq 0$. Reflection thus gives $H_N^{(2)}=0$. Splitting the interval $1,\dots,p-1-m$ into even and odd indices and applying the same bijection $j\mapsto p-2j$ gives

$$-H_m^{(2)} = H_m^{(2)}/4 - H_d^{(2)}/4, \quad H_d^{(2)} = 5H_m^{(2)}.$$

Consequently $\bar{C}_n^{\mathrm{ref}} = -H_n^{(2)}$, $\bar{C}^{\mathrm{odd}} = 5H_m^{(2)}/2$. The exact sixth harmonic carry is

$$C_d = 2k_H + C_m + 2C_N^{\mathrm{ref}} - 2C_m^{\mathrm{ref}} - C^{\mathrm{odd}}, \quad H_d = h_{\mathbb{Q}} + pC_d,$$

with reduction $q^2 + 3r^2/4 - 5H_m^{(2)}/6$. For the endpoint product set

$$P_d^{(2)} = \sum_{i<j\le d} \frac{1}{ij}, \qquad \prod_{j=1}^{d}(1-\eta/j) = 1 - \eta H_d + \eta^2 P_d^{(2)} - \eta^3 P_d^{(3)}(\eta),$$

where $P_d^{(3)}(\eta) = \sum_{r=3}^{d}(-\eta)^{r-3}\sum_{i_1<\cdots<i_r\le d} 1/(i_1\cdots i_r)$. All terms are integral at $\eta = p/3$. The recurrence endpoint $J_{\mathbb{Q}}(1) = (m_*-1)/\eta_*$ therefore gives

$$k_e = C_d - P_d^{(2)}/3 + pP_d^{(3)}(p/3)/9.$$

The relation $2P_d^{(2)} = H_d^2 - H_d^{(2)}$ shows explicitly that its $H_m^{(2)}$ terms cancel:

$$\bar{k}_e = q^2 + 3r^2/4 - \frac{5}{6}H_m^{(2)} - \frac{1}{6}(h^2 - 5H_m^{(2)}) = q^2/3 - qr + 3r^2/8.$$

For k_0, retain the rational phases

$$\kappa_\phi = 4^{-p} + 4^p - (11/4)^p, \quad \beta_\phi = 4^{-p} + 2\cdot 4^p,$$

$$A_e = 2e_{\mathbb{Q}}(4) - e_{\mathbb{Q}}(1/4) - e_{\mathbb{Q}}(1) - 3C_{\mathbb{Q}}^{(5)}, \quad D_e = e_{\mathbb{Q}}(1/4) - e_{\mathbb{Q}}(4).$$

Their actual initial weighted trace is

$$F_{0,\mathbb{Q}} = \frac{5^{p-1}}{5}\{(17-4\kappa_\phi)A_e + 4\beta_\phi D_e\}.$$

Expand $a^p = a(1+pQ_a)$ at $a=2,5,11$, using exact inverses of these units. Then

$$\kappa_\phi = 3/2 + p\kappa_1, \quad \beta_\phi = 33/4 + p\beta_1, \quad \bar{\kappa}_1 = 13q - 11j/4, \quad \bar{\beta}_1 = 31q/2.$$

The entropy endpoints are

$$\mathcal{E}(1) = -Q_2, \quad \mathcal{E}(2) = -3Q_3/2, \quad \mathcal{E}(1/2) = (Q_2 - 3Q_3/4)/2^{p-1}.$$

Thus

$$A_e + 3D_e = 3(Q_2 - C_{\mathbb{Q}}^{(5)}) + pr_{1,\mathbb{Q}},$$

$$r_{1,\mathbb{Q}} = -a_{\mathbb{Q}}(2) + 2a_{\mathbb{Q}}(1/2) - k_H - Q_2(2Q_2 - 3Q_3/2)/2^{p-1}.$$

This integral recipe proves the actual quotient division. Its residue, using the rational R values in (70), is

$$-a(2) + 2a(1/2) - 5q^2/2 + 3qr/2 = R(-2)/2 - 2q^2 + 3qr/2 = r_1.$$

Expansion of $F_{0,\mathbb{Q}}$ now gives, exactly,

$$k_0 = \frac{11}{5}r_{1,\mathbb{Q}} + \frac{33}{5}Q_5(Q_2 - C_{\mathbb{Q}}^{(5)}) + \frac{11p}{5}Q_5 r_{1,\mathbb{Q}} - \frac{4}{5}5^{p-1}\kappa_1 A_e + \frac{4}{5}5^{p-1}\beta_1 D_e.$$

Reducing $A_e = -9r/4 - 3C$, $D_e = q + 3r/4$ gives precisely $\bar{k}_0 = 11r_1/5 + \Phi$ in (152).

The fifth entropy carry. Work first in the integral quadratic algebra $\mathbb{Z}_{(p)}[u]/(5u^2-1)$. Put

$$U_2 = 2^{p-1} = 1 + pQ_2, \quad W_5 = 5^{p-1} = 1 + pQ_5, \quad J = \frac{5Q_5 - 8Q_2 - 4pQ_2^2}{10W_5},$$

$$H_{\mathbb{Q}} = Q_5/2 - 4Q_2/5, \qquad J_1 = \frac{-2Q_2^2/5 - Q_5 H_{\mathbb{Q}}}{W_5}.$$

Thus $J = H_{\mathbb{Q}} + pJ_1$ exactly. For $E_1 = \mathcal{E}(u)$, $E_2 = \mathcal{E}(2u)$, $O_1 = \mathcal{O}(u)$, $O_2 = \mathcal{O}(2u)$, $r_u = u^p$, $s_u = 2U_2 r_u$, define

$$\begin{aligned} Q_{\mathrm{cub}} &= -E_2^2 - O_2^2 - 2E_1E_2 + 2s_uE_1O_2 + 2O_1O_2 - 2r_uE_2O_2 - 2s_uE_2O_1, \\ R_{\mathrm{cub}} &= E_1(E_2^2 + O_2^2) - 2E_2O_1O_2. \end{aligned}$$

The exact entropy split $(1 \pm t)^p = 1 \pm t^p - p\mathcal{E}(t) \pm p\mathcal{O}(t)$ gives the norm equations

$$E_1 + r_uO_1 = J + \frac{p}{2}(E_1^2 - O_1^2), \quad E_2 + s_uO_2 = J + \frac{p}{2}(E_2^2 - O_2^2).$$

The cubic factorizations

$$1 + 11u/5 = (1-u)(1+2u)^2, \qquad 1 - 11u/5 = (1+u)(1-2u)^2$$

then give, by multiplying the two entropy splits,

$$\begin{aligned} \mathcal{E}(-11u/5) = {} & -4U_2Q_2/(5W_5) + (4 - 4U_2/(5W_5) - 1/U_2)E_2 \\ & + (1 + 4U_2^2/(5W_5) - 4U_2)E_1 + (1/U_2 - 2 + 4U_2)J \\ & + p\{Q_{\mathrm{cub}} + (1/(2U_2) - 1)(E_2^2 - O_2^2) + 2U_2(E_1^2 - O_1^2)\} + p^2R_{\mathrm{cub}}. \end{aligned}$$

This is an equality in characteristic zero with its full p^2 term. Subtract the zeroth expression $11(E_2 - E_1)/5 - 16Q_2/5 + 3Q_5/2$; use $U_2 - 1 = pQ_2, W_5 - 1 = pQ_5, J - H_{\mathbb{Q}} = pJ_1$. The exact quotient is the integral element

$$\begin{aligned} F_5 = {} & (Q_2/U_2 - 4(Q_2 - Q_5)/(5W_5))E_2 \\ & + (4(2Q_2 - Q_5 + pQ_2^2)/(5W_5) - 4Q_2)E_1 \\ & + 3J_1 + (4Q_2 - Q_2/U_2)J - 4Q_2(Q_2 - Q_5)/(5W_5) \\ & + Q_{\mathrm{cub}} + (1/(2U_2) - 1)(E_2^2 - O_2^2) + 2U_2(E_1^2 - O_1^2) + pR_{\mathrm{cub}}. \end{aligned}$$

All scalar denominators are units. Every term is even under $u \mapsto -u$: E is even, O, r_u, s_u are odd. It therefore descends to an integral rational number. Inserting $e_{\mathbb{Q}}(t^2) = \mathcal{E}(t) + pa_{\mathbb{Q}}(t)$ proves

$$k_5 = F_5 + a_{\mathbb{Q}}(-11u/5) - 11(a_{\mathbb{Q}}(2u) - a_{\mathbb{Q}}(u))/5.$$

For its residue set $E = E(u)$, $H = k/2 - 4q/5$, $E_2 = E + C$. The reduced norm equations are $E + u^pO_1 = H$, $E + C + 2u^pO_2 = H$, with $(u^p)^2 = 1/5$. Use these equations to eliminate u^pO_1, u^pO_2 in $\bar{F}_5$. Its complete ordinary product is

$$\begin{aligned} -22q^2/5 + 47qk/10 - 3k^2/2 + (q+4k)C/5 - 11qE/5 - 33E^2/8 + 33EH/4 + 11EC/4 \\ -45H^2/8 - 35HC/4 - 9C^2/8. \end{aligned}$$

Now substitute $E = C/5 - 6q/5 + k/2$ from the fifth entropy bridge. In the column order $q^2, qk, k^2, qC, kC, C^2$, expansion yields

$$(-169/50,\ 24/5,\ -15/8,\ 103/25,\ -11/5,\ -37/50).$$

This proves the k_5 row of (152), with both weight-two endpoints retained until its final substitution.

The conic coefficient carry, including its extreme. Write $g_N(a,x)=\sum_{j=0}^{N-1}a^jx^j$, and define the finite inverse jet

$$Q_N(x)=\frac{1}{15}\{g_N(1/4,x)/4-4g_N(4,x)\}.$$

It agrees with $1/A(x)$ through degree $N-1$: multiplying by A and using $(1-ax)g_N(a,x)=1-a^Nx^N$ proves this statement exactly. The cleared conic log equation follows from $1-V=s(4x-11)^2/25$, $AV'=-3(4x-11)V$:

$$2A\mathcal{L}'_{\mathbb{Q}}-3(4x-11)=-3\,5^{-(p-1)}s^N(4x-11)^p.$$

Take the coefficient of degree $N-1$ after multiplying by Q_N. Every omitted inverse-jet tail starts at degree N, so it cannot contribute. With

$$\mathcal{R}_{\mathbb{Q}}=-3\,5^{-(p-1+N)}[x^{N-1}]\{Q_N(x)(x+1)^N(4x-11)^p\},$$

the two unit-pole contributions give exactly

$$c_{\log}=-U_2-\frac{1}{2U_2}-\frac{1}{2}\mathcal{R}_{\mathbb{Q}}.$$

Define the integer row

$$b_{m,k}=\begin{cases}1 & m=k=0,\\ \binom{m-1}{k-1} & m,k\ge 1,\\ 0 & \text{otherwise,}\end{cases}\qquad I_{N,k}=\sum_{m=0}^{N-1}b_{m,k}5^{-m}(1-4^m/4^N).$$

Expanding $(4x-11)^p$ in the coefficient above gives the exact finite formula

$$\mathcal{R}_{\mathbb{Q}}=\sum_{k=0}^{N-1}\binom{p}{k}(-11/5)^{p-k}(4/5)^kI_{N,k}.$$

For completeness the inverse-jet coefficient in this expansion is

$$I_{N,k}=-15\,5^{-N}[x^{N-1-k}]\{Q_N(x)(x+1)^N\}\quad(k<N).$$

This coefficient equality follows from Pascal's identity: both sides have extreme $1-4^{-N}$ at $k=0$, and satisfy $I_{N+1,k+1}=(I_{N,k}+I_{N,k+1})/5$. Terms $k\ge N$ have negative requested degree and are zero. The term $m=N$ in the row, if appended, is also zero because $1-4^N/4^N=0$; its absence removes no nonzero endpoint.

The $k=0$ term is divisible by p, since $I_{N,0}=1-U_2^{-1}=pQ_2/U_2$. Every intermediate $\binom{p}{k}$, $1\le k<N<p$, is divisible by p in the integers. Thus the actual divided residue is integral:

$$\mathcal{R}_{\mathbb{Q}}/p=(-11/5)^pQ_2/U_2+\sum_{k=1}^{N-1}\frac{\binom{p}{k}}{p}(-11/5)^{p-k}(4/5)^kI_{N,k}.$$

The displayed reciprocal equation proves

$$k_c=-Q_2+Q_2/(2U_2)-\mathcal{R}_{\mathbb{Q}}/(2p).$$

To reduce its intermediate sum use $\binom{p}{k}/p=(-1)^{k-1}/k$, and reverse the finite k,m sums. For $m\ge 1$,

$$\sum_{k=1}^{m}\binom{m-1}{k-1}\frac{(4/11)^k}{k}=\frac{(1+4/11)^m-1}{m};$$

this is the integrated binomial expansion, using $\binom{m-1}{k-1}/k=\binom{m}{k}/m$. Terms $k>m$ vanish, so the upper $N-1$ limit is unchanged. Multiplying by $5^{-m}(1-4^m)$ and summing gives

$$L_N(3/11)-L_N(12/11)+L_N(4/5)-L_N(1/5)=:B_N.$$

Its $m=0$ term is zero and its $m=N$ term is zero by $4^N=1$; this explains the full endpoint shift from $N-1$ to N. Hence

$$\overline{\mathcal{R}_{\mathbb{Q}}/p}=\frac{11}{5}(-q+B_N),\qquad \bar{k}_c=\frac{3}{5}q-\frac{11}{10}B_N.$$

Since $L_N(t^2)=2E(t)$, $B_N=2(C-C_w)$. This proves the conic digit in (152) for either root and both Frobenius classes.

Why the complete weighted carry is integral. Put $a=1/4, b=4, c=11/4$, $z=x^{-1}$, and $U(z)=(1-cz)/((1-az)(1-bz))$. Then $R_w/A=-U(z)$. For any unit v the exact binomial expansion gives

$$(1-vz)^p=1-v^pz^p-pL_v(z),\qquad L_v(z)=\sum_{i=1}^{p-1}(-1)^{i-1}\binom{p}{i}v^iz^i/p.$$

Each coefficient of L_v is integral and reduces to v^i/i. Products and inverses of these three power series have integral coefficients, because their constant terms are 1. Below degree $2p$, divide their difference from the Frobenius zeroth series by p before reduction. The exact first residue is

$$F_1=L_a^{(0)}+L_b^{(0)}-L_c^{(0)}+z^p\{7L_a^{(0)}/4+11L_b^{(0)}/2-17L_c^{(0)}/4\},\quad L_v^{(0)}=\sum_{i=1}^{p-1}v^iz^i/i.$$

Indeed the coefficient of z^p in the zeroth series is $a^p+b^p-c^p$, reducing to $3/2$. Thus the respective shifted coefficients are $3/2+a=7/4$, $3/2+b=11/2$, and $a+b=17/4$. The rational comparison has upper exponent at most $p+N<2p$; the infinity coefficient in its weighted trace therefore sees only these degrees. Its zeroth series gives precisely $F_{0,\mathbb{Q}}$ above, including the factor 5^{p-1}. The remaining quotient is integral coefficientwise and reduces to

$$\bar{k}_w=-3\tau_1(d_xF_1S)=T_w.$$

The finite integration of the four moving endpoint polynomials, with their origin zero, gives the displayed H_{mov} formula. Their exact endpoint substitutions are the six formulas in (101); insertion gives the following coefficient packet before any rational S value is used:

$$\begin{aligned}-T_w=&\frac{-4(S_{12}-S_{\frac12,1})+22(S_{11}-S_{\frac12,\frac12})}{10}\\&+\frac{11(S_{22}-S_{11})+7(S_{21}-S_{1,\frac12})}{10}\\&-\frac{3}{2}T_5+\frac{2}{5}\mathcal{G}(1/5)-\frac{11}{10}\mathcal{G}(3/11)\\&-\frac{D_e}{5}(33j-138q-\ell(16)).\end{aligned}$$

Here each S_{ab} means the original $S(a,b)$. This is (153); the seven values in (70) finish its rational substitution.

4.6 Whole-source cancellation

We print the densities used in the source collection. They give explicit finite definitions for the intermediate residues as well as a way to check that the same scalar from the rational moment is being evaluated. Let

$$Q_H=\frac{H}{A_*}=\frac{4x-1}{125}H_0(V)/V,\quad y_q=(18-12x)Q_H-\frac{3H}{x},\quad \chi=\mathcal{P}-a/3,\quad \chi_0=\mathcal{P}-(a+h)/3.$$

Q_H is a polynomial since $H_0(0) = 0$. The Euler bracket is defined from its original finite polynomials, before any collection. With an independent variable t, set

$$\begin{aligned}
L_0(t) &= q + E(t),\\
K_h(t) &= \frac{R(1-t) + R(1+t) + q^2}{2} - \frac{R((1-t)/2) + R((1+t)/2)}{6} + (2h/3 - q)L_0(t),\\
B_h(t) &= -\frac{2}{3}K_h(t) - \frac{1}{3}(L_0(t) - h)^2 + \frac{h^2}{3},\\
B_Q(z) &= \operatorname{inv}_2 B_h(z), \qquad Q_b(x) = B_Q(V(x^{-1})), \qquad G_{\mathrm{KQ}} = \Omega(LQ_b).
\end{aligned}$$

The last definition is made over $\mathbb{F}_p$: all coefficients of B_h, B_Q, V and L belong to $\mathbb{F}_p$. Applying ι commutes with finite products, coefficient extraction and division by the nonzero endpoint constants in Ω. Thus the calculation over K below evaluates $\iota(G_{\mathrm{KQ}})$. Here B_h is even, as is seen by interchanging t and $-t$ in its displayed formula. Moreover $K_h(1) = K_h(-1) = 0$, $L_0(1) = L_0(-1) = 0$ by (86); therefore $B_Q(0) = 0$. The original reciprocal cubic is $V(x^{-1}) = (4x^{-1} - 1)^2(4 - x^{-1})/125$. Define the six-term completion

$$\begin{aligned}
\mathcal{C}_H = &-\frac{5}{3}eQ_H - \frac{2}{3}(D + \mathcal{P}^\vee)H(a^\vee)' + \frac{2}{9}\big(a^\vee + h - (1 + x^N)(a + h)\big)H(a^\vee)'\\
&- \frac{1}{9}\big(a + x^N(a + h)\big)y_q - \frac{1}{3}\mathcal{P}^\vee y_q - \frac{2}{5}\varphi D_s\left(\frac{H}{2x} + \frac{Ha'}{3}\right).
\end{aligned} \tag{180}$$

All reciprocal terms in this formula are retained. Put

$$\mathcal{G}_B = -L\mathcal{K}\chi_0 + (31q - 4qx)\mathcal{K} - 2hA_*\mathcal{K}\chi_0', \tag{181}$$

$$\mathcal{G}_2 = L\left(\frac{5}{24}x^pZ_{52} + \frac{2}{3}De\right) + s^N\left(\frac{5h}{4}x^pR_{52} - \frac{10h}{3}e\right). \tag{182}$$

For the proper metric, with P_M from (103), define

$$\mathcal{M}_{\mathrm{pr}} = -\frac{x^{-p-1}}{250\,5^N}(x + 1)^N(4 - 11x)^{p-1}(4x - 11)(11x^2 + 4x + 8)P_M(V).$$

The other metric is

$$\begin{aligned}
\mathcal{M} = &-2C_M(1 - \tfrac{2}{3}x^{-p})\mathcal{K}' - 2C_M\left(\frac{5}{24}x^p(4x^p - 11) + \frac{\varphi}{9}\right)a'\\
&- \frac{25}{9}C_M(a^\vee)' + \frac{55}{12}x^pC_M(a^\vee)' - \frac{\varphi C_M}{3x}.
\end{aligned}$$

Set

$$P_C = \frac{x^{N-1}(4x - 11)P_M(V)}{75} + \frac{2}{5}H\beta_D, \qquad \mathcal{J}_{\mathrm{met}} = \frac{3}{10}(4 - 33x^{-p} + 72x^{-2p} - 16x^{-3p})x^2P_C + A_*\mathcal{M}.$$

The three negative conic phases and the full metric occur together. Finally put

$$\begin{aligned}
\Phi_H &= -x^{-N}\mathcal{P}^\vee + \frac{1}{3}\big(x^{-N}(a^\vee + h) - (a + h)\big),\\
\Sigma_H &= -\frac{A_*}{25}(-11x^2 - 4x - 8)(4 - 11x)^{p-1}x^{-p-N}s^N\beta_D,\\
\mathcal{E}_H &= \frac{2}{5}A_*H\varphi D_s\left(\frac{1}{2x} + \frac{a'}{3}\right).
\end{aligned}$$

These definitions clear the original logarithmic denominator $2x(x + 1)(4 - 11x)$: $2x(x + 1)(4 - 11x)\Sigma_H = A_*(-2\varphi D_s/5)(-11x^2 - 4x - 8)$.

For clarity, the original low-H density can be defined without any auxiliary kernel name. Put

$$V_0 = -\mathcal{K}\chi + \frac{5}{24}x^pZ_{52}, \qquad \mathcal{B}_{\mathrm{met}} = -A_*B_0C_M' + 3hs^NB_0,$$

and

$$\mathcal{H}_{\mathrm{low}} = -A_*HV_0' + H\left(-L\mathcal{K} + \frac{A_*\mathcal{K}}{2x}\right) + C_BB_0L - \mathcal{B}_{\mathrm{met}} + A_*\mathcal{C}_H. \tag{183}$$

Its whole attachment is proved in Section 6.10: $\mathcal{H}_{\mathrm{low}} = A_*(\mathscr{R} + 5x^p a l_y/24)$, where $\mathscr{R}$ is (257). The retained second summand is a polynomial with exponents at least p, so its residue vanishes. The full source is kept until that support argument is applied.

5 The original rational chi and its bounded conic kernel

Let $p \geq 19$ be prime, $R = \mathbb{Z}_{(p)}$, $N = (p-1)/2$, and $d = \lfloor p/6 \rfloor$. All rational divisions in this section precede reduction $R \to \mathbb{F}_p$. Put

$$c_n = \frac{(6n)!}{(n!)^3(3n)!\,1728^n}, \qquad F_N(z) = \sum_{n=0}^{N} c_n z^n, \qquad F(z) = \sum_{n=0}^{d} \bar{c}_n z^n.$$

We use the already defined

$$A_*(x) = (4-x)(4x-1), \quad s(x) = \frac{x+1}{5}, \quad V(x) = \frac{(4x-1)^2(4-x)}{125}, \quad Q(x) = V(1/x).$$

For a Laurent polynomial G, write $G^\vee(x) = G(1/x)$. Define the actual half-cutoff rational source and its shifted polynomial by

$$\begin{aligned} f_{\mathbb{Q}} &= F_N(V), & \chi_{N,\mathbb{Q}} &= \frac{f_{\mathbb{Q}} - x^N f_{\mathbb{Q}}^\vee}{p}, \\ \mathcal{H}_{\mathbb{Q}} &= x^p \chi_{N,\mathbb{Q}} = \frac{x^p f_{\mathbb{Q}} - x\ \mathrm{rev}_{3N} f_{\mathbb{Q}}}{p}, & \mathrm{rev}_m G &= x^m G(1/x). \end{aligned} \tag{184}$$

The second equality uses $p + N = 3N + 1$, not a reduction of exponents. For $n < p$,

$$v_p(c_n) = \left\lfloor \frac{6n}{p} \right\rfloor - \left\lfloor \frac{3n}{p} \right\rfloor \geq 0, \qquad v_p(c_n) \geq 1 \quad (n > d).$$

Here $6n < p^2$, $3n < p^2$, $n!$ is a unit, and $p \nmid 1728$; thus these are the complete factorial valuations. In particular $\bar{F}_N = F$ and $\deg F = d$, because $6d < p$ makes $\bar{c}_d \neq 0$. The reciprocal calculation in Section 2.7 proves $F(V) = x^N F(Q)$. Hence the numerator of $\chi_{N,\mathbb{Q}}$ belongs to $pR[x, x^{-1}]$, and division in $\mathbb{Q}[x, x^{-1}]$ gives $\chi_{N,\mathbb{Q}} \in R[x, x^{-1}]$. No coefficientwise division is performed in characteristic p. Since $\deg f_{\mathbb{Q}} \leq 3N$, its reciprocal weighted by x^N has support in $[-2N, N]$. Consequently

$$\operatorname{supp} \chi_{N,\mathbb{Q}} \subseteq [-(p-1), 3N], \qquad \mathcal{H}_{\mathbb{Q}} \in R[x], \qquad \deg \mathcal{H}_{\mathbb{Q}} \leq p + 3N. \tag{185}$$

Set

$$f = F(V) = \bar{f}_{\mathbb{Q}}, \qquad \chi_N = \overline{\chi_{N,\mathbb{Q}}}, \qquad \mathcal{H} = \overline{\mathcal{H}_{\mathbb{Q}}} = x^p \chi_N.$$

The symbol $\mathcal{H}$ is this shifted original chi polynomial. It is distinct from the native harmonic polynomial $H = H_0(V)$.

The literal moment retains the cutoff $e = p - 1 - d$. We record its exact comparison with the half cutoff. In the two prime classes $p = 6d + 1, 6d + 5$, respectively, $e = 5d, 5d + 4$, so

$$N \leq e < p, \qquad e < 5p/6.$$

For $N < n \leq e$, $3n > 3p/2$ and $6n < 5p$. If $3n < 2p$, then $3p < 6n < 4p$, and the full valuation is $\lfloor 6n/p \rfloor - \lfloor 3n/p \rfloor = 3 - 1 = 2$. If $3n \geq 2p$, then $2p \leq 3n < 5p/2$, and it is $4 - 2 = 2$. Thus, with the same weight N,

$$\chi_{e,\mathbb{Q}} := \frac{F_e(V) - x^N F_e(Q)}{p}, \qquad \chi_{e,\mathbb{Q}} - \chi_{N,\mathbb{Q}} = \sum_{n=N+1}^{e} \frac{c_n}{p}(V^n - x^N Q^n) \in pR[x, x^{-1}]. \tag{186}$$

The factors V, Q have integral coefficients since 125 is a unit, and $v_p(c_n/p) = 1$ on the entire displayed tail. Consequently $\overline{\chi_{e,\mathbb{Q}}} = \chi_N$, including every Laurent coefficient. Also, for $H_m(z) = \sum_{n=1}^{m} c_n z^n/n$,

$$H_e - H_d = \sum_{n=d+1}^{e} \frac{c_n}{n} z^n \in pR[z], \qquad \overline{H_e(V)} = H_0(V) = H.$$

Here $v_p(c_n) \geq 1$ for $n > d$, and each $1 \leq n \leq e < p$ is a unit, so the harmonic division loses no precision.

For a unit $c \in R^\times$, the endpoint divided difference is $\mathrm{DD}_c\, G = (G - G(c))/(x - c)$. On every Laurent monomial it is the finite formula

$$\mathrm{DD}_c(x^j) = \begin{cases} \displaystyle\sum_{r=0}^{j-1} x^{j-1-r} c^r, & j > 0, \\ 0, & j = 0, \\ \displaystyle -\sum_{r=0}^{m-1} x^{-1-r} c^{r-m}, & j = -m < 0. \end{cases}$$

This proves that it maps $R[x, x^{-1}]$ to itself, preserves $pR[x, x^{-1}]$, and commutes with coefficientwise reduction. Both endpoints $1/4, 4$ and their inverses are units. Define $\mathrm{WD} = 2\,\mathrm{DD}_{1/4} + \mathrm{DD}_4$. The actual rational eta source and its reduction are therefore

$$P_{\eta,\mathbb{Q}} = \mathrm{WD}(\chi_{e,\mathbb{Q}} + H_e(V)) - \frac{H_e(V)}{2x}, \qquad \overline{P_{\eta,\mathbb{Q}}} = \mathrm{WD}(\chi_N + H) - \frac{H}{2x}. \tag{187}$$

All endpoint quotients in this equality are formed over the integral rational Laurent ring before reduction. Thus the chi in the conic proof below is the reduction of the literal full-cutoff source.

5.1 The rational self norm and the original source equation

Define the integral polynomial coefficients

$$c_2 = \frac{4}{125}(x+1)A_*^2, \qquad c_1 = \frac{2}{125}A_*^2 + \frac{4}{125}(x+1)A_*(17 - 8x) = \frac{1}{2}c_2', \qquad c_0 = -5V.$$

For Laurent polynomials a, b define

$$\begin{aligned} \mathcal{C}(a,b) &= c_2(ab'' + ba'' - a'b') + c_1(ab' + ba') + c_0 ab, \\ r_0 &= \frac{1}{2}(c_1/x - c_2/x^2), \\ r_1 &= c_2/(2x), \\ \mu &= c_2/(4x^2), \\ \Psi(a) &= r_0 a^2 + r_1 aa', \\ \Psi_\times(a,b) &= 2r_0 ab + r_1(ab' + ba'). \end{aligned}$$

All fixed denominators are powers of 2 and 5, so these expressions preserve $R[x, x^{-1}]$. Differentiation and reciprocal substitution also preserve it: the coefficient of x^j in G' is $(j+1)G_{j+1}$, and that in $G^\vee$ is G_{-j}.

We first obtain two powers of p in the whole rational self norm. Define

$$\begin{aligned} &\mathcal{D}_{\mathrm{nat}}(P) = 72z(1-z)(2PP'' - (P')^2) + 72(2-3z)PP' - 10P^2, \\ &\mathcal{N}_{\mathrm{rat}}(z) = \frac{z}{2}\mathcal{D}_{\mathrm{nat}}(F_N)(z), \qquad q_t = 1 - t^2, \qquad G(t) = F_N(q_t), \\ &\mathcal{D}_0(G) = 9q_t(2GG'' - (G')^2) - 36tGG' - 5G^2, \qquad \mathcal{W}(t) = q_t \mathcal{D}_0(G). \end{aligned}$$

The derivative rules

$$G' = -2tF_N'(q_t), \quad G'' = -2F_N'(q_t) + 4t^2F_N''(q_t), \quad G''' = 12tF_N''(q_t) - 8t^3F_N'''(q_t)$$

give

$$2\mathcal{D}_0(G) = \mathcal{D}_{\mathrm{nat}}(F_N)(q_t), \qquad \mathcal{W} = \mathcal{N}_{\mathrm{rat}}(q_t). \tag{188}$$

For example $2GG'' - (G')^2 = -4GF_N'(q_t) + 4t^2\{2GF_N''(q_t) - F_N'(q_t)^2\}$. Substitution into $2\mathcal{D}_0$ gives its first coefficient $72q_t(1-q_t)$, its GF_N' coefficient $72(2-3q_t)$, and its G^2 coefficient -10, as stated.

The coefficient recurrence

$$72(n+1)^3c_{n+1} = (2n+1)(6n+1)(6n+5)c_n$$

shows that the full cutoff satisfies

$$\mathcal{L}(F_N) = -p(9p^2-4)c_Nz^N,$$
$$\mathcal{L}(P) = 72z^2(1-z)P''' + (216z - 324z^2)P'' + (72 - 226z)P' - 5P.$$

Indeed every coefficient below degree N cancels by the recurrence; the last coefficient is $-(2N+1)(6N+1)(6N+5)c_N = -p(3p-2)(3p+2)c_N$. There are no coefficients above N. Define the transported third-order expression

$$\mathcal{J}_0(G) = 36q_t^2G''' - 216tq_tG'' + (-92q_t + 144t^2)G' + 20tG.$$

The three displayed chain derivatives give $\mathcal{J}_0(F_N(q_t)) = -4t\mathcal{L}(F_N)(q_t)$. Differentiation of $\mathcal{D}_0$ gives

$$\mathcal{D}_0' = 18q_tGG''' - 72tGG'' - 18t(G')^2 - 46GG'.$$

Multiplying by q_t and adding $q_t'\mathcal{D}_0$ cancels the $(G')^2$ terms and gives

$$\mathcal{W}' = 18q_t^2GG''' - 108tq_tGG'' + (-46q_t + 72t^2)GG' + 10tG^2 = \frac{1}{2}G\mathcal{J}_0(G).$$

Therefore the exact rational derivative source is

$$\mathcal{W}' = 2p(9p^2-4)c_N\, tq_t^NG. \tag{189}$$

We have $\deg G = 2N$, $\deg \mathcal{W} \le 4N+2 = 2p$; $\mathcal{W}$ is even and $\mathcal{W}(1) = 0$. The top coefficient of q_t^NG is $(-1)^Nc_N(-1)^N = c_N$. Comparing degree $2p-1 = 4N+1$ in (189) gives, over $\mathbb{Q}$,

$$[t^{2p}]\mathcal{W} = (9p^2-4)c_N^2. \tag{190}$$

The cancellation of $2p$ in this calculation is rational.

The terminating square already proved in Section 2.8 gives $\bar{G} = Y_d(t)^2$, where $Y_d = P_d$ and

$$36(1-t^2)Y_d'' - 72tY_d' - 5Y_d = 0.$$

Substituting the derivatives of Y_d^2 gives

$$\mathcal{D}_0(Y_d^2) = Y_d^3\{36(1-t^2)Y_d'' - 72tY_d' - 5Y_d\} = 0.$$

Thus $\mathcal{W}/p$ has integral coefficients. At the half cutoff

$$v_p(c_N) = \left\lfloor \frac{3p-3}{p} \right\rfloor - \left\lfloor \frac{3p-3}{2p} \right\rfloor = 2 - 1 = 1.$$

Write $a_N = c_N/p \in R$. The exact source and top coefficient after the first division are

$$(\mathcal{W}/p)' = 2p(9p^2-4)a_N\, tq_t^NG, \qquad [t^{2p}](\mathcal{W}/p) = p(9p^2-4)a_N^2.$$

Both reduce to zero. If $U = \overline{\mathcal{W}/p}$, then $U' = 0$ means $jU_j = 0$ coefficientwise. As $\deg U \le 2p$, its only possible powers are $1, t^p, t^{2p}$. Evenness kills t^p, the top coefficient kills t^{2p}, and $U(1) = 0$ kills the constant. It follows that

$$\mathcal{W}/p^2 \in R[t].$$

The inverse of substitution $z \mapsto 1 - t^2$, on its image, is

$$\operatorname{qi}\left(\sum_i u_{2i} t^{2i}\right) = \sum_i u_{2i}(1-z)^i.$$

It selects even coefficients and composes with $1 - z$; it divides no coefficient. Equation (188) consequently gives

$$\mathcal{N}_{\mathrm{rat}}/p^2 \in R[z]. \tag{191}$$

The otherwise invisible top coefficient after the second division is $-4\bar{a}_N^2$; it was retained in the first-division argument.

The pullback calculation uses the polynomial identities

$$V' = -\frac{3}{125}(4x-1)(4x-11), \quad V'' = \frac{144-96x}{125}, \quad 1 - V = \frac{(x+1)(4x-11)^2}{125},$$

$$c_2(V')^2 = 36V^2(1-V), \qquad 2c_2V'' + 2c_1V' = 36V(2-3V), \qquad c_0 = -5V.$$

For the first identity the two factorizations on either side are

$$\frac{36}{125^3}(x+1)(4-x)^2(4x-1)^4(4x-11)^2.$$

For the second, substituting $A_* = (4-x)(4x-1)$ and $A_*' = 17 - 8x$ reduces it, after the common factor $24A_*/125^2$ is removed polynomially, to

$$(x+1)A_*(48-32x) - \frac{(4x-1)(4x-11)}{2}\{A_* + 2(x+1)(17-8x)\} = \frac{3}{2}(4x-1)(250 - 3(4x-1)^2(4-x)).$$

Both sides expand to $288x^4 - 1368x^3 + 918x^2 + (2559/2)x - 357$. Thus the product and chain rules give the whole self norm

$$\mathcal{C}(f_{\mathbb{Q}}, f_{\mathbb{Q}}) = \mathcal{N}_{\mathrm{rat}}(V) = p^2 K_{\mathbb{Q}}, \qquad K_{\mathbb{Q}} = (\mathcal{N}_{\mathrm{rat}}/p^2)(V) \in R[x].$$

Here is the reciprocal covariance calculation before reduction. The coefficient polynomials obey

$$c_2^\vee = x^{-5}c_2, \qquad c_1^\vee = \frac{5}{2}x^{-4}c_2 - x^{-3}c_1, \qquad c_0 - xc_0^\vee = \frac{c_1}{x} - \frac{5c_2}{4x^2}.$$

The first follows from $A_*^\vee = x^{-2}A_*$; differentiation gives the second since $c_1 = c_2'/2$. For the third, both expressions are

$$\frac{1}{25}\{16x^3 - 72x^2 + 37x - 37 + 72x^{-1} - 16x^{-2}\}.$$

For $w = x^N f_{\mathbb{Q}}^\vee$, the two derivatives are

$$w' = x^N\{Nx^{-1}f_{\mathbb{Q}}^\vee - x^{-2}(f_{\mathbb{Q}}')^\vee\},$$
$$w'' = x^N\{N(N-1)x^{-2}f_{\mathbb{Q}}^\vee + 2(1-N)x^{-3}(f_{\mathbb{Q}}')^\vee + x^{-4}(f_{\mathbb{Q}}'')^\vee\}.$$

In $x^{-2N}\mathcal{C}(w,w)$ the coefficients of $2f^\vee(f'')^\vee - ((f')^\vee)^2$, $f^\vee(f')^\vee$, and $(f^\vee)^2$, respectively, are

$$c_2x^{-4}, \qquad (4-2N)c_2x^{-3} - 2c_1x^{-2}, \qquad N(N-2)c_2x^{-2} + 2Nc_1x^{-1} + c_0.$$

Subtracting the coefficients of $x\mathcal{C}(f_{\mathbb{Q}}, f_{\mathbb{Q}})^\vee$ leaves $0, -pc_2x^{-3}$, and

$$pc_1x^{-1} + \{p(N-1) - p^2/4\}c_2x^{-2}, \qquad p = 2N+1.$$

These are precisely the coefficients of $x^{-2N}\{2p\Psi(w) - p^2\mu w^2\}$. Hence

$$\mathcal{C}(w,w) = x^p\mathcal{C}(f_{\mathbb{Q}}, f_{\mathbb{Q}})^{\vee} + 2p\Psi(w) - p^2\mu w^2.$$

Writing $w = f_{\mathbb{Q}} - p\chi_{N,\mathbb{Q}}$, bilinearity gives

$$\mathcal{C}(w,w) = \mathcal{C}(f_{\mathbb{Q}}, f_{\mathbb{Q}}) - 2p\mathcal{C}(f_{\mathbb{Q}}, \chi_{N,\mathbb{Q}}) + p^2\mathcal{C}(\chi_{N,\mathbb{Q}}, \chi_{N,\mathbb{Q}}),$$

and direct expansion gives $\Psi(w) = \Psi(f_{\mathbb{Q}}) - p\Psi_{\times}(f_{\mathbb{Q}}, \chi_{N,\mathbb{Q}}) + p^2\Psi(\chi_{N,\mathbb{Q}})$. Substitution yields

$$2p\{\mathcal{C}(f_{\mathbb{Q}}, \chi_{N,\mathbb{Q}}) + \Psi(f_{\mathbb{Q}})\} = p^2\{K_{\mathbb{Q}} - x^pK_{\mathbb{Q}}^{\vee} + E_{\mathbb{Q}}\},$$

where the complete integral remainder is

$$E_{\mathbb{Q}} = \mathcal{C}(\chi_{N,\mathbb{Q}}, \chi_{N,\mathbb{Q}}) + 2\Psi_{\times}(f_{\mathbb{Q}}, \chi_{N,\mathbb{Q}}) - 2p\Psi(\chi_{N,\mathbb{Q}}) + \mu w^2 \in R[x, x^{-1}].$$

Canceling the nonzero rational $2p$ proves that the expression in braces on the left belongs to $pR[x, x^{-1}]$. Reduction therefore gives the actual source, together with its self equation:

$$\mathcal{C}(f, \chi_N) + \Psi(f) = 0, \qquad \mathcal{C}(f,f) = 0. \tag{192}$$

Since $(x^p)' = 0$, multiplication of the first equation by $2x^{p+2}$ gives the polynomial equation

$$2x^2\mathcal{C}(f, \mathcal{H}) + x^p\{(xc_1 - c_2)f^2 + xc_2ff'\} = 0. \tag{193}$$

The embedding of polynomials in Laurent polynomials is injective, so this denominator clearing creates no extra premise.

5.2 Conic transport, division and the odd kernel

Use a new variable t and define

$$B_c(t) = 5t^2 - 1, \quad T_c(t) = 4t^3 - 3t, \quad A_c(t) = (1-t^2)(4t^2-1) = -4t^4 + 5t^2 - 1.$$

Then

$$s(B_c) = t^2, \quad A_*(B_c) = 25A_c, \quad V(B_c) = 1 - T_c^2, \quad B_c' = 10t, \quad B_c'' = 10.$$

For clarity the finite proper polynomials used here are

$$Y_d(T) = \sum_{i=0}^{d} \frac{(-d)_i(d+1)_i}{(i!)^2}\left(\frac{1-T}{2}\right)^i, \qquad n_d(T) = -\sum_{i=0}^{d}(H_d - H_i)\frac{(-d)_i(d+1)_i}{(i!)^2}\left(\frac{1-T}{2}\right)^i.$$

Here $(a)_i = \prod_{j=0}^{i-1}(a+j)$ and $H_i = \sum_{j=1}^{i} 1/j$, with $H_0 = 0$. Their denominators are units because $i \le d < p$. Set

$$y = Y_d(T_c), \quad m = n_d(T_c)/3, \quad \mathcal{H}_c = \mathcal{H}(B_c), \quad v_c = -5(1-t^2)(4t^2-1)^2, \quad E_c = 5tA_cB_c^{p-1}.$$

The square identity and proper Wronskian (102) give

$$f(B_c) = y^2, \qquad A_c(ym' - y'm) = 1 - y^2. \tag{194}$$

For the second identity, $T_c' = 3(4t^2-1)$ and $1 - T_c^2 = A_c(4t^2-1)$; multiplying the original proper Wronskian by the derivative of T_c divided by 3 gives exactly its displayed scale. Also $y(1) = 1$, so $y \neq 0$. The leading coefficient of Y_d is $(2d)!/(2^d(d!)^2) \neq 0$, since $2d < p$. Thus

$$\deg y = 3d, \qquad [t^{3d}]y \neq 0, \qquad \deg m \le 3d - 3.$$

Define the conic bilinear expression

$$\mathcal{C}_c(F,G) = A_c^2(FG'' + GF'' - F'G') + A_cA_c'(FG' + GF') + v_cFG.$$

The exact coefficient transport is

$$c_2(B_c) = 100t^2A_c^2 = (B_c')^2A_c^2, \quad c_1(B_c) = 10A_c^2 + 10tA_cA_c' = B_c''A_c^2 + B_c'A_cA_c', \quad c_0(B_c) = v_c.$$

In fact $A_*(B_c) = 25A_c$, $B_c + 1 = 5t^2$, and $17 - 8B_c = 25 - 40t^2$ substitute into c_2, c_1, c_0. For c_1 this gives

$$10A_c^2 + 4t^2A_c(25 - 40t^2) = 10A_c^2 + 10tA_c(-16t^3 + 10t).$$

If $F = a(B_c)$, $G = b(B_c)$, then

$$F' = B_c'a'(B_c), \quad F'' = (B_c')^2a''(B_c) + B_c''a'(B_c)$$

and the same formulas hold for G. Hence

$$\begin{aligned}\mathcal{C}_c(F,G) &= A_c^2(B_c')^2(ab'' + ba'' - a'b')(B_c)\\ &\quad + \{A_c^2B_c'' + A_cA_c'B_c'\}(ab' + ba')(B_c) + c_0(B_c)(ab)(B_c)\\ &= \mathcal{C}(a,b)(B_c).\end{aligned}$$

In particular $\mathcal{C}_c(y^2, y^2) = 0$.

In characteristic p,

$$E_c' = 5(A_c + tA_c')B_c^{p-1} - 5tA_cB_c^{p-2}B_c'.$$

Multiplication, using $B_c^{p-1}B_c = B_c^p$ and $B_c^{p-2}B_c^2 = B_c^p$, gives

$$\begin{aligned}2B_c^2A_cE_c' &= B_c^p\{10A_c(A_c + tA_c')B_c - 100t^2A_c^2\}\\ &= B_c^p\{B_cc_1(B_c) - c_2(B_c)\},\\ 2B_c^2A_cE_cB_c' &= 100t^2A_c^2B_c^{p+1} = B_c^pB_cc_2(B_c).\end{aligned}$$

Substitute these into the pullback of (193). The first forcing term multiplies y^4, the second multiplies $y^2(y^2)'$, and cancellation of the nonzero polynomial $2B_c^2$ gives

$$\mathcal{C}_c(y^2, \mathcal{H}_c) + A_c\{E_c'y^4 + E_cy^2(y^2)'\} = 0. \tag{195}$$

Put

$$M_c = A_c(y\mathcal{H}_c' - 2y'\mathcal{H}_c) + E_cy^3.$$

Its derivative is

$$M_c' = A_c'(y\mathcal{H}_c' - 2y'\mathcal{H}_c) + A_c(y\mathcal{H}_c'' - y'\mathcal{H}_c' - 2y''\mathcal{H}_c) + E_c'y^3 + 3E_cy^2y'.$$

Consequently

$$\begin{aligned}yM_c' - y'M_c &= A_c'(y^2\mathcal{H}_c' - 2yy'\mathcal{H}_c)\\ &\quad + A_c\{y^2\mathcal{H}_c'' - 2yy'\mathcal{H}_c' + (2(y')^2 - 2yy'')\mathcal{H}_c\}\\ &\quad + E_c'y^4 + 2E_cy^3y'.\end{aligned}$$

Multiplication by A_cy^2, followed by the derivatives $(y^2)' = 2yy'$, $(y^2)'' = 2(y')^2 + 2yy''$, gives

$$A_cy^2(yM_c' - y'M_c) = y^2\{\mathcal{C}_c(y^2, \mathcal{H}_c) + A_c(E_c'y^4 + E_cy^2(y^2)')\} - \mathcal{H}_c\mathcal{C}_c(y^2, y^2) = 0.$$

Since $A_c \neq 0$ and $y \neq 0$, $yM_c' - y'M_c = 0$. The proper Wronskian supplies the polynomial quotient

$$b = M_c(y + A_cm') - A_cM_c'm, \qquad M_c = yb, \qquad b' = 0. \tag{196}$$

To verify the middle equality, multiply the definition by y:

$$yb = M_c\{y^2 + A_cym'\} - A_cyM_c'm = M_c\{1 + A_cy'm\} - A_cm\,y'M_c = M_c.$$

Substitution of $M_c = yb$ in its Wronskian gives $y^2b' = 0$, hence $b' = 0$. There is also a genuine polynomial divisor $A_c \mid b$. Indeed, with

$$L_c = y\mathcal{H}_c' - 2y'\mathcal{H}_c + 5tB_c^{p-1}y^3, \qquad M_c = A_cL_c,$$

the proper identity gives

$$b = A_c\{yL_c + (ym' - y'm)b\}.$$

The difference between the right side and b is $yM_c + (1 - y^2)b - b = 0$. This division holds at every zero of y as a polynomial identity.

5.3 The terminal slot that fixes the normalization

Write

$$\epsilon_6 = \begin{cases} 1, & p = 6d+1, \\ -1, & p = 6d+5. \end{cases}$$

The sign ϵ_6 is distinct from $\epsilon = 5^N$ in the finite logarithm recipes. We now compute the actual top coefficient, without adding a kernel normalization hypothesis. For the finite parameter products $A_n(\tau), D_n(\tau)$ and $C_n(\tau) = A_n(\tau)/D_n(\tau)$ defined in the moment section, set

$$\tau_0 = -p/6, \qquad J(z) = \sum_{n=0}^{N} \overline{C_n'(\tau_0)} z^n.$$

Every $D_n(\tau_0)$ is a unit because its reduction is $\prod_{j=0}^{n-1} 72(j+1)^3$, with $n < p$. The quotient Taylor formula there gives, for $n > d$,

$$c_n = C_n(0) = C_n(\tau_0) + \frac{p}{6} C_n'(\tau_0) + p^2 r_n = \frac{p}{6} C_n'(\tau_0) + p^2 r_n, \qquad r_n \in R.$$

The terminating coefficient $C_n(\tau_0)$ is zero for $n > d$. Thus $\bar{a}_N = J_N/6$. The full first integral (57) is

$$z(FJ' - F'J) = F\{(1-z)^N - F\}, \qquad \deg J \le N.$$

At degree $d+N$, the coefficient of F^2 is zero because $2d < d+N$. The other two sides have coefficients $\bar{c}_d(N-d)J_N$ and $\bar{c}_d(-1)^N$. Canceling the nonzero leading coefficient $\bar{c}_d$ gives

$$(N-d)J_N = (-1)^N.$$

If $p = 6d+1$, then $N = 3d$ and $N - d = 2d = -1/3$ in $\mathbb{F}_p$. If $p = 6d+5$, then $N = 3d+2$ and $N - d = 2d + 2 = 1/3$. Therefore

$$J_N = -3\epsilon_6(-1)^N, \qquad \bar{a}_N = -\frac{\epsilon_6(-1)^N}{2}. \tag{197}$$

All divisions here use the nonzero field constants $2, 3, 6$.

The rational shifted top coefficient in (184) is

$$[x^{p+3N}]\mathcal{H}_{\mathbb{Q}} = a_N(-16/125)^N.$$

Indeed V has degree 3 and leading coefficient $-16/125$, whereas the reflected term has degree at most $3N+1 < p+3N$. After substitution $B_c = 5t^2 - 1$, degree $5p - 3 = 2(p+3N)$ has coefficient

$$a_N(-16/125)^N 5^{p+3N} = a_N 5^p (-16)^N.$$

This equality is first an equality of integral rational coefficients. Reduction uses $5^p = 5$ and

$$(-16)^N = (-1)^N 4^{2N} = (-1)^N 4^{p-1} = (-1)^N.$$

Together with (197), it proves

$$\deg \mathcal{H}_c \le h_c := 5p - 3, \qquad [t^{h_c}]\mathcal{H}_c = -5\epsilon_6/2. \tag{198}$$

Let $n_c = 3d$ and $\theta = t\, d/dt$. Multiplying $M_c = yb$ by t gives

$$tyb = A_c I + tE_c y^3, \qquad I = y\theta\mathcal{H}_c - 2(\theta y)\mathcal{H}_c.$$

We have $\deg I \le n_c + h_c$ and

$$\deg E_c \le 5 + 2(p-1) = 2p + 3, \qquad \deg(tE_c y^3) \le 2p + 4 + 3n_c < n_c + h_c + 4.$$

The last strict inequality follows from $2n_c = 6d \le p-1$: the difference between its right side and left side is $3p-3-2n_c \ge 2p-2 > 0$. As $\deg A_c = 4$, this gives $\deg(tyb) \le n_c + h_c + 4$. If $b \neq 0$, $\deg(tyb) = 1 + n_c + \deg b$; if $b = 0$ the bound is immediate. Hence $\deg b \le h_c + 3 = 5p$. At degree $n_c + h_c + 4$ the strict forcing bound gives

$$\begin{aligned} y_{n_c} b_{h_c+3} &= -4 I_{n_c+h_c} \\ &= -4 y_{n_c} (\mathcal{H}_c)_{h_c} (h_c - 2n_c). \end{aligned}$$

The leading coefficient $y_{n_c} \neq 0$ cancels. In the two prime classes $h_c = -3$ and $2n_c = 6d = -1$ or -5, so $h_c - 2n_c = -2\epsilon_6$. Equation (198) now gives

$$b_{5p} = -4(-5\epsilon_6/2)(-2\epsilon_6) = -20.$$

The polynomials $A_c, y^2, \mathcal{H}_c$ are even, and E_c is odd. Multiplying $M_c = yb$ by y yields

$$y^2 b = A_c\{y^2 \mathcal{H}_c' - (y^2)' \mathcal{H}_c\} + E_c y^4.$$

The right side is odd. Replacing t by $-t$ and canceling $y^2 \neq 0$ proves $b(-t) = -b(t)$. Since $A_c \mid b$, $b(1) = b(1/2) = 0$. These endpoints use $A_c(1) = 0$ and $4(1/2)^2 - 1 = 0$. We have thus derived every item of the bounded kernel data:

$$b' = 0, \quad \deg b \le 5p, \quad b(-t) = -b(t), \quad b(1) = b(1/2) = 0, \quad b_{5p} = -20. \tag{199}$$

Coefficientwise $b' = 0$ says $b_j = 0$ unless $p \mid j$. Write $b = U(t^p)$, $\deg U \le 5$. As p is odd, oddness kills its even coefficients. Thus

$$U(t) = a_1 t + a_3 t^3 + a_5 t^5, \qquad a_5 = -20.$$

The two endpoint equations, since 1 and $1/2$ are in the prime field, are

$$a_1 + a_3 + a_5 = 0, \qquad 16a_1 + 4a_3 + a_5 = 0.$$

The second is $32U(1/2) = 0$. Subtraction gives $15a_1 + 3a_3 = 0$, hence $a_3 = -5a_1$; the first gives $a_5 = 4a_1$. Consequently

$$a_1 = -5, \quad a_3 = 25, \quad a_5 = -20, \qquad U(t) = 5tA_c(t).$$

Frobenius fixes the coefficients of A_c and 5, so $U(t^p) = 5t^p A_c(t)^p$. The normalized original numerator is therefore

$$A_c(y\mathcal{H}_c' - 2y'\mathcal{H}_c) + E_c y^3 = 5t^p A_c^p y. \tag{200}$$

5.4 Exact half-chi descent and the three-phase remainder

Multiply (200) by y and use $y^2 = f(B_c)$, $\mathcal{H}_c = \mathcal{H}(B_c)$. The derivative of B_c gives

$$10tA_c(f\mathcal{H}' - f'\mathcal{H})(B_c) + 5tA_c B_c^{p-1} f(B_c)^2 = 5t^p A_c^p f(B_c).$$

Cancel the nonzero polynomial $5tA_c$:

$$2(f\mathcal{H}' - f'\mathcal{H})(B_c) = t^{p-1} A_c^{p-1} f(B_c) - B_c^{p-1} f(B_c)^2.$$

Here $A_*(B_c) = 25A_c$, $s(B_c) = t^2$, $2N = p-1$, and $25^{p-1} = 1$. Polynomial substitution by B_c is injective: for a nonzero polynomial of degree j, the pullback has degree $2j$ and nonzero leading coefficient multiplied by 5^j. Thus the exact shifted original Wronskian is

$$\boxed{2(f\mathcal{H}' - f'\mathcal{H}) = A_*^{p-1} s^N f - x^{p-1} f^2.} \tag{201}$$

Dividing the Laurent identity by the invertible monomial x^p and the nonzero constant 2 gives

$$f\chi_N' - f'\chi_N = f\mathcal{R} - \frac{f^2}{2x}, \qquad \mathcal{R} = \frac{1}{2}x^{-p}A_*^{p-1}s^N. \tag{202}$$

No value of f, at any field point, is divided out.

We use the original finite $\mathcal{P}$ in (94), with all its three ranges retained, and put

$$P_{\text{num}} = x^p\mathcal{P} = x^{2p}A_+ + x^pB_\chi + C_-.$$

Here is the finite derivative calculation needed for its source. The full logarithm satisfies $\ell'(z) = \sum_{i=0}^{p-2} z^i = (1-z^{p-1})/(1-z)$, where the quotient denotes the polynomial geometric sum. If

$$S_W = \sum_{j=1}^{N}(4^j-1)s^{j-1}, \qquad W' = \frac{\epsilon}{75}S_W, \qquad \epsilon = 5^N,$$

then $4^N = 1$, $A_* = -25(1-s)(1-4s)$ give

$$\begin{aligned} A_*S_W &= -25\{4(1-s)(1-(4s)^N) - (1-4s)(1-s^N)\} \\ &= 75(s^N-1). \end{aligned}$$

Also $4^{p-1} = 1$ gives

$$A_*\ell'(x/4) = 4(4x-1)(1-x^{p-1}), \qquad A_*\ell'(4x) = -(4-x)(1-x^{p-1}).$$

Differentiating the full recipes in (94), including their chain factors $1/4$ and 4, therefore gives

$$\begin{aligned} A_*A_+' &= 2(1-s^N), \\ A_*B_\chi' &= \frac{17}{2}(s^N-1) + \frac{2}{15}\{4(4x-1)+(4-x)\}(1-x^{p-1}) \\ &= \frac{17}{2}(s^N-1) + 2x - 2x^p, \\ A_*C_-' &= 2(1-s^N) + \frac{2}{15}\{-4(4-x)-(4x-1)\}(1-x^{p-1}) \\ &= 2x^{p-1} - 2s^N. \end{aligned}$$

Since the derivatives of x^p and x^{-p} vanish, addition yields

$$A_*\mathcal{P}' = \frac{1}{2}s^N(17-4x^p-4x^{-p}) - \frac{17}{2} + 2x + 2x^{-1} = \frac{1}{2}s^N(17-4x^p-4x^{-p}) - \frac{A_*}{2x}.$$

As $A_*^p = x^p(17-4x^p-4x^{-p})$, cancellation of the nonzero polynomial A_* in the Laurent domain proves $\mathcal{P}' = \mathcal{R} - 1/(2x)$. Multiplication by x^p gives the polynomial source

$$2P_{\text{num}}' = A_*^{p-1}s^N - x^{p-1}. \tag{203}$$

The N-th coefficient of W is zero because $4^N = 1$. Thus $\deg A_+ \le N-1$, whereas the full logarithms give $\deg B_\chi, \deg C_- \le p-1$. It follows that

$$\deg P_{\text{num}} \le \max(2p+N-1,\, 2p-1,\, p-1) = 2p+N-1. \tag{204}$$

The same C_B as in (103) has pullback

$$C_B(B_c) = \frac{t^p}{5}Y_d(T_c)n_d(T_c) = \frac{3}{5}t^pym.$$

Since $(t^p)' = 0$, its proper Wronskian gives

$$A_c\{y^2(C_B(B_c))' - (y^2)'C_B(B_c)\} = \frac{3}{5}t^py^2A_c(ym'-y'm) = \frac{3}{5}t^py^2(1-y^2).$$

The left side is $10tA_c(fC_B' - f'C_B)(B_c)$. Multiplying by 5, canceling $t \neq 0$, and using $25A_c = A_*(B_c)$ therefore gives, after injective descent,

$$2A_*(fC_B' - f'C_B) = 3s^N f(1-f). \tag{205}$$

Now define the polynomial remainder

$$G_\chi = \mathcal{H} - fP_{\text{num}} - \frac{1}{3}A_*^p C_B.$$

The derivative of A_*^p is zero. Equations (201), (203), and (205) give the complete cancellation

$$\begin{aligned} 6A_*(fG_\chi' - f'G_\chi) &= 3A_*\{A_*^{p-1}s^N f - x^{p-1}f^2\} \\ &\quad - 3A_* f^2\{A_*^{p-1}s^N - x^{p-1}\} \\ &\quad - 2A_*^{p+1}(fC_B' - f'C_B) \\ &= 3A_*^p s^N f(1-f) - 3A_*^p s^N f(1-f) = 0. \end{aligned}$$

Since $6A_* \neq 0$, $fG_\chi' - f'G_\chi = 0$. This proof remains valid after any field extension of $\mathbb{F}_p$.

Set $M = G_\chi(B_c)$. Its pulled-back Wronskian and $f(B_c) = y^2$ give

$$y^2M' - (y^2)'M = 0, \qquad yM' - 2y'M = 0.$$

We give the square division explicitly. Put $L = y + A_c m'$; the proper identity is $yL = 1 + A_c y' m$. Define

$$b_1 = ML - \frac{1}{2}A_c m M', \qquad b_2 = b_1 L - A_c m b_1'.$$

Then

$$yb_1 = M(1 + A_c y' m) - \frac{1}{2}A_c m\, yM' = M.$$

Substitution into $yM' - 2y'M = 0$ gives

$$0 = y\{yb_1' - y'b_1\}, \qquad yb_1' - y'b_1 = 0.$$

Thus the same proper identity gives $yb_2 = b_1$, as follows:

$$yb_2 = b_1(1 + A_c y' m) - A_c m\, yb_1' = b_1.$$

Consequently

$$G_\chi(B_c) = y^2 b_2, \qquad b_2' = 0. \tag{206}$$

For the derivative assertion substitute $b_1 = yb_2$ into its Wronskian to obtain $y^2 b_2' = 0$. All quotients are polynomial constructions; the only scalar inverse introduced is $1/2$.

Both $G_\chi(B_c)$ and $y^2 = f(B_c)$ are even. Canceling $y^2 \neq 0$ in the reflected equation gives $b_2(-t) = b_2(t)$. The full degree bounds are

$$\begin{aligned} \deg \mathcal{H}(B_c) &\leq 5p - 3, \\ \deg P_{\text{num}}(B_c) &\leq 2(2p + N - 1) = 5p - 3, \\ \deg C_B(B_c) &\leq p + 3d + 3(d-1) = p + 6d - 3, \\ \deg(A_*^p C_B)(B_c) &\leq 4p + (p + 6d - 3) = 6d + 5p - 3. \end{aligned}$$

Since $\deg y^2 = 6d$, these imply $\deg M \leq 6d + 5p - 3$. If $b_2 \neq 0$, the product in (206) has exact degree $6d + \deg b_2$; if $b_2 = 0$ the bound is immediate. Hence

$$b_2' = 0, \qquad b_2(-t) = b_2(t), \qquad \deg b_2 \leq 5p - 3. \tag{207}$$

Writing $b_2 = U(t^p)$, the degree bound gives $\deg U \leq 4$. Evenness and oddness of p kill the two odd coefficients, so

$$b_2 = a_0 + a_2 t^{2p} + a_4 t^{4p}.$$

The descent polynomial $Q_\chi = a_0 + a_2 s^p + a_4 s^{2p}$ satisfies $Q_\chi(B_c) = b_2$. Injectivity of substitution gives $G_\chi = fQ_\chi$. Since $5^p = 5$,

$$s^p = (x^p + 1)/5, \quad Q_\chi = \alpha_1 x^{2p} + \beta_1 x^p + \gamma_1,$$

where

$$\alpha_1 = a_4/25, \qquad \beta_1 = a_2/5 + 2a_4/25, \qquad \gamma_1 = a_0 + a_2/5 + a_4/25.$$

Returning from the shifted polynomial to the same original rational chi therefore proves

$$\boxed{\chi_N = f\mathcal{P} + \zeta C_B + f(\alpha_1 x^p + \beta_1 + \gamma_1 x^{-p}), \qquad \zeta = \frac{x^{-p}A_*^p}{3} = \frac{17 - 4x^p - 4x^{-p}}{3}.} \tag{208}$$

This equality holds as a complete Laurent polynomial, with the phases multiplied by f. The notation χ_{par} in the following section denotes exactly $\chi_N = \overline{(F_N(V) - x^N F_N(Q))/p}$; the three constants are determined in the following scalar attachment, using its full middle coefficient and its two endpoints. All preceding statements remain valid under the injective prime-field map to any field of characteristic p. The finite $\mathcal{P}$ can be evaluated with either chosen root $u^2 = 1/5$; its derivative, degree bounds and the conic normalization used no root-specific extra premise.

6 Attaching the original moment scalar to its whole source

Throughout this section $p \geq 19$ is prime, $N = (p-1)/2$, $d = \lfloor p/6 \rfloor$, and $R = \mathbb{Z}_{(p)}$. Reduction $R \to \mathbb{F}_p$ is denoted by a bar. All quotients by p in the following rational calculation are formed in $\mathbb{Q}$, and are reduced only after their membership in R has been established. The letters $Y, n, h, q, E, O, L_0, K_h$ have the finite definitions in the native section: $Y = P_d$, n is its proper polynomial, $h = H_d$, $q = q_2$, $L_0 = q + E$, and

$$K_h = \frac{R(1-t) + R(1+t) + q^2}{2} - \frac{R((1-t)/2) + R((1+t)/2)}{6} + (2h/3 - q)L_0.$$

The finite dilogarithm $R(t)$ in this display is a polynomial; it is distinct from the local ring R.

For an even polynomial $P(t)$, write $\mathrm{qi}\, P(z)$ for the unique polynomial satisfying $\mathrm{qi}\, P(1-t^2) = P(t)$. Uniqueness follows because substitution $z \mapsto 1 - t^2$ is injective, and existence follows by expressing an even polynomial in powers of t^2. Put

$$V(x) = \frac{(4x-1)^2(4-x)}{125}, \quad F_{\mathrm{rev}}(x) = \frac{132x^3 - 81x^2 + 12x}{125}, \quad \mathcal{D}(G) = \frac{G(V) - G(0)}{V} F_{\mathrm{rev}}.$$

The fraction denotes the polynomial $(G.\operatorname{div} z)(V)$, so it is defined at both zeros of V. The original projections and Euler value are

$$I_j(G) = [x^{N+jp+1}]\mathcal{D}(G), \qquad Q_O(P) = I_0(\mathrm{qi}\, P), \qquad Q_H(P) = \left(\sum_{\substack{i \geq 1 \\ p \nmid i}} \frac{[z^i]\,\mathrm{qi}\, P}{i} z^i \right) (4/125). \tag{SSA1}$$

Every operation in (SSA1) is R-linear on integral inputs: the only fixed denominator in $\mathcal{D}$ is 125, and only indices prime to p are inverted in Q_H. For $\deg P \leq 2d$, $\deg \mathrm{qi}\, P \leq d$, and $\deg \mathcal{D}(\mathrm{qi}\, P) \leq 3d \leq N$. Hence

$$Q_O(P) = 0, \qquad I_j(\mathrm{qi}\, P) = 0 \quad (j \geq 0). \tag{SSA2}$$

This degree statement holds over $\mathbb{Q}$ as well as over $\mathbb{F}_p$.

Let $Y_1, Y_5, u = (Y_1 - Y_5)/(4p)$, $G_1 = \mathrm{qi}(Y_1 u)$ and $G_2 = \mathrm{qi}(u^2)$ be the rational polynomials in the moment calculation. With the same rational r_* used there,

$$\Delta_U = -2I_0(G_1) - \frac{1}{2}Q_H(Y_1^2) - r_*, \qquad S = 2\overline{\Delta_U/p} - 12\overline{I_0(G_2)} - \frac{15}{2}\overline{I_1(G_2)} - \mathcal{B}_p. \tag{SSA3}$$

This is exactly (55); in particular its I_1 term has not been discarded. Since all Euler indices of $\mathrm{qi}(Y_1^2)$ are at most $d < p$, its Euler value in (SSA3) equals the ordinary primitive with zero constant term. The proof of integrality of Δ_U/p in the moment section applies to the rational number in (SSA3), rather than to a lift of its residue.

6.1 The low carry and the proper correction

Put $Y_{\mathrm{rat}} = P_d$ over $\mathbb{Q}$, and let $\widehat{R}_{\mathrm{rat}}$ be the rational polynomial whose field reduction is

$$\widehat{R} = Y(L_0 - h) - t^p n. \tag{SSA4}$$

Its exact rational definition is

$$Q_2^{\mathbb{Z}} = \frac{2^{p-1}-1}{p}, \qquad E_{\mathrm{rat}} = \sum_{i=1}^{N} \frac{t^{2i}}{2i}, \qquad H_d^{\mathbb{Q}} = \sum_{i=1}^{d} \frac{1}{i},$$

$$\widehat{R}_{\mathrm{rat}} = Y_{\mathrm{rat}}(Q_2^{\mathbb{Z}} + E_{\mathrm{rat}} - H_d^{\mathbb{Q}}) - t^p n_{\mathrm{rat}},$$

where n_{rat} is the rational proper polynomial. Fermat's identity $2^{p-1} \equiv 1 \pmod p$ makes $Q_2^{\mathbb{Z}}$ an integer, whose reduction is the specified q. The logarithmic and harmonic denominators are units: $1 \le 2i \le p-1$ and $1 \le i \le d < p$, respectively. Thus this is the canonical rational hat used in the endpoint quotient γ below. The two actual carries are defined by the exact rational divisions

$$L_{\mathrm{rat}} = \frac{Y_1 - Y_{\mathrm{rat}}}{p}, \qquad T_{\mathrm{rat}} = \frac{6u - \widehat{R}_{\mathrm{rat}}}{p}. \tag{SSA5}$$

Here is the coefficient and hat comparison proving both divisions in (SSA5) integral. Put $m = p-1-d$, $z = (1-t)/2$, and define the finite parameter products

$$A_j(s) = \prod_{i=0}^{j-1} (6i+1+6s)(6i+5+6s), \qquad D_j(s) = 36^j \prod_{i=0}^{j-1} (i+1)(i+1+s).$$

Then the actual nodes are

$$Y_k(t) = \sum_{j=0}^{m} \frac{A_j(-kp/6)}{D_j(-kp/6)} z^j, \qquad k = 1, 5.$$

For every $0 \le j \le m < p$, the denominator reduces to $36^j (j!)^2 \ne 0$. Thus its inverse belongs to R. Write $p = 6d + r$, $r = 1$ or 5. The exact terminating factors are

	Y_1	Y_5	
$r = 1$	$6d + 1 - p = 0$	$6m + 5 - 5p = 0,$	$m = 5d$
$r = 5$	$6d + 5 - p = 0$	$6m + 1 - 5p = 0,$	$m = 5d + 4.$

Consequently $\deg Y_1 \le d$, $\deg Y_5 \le m$, and $Y_1(1) = Y_5(1) = 1$, since $z = 0$ at that endpoint. For the low node the numerator factors, in the two classes, are

$r = 1$	$(6i + 1 - p)(6i + 5 - p) = 36(i - d)(i - d + 2/3)$
$r = 5$	$(6i + 1 - p)(6i + 5 - p) = 36(i - d)(i - d - 2/3).$

Since $i + d + 1 - p/3 = i - d + 2/3$ in the first class and $i - d - 2/3$ in the second, both rows give the fixed family

$$f_{d,j}(\mu) = \frac{\prod_{i=0}^{j-1} (i-d)(i+d+1+2\mu)}{\prod_{i=0}^{j-1} (i+1)(i+1+\mu)}, \qquad \frac{A_j(-p/6)}{D_j(-p/6)} = f_{d,j}(-p/6).$$

Its denominators at 0 and $-p/6$ are units for $0 \le j \le d < p$. Clearing those denominators in the first Taylor identity gives

$$f_{d,j}(-p/6) = f_{d,j}(0) - \frac{p}{6} f'_{d,j}(0) + p^2 F_j, \qquad F_j \in R.$$

For clarity this is a quotient Taylor identity. If a, b are its numerator and denominator polynomials and $s_0 = -p/6$, write $a(s_0) = a(0) + s_0 a'(0) + s_0^2 a_2$ and $b(s_0) = b(0) + s_0 b'(0) + s_0^2 b_2$. Then its exact second remainder is

$$\frac{s_0^2\{a_2 b(0)^2 - a(0)b_2 b(0) - (a'(0)b(0) - a(0)b'(0))b'(0) - s_0(a'(0)b(0) - a(0)b'(0))b_2\}}{b(s_0)b(0)^2}.$$

Each numerator coefficient is integral and each denominator is a unit; $s_0^2 = p^2/36$, and 36 is a unit. Let

$$A_{\mathrm{rat}}(t) = \sum_{j=0}^{d} f'_{d,j}(0) z^j, \qquad F_{\mathrm{rat}}(t) = \sum_{j=0}^{d} F_j z^j.$$

Since $\sum_{j=0}^{d} f_{d,j}(0) z^j = P_d(t)$, this proves

$$Y_1 = Y_{\mathrm{rat}} - \frac{p}{6} A_{\mathrm{rat}} + p^2 F_{\mathrm{rat}}, \quad L_{\mathrm{rat}} = -A_{\mathrm{rat}}/6 + p F_{\mathrm{rat}} \in R[t], \quad \deg A_{\mathrm{rat}}, \deg F_{\mathrm{rat}} \le d.$$

For the divided node, the polynomial $A_j(s)D_j(b) - A_j(b)D_j(s)$ vanishes at $s = b$, so it is divisible by $s - b$ over $R[b, s]$. Evaluation of its polynomial quotient at $s = -p/6,\ b = -5p/6$, followed by division by the two unit denominators, proves

$$R_\Delta = \frac{Y_1 - Y_5}{2p/3} = 6u \in R[t], \qquad \deg R_\Delta \le m, \qquad R_\Delta(1) = 0.$$

This quotient cancels $s - b = 2p/3$ in characteristic zero.

We must compare its entire field polynomial with the hat before dividing their difference by p. Define, over $\mathbb{Q}$,

$$\mathcal{G}_s(P) = 36(1 - t^2)P'' - 72(1 + s)tP' - (1 + 6s)(5 + 6s)P.$$

The coefficient recurrence of the two products is

$$36(j + 1)(j + 1 + s)a_{j+1} = (6j + 1 + 6s)(6j + 5 + 6s)a_j.$$

Inserting it in the differential equation in z, and using $d/dt = -\frac{1}{2} d/dz$, gives $\mathcal{G}_{-kp/6}(Y_k) = 0$; the highest row is zero because of the terminating factor in the table. The two rational operator changes are

$$\mathcal{G}_{-5p/6}(P) - \mathcal{G}_{-p/6}(P) = 24p\{2tP' + (1 - p)P\},$$

$$\mathcal{G}_{-p/6}(P) = 36\mathcal{L}_d P + 12p(tP' - dP), \qquad \mathcal{L}_d = (1 - t^2)\frac{d^2}{dt^2} - 2t\frac{d}{dt} + d(d + 1).$$

The latter identity keeps the rational constant coefficient: for $r = 1$, $(1 - p)(5 - p) = 36d^2 - 24d$, and for $r = 5$ it is $36d^2 + 24d$. These are respectively $12pd - 36d(d + 1)$, with $p = 6d + 1$ and $p = 6d + 5$. The derivative coefficient is $-72(1 - p/6) = -72 + 12p$ in both classes. Thus the asserted low operator holds in both classes. Apply it to $Y_1 - Y_5 = (2p/3)R_\Delta$ and cancel the nonzero rational factor $24p$. This gives exactly

$$\mathcal{L}_d R_\Delta + \frac{p}{3}(tR'_\Delta - dR_\Delta) = 2tY'_5 + (1 - p)Y_5. \tag{SSA5a}$$

Reduction of both node coefficient recipes gives Y: their denominator residues are the same units, and all coefficients above d have the zero numerator factor at $i = d$. Therefore $\mathcal{L}_d \bar{R}_\Delta = 2tY' + Y$.

The full finite field hat has this same image. Indeed $q_t E' = t - t^p$, with $q_t = 1 - t^2$, gives $q_t E'' - 2tE' = 1$ in the field. The product rule and $\mathcal{L}_d n = -2Y'$ then give

$$\mathcal{L}_d\{Y(E + q - h) - t^p n\} = Y + 2Y'(t - t^p) + 2t^p Y' = 2tY' + Y.$$

Its endpoint is zero since $E(1) = -q,\ Y(1) = 1,\ n(1) = -h$. Its initial degree bound is $p + d - 1$. If its degree e exceeded $m = p - 1 - d$, its leading Legendre coefficient would be $-(e - d)(e + d + 1)[t^e]\widehat{R}$, at degree $e > d$. In this range

$$p - 2d \le e - d \le p - 1, \qquad 1 \le e + d + 1 - p \le 2d < p.$$

Both factors are units in the field, contradicting the degree d of its image. Hence $\deg \widehat{R} \le m < p$. The difference $\bar{R}_\Delta - \widehat{R}$ has zero Legendre image, zero endpoint and degree $< p$. In $z = (1-t)/2$, its coefficients satisfy

$$(j+1)^2 b_{j+1} = (j-d)(j+d+1)b_j, \qquad b_0 = 0.$$

For $0 \le j \le p-2$, the coefficient $(j+1)^2$ is a unit, so induction makes every coefficient zero. Consequently $\bar{R}_\Delta = \widehat{R}$. The rational hat is integral by its finite recipe and has degree at most $p+d-1$; thus coefficientwise $R_\Delta - \widehat{R}_{\mathrm{rat}} \in pR[t]$. This proves $T_{\mathrm{rat}} \in R[t]$ in (SSA5), including all rational-hat coefficients above degree m. No inverse of p is used in the coefficient field. The node coefficient recurrence and its endpoint, already derived in the moment section, give

$$\begin{aligned} \bar{L}_{\mathrm{rat}} &= -A_v/6, \quad \mathcal{L}_d A_v = 2tY' - 2dY, \\ A_v(1) &= 0, \quad \deg A_v \le d, \\ \mathcal{L}_d &= (1-t^2)\frac{d^2}{dt^2} - 2t\frac{d}{dt} + d(d+1). \end{aligned} \tag{SSA6}$$

Equivalently, $A_v = -2\,\partial_\eta(a_d(\eta,t)/m_d(\eta))|_{\eta=0}$, with the family and metric in (108). To identify this with the literal coefficient derivative above, let $F_d(\mu,t) = \sum_{j=0}^{d} f_{d,j}(\mu)((1-t)/2)^j$. Its coefficient recurrence gives

$$\mathcal{L}_d F_d + 2\mu(dF_d - tF_d') = 0, \qquad F_d(\mu,1) = 1.$$

Putting $\eta = -2\mu$ gives the normalized family's same differential equation. Over $\mathbb{Q}(\mu)$, a difference of degree $e < d$ has nonzero top multiplier $(d-e)(d+e+1+2\mu)$; the endpoint fixes the remaining multiple of the degree-d solution. Hence $F_d(\mu,t) = a_d(-2\mu,t)/m_d(-2\mu)$, so $\bar{A}_{\mathrm{rat}} = A_v$. Differentiating $\mathcal{L}_d Y_\eta + \eta(tY_\eta' - dY_\eta) = 0$ gives (SSA6); the identity $Y_\eta(1) = 1$ gives its endpoint.

Expanding the original rational Δ_U using $6u = \widehat{R}_{\mathrm{rat}} + pT_{\mathrm{rat}}$ gives

$$\begin{aligned} \Delta_U &= D_{\mathrm{hat}} + pD_{\mathrm{next}} + p^2 D_{\mathrm{high}}, \\ D_{\mathrm{hat}} &= -\frac{1}{3}Q_O(Y_{\mathrm{rat}}\widehat{R}_{\mathrm{rat}}) - \frac{1}{2}Q_H(Y_{\mathrm{rat}}^2) - r_*, \\ D_{\mathrm{next}} &= -\frac{1}{3}\{Q_O(Y_{\mathrm{rat}}T_{\mathrm{rat}}) + Q_O(L_{\mathrm{rat}}\widehat{R}_{\mathrm{rat}})\} - Q_H(Y_{\mathrm{rat}}L_{\mathrm{rat}}), \\ D_{\mathrm{high}} &= -\frac{1}{3}Q_O(L_{\mathrm{rat}}T_{\mathrm{rat}}) - \frac{1}{2}Q_H(L_{\mathrm{rat}}^2). \end{aligned} \tag{SSA7}$$

Indeed the product term has coefficients $1, p, p, p^2$, and the square has coefficients $1, 2p, p^2$. Both D_{next} and D_{high} belong to R by (SSA1). Consequently

$$D_{\mathrm{hat}}/p = \Delta_U/p - D_{\mathrm{next}} - pD_{\mathrm{high}} \in R, \quad \overline{\Delta_U/p} = \overline{D_{\mathrm{hat}}/p} - \frac{1}{3}Q_O(Y\bar{T}_{\mathrm{rat}}) + \frac{1}{18}Q_O(A_v\widehat{R}) + \frac{1}{6}Q_H(YA_v). \tag{SSA8}$$

The $p^2 D_{\mathrm{high}}$ term disappears only after division by p.

We give the polynomial reconstruction of $\bar{T}_{\mathrm{rat}}$ needed in (SSA8). Set $q_t = 1-t^2$, and define

$$\mathcal{H}_Y(C) = q_t(YC' - 2Y'C), \qquad \mathcal{N} = A_v n + q_t Y n(A_v n' - A_v' n), \qquad C_v = \tfrac{1}{2}(\mathcal{N} \bmod Y^2). \tag{SSA9}$$

The remainder is the unique polynomial of degree $< 2d$ in the division by Y^2; the leading coefficient of Y is nonzero. The Wronskian (102) and the equations $\mathcal{L}_d Y = 0$, $\mathcal{L}_d n = -2Y'$, (SSA6) give

$$\begin{aligned} \mathcal{H}_Y(\mathcal{N}) - 2A_v - 2tnY^2 &= Y^3 T_v, \\ T_v &= -3q_t A_v n' - 2q_t tnn' - 2Y(A_v + tn) + 2dq_t n^2 + q_t A_v' n. \end{aligned} \tag{SSA10}$$

These identities follow by the product rule with the following substitutions, which fix every second derivative:

$$q_t Y'' = 2tY' - d(d+1)Y, \quad q_t n'' = 2tn' - d(d+1)n - 2Y', \quad q_t A_v'' = 2tA_v' - d(d+1)A_v + 2tY' - 2dY.$$

If $\mathcal{N} = Y^2 D_v + 2C_v$, then $\mathcal{H}_Y(Y^2 D_v) = q_t Y^3 D_v'$. Thus

$$\mathcal{H}_Y(C_v) - A_v - tnY^2 = \tfrac{1}{2} Y^3 (T_v - q_t D_v').$$

The left side has degree at most $3d$. It is therefore a constant multiple of Y^3. At $t = 1$, $Y = 1$, $n = -h$, and $A_v = 0$; its value is h. We obtain

$$\mathcal{H}_Y(C_v) = tnY^2 + A_v + hY^3, \qquad \deg C_v < 2d. \tag{SSA11}$$

For the divisibility used next put $a_v = q_t n' + Y$ and $v_v = -n(1 + Y a_v)$. The Wronskian gives

$$Y a_v - q_t Y' n = 1, \qquad q_t Y' v_v + Y^2 a_v^2 = 1,$$

and direct multiplication gives the representative

$$-A_v v_v - Y q_t A_v' v_v^2 - \mathcal{N} = Y^2 (A_v n - 2 q_t A_v' n^2 a_v - Y q_t A_v' n^2 a_v^2). \tag{SSA12}$$

The division in (SSA9) gives directly

$$2C_v - nA_v = Y\{q_t n (A_v n' - A_v' n) - Y D_v\}.$$

Thus $Y \mid (2C_v - nA_v)$, and the quotient is a polynomial even at every root of Y. Define

$$C_{\mathrm{div}} = (2C_v - nA_v)/Y, \qquad \deg C_{\mathrm{div}} \le d - 1. \tag{SSA13}$$

This is polynomial division by the nonzero polynomial Y; no value $Y(t)$ is inverted.

Define the adjacent primitive by the finite Legendre sum

$$B_{\mathrm{adj}}(t) = -\sum_{j=0}^{d-1} \frac{P_{j+1}(t) P_j(t)}{j+1}.$$

The ladder equations $q_t P_j' = j(P_{j-1} - tP_j)$ and $q_t P_{j-1}' = j(tP_{j-1} - P_j)$ give

$$q_t (P_j P_{j-1})' = j(P_{j-1}^2 - P_j^2).$$

Thus the derivative of the displayed sum telescopes, and its endpoint is the sum of $-1/(j+1)$. This proves

$$q_t B_{\mathrm{adj}}' = Y^2 - 1, \qquad B_{\mathrm{adj}}(1) = -h, \qquad \deg B_{\mathrm{adj}} \le 2d - 1. \tag{SSA14}$$

Equivalently it is the Euler primitive of $t(Y^2 - 1)/q_t$, with zero constant term: the numerator vanishes at $1, -1$, its quotient is polynomial, and the integration indices are $< p$. Since Yn and B_{adj} are odd and agree at 1, they agree at -1 too. Hence

$$M_v' = (Yn - B_{\mathrm{adj}})/q_t, \quad M_v(0) = 0, \quad \deg M_v \le 2d - 2. \tag{SSA15}$$

All integration indices in (SSA15) are nonzero modulo p. This M_v is the low carry primitive; it is distinct from the proper metric primitive defining C_M.

The product rule, (SSA14), (SSA15), and the Wronskian give

$$\mathcal{H}_Y(Y^2 M_v - Y n B_{\mathrm{adj}}) = Y(Yn - B_{\mathrm{adj}}). \tag{SSA16}$$

The argument $Y^2 M_v - Y n B_{\mathrm{adj}}$ initially has degree at most $4d - 2 < p$. If its degree e exceeded $2d$, the top coefficient of its Hermite image would be

$$-(e - 2d)[t^d] Y \, [t^e](Y^2 M_v - Y n B_{\mathrm{adj}})$$

at degree $d + e + 1 > 3d - 1$. It cannot vanish because $0 < e - 2d < p$. This contradicts the degree of the right side of (SSA16). Thus the polynomial has degree at most $2d$, and (SSA2) proves

$$Q_O(Y^2 M_v) = Q_O(Y n B_{\mathrm{adj}}). \tag{SSA17}$$

Put $c = q - h$, $U = t^p$, $T = t^{p-1}$, $W = t^{p-2}$, and

$$\begin{aligned} Z_v &= U(6cn - C_{\mathrm{div}}) + 6Y(M_v + K_h) - 6n(B_{\mathrm{adj}} + O) - A_v(L_0 - h), \\ P_v &= YZ_v/6. \end{aligned} \tag{SSA18}$$

The derivative identities of the explicitly defined finite logarithms are

$$q_t O' = 1 - t^{p-1}, \quad 6q_t K_h' = (6c + 2h)t^p - 6O - 4t(L_0 - h). \tag{SSA19}$$

Using $YC_{\mathrm{div}} = 2C_v - nA_v$ and (SSA4), expansion of (SSA18) gives the full polynomial identity

$$6P_v + A_v\widehat{R} = 6c\,t^pYn - 2t^pC_v + 6Y^2(M_v + K_h) - 6Yn(B_{\mathrm{adj}} + O). \tag{SSA20}$$

In particular the derivative-invisible t^p term is retained.

For completeness, its differential reconstruction can be checked without choosing roots of Y. Put $G_v = cU - B_{\mathrm{adj}} - O$. Then $q_tG_v' = T - Y^2$, $U' = 0$, $T' = -W$, and $U = tT$. Applying $\mathcal{H}_Y$ to (SSA20) before division by Y gives

$$\mathcal{H}_Y(YZ_v) = 6YG_v + 6q_tTY^2n - 2UA_v - 4tY^2\widehat{R} - \mathcal{H}_Y(\widehat{R}A_v). \tag{SSA21}$$

The two product identities used to differentiate (SSA21) are

$$\begin{aligned} Y(\mathcal{H}_Y(\widehat{R}A_v))' - Y'\mathcal{H}_Y(\widehat{R}A_v) &= Y^2(\widehat{R}\mathcal{L}_dA_v + A_v\mathcal{L}_d\widehat{R}) - 2Y\widehat{R}A_v\mathcal{L}_dY \\ &\quad + 2q_t(Y\widehat{R}' - Y'\widehat{R})(YA_v' - Y'A_v), \\ Y(\mathcal{H}_Y(YZ_v))' - Y'\mathcal{H}_Y(YZ_v) &= Y^3\mathcal{L}_dZ_v - Y^2Z_v\mathcal{L}_dY. \end{aligned} \tag{SSA22}$$

Here

$$\mathcal{L}_d\widehat{R} = 2tY' + Y, \qquad q_t(Y\widehat{R}' - Y'\widehat{R}) = tY^2 - U.$$

Substitution of these two rows and (SSA6), (SSA11), (SSA15), (SSA19) into (SSA21)–(SSA22) gives

$$\begin{aligned} \mathcal{L}_dZ_v = {} & -(2tA_v' + A_v) - 6Y - 10t\widehat{R}' + 2(d-2)\widehat{R} + 6TY + 12q_tTn' \\ & - 6q_tWn - 12Un. \end{aligned} \tag{SSA23}$$

Here is the full cancellation certificate. Write

$$\begin{aligned} H_v &= \mathcal{H}_Y(\widehat{R}A_v), \\ W_A &= YA_v' - Y'A_v, \\ S_v &= \text{the right side of (SSA23)}, \\ V_v &= \text{the right side of (SSA21)}. \end{aligned}$$

Define the five polynomial differences

$$\begin{aligned} \varepsilon_g &= YH_v' - Y'H_v - Y^2\{\widehat{R}(2tY' - 2dY) + A_v(2tY' + Y)\} \\ &\quad - 2q_t(Y\widehat{R}' - Y'\widehat{R})W_A, \\ \varepsilon_G &= q_tG_v' - (T - Y^2), \\ \varepsilon_R &= q_t(Y\widehat{R}' - Y'\widehat{R}) - (tY^2 - U), \\ \varepsilon_n &= q_t(Yn' - Y'n) - (1 - Y^2), \qquad \varepsilon_U = U - tT. \end{aligned}$$

Each is zero by (SSA22), the two displayed hat rows, the Wronskian, and $U = tT$. Expanding V_v' with the product rule gives the exact polynomial identity

$$\begin{aligned} q_t(YV_v' - Y'V_v) - q_tY^3S_v = {} & -q_t\varepsilon_g + 6Y^2\varepsilon_G + (6tY^2 - 2q_tW_A)\varepsilon_R \\ & - 6q_tTY^2\varepsilon_n + (12q_tY^3n - 6tY^2)\varepsilon_U = 0. \end{aligned}$$

For instance this expansion differentiates the explicit T using $T' = -W$; it never differentiates U as a nonzero term. Equation (SSA21) replaces V_v by $\mathcal{H}_Y(YZ_v)$, and the second row of (SSA22) then turns its left side into $q_tY^3(\mathcal{L}_dZ_v - S_v)$. The polynomial q_tY^3 is nonzero because $q_t = 1 - t^2 \neq 0$ and $Y(1) = 1$. Cancellation in the polynomial domain proves (SSA23) without inverting a value at a root of Y. This certificate keeps all three upper slots U, T, W.

We now derive the rational forcing before reducing it. In the next displays $Y = Y_{\mathrm{rat}}$, $n = n_{\mathrm{rat}}$, $A = A_{\mathrm{rat}}$, $F = F_{\mathrm{rat}}$, $R_\Delta = 6u$, and the hat are all rational polynomials. In particular the derivative of t^p is pt^{p-1}. Put $U = t^p,\ T = t^{p-1},\ W = t^{p-2}$, and define

$$B_{\mathrm{hat}} = -TY - 2q_tTn' - (p-1)q_tWn + 2Un.$$

The complete rational even logarithm gives

$$q_tE' = t - U, \qquad q_tE'' - 2tE' = 1 - pT.$$

For the power product, retaining both its derivatives,

$$\mathcal{L}_d(Un) = U\mathcal{L}_dn + 2pq_tTn' + p(p-1)q_tWn - 2pUn.$$

The rational equations $\mathcal{L}_dY = 0$, $\mathcal{L}_dn = -2Y'$ and the product rule therefore yield

$$\begin{aligned}\mathcal{L}_d\widehat{R}_{\mathrm{rat}} &= Y(1-pT) + 2Y'(t-U) + 2UY' \\ &\quad - 2pq_tTn' - p(p-1)q_tWn + 2pUn \\ &= 2tY' + Y + pB_{\mathrm{hat}}.\end{aligned}$$

The rational low Taylor row proved above, together with $Y_1 - Y_5 = (2p/3)R_\Delta$, gives

$$Y_5 = Y - p(A/6 + 2R_\Delta/3) + p^2F.$$

Insert it in (SSA5a) and use $R_\Delta = \widehat{R}_{\mathrm{rat}} + pT_{\mathrm{rat}}$. Before cancellation, the two sides become

$$\begin{aligned}&2tY' + Y + pB_{\mathrm{hat}} + p\mathcal{L}_dT_{\mathrm{rat}} + \frac{p}{3}(tR'_\Delta - dR_\Delta), \\ &2tY' + Y - pY - \frac{p}{6}(2tA' + A) - \frac{2p}{3}(2tR'_\Delta + R_\Delta) \\ &\qquad + p^2(2tF' + F + A/6 + 2R_\Delta/3) - p^3F.\end{aligned}$$

Cancel $2tY' + Y$, then cancel the nonzero rational factor p. Collection of the coefficients $-4/3 - 1/3$ of tR'_Δ and $-2/3 + d/3$ of R_Δ gives the exact identity

$$\begin{aligned}\mathcal{L}_dT_{\mathrm{rat}} = &-\frac{2tA' + A}{6} - Y - \frac{5}{3}tR'_\Delta + \frac{d-2}{3}R_\Delta - B_{\mathrm{hat}} \\ &+ p(2tF' + F + A/6 + 2R_\Delta/3) - p^2F. \qquad \text{(SSA23a)}\end{aligned}$$

Every polynomial in this equality lies in $R[t]$. Only now reduce: $\bar{A} = A_v$, $\bar{R}_\Delta = \widehat{R}$, $\overline{p-1} = -1$. Thus

$$-6\bar{B}_{\mathrm{hat}} = 6TY + 12q_tTn' - 6q_tWn - 12Un,$$

and the explicit p and p^2 terms disappear at this final step. Equation (SSA23a) consequently gives $\mathcal{L}_d\bar{T}_{\mathrm{rat}} = \mathcal{L}_dZ_v/6$. The degree bound $\deg \bar{T}_{\mathrm{rat}} \leq p + d - 1$ follows from the coefficientwise division of $R_\Delta - \widehat{R}_{\mathrm{rat}}$, whose two degrees were bounded above; division by a scalar does not add exponents. Its endpoint is $\bar{T}_{\mathrm{rat}}(1) = -\bar{\gamma}$, where the fixed integral endpoint is

$$\gamma = \frac{\sum_{i=1}^{N}(2i)^{-1} + q_2^{\mathbb{Z}}}{p} = -T_{\mathrm{rat}}(1) \in R. \qquad \text{(SSA24)}$$

The equality uses $Y_1(1) = Y_5(1) = 1$ and the actual rational hat endpoint, so it fixes this carry before reduction. Equation (SSA18) has degree at most $p + d - 1$, and, since $K_h(1) = 0,\ L_0(1) = 0,\ O(1) = q$,

$$Z_v(1) = -2C_v(1) + 6M_v(1).$$

Set $\kappa_v = -\bar{\gamma} + C_v(1)/3 - M_v(1)$. The difference $\bar{T}_{\mathrm{rat}} - Z_v/6 - \kappa_v Y$ has zero endpoint, zero Legendre image, and degree at most $p + d - 1$. Its top coefficient would be multiplied by $(e - d)(e + d + 1)$; for $p - 1 - d < e \le p + d - 1$ these factors have the explicit unit ranges

$$p - 2d \le e - d \le p - 1, \qquad 1 \le e + d + 1 - p \le 2d < p.$$

It therefore has degree at most $p - 1 - d < p$. In the coordinate $z = (1 - t)/2$, the coefficient recurrence of a zero Legendre image is

$$(j + 1)^2 b_{j+1} = (j - d)(j + d + 1)b_j.$$

Its constant b_0 is the endpoint at $t = 1$, hence zero; induction for $0 \le j < p - 1$ proves that every coefficient is zero. We have proved

$$\bar{T}_{\mathrm{rat}} = Z_v/6 + \kappa_v Y, \quad Q_O(Y\bar{T}_{\mathrm{rat}}) = Q_O(P_v). \tag{SSA25}$$

The second equality uses (SSA2); the endpoint carry has not been assumed zero.

6.2 The balanced Green numerator and the original norm

Use $Y_{\mathbb{Q}} = a_d(p/3, t)/m_d(p/3)$ and $J_{\mathbb{Q}} = b_d(p/3, t)$ from (108)–(109). Write

$$m_d(\eta) = \prod_{j=1}^{d}(1 - \eta/j), \qquad \sigma_{\mathbb{Q}} = \partial_\eta a_d(\eta, t)|_{\eta=0}, \qquad N_{\mathbb{Q}} = \partial_\eta b_d(\eta, t)|_{\eta=0}.$$

For every recurrence step $0 \le j \le d - 2$, the denominator $j + 2$ lies in $\{2, \dots, d\} \subset R^\times$. The initial polynomials and the recurrence therefore give $a_d, b_d \in R[\eta, t]$, with t-degrees at most $d, d-1$, respectively, for every coefficient of η. In particular, put $P_j = a_j(0, t)$, $n_j = b_j(0, t)$, and $\sigma_j = \partial_\eta a_j|_0,\ \nu_j = \partial_\eta b_j|_0$. Differentiation of the finite recurrence gives the complete jet recurrences

$$\begin{aligned} \sigma_0 = 0, \quad \sigma_1 = -t, \qquad \nu_0 = \nu_1 = 0, \\ (j + 2)\sigma_{j+2} = (2j + 3)t\sigma_{j+1} - (j + 1)\sigma_j - tP_{j+1} + P_j, \\ (j + 2)\nu_{j+2} = (2j + 3)t\nu_{j+1} - (j + 1)\nu_j - tn_{j+1} + n_j \qquad (0 \le j \le d - 2). \end{aligned}$$

Thus $\sigma_{\mathbb{Q}} = \sigma_d$ and $N_{\mathbb{Q}} = \nu_d$, with no parameter denominators beyond the specified units. Their constant coefficients are Y_{rat} and n_{rat}. Indeed the first satisfies the Legendre equation and endpoint 1, and the second satisfies the proper equation and endpoint $-H_d^{\mathbb{Q}}$; the endpoint-zero coefficient recurrence used above identifies them over $\mathbb{Q}$. The proper parameter equation is

$$q_t b_d'' - (2 + \eta)tb_d' + (d + 1)(d - \eta)b_d = -2a_d'.$$

The endpoint identity is $\eta b_j(\eta, 1) = m_j(\eta) - 1$. It holds for $j = 0, 1$; multiplying the next recurrence by η gives, on inserting the two preceding identities,

$$(2j + 3 - \eta)(m_{j+1} - 1) - (j + 1 - \eta)(m_j - 1) = (j + 2)(m_{j+2} - 1),$$

where $m_{j+1} = (j + 1 - \eta)m_j/(j + 1)$ and $m_{j+2} = (j + 2 - \eta)m_{j+1}/(j + 2)$. This proves the endpoint identity by induction. Its coefficient of η is $b_d(0, 1) = m_d'(0) = -H_d^{\mathbb{Q}}$. There are therefore polynomials a_2, b_2, m_2 over R such that

$$\begin{aligned} a_d(\eta, t) &= Y_{\mathrm{rat}} + \eta\sigma_{\mathbb{Q}} + \eta^2 a_2(\eta, t), \\ b_d(\eta, t) &= n_{\mathrm{rat}} + \eta N_{\mathbb{Q}} + \eta^2 b_2(\eta, t), \\ m_d(\eta) &= 1 - \eta H_d^{\mathbb{Q}} + \eta^2 m_2(\eta). \end{aligned}$$

Each remainder is explicitly the sum of the parameter coefficients of orders $k \geq 2$, multiplied by η^{k-2}; these are finite sums because the recurrence families are polynomials. Put $a_{\mathbb{Q}} = \sigma_{\mathbb{Q}} + H_d^{\mathbb{Q}} Y_{\mathrm{rat}}$. At $\eta_* = p/3$, the value of m_d reduces to 1, so it is a unit. Subtracting $(Y_{\mathrm{rat}} + \eta_* a_{\mathbb{Q}}) m_d(\eta_*)$ from $a_d(\eta_*, t)$ gives the exact normalized Taylor formula

$$Y_{\mathbb{Q}} = Y_{\mathrm{rat}} + \frac{p}{3} a_{\mathbb{Q}} + \frac{p^2}{9} R_Y, \qquad R_Y = \frac{a_2(\eta_*, t) - Y_{\mathrm{rat}} m_2(\eta_*) + H_d^{\mathbb{Q}} a_{\mathbb{Q}} - \eta_* a_{\mathbb{Q}} m_2(\eta_*)}{m_d(\eta_*)} \in R[t].$$

The corresponding proper Taylor formula is

$$J_{\mathbb{Q}} = n_{\mathrm{rat}} + \frac{p}{3} N_{\mathbb{Q}} + \frac{p^2}{9} b_2(\eta_*, t).$$

Consequently the first increments

$$Y_{\mathbb{Q}} = Y_{\mathrm{rat}} + p L_b, \qquad J_{\mathbb{Q}} = n_{\mathrm{rat}} + p N_b$$

are the integral polynomials

$$L_b = \frac{a_{\mathbb{Q}}}{3} + \frac{p}{9} R_Y, \qquad N_b = \frac{N_{\mathbb{Q}}}{3} + \frac{p}{9} b_2(\eta_*, t).$$

Their degrees are at most $d, d-1$. Here $3, 9$ are units because $p \geq 19$. Define the field jets

$$\sigma = \bar{\sigma}_{\mathbb{Q}}, \qquad a_{\mathrm{jet}} = \bar{a}_{\mathbb{Q}} = \partial_\eta (a_d/m_d)|_{\eta=0} \bmod p, \qquad N_v = \bar{N}_{\mathbb{Q}}.$$

In particular $\bar{N}_b = N_v/3$; N_v is defined by the actual proper family, before any Hermite comparison.

The differentiated equation for the scaled family is explicit. Its parameter equation, obtained from the ladder equations, is

$$q_t \partial_t^2 a_d - (2 - \eta) t \partial_t a_d + d(d + 1 - \eta) a_d = 0.$$

Taking the coefficient of η and reducing gives

$$\mathcal{L}_d \sigma = -tY' + dY, \qquad \sigma(1) = m_d'(0) \bmod p = -h, \qquad \deg \sigma \leq d.$$

The equation (SSA6) for A_v gives

$$\mathcal{L}_d \{2(\sigma + hY) + A_v\} = 0, \qquad \{2(\sigma + hY) + A_v\}(1) = 0.$$

This polynomial has degree at most $d < p$. In the coordinate $z = (1 - t)/2$, its coefficients obey $(j+1)^2 b_{j+1} = (j - d)(j + d + 1) b_j$. The endpoint is $b_0 = 0$, and every $j + 1$ for $0 \leq j \leq p - 2$ is a unit, so it vanishes. Thus

$$\sigma = -A_v/2 - hY, \qquad a_{\mathrm{jet}} = -A_v/2, \qquad \bar{L}_b = -A_v/6.$$

We now differentiate the metric, rather than infer the jet source from divisibility. The full parameter metric before normalization is

$$q_t (a_d b_d' - a_d' b_d) - \eta t a_d b_d = m_d - a_d^2.$$

Its coefficient of η, reduced to the field, is

$$q_t \{Y N_v' - Y' N_v + \sigma n' - \sigma' n\} - tYn = -h - 2Y\sigma.$$

Substitute $\sigma = a_{\mathrm{jet}} - hY$ and use $q_t(Yn' - Y'n) = 1 - Y^2$. The left side becomes

$$q_t \{Y N_v' - Y' N_v + a_{\mathrm{jet}} n' - a_{\mathrm{jet}}' n\} - h(1 - Y^2) - tYn,$$

while the right side is $-h - 2Ya_{\mathrm{jet}} + 2hY^2$. Canceling the constant $-h$ and one hY^2 proves

$$q_t\{YN_v' - Y'N_v + a_{\mathrm{jet}}n' - a_{\mathrm{jet}}'n\} - tYn = hY^2 - 2Ya_{\mathrm{jet}}. \tag{SSA26a}$$

It is also the direct derivative at $\eta = 0$ of the normalized identity

$$q_t\{(a_d/m_d)b_d' - (a_d/m_d)'b_d\} - \eta t(a_d/m_d)b_d = 1 - m_d(a_d/m_d)^2,$$

since $m_d(0) = 1$ and $m_d'(0) = -H_d^{\mathbb{Q}}$.

Let $C_{\mathrm{met}} = YN_v + nA_v/2$. The product rule, followed by $a_{\mathrm{jet}} = -A_v/2$, gives all terms of its Hermite source:

$$\begin{aligned}\mathcal{H}_Y(C_{\mathrm{met}}) &= q_t\{Y^2N_v' - YY'N_v + (Yn'A_v + YnA_v')/2 - Y'nA_v\}\\ &= Yq_t\{YN_v' - Y'N_v - (A_vn' - A_v'n)/2\} + A_vq_t(Yn' - Y'n)\\ &= Y\{tYn + hY^2 + YA_v\} + A_v(1 - Y^2)\\ &= tnY^2 + hY^3 + A_v.\end{aligned}$$

The two Y^2A_v terms cancel in the last line. Moreover $\deg C_{\mathrm{met}} \le 2d - 1 < 2d$, because $\deg N_v, \deg n \le d - 1$ and $\deg Y, \deg A_v \le d$. Thus its source equals the source of C_v in (SSA11). For completeness, if $D = C_{\mathrm{met}} - C_v$, then $\mathcal{H}_Y D = 0$ and $\deg D < 2d$. The second Bezout identity in (SSA12) makes q_tY' invertible modulo Y^2, hence modulo Y. Reducing $q_t(YD' - 2Y'D) = 0$ modulo Y, and using $2 \neq 0$, gives $Y \mid D$. Write $D = YP$ and cancel the nonzero Y; the resulting equation is $q_t(YP' - Y'P) = 0$. Reduction modulo Y with the same Bezout identity gives $Y \mid P$. Hence $Y^2 \mid D$; the bound $\deg D < 2d = \deg Y^2$ forces $D = 0$. We have therefore identified the actual proper jet:

$$C_v = YN_v + nA_v/2, \qquad N_v = \frac{2C_v - nA_v}{2Y} = C_{\mathrm{div}}/2. \tag{SSA26}$$

The quotient in this display is the polynomial quotient (SSA13). It has now been identified with the derivative of the specified rational family without inverting any value $Y(t)$.

Write

$$c_i = \binom{2i}{i}/4^i, \qquad L_c(z) = \sum_{i=1}^{N} \frac{c_iz^i}{2i}, \qquad L_2(z) = \sum_{i=1}^{N} \frac{c_iz^i}{2i} \sum_{j=0}^{i-1} \frac{1}{2j+1}.$$

Their denominators are units, since $2i < p$ and $2j + 1 < p$. We need their exact rational relation to the logarithm in the specified hat, including its endpoint and second carry. Define

$$L_f(z) = \sum_{i=1}^{N} \frac{(-1)^i\binom{N}{i}}{2i} z^i, \qquad s_i = \sum_{j=0}^{i-1} \frac{1}{2j+1}.$$

Put $s_0 = 0,\ r_0 = 0$, and, for $0 \le i < N$, set

$$r_{i+1} = r_i + \frac{s_i}{2i+1} - \frac{p\,r_i}{2i+1}.$$

Every denominator $2i + 1$ in this recurrence satisfies $1 \le 2i + 1 \le 2N - 1 = p - 2$, hence is a unit. Multiplying $1 - ps_i + p^2r_i$ by $1 - p/(2i + 1)$ gives

$$1 - p\left(s_i + \frac{1}{2i+1}\right) + p^2\left(r_i + \frac{s_i}{2i+1} - \frac{p\,r_i}{2i+1}\right).$$

Induction therefore proves

$$\prod_{j=0}^{i-1}\left(1 - \frac{p}{2j+1}\right) = 1 - ps_i + p^2r_i, \qquad r_i \in R \quad (0 \le i \le N).$$

Since $N=(p-1)/2$, the exact coefficient product is

$$(-1)^i\binom{N}{i}=\prod_{j=0}^{i-1}\frac{2j+1-p}{2(j+1)}$$
$$=c_i\prod_{j=0}^{i-1}\left(1-\frac{p}{2j+1}\right)=c_i(1-ps_i+p^2r_i),\qquad 1\le i\le N.$$

Here $\prod_{j=0}^{i-1}(2j+1)/(2(j+1))=\binom{2i}{i}/4^i=c_i$. The denominators $2i\le p-1$ and 4^i are units as well. Consequently the specified full remainder

$$L_{\mathrm{rem}}(z)=\sum_{i=1}^{N}\frac{c_ir_i}{2i}z^i\in R[z]$$

gives the exact polynomial equality

$$L_f=L_c-pL_2+p^2L_{\mathrm{rem}}. \tag{SSA26b}$$

No coefficient at the terminal index $i=N$ has been removed.

The rational pullback identity can also be checked without reduction. For $w=1-t^2=q_t$,

$$\frac{d}{dt}L_f(1-t^2)=-t\sum_{i=1}^{N}(-1)^i\binom{N}{i}(1-t^2)^{i-1}=t\,\frac{1-t^{2N}}{1-t^2}=t\sum_{j=0}^{N-1}t^{2j}.$$

The geometric quotient here denotes the polynomial in the last expression. It equals E'_{rat}. At $t=1$, $L_f(0)=0$ and $E_{\mathrm{rat}}(1)=H_N^{\mathbb{Q}}/2$; their difference has derivative zero over $\mathbb{Q}$, hence is that constant:

$$E_{\mathrm{rat}}(t)=L_f(1-t^2)+H_N^{\mathbb{Q}}/2. \tag{SSA26c}$$

With the actual integral endpoint quotient γ from (SSA24), namely $p\gamma=H_N^{\mathbb{Q}}/2+Q_2^{\mathbb{Z}}$, this gives

$$E_{\mathrm{rat}}+Q_2^{\mathbb{Z}}=L_c(q_t)+p\{\gamma-L_2(q_t)\}+p^2L_{\mathrm{rem}}(q_t).$$

Define the rational central hat

$$\widehat{R}_c=Y_{\mathrm{rat}}\{L_c(q_t)-H_d^{\mathbb{Q}}\}-t^pn_{\mathrm{rat}}.$$

Substituting the preceding logarithm equality in the specified rational hat from (SSA4) proves

$$\widehat{R}_{\mathrm{rat}}=\widehat{R}_c+pY_{\mathrm{rat}}\{\gamma-L_2(q_t)\}+p^2Y_{\mathrm{rat}}L_{\mathrm{rem}}(q_t). \tag{SSA26d}$$

The endpoint carry is still present. At $t=1$, the three logarithms vanish, $Y_{\mathrm{rat}}=1$ and $n_{\mathrm{rat}}=-H_d^{\mathbb{Q}}$, so $\widehat{R}_c(1)=0$ and $\widehat{R}_{\mathrm{rat}}(1)=p\gamma$. Their parities are d, since p is odd and n_{rat} has parity $d-1$; thus the other endpoint is $\widehat{R}_c(-1)=0$ and $\widehat{R}_{\mathrm{rat}}(-1)=(-1)^dp\gamma$. Reduction of (SSA26d), using integrality of every displayed polynomial, gives $\widehat{R}=Y\{\bar{L}_c(q_t)-h\}-t^pn$; the preceding logarithm equality also gives $\bar{L}_c(q_t)=E+q=L_0$.

The balanced Green functional is

$$\mathcal{F}(A,B,L)=-\frac{1}{3}Q_O\{A^2L(1-t^2)-t^pAB\}-\frac{1}{2}Q_H(A^2)-r_*.$$

First put $D_c=\mathcal{F}(Y_{\mathrm{rat}},n_{\mathrm{rat}},L_c)$. Because (SSA2) gives the exact rational equality $Q_O(Y_{\mathrm{rat}}^2)=0$, its definition is also

$$D_c=-\frac{1}{3}Q_O(Y_{\mathrm{rat}}\widehat{R}_c)-\frac{1}{2}Q_H(Y_{\mathrm{rat}}^2)-r_*.$$

Multiplying (SSA26d) by Y_{rat} and applying Q_O therefore gives, before division or reduction,

$$\begin{aligned} D_{\mathrm{hat}} &= D_c - \frac{p\gamma}{3}Q_O(Y_{\mathrm{rat}}^2) + \frac{p}{3}Q_O(Y_{\mathrm{rat}}^2 L_2(q_t)) \\ &\quad - \frac{p^2}{3}Q_O(Y_{\mathrm{rat}}^2 L_{\mathrm{rem}}(q_t)) \\ &= D_c + \frac{p}{3}Q_O(Y_{\mathrm{rat}}^2 L_2(q_t)) - \frac{p^2}{3}Q_O(Y_{\mathrm{rat}}^2 L_{\mathrm{rem}}(q_t)). \end{aligned}$$

The endpoint term vanishes by its low-degree projection, not by an assumption on the value of γ. Both retained pairings lie in R, because their polynomials lie in $R[t]$ and (SSA1) is integral on such inputs. Since $D_{\mathrm{hat}}/p \in R$ was proved in (SSA8), the exact rational division now gives

$$\frac{D_c}{p} = \frac{D_{\mathrm{hat}}}{p} - \frac{1}{3}Q_O(Y_{\mathrm{rat}}^2 L_2(q_t)) + \frac{p}{3}Q_O(Y_{\mathrm{rat}}^2 L_{\mathrm{rem}}(q_t)) \in R. \tag{SSA26e}$$

Only after this membership has been established do we reduce. The last term is an integral p-multiple; hence

$$\overline{D_{\mathrm{hat}}/p} = \overline{D_c/p} + \frac{1}{3}Q_O(Y^2 \bar{L}_2(q_t)).$$

Next put $D_b = \mathcal{F}(Y_{\mathbb{Q}}, J_{\mathbb{Q}}, L_c + J_{\mathbb{Q}}(1))$. Its endpoint constant contributes $-J_{\mathbb{Q}}(1)Q_O(Y_{\mathbb{Q}}^2)/3 = 0$: $\deg Y_{\mathbb{Q}} \le d$, so (SSA2) applies over $\mathbb{Q}$. Expanding the remaining polynomial functional at $(Y_{\mathrm{rat}}, n_{\mathrm{rat}})$ gives the exact identity

$$D_b = D_c + p\delta_b + p^2\epsilon_b, \tag{SSA26f}$$

where

$$\delta_b = -\frac{1}{3}Q_O\{2Y_{\mathrm{rat}}L_bL_c(q_t) - t^p(L_b n_{\mathrm{rat}} + Y_{\mathrm{rat}}N_b)\} - Q_H(Y_{\mathrm{rat}}L_b).$$

The complete second remainder is

$$\epsilon_b = -\frac{1}{3}Q_O\{L_b^2 L_c(q_t) - t^p L_b N_b\} - \frac{1}{2}Q_H(L_b^2).$$

For the product in the first Q_O, the coefficients of $1, p, p^2$ are respectively

$$Y_{\mathrm{rat}}^2 L_c(q_t) - t^p Y_{\mathrm{rat}} n_{\mathrm{rat}}, \quad 2Y_{\mathrm{rat}}L_bL_c(q_t) - t^p(L_b n_{\mathrm{rat}} + Y_{\mathrm{rat}}N_b), \quad L_b^2 L_c(q_t) - t^p L_b N_b.$$

The square in Q_H has coefficients $Y_{\mathrm{rat}}^2, 2Y_{\mathrm{rat}}L_b, L_b^2$. These are all the terms, and $\delta_b, \epsilon_b \in R$ by the integrality of L_b, N_b, L_c and (SSA1). Consequently

$$D_b/p = D_c/p + \delta_b + p\epsilon_b \in R.$$

This rational equality justifies the division before reduction.

Now use (SSA26), $\bar{L}_b = -A_v/6$, $\bar{N}_b = N_v/3$, and the reduction of the central hat. The entire first product coefficient is

$$\begin{aligned} 2Y\bar{L}_b\bar{L}_c(q_t) - t^p(\bar{L}_b n + Y\bar{N}_b) &= -YA_v\bar{L}_c(q_t)/3 + t^p A_v n/6 - t^p Y N_v/3 \\ &= -\{t^p C_v + A_v\widehat{R} + hYA_v\}/3. \end{aligned}$$

Indeed inserting $C_v = YN_v + nA_v/2$ and $\widehat{R} = Y\{\bar{L}_c(q_t) - h\} - t^p n$ in the second line leaves exactly the three terms in the first line. The last summand has zero Q_O: $\deg(YA_v) \le 2d$, so (SSA2) applies. The Euler coefficient is $-Q_H(Y\bar{L}_b) = Q_H(YA_v)/6$. Thus

$$\bar{\delta}_b = \frac{1}{9}Q_O(t^p C_v + A_v\widehat{R}) + \frac{1}{6}Q_H(YA_v).$$

Combining the reduced forms of (SSA26e) and (SSA26f) proves

$$\overline{D_{\mathrm{hat}}/p} + \frac{1}{9}Q_O(t^p C_v + A_v\widehat{R}) + \frac{1}{6}Q_H(YA_v) = \overline{D_b/p} + \frac{1}{3}Q_O(Y^2 L_2(1-t^2)). \tag{SSA27}$$

Here L_2 denotes its field reduction. Both p^2 errors have survived the rational identities; they disappear as integral p-multiples only after division and the final residue map.

Substitution of (SSA25), (SSA27) into (SSA8), followed by (SSA20) and (SSA17), gives, term by term,

$$\begin{aligned} 2\overline{\Delta_U/p} = 2\overline{D_b/p} &+ \frac{2}{3}Q_O(Y^2L_2(1-t^2)) \\ &- \frac{2c}{3}Q_O(t^pYn) - \frac{2}{3}Q_O(Y^2K_h) + \frac{2}{3}Q_O(YnO). \end{aligned} \tag{SSA28}$$

Indeed (SSA8) minus twice the left correction in (SSA27) leaves $-2Q_O(P_v)/3 - Q_O(A_v\widehat{R})/9 - 2Q_O(t^pC_v)/9$. Dividing (SSA20) by 9 cancels the C_v terms, and its $Y^2M_v - YnB_{\mathrm{adj}}$ terms vanish by (SSA17).

Now replace the central logarithm by the finite binomial logarithm already defined above. Equation (SSA26b) gives $L_c = L_f + pL_2 - p^2L_{\mathrm{rem}}$ with the same full integral remainder. Let $D_f = \mathcal{F}(Y_{\mathbb{Q}}, J_{\mathbb{Q}}, L_f + J_{\mathbb{Q}}(1))$. Then

$$D_f = D_b + \frac{p}{3}Q_O(Y_{\mathbb{Q}}^2L_2(1-t^2)) - \frac{p^2}{3}Q_O(Y_{\mathbb{Q}}^2L_{\mathrm{rem}}(1-t^2)), \quad D_f/p \in R. \tag{SSA29}$$

By $u \equiv \widehat{R}/6$, the original norm in (SSA3) is

$$12\overline{I_0(G_2)} + \frac{15}{2}\overline{I_1(G_2)} = \frac{1}{3}Q_O(\widehat{R}^2) + \frac{5}{24}I_1(\mathrm{qi}\,\widehat{R}^2). \tag{SSA30}$$

Thus (SSA28)–(SSA30) give the SAME scalar, with no support restriction on its square:

$$S = 2\overline{D_f/p} + \frac{1}{3}Q_O(\mathscr{R}_f) - \frac{5}{24}I_1(\mathrm{qi}\,\widehat{R}^2) - \mathcal{B}_p, \quad \mathscr{R}_f = -2c\,t^pYn - 2Y^2K_h + 2YnO - \widehat{R}^2. \tag{SSA31}$$

6.3 The finite Cauchy window

The passage from D_f to the common trace can be made entirely over $\mathbb{Q}$. Put $\eta = p/3$ and

$$c_j(\eta) = \frac{1}{2j+1}\prod_{r=0}^{j}\left(1 - \frac{\eta}{2r+1}\right)^{-1}, \qquad \mu(t^{2j}) = c_j(\eta), \qquad \mu(t^{2j+1}) = 0. \tag{SSA32}$$

No denominator $2r+1-\eta$ is zero: such an equality would make the prime $p \geq 19$ divisible by 3. These numbers need not all be p-integral; the following identities are rational identities. The recurrence

$$(2j+1)c_j(\eta) = (2j+3-\eta)c_{j+1}(\eta)$$

gives $\mu(q_tP' - (2-\eta)tP) = 0$. Applying it to AB' and $A'B$ proves self-adjointness of $\Lambda_\eta = q_t\,d^2/dt^2 - (2-\eta)t\,d/dt$. Since $\Lambda_\eta Y_{\mathbb{Q}} = -d(d+1-\eta)Y_{\mathbb{Q}}$, comparison with its action on t^j, successively for $0 \leq j < d$, gives $\mu(Y_{\mathbb{Q}}t^j) = 0$. The factor $(d-j)(d+j+1-\eta)$ used at that step is nonzero over $\mathbb{Q}$.

Define the polynomial Cauchy operator by

$$\mathcal{C}_\eta(t^\ell) = \sum_{r=0}^{\ell-1}\mu(t^r)t^{\ell-1-r}.$$

Its product identity is $\mathcal{C}_\eta(tP) = t\mathcal{C}_\eta(P) + \mu(P)$. Applying the three-term recurrence (108), whose lower moment vanishes as just proved, gives $J_{\mathbb{Q}} = -m\,\mathcal{C}_\eta(Y_{\mathbb{Q}})$, where $m = m_d(\eta)$. The identity

$$\mathcal{C}_\eta(Y_{\mathbb{Q}}^2) = Y_{\mathbb{Q}}\mathcal{C}_\eta(Y_{\mathbb{Q}})$$

follows by expanding $(Y(t)-Y(s))/(t-s)$: every omitted term contains a moment $\mu(Y(s)s^j)$, $j < d$. Consequently

$$Y_{\mathbb{Q}}J_{\mathbb{Q}} = -m\mathcal{C}_\eta(Y_{\mathbb{Q}}^2). \tag{SSA33}$$

For a polynomial P of degree at most $D = 2d < p$, set

$$W_{\mathrm{win}}(t) = \sum_{j=0}^{N-1} c_j(\eta) t^{2(N-j)}, \qquad \mathcal{M}(P) = \sum_{s=0}^{p-1} \mu(Pt^s) t^{p-1-s},$$

and

$$\mathcal{T}(P) = \sum_{\ell=0}^{D} [t^\ell] P \sum_{r=0}^{\ell} \mu(t^{p-1+r}) t^{\ell-r}.$$

For a monomial t^ℓ, splitting the indices of $\sum_{s=0}^{p-1} \mu(t^{\ell+s}) t^{p-1-s}$ into $\ell + s < p$ and $\ell + s \geq p$, and then restoring the missing indices $0, \ldots, \ell - 1$, gives

$$t^p \mathcal{C}_\eta(P) + \mathcal{M}(P) = P W_{\mathrm{win}} + \mathcal{T}(P). \tag{SSA34}$$

Split $\mathcal{M}$ at $s = p - 1 - D$:

$$\mathcal{M}_{\mathrm{low}}(P) = \sum_{s=0}^{p-2-D} \mu(Pt^s) t^{p-1-s}, \quad \mathcal{M}_{\mathrm{high}}(P) = \sum_{s=0}^{D} \mu(Pt^{p-1-D+s}) t^{D-s}.$$

Both $\mathcal{M}_{\mathrm{high}}$ and $\mathcal{T}$ have degree at most $2d$, so (SSA2) kills their Q_O, regardless of their rational denominators. Taking $P = Y_{\mathbb{Q}}^2$ in (SSA33)–(SSA34) gives

$$Q_O(t^p Y_{\mathbb{Q}} J_{\mathbb{Q}}) = G_{\mathrm{win}} - U_{\mathrm{win}}, \quad G_{\mathrm{win}} = m Q_O(\mathcal{M}_{\mathrm{low}}(Y_{\mathbb{Q}}^2)), \quad U_{\mathrm{win}} = m Q_O(Y_{\mathbb{Q}}^2 W_{\mathrm{win}}). \tag{SSA35}$$

Define the coupled numerator

$$D_{\mathrm{cc}} = G_{\mathrm{win}}/3 - \frac{1}{2} Q_H(Y_{\mathbb{Q}}^2) - r_*, \quad L_{\mathrm{pair}} = Q_O\{Y_{\mathbb{Q}}^2 (L_f(1 - t^2) + m W_{\mathrm{win}})\}.$$

Substitution of (SSA35) in the definition of D_f, and $Q_O(Y_{\mathbb{Q}}^2) = 0$, yield the exact identity

$$D_f = D_{\mathrm{cc}} - L_{\mathrm{pair}}/3. \tag{SSA36}$$

To justify the next division, expand the integral window weights. For $j < N$, put

$$D_j(z) = \prod_{r=0}^{j} (1 - z/(2r+1)), \quad m(z) = \prod_{i=1}^{d} (1 - z/i), \quad a_j = \sum_{r=0}^{j} \frac{1}{2r+1} - H_d.$$

The polynomial $m(z) - D_j(z)(1 + z a_j)$ has zero constant and linear coefficients, so it is z^2 times an integral polynomial. Since $D_j(p/3) \equiv 1$, its reciprocal is in R, and

$$m c_j(p/3) = \frac{1}{2j+1} + \frac{p\, a_j}{3(2j+1)} + p^2 b_j, \qquad b_j \in R. \tag{SSA37}$$

Put $j = N - r$, $1 \leq r \leq N$. The rational identity $(p - 2r)^{-1} = -(2r)^{-1} - p/(4r^2) + p^2/(4r^2(p - 2r))$ and odd-index reflection give

$$a_{N-r} = q_2^{\mathbb{Z}} - H_d + \tfrac{1}{2} H_{r-1} + p\, a_r^{\mathrm{rem}}, \quad a_r^{\mathrm{rem}} \in R.$$

Reflection here is the equality $(p - 2i)^{-1} = -(2i)^{-1} + p/(2i(p - 2i))$, summed over the complementary odd indices; its only denominators are $2i, p - 2i$, both units. The half-range harmonic carry that identifies the constant with $q_2^{\mathbb{Z}}$ is the rational carry proved in the moment endpoint calculation. Combining this equality with (SSA37) gives

$$m c_{N-r}(p/3) = -\frac{1}{2r} - p A_r + p^2 b_r^{\mathrm{rem}}, \quad A_r = \frac{1}{4r^2} + \frac{q_2^{\mathbb{Z}} - H_d}{6r} + \frac{H_{r-1}}{12r}, \quad b_r^{\mathrm{rem}} \in R. \tag{SSA38}$$

An explicit remainder is

$$b_{N-r} + \frac{1}{4r^2(p-2r)} + \frac{a_{N-r}}{6r(p-2r)} - \frac{a_r^{\mathrm{rem}}}{6r}.$$

No coefficient with index $j = N$, whose odd denominator would be p, has entered the window.

The binomial identity

$$L_f(1-t^2) = \sum_{r=1}^{N} \frac{t^{2r}}{2r} - \frac{H_N}{2} \tag{SSA39}$$

is exact over $\mathbb{Q}$: differentiation gives on either side $t\sum_{j=0}^{N-1} t^{2j}$, and evaluation at $t = 1$ fixes the constant. Equations (SSA38)–(SSA39) imply

$$L_{\mathrm{pair}} = -pQ_O(Y_{\mathbb{Q}}^2 V_2) + p^2 Q_O(Y_{\mathbb{Q}}^2 V_{\mathrm{rem}}), \quad V_2 = \sum_{r=1}^{N} A_r t^{2r}, \quad V_{\mathrm{rem}} \in R[t]. \tag{SSA40}$$

The discarded constant is a multiple of $Y_{\mathbb{Q}}^2$ and has zero Q_O by (SSA2). For its field reduction define

$$T_b = \frac{R(t) + R(-t)}{2} - \frac{S_2(t)}{6} - \frac{h}{3} L_0(t),$$

where $S_2(t) = R((1-t)/2) + R((1+t)/2)$. The coefficient recipe for T_b, or differentiation of the three displayed finite sums, gives the same cleared derivative as (SSA38):

$$q_t t\, \bar{V}_2' = q_t E + \frac{t^2}{3}(L_0 - h) + \frac{h}{3} t^{p+1} = q_t t\, T_b'. \tag{SSA41}$$

Both polynomials have degree $< p$; a polynomial of degree $< p$ with zero derivative is constant. Thus $Q_O(Y^2 \bar{V}_2) = Q_O(Y^2 T_b)$, without needing to assign that constant. It follows that

$$L_{\mathrm{pair}}/p \in R, \quad \overline{L_{\mathrm{pair}}/p} = -Q_O(Y^2 T_b), \quad D_{\mathrm{cc}}/p \in R, \quad \overline{D_f/p} = \overline{D_{\mathrm{cc}}/p} + Q_O(Y^2 T_b)/3. \tag{SSA42}$$

Integrality of D_{cc}/p follows from (SSA36), (SSA29), (SSA40); it does not require the separate moments in $\mathcal{M}_{\mathrm{low}}$ to be integral.

Put

$$D_{\mathrm{univ}} = \tfrac{1}{2}\{R(t) + R(-t) - R(1-t) - R(1+t) - E(t)^2\}.$$

Expansion of its definition and that of K_h, T_b gives

$$6T_b = 6K_h + 3(L_0 - h)^2 - 3h^2 + 6D_{\mathrm{univ}}. \tag{SSA43}$$

The $R((1 \pm t)/2)$ terms on both sides have coefficient -1. After $L_0 = q + E$ is expanded, the E^2, qE, hE coefficients are $0, 0, -2h$ on both sides; their constant parts agree as well. Combining (SSA43) with (SSA4) and (SSA31) gives the exact polynomial row

$$2Y^2 T_b + \mathscr{R}_f = 2Y^2 D_{\mathrm{univ}} - h^2 Y^2 + 2Yn(O + t^p E) - t^{2p} n^2. \tag{SSA44}$$

Indeed $-\widehat{R}^2$ contributes $-Y^2(L_0 - h)^2 + 2t^p Yn(L_0 - h) - t^{2p} n^2$; the term $-2c\, t^p Yn$ changes its cross factor to $2t^p YnE$, since $c = q - h$.

The upper proper square must still be restored. If $B_{\mathrm{pr}}(z) = \mathrm{qi}(n^2)$, then

$$\mathrm{qi}(t^{2p} n^2) = (1 - z^p) B_{\mathrm{pr}}(z), \quad \deg B_{\mathrm{pr}} \le d - 1, \quad B_{\mathrm{pr}}(0) = h^2. \tag{SSA45}$$

For any polynomials A, B, $\mathcal{D}(AB) = A(V)\mathcal{D}(B) + B(0)\mathcal{D}(A)$. When $A = z^p$, the first term has exponents $kp + r$, $0 \le r \le 3 \deg B \le N$, and has no coefficient at $N + jp + 1$. Thus $I_j(z^p B) = B(0) I_j(z^p)$. To compute the two remaining universal coefficients put

$$g_- = \sum_{i=0}^{p-1} 4^{p-1-i} x^i, \qquad g_+ = \sum_{i=0}^{p-1} 4^i x^i.$$

They satisfy $(4-x)g_- = 4-x^p$, $(4x-1)g_+ = 4x^p-1$. Multiplication by V verifies

$$\begin{aligned}125\mathcal{D}(z^p) &= 32x^{2p+2}g_- - 16x^{p+2}g_- + 2x^2 g_- \\ &\quad + 16x^{2p+2}g_+ - 68x^{p+2}g_+ + 16x^2 g_+ \\ &\quad - 48x^{3p+1} + 216x^{2p+1} - 99x^{p+1} + 12x. \end{aligned} \tag{SSA46}$$

Here $[x^{N-1}]g_- = 4$ and $[x^{N-1}]g_+ = 1/4$, since $4^N = 1$. At $N+1$ only $2x^2g_- + 16x^2g_+$ contributes, giving $2\cdot 4 + 16/4 = 12$; at $N+p+1$ only $-16x^{p+2}g_- - 68x^{p+2}g_+$ contributes, giving $-16\cdot 4 - 68/4 = -81$. Hence

$$Q_O(t^{2p}n^2) = -12h^2/125, \qquad I_1(\mathrm{qi}(t^{2p}n^2)) = 81h^2/125. \tag{SSA47}$$

Define the full crossed square $\Xi = \mathrm{qi}(\widehat{R}^2 - t^{2p}n^2)$, and $\mathcal{Q}_{\mathrm{pr}} = Yn(O + t^pE)$. Using (SSA2), (SSA42), (SSA44), (SSA47) in (SSA31) gives

$$S = 2\overline{D_{\mathrm{cc}}/p} + \frac{2}{3}Q_O(Y^2 D_{\mathrm{univ}} + \mathcal{Q}_{\mathrm{pr}}) - \frac{5}{24}I_1(\Xi) - \frac{103}{1000}h^2 - \mathcal{B}_p. \tag{SSA48}$$

The constant is explicitly

$$-\frac{1}{3}(-12/125)h^2 - \frac{5}{24}(81/125)h^2 = (4/125 - 27/200)h^2 = -103h^2/1000.$$

This explains why the original I_1 norm cannot be replaced by its low part.

6.4 The rational common trace, before reduction

Use the rational objects $F_{\mathbb{Q}}, S_{\mathbb{Q}}, E_{\mathbb{Q}}, \delta_{\mathbb{Q}}, \sigma_{\mathbb{Q}}, H_{\mathbb{Q},c}, \mathcal{L}_{\mathbb{Q}}$, and the trace functionals $\tau_m, \kappa_{\mathbb{Q}}, \lambda_{\mathbb{Q}}, \xi_{\mathbb{Q}}, t_{\mathbb{Q}}$ defined in (111), (123). In this subsection the letter A is the rational polynomial $(4-x)(4x-1)$. Put

$$W = 3\,5^{p-1}\{x(4x-11)\}^p, \quad W_1 = 3\,5^{p-1}\{x(4x-11)\}^{p-1}(8x-11).$$

Then $W' = pW_1$ and

$$\lambda_{\mathbb{Q}}(F) = \tau_{p+1}((AW_1 - WA')F).$$

Write $\mathrm{res}_\infty F = -[x^{-1}]F$ for a Laurent polynomial, with the same convention for its rational expansion at infinity. The coefficient of a derivative is zero; therefore

$$\kappa_{\mathbb{Q}}(H'S) + \kappa_{\mathbb{Q}}(HS') = -p\lambda_{\mathbb{Q}}(HS). \tag{SSA49}$$

Indeed differentiate WHS/A^p; its coefficient of x^{-1} is zero, and collecting $pW_1/A^p - pWA'/A^{p+1}$ gives (SSA49). The rational Euler source is

$$AH'_{\mathbb{Q},c} = d_x(F_{\mathbb{Q}}^\vee - 1), \qquad d_x = 3x^{-1}(4-11x).$$

For the weighted trace $\mathcal{T}_w(F) = \tau_{p+1}(W d_x F)$, it gives

$$\mathcal{T}_w(F_{\mathbb{Q}}^\vee S_{\mathbb{Q}}) = \mathcal{T}_w(S_{\mathbb{Q}}) + \kappa_{\mathbb{Q}}(H'_{\mathbb{Q},c}S_{\mathbb{Q}}). \tag{SSA50}$$

Here and below $F^\vee(x) = F(x^{-1})$.

The derivative counterterm in (177) is the actual rational Laurent polynomial

$$B_{\mathrm{act}} = \frac{x^{-p}s^N(1-x^{-N})A^{p-1}}{2\,5^{\,2(p-1)}}, \qquad s = (x+1)/5, \qquad S'_{\mathbb{Q}} = B_{\mathrm{act}} + p\sigma_{\mathbb{Q}}. \tag{SSA51}$$

It is the same integral $\sigma_{\mathbb{Q}}$ used in $C_{\mathbb{Q}}$, rather than an independently chosen primitive. Put

$$T_{\mathrm{pr}} = 5^{-(p-1)}(4x-11)^p C_{B,\mathbb{Q}}^\vee.$$

The rational balanced numerator is $\zeta_{\mathbb{Q}}(C_{B,\mathbb{Q}} + x^N C^{\vee}_{B,\mathbb{Q}}) + x^N F^{\vee}_{\mathbb{Q}} S_{\mathbb{Q}} = pE_{\mathbb{Q}}$. Cancellation of A^{p-1} in its weighted trace, followed by the cubic coordinate substitution $x = 5t^2 - 1$, gives

$$pB^{\mathrm{tr}}_{\mathbb{Q}} = Q_O(t^p Y_{\mathbb{Q}} J_{\mathbb{Q}}) + t_{\mathbb{Q}}(T_{\mathrm{pr}}) + \mathcal{T}_w(F^{\vee}_{\mathbb{Q}} S_{\mathbb{Q}}). \tag{SSA52}$$

The cancellation is exact over $\mathbb{Q}$: $\zeta_{\mathbb{Q}} = x^{-p}A^p/(3\,5^{2(p-1)})$; its positive term has weight $5^{-(p-1)}x^{-N}(4x-11)^p C_{B,\mathbb{Q}}$, and its reflected term has weight $5^{-(p-1)}(4x-11)^p C^{\vee}_{B,\mathbb{Q}} = T_{\mathrm{pr}}$. The first weight is the coefficient expression (SSA1) for $Q_O(t^p Y_{\mathbb{Q}} J_{\mathbb{Q}})$. The third term uses $x^{-N}x^N = 1$. Thus this transformation keeps the literal powers A^p and 5^{p-1}; they have not yet been replaced by their residues.

Combining (SSA35), (SSA52) with the definition of D_{cc} gives

$$3D_{\mathrm{cc}} = \mathcal{A}_b + pB^{\mathrm{tr}}_{\mathbb{Q}}, \quad \mathcal{A}_b = U_{\mathrm{win}} - t_{\mathbb{Q}}(T_{\mathrm{pr}}) - \tfrac{3}{2}H_{\mathbb{Q},0}(4/125) - 3r_* - \mathcal{T}_w(F^{\vee}_{\mathbb{Q}} S_{\mathbb{Q}}). \tag{SSA53}$$

Use (SSA50) to replace the last trace. Then (SSA49), (SSA51) replace $-\kappa_{\mathbb{Q}}(H'_{\mathbb{Q},c}S_{\mathbb{Q}})$ by $\kappa_{\mathbb{Q}}(H_{\mathbb{Q},c}B_{\mathrm{act}}) + p\{\kappa_{\mathbb{Q}}(H_{\mathbb{Q},c}\sigma_{\mathbb{Q}}) + \lambda_{\mathbb{Q}}(H_{\mathbb{Q},c}S_{\mathbb{Q}})\}$. Consequently

$$\mathcal{A}_b = q_* + pC_{\mathbb{Q}}, \quad q_* = U_{\mathrm{win}} - t_{\mathbb{Q}}(T_{\mathrm{pr}}) - \tfrac{3}{2}H_{\mathbb{Q},0}(4/125) - 3r_* + \kappa_{\mathbb{Q}}(H_{\mathbb{Q},c}B_{\mathrm{act}}) - \mathcal{T}_w(S_{\mathbb{Q}}). \tag{SSA54}$$

This is the full common-source term; omitting it would change the scalar attached to the balanced computation.

Here are the rational endpoint and metric rows needed to evaluate q_*. The finite conic logarithm satisfies

$$2A\mathcal{L}'_{\mathbb{Q}} - 3(4x-11) = -3\,5^{-(p-1)}s^N(4x-11)^p. \tag{SSA55}$$

Multiplying by $\frac{1}{2}(1-x^{-N})H_{\mathbb{Q},c}$, and canceling the literal A^{p-1} in $\kappa_{\mathbb{Q}}(H_{\mathbb{Q},c}B_{\mathrm{act}})$, gives

$$\begin{aligned}\kappa_{\mathbb{Q}}(H_{\mathbb{Q},c}B_{\mathrm{act}}) = {} & -\operatorname{res}_{\infty}(H_{\mathbb{Q},c}\mathcal{L}'_{\mathbb{Q}}) + \operatorname{res}_{\infty}(x^{-N}H_{\mathbb{Q},c}\mathcal{L}'_{\mathbb{Q}}) \\ & - \tfrac{1}{2}\,t_V(H_{\mathbb{Q},c}) + \tfrac{1}{2}\,t_V(x^{-N}H_{\mathbb{Q},c}), \\ t_V(F) = {} & \tau_1((33-12x)F).\end{aligned} \tag{SSA56}$$

The endpoint Euler value is $t_V(H_{\mathbb{Q},c}) = -3H_{\mathbb{Q},0}(4/125)$. It is the partial-fraction coefficient of the cubic: $V'/V = (33-12x)/A$. The shifted inverted polynomial $x^{-N}H_{\mathbb{Q},c}$ has exponents at most $-N$. Since $(33-12x)/A$ starts with x^{-1} at infinity, their product has exponents at most $-N-1 < -1$, and its x^{-1} coefficient is zero. Thus the last line of (SSA56) is $3H_{\mathbb{Q},0}(4/125)/2$. Integration by parts and the rational Euler source also give

$$-\operatorname{res}_{\infty}(H_{\mathbb{Q},c}\mathcal{L}'_{\mathbb{Q}}) = t_{\mathbb{Q}}\{\mathcal{L}_{\mathbb{Q}}(F^{\vee}_{\mathbb{Q}} - 1)\}. \tag{SSA57}$$

The exact reflection defect $F_{\mathbb{Q}} - x^N F^{\vee}_{\mathbb{Q}} = p\delta_{\mathbb{Q}}$ implies

$$\begin{aligned}-t_{\mathbb{Q}}(x^{-N}F_{\mathbb{Q}}\mathcal{L}_{\mathbb{Q}}) + t_{\mathbb{Q}}\{\mathcal{L}_{\mathbb{Q}}(F^{\vee}_{\mathbb{Q}} - 1)\} \\ = -pP^{\mathrm{tr}}_{\mathbb{Q}} - t_{\mathbb{Q}}(\mathcal{L}_{\mathbb{Q}}).\end{aligned} \tag{SSA58}$$

This follows by multiplying the defect by $x^{-N}\mathcal{L}_{\mathbb{Q}}$; both reflected terms cancel. The complete logarithmic pair has

$$L_{\mathrm{pair}} = U_{\mathrm{win}} + t_{\mathbb{Q}}(x^{-N}F_{\mathbb{Q}}\mathcal{L}_{\mathbb{Q}}). \tag{SSA59}$$

For this equality use $\mathcal{L}_{\mathbb{Q}} = e_{\mathbb{Q}}(1-V)$, then invert the coefficient defining I_0; the constant $e_{\mathbb{Q}}(1)$ contributes $Q_O(Y^2_{\mathbb{Q}}) = 0$.

To give the rational proper metric completely, put

$$H_{\mathbb{Q},0}(z) = \sum_{i=1}^{d} \frac{[z^i]\,\mathrm{qi}(Y^2_{\mathbb{Q}})}{i} z^i, \quad U_{\mathbb{Q},g}(t) = 2H_{\mathbb{Q},0}(1-t^2) + 4tY_{\mathbb{Q}}(t)J_{\mathbb{Q}}(t) - 4J_{\mathbb{Q}}(1).$$

The polynomial $U_{\mathbb{Q},g}$ is even. Its values at both $t=1$ and $t=-1$ are zero, because $Y_{\mathbb{Q}}(1)=1$ and $tY_{\mathbb{Q}}J_{\mathbb{Q}}$ is even. Thus $U_{\mathbb{Q},g}/(1-t^2)$ is an even integral polynomial of degree at most $2d-2$. Define, coefficientwise,

$$M_{\mathbb{Q},g}(t)=\sum_{i=0}^{2d-2}\frac{[t^i]\{U_{\mathbb{Q},g}/(1-t^2)\}}{i+1}t^{i+1},\quad P_{M,\mathbb{Q}}=\mathrm{qi}(M_{\mathbb{Q},g}/t),\quad C_{\mathbb{Q},g}=\mathrm{ds}\{t^pM_{\mathbb{Q},g}(T_c)/5\}.$$

Only denominators $i+1\le 2d-1<p$ are used, so all three polynomials have coefficients in R. The primitive is odd and zero at the origin, and

$$(1-t^2)M'_{\mathbb{Q},g}=U_{\mathbb{Q},g},\qquad C_{\mathbb{Q},g}=s^{N+1}(4x-11)P_{M,\mathbb{Q}}(V)/25.$$

For the second equality, substitute $M_{\mathbb{Q},g}(T_c)=T_cP_{M,\mathbb{Q}}(1-T_c^2)$; then $t^pT_c=t^{p+1}(4t^2-3)$, $t^{p+1}=s^{N+1}$, and $4t^2-3=(4x-11)/5$. Reduction gives $M_{\mathbb{Q},g}\mapsto M$, $P_{M,\mathbb{Q}}\mapsto P_M$, and $C_{\mathbb{Q},g}\mapsto 2C_M$: the rational descent uses $1/5$, whereas the field carrier in (103) uses $1/10$.

The descended source is

$$S_g=2H_{\mathbb{Q},c}+4T_{B,\mathbb{Q}}-4J_{\mathbb{Q}}(1),\qquad T_{B,\mathbb{Q}}=\mathrm{qi}(tY_{\mathbb{Q}}J_{\mathbb{Q}})(V)^{\vee}.$$

Write $c=4x-11$, $d_0=4-11x$, $k=5^{-(p-1)}$, and $a_p=1/(250\,5^{N+p-1})$. The chain rule in the first primitive identity above, using $1-T_c^2=(1-t^2)(4t^2-1)^2$ and $T_c'=3(4t^2-1)$, gives, after reflection,

$$10sA(C^{\vee}_{\mathbb{Q},g})'=-15s^{N+1}x^{-N}S_g-pAx^{-1}C^{\vee}_{\mathbb{Q},g}.$$

The unreflected equation has right side $15s^{N+1}\,\mathrm{qi}(U_{\mathbb{Q},g})(V)+pAC_{\mathbb{Q},g}$. For example its pullback follows by differentiating $5C_{\mathbb{Q},g}(5t^2-1)=t^pM_{\mathbb{Q},g}(T_c)$; multiplication by $1-t^2$ and $4t^2-1$ clears the two cubic factors. Reflection uses $s^{\vee}=s/x$ and $A^{\vee}=A/x^2$, and differentiating an inverse argument introduces $-x^{-2}$; these give the displayed signs and powers.

Let $J=P_{M,\mathbb{Q}}(V(x^{-1}))$. The compact formula gives

$$kC^{\vee}_{\mathbb{Q},g}=2a_px^{-N-2}(x+1)^{N+1}d_0J.$$

Define the two full Laurent metric densities

$$\begin{aligned}m_{\infty}&=a_px^{-N-3}(x+1)^Nc^{p-1}d_0(8x^2+4x+11)J,\\ m_{\mathrm{orig}}&=-a_px^{-p-1}(x+1)^Nd_0^{p-1}c(11x^2+4x+8)P_{M,\mathbb{Q}}(V).\end{aligned}$$

Because $8x^2+4x+11=8x(x+1)-c$,

$$10s\,m_{\infty}=k(40s\,c^{p-1}-x^{-1}c^p)C^{\vee}_{\mathbb{Q},g}.$$

Let $S_{\mathrm{nat}}=\mathrm{qi}(U_{\mathbb{Q},g})$ and $\rho_g=\mathcal{D}(S_{\mathrm{nat}})^{\vee}$. Its constant coefficient is zero, so the density identity $F_{\mathrm{rev}}/V=3x(11x-4)/A$ implies $A\rho_g=-3cS_g$. Retaining the rational derivative $(c^p)'=4pc^{p-1}$, the preceding identities and (SSA55) give every term of

$$\begin{aligned}(kc^pC^{\vee}_{\mathbb{Q},g})'&=pk\{4c^{p-1}-x^{-1}c^p/(10s)\}C^{\vee}_{\mathbb{Q},g}-\frac{3ks^Nc^p}{2A}x^{-N}S_g\\&=pm_{\infty}+x^{-N}\{S_g\mathcal{L}'_{\mathbb{Q}}+\rho_g/2\}.\end{aligned}$$

Indeed $\mathcal{L}'_{\mathbb{Q}}=3c/(2A)-3ks^Nc^p/(2A)$ and $\rho_g/2=-3cS_g/(2A)$, so the terms without k cancel. Taking coefficients of x^{-1}, the derivative has coefficient zero. Also

$$[x^{-1}]x^{-N}\rho_g=[x^{1-N}]\mathcal{D}(S_{\mathrm{nat}})=0,$$

since $N\ge 9$ and $\mathcal{D}(S_{\mathrm{nat}})$ is a polynomial. Direct reflection of the two metric densities gives $m_{\infty}=-x^{-2}m^{\vee}_{\mathrm{orig}}$. Thus

$$\mathrm{res}_{\infty}\,m_{\infty}=[x^{-1}]m_{\mathrm{orig}}=M^{\mathrm{pr}}_{\mathbb{Q}}.$$

These coefficient identities prove the full finite-log residue and its proper trace:

$$\operatorname{res}_\infty(x^{-N}S_g\mathcal{L}'_{\mathbb{Q}}) = -pM^{\mathrm{pr}}_{\mathbb{Q}}, \qquad t_{\mathbb{Q}}(T_{\mathrm{pr}}) = -2\operatorname{res}_\infty(x^{-N}T_{B,\mathbb{Q}}\mathcal{L}'_{\mathbb{Q}}). \tag{SSA60}$$

The first equality follows by taking $\operatorname{res}_\infty$ in the displayed product derivative. The second follows from (SSA55) multiplied by $x^{-N}T_{B,\mathbb{Q}}$; the term $3x^{-N}(4x-11)T_{B,\mathbb{Q}}/A$ has exponents at most $-N-1<-1$ at infinity and contributes zero. If $C_f = \operatorname{res}_\infty(x^{-N}\mathcal{L}'_{\mathbb{Q}})$, then expanding S_g in (SSA60) gives

$$\operatorname{res}_\infty(x^{-N}H_{\mathbb{Q},c}\mathcal{L}'_{\mathbb{Q}}) - t_{\mathbb{Q}}(T_{\mathrm{pr}}) = -pM^{\mathrm{pr}}_{\mathbb{Q}}/2 + 2J_{\mathbb{Q}}(1)C_f. \tag{SSA61}$$

Every number in (SSA60)–(SSA61) is the actual rational coefficient of its displayed finite carrier. Their integrality follows from the proper coefficient recipe, which uses only indices $<p$; in particular $M^{\mathrm{pr}}_{\mathbb{Q}} \in R$.

Inserting (SSA56)–(SSA61) in (SSA54), the $3H_{\mathbb{Q},0}(4/125)/2$ terms cancel. What remains is

$$q_* = V_{\mathrm{bal}} + L_{\mathrm{pair}} - pP^{\mathrm{tr}}_{\mathbb{Q}} - pM^{\mathrm{pr}}_{\mathbb{Q}}/2, \tag{SSA62}$$

where, with precisely the convention (150),

$$V_{\mathrm{bal}} = -t_{\mathbb{Q}}(\mathcal{L}_{\mathbb{Q}}) - \mathcal{T}_w(S_{\mathbb{Q}}) - 3r_* + 2J_{\mathbb{Q}}(1)C_f.$$

Equations (SSA53), (SSA54), (SSA62) are rational equalities. Since $D_{\mathrm{cc}}/p, L_{\mathrm{pair}}/p, C_{\mathbb{Q}}, B^{\mathrm{tr}}_{\mathbb{Q}}$, $P^{\mathrm{tr}}_{\mathbb{Q}}, M^{\mathrm{pr}}_{\mathbb{Q}}$ are integral, they also prove $V_{\mathrm{bal}}/p \in R$. Dividing first, and only then using (SSA42), gives

$$3\overline{D_{\mathrm{cc}}/p} = \overline{V_{\mathrm{bal}}/p} + \bar{C}_{\mathbb{Q}} + \bar{B}^{\mathrm{tr}}_{\mathbb{Q}} - Q_O(Y^2T_b) - \bar{P}^{\mathrm{tr}}_{\mathbb{Q}} - \bar{M}^{\mathrm{pr}}_{\mathbb{Q}}/2. \tag{SSA63}$$

All precision changes in this subsection have therefore occurred inside R.

6.5 The bracket, marked jet and Green adjoint

We now work in $\mathbb{F}_p$, then embed it injectively in the field K of the native calculation. Define in the prime field

$$\begin{aligned} B_h(t) &= -\frac{2}{3}K_h(t) - \frac{1}{3}(L_0(t)-h)^2 + \frac{h^2}{3}, \\ B_Q &= \mathrm{qi}\, B_h, \quad Q_b(x) = B_Q(V(x^{-1})), \\ G_{\mathrm{KQ}} &= \Omega_{\mathbb{F}_p}((33-12x)Q_b). \end{aligned} \tag{SSA64}$$

Here

$$\Omega_F(U) = [x^{-1}]_0(U/A_*), \qquad A_* = (4-x)(4x-1).$$

The reciprocal A_*^{-1} has coefficients in $\mathbb{F}_p$ and exists at the origin because $A_*(0) = -4 \neq 0$. Thus $G_{\mathrm{KQ}} \in \mathbb{F}_p$, and $\iota(G_{\mathrm{KQ}}) = \Omega_K(LQ_b)$. There is no choice of a K-valued relift in this definition. Evenness follows from $t \mapsto -t$; its endpoints are $K_h(\pm1) = L_0(\pm1) = 0$, so $B_Q(0) = 0$.

Insert (SSA63) into (SSA48). The D_{univ} term cancels against the last term of (SSA43); the residual $h^2Q_O(Y^2)$ is zero by (SSA2). The two remaining bracket projections are

$$-\frac{2}{3}Q_O(Y^2K_h) - \frac{1}{3}Q_O(Y^2(L_0-h)^2) = I_0(\mathrm{qi}(Y^2)B_Q). \tag{SSA65}$$

This uses multiplicativity of qi on even polynomials and $Q_O(Y^2) = 0$, so it is exact. For any polynomial B with $B(0) = 0$, the reciprocal density is

$$\mathcal{D}^\vee(B) = (B.\operatorname{div} z)(V(x^{-1}))F_{\mathrm{rev}}(x^{-1}), \quad A_*\mathcal{D}^\vee(B) = B(V(x^{-1}))L. \tag{SSA66}$$

The native square reciprocity and Euler source are

$$f = x^N f^\vee, \qquad A_*H' = (f-1)L, \qquad H = H_0(V).$$

Consequently

$$f\,\mathcal{D}^{\vee}(B) = B(V(x^{-1}))H' + \mathcal{D}^{\vee}(B).$$

The coefficient defining $I_0(\mathrm{qi}(Y^2)B)$ is $[x^{-1}]f\mathcal{D}^{\vee}(B)$. The coefficient of $(HB(V(x^{-1})))'$ is zero. Taking it in the preceding identity proves

$$I_0(\mathrm{qi}(Y^2)B_Q) = G_{\mathrm{KQ}} - [x^{-1}]HQ_b'. \tag{SSA67}$$

Thus the derivative of the full bracket remains beside its value.

Put

$$T_{\mathrm{ret}} = \frac{2}{3}\{\overline{V_{\mathrm{bal}}/p} + \bar{C}_{\mathbb{Q}} + \bar{B}^{\mathrm{tr}}_{\mathbb{Q}} - \bar{P}^{\mathrm{tr}}_{\mathbb{Q}}\} + \frac{2}{3}Q_O(\mathcal{Q}_{\mathrm{pr}}) - \frac{103}{1000}h^2.$$

The proper metric number is the coefficient of the full carrier already defined above (209):

$$\bar{M}^{\mathrm{pr}}_{\mathbb{Q}} = [x^{-1}]\mathcal{M}_{\mathrm{pr}}. \tag{SSA68}$$

Indeed its rational residue is the coefficient of x^p in $(x+1)^N(4-11x)^{p-1}(4x-11)(11x^2+4x+8)P_{M,\mathbb{Q}}(V)$, with factor $-1/(250\,5^{N+p-1})$. Reducing $P_{M,\mathbb{Q}}$ gives P_M, and $5^{p-1}=1$ in $\mathbb{F}_p$; hence multiplication by x^{-p-1} gives exactly $\mathcal{M}_{\mathrm{pr}}$. Every denominator is a unit in R. This equality is reduced from that rational metric, not inferred from a field differential equation. Equations (SSA48), (SSA63), (SSA65)–(SSA68) now give

$$S = T_{\mathrm{ret}} + G_{\mathrm{KQ}} - [x^{-1}]HQ_b' - \tfrac{1}{3}[x^{-1}]\mathcal{M}_{\mathrm{pr}} - \tfrac{5}{24}I_1(\Xi) - \mathcal{B}_p. \tag{SSA69}$$

The pairing kernel is the full finite Laurent polynomial

$$\begin{aligned}\mathcal{K} = \tfrac{1}{6}\{&35\ell(x/4) + 92x^{-p}\ell(x/4) + 368x^{-2p}\ell(x/4)\\ &- 5\ell(4x) - \tfrac{29}{4}x^{-p}\ell(4x) - \tfrac{29}{16}x^{-2p}\ell(4x)\}.\end{aligned}$$

Its three bands have exponents respectively $[1,p-1]$, $[1-p,-1]$, $[1-2p,-p-1]$. Hence $[x^0]\mathcal{K}=0$. Differentiation gives

$$A_*\mathcal{K}' = K_{\mathrm{der}} = (1-x^{p-1})\left\{20x + \frac{15}{2} + \left(\frac{113x}{2}+4\right)x^{-p} + \left(\frac{1953x}{8} - \frac{113}{2}\right)x^{-2p}\right\}. \tag{SSA70}$$

For example $A_*\{\ell(x/4)\}' = (4x-1)(1-x^{p-1})$, whereas $A_*\{\ell(4x)\}' = -4(4-x)(1-x^{p-1})$; the derivatives of x^{-p}, x^{-2p} vanish. Substitution of the six coefficients gives (SSA70). The pairing in (SSA3) is therefore $\mathcal{B}_p = [x^{-1}]\mathcal{K}B_p'$, with the same canonical remainder B_p as the moment calculation. To see the identification directly on each monomial b_ix^i, its contribution is $ib_i[x^{-i}]\mathcal{K}$; the three displayed bands give the three moment kernel slots, including their signs and their i mod p derivative factor.

Let $\mathcal{G}$ be the original Green primitive and $J_H(x) = H_J(V(x))$ the moment parameter jet. Its source and the canonical derivative give

$$A_*J_H' = 2\Sigma L, \qquad B_p' = \mathcal{G}' + 2\frac{\mathcal{P}-\mathcal{P}(1/4)}{x-1/4} + \frac{\mathcal{P}-\mathcal{P}(4)}{x-4} + (h-c_{\mathrm{res}})x^{-1} - J_H'/6. \tag{SSA71}$$

The central term has zero pairing because $\mathcal{K}_0 = 0$. The two divided differences are Laurent polynomials regular at their nonzero endpoints: writing $\mathcal{P} = x^{-r}P_+$ reduces each numerator to an ordinary polynomial vanishing at the indicated endpoint, with only powers of x in its denominator. Multiplication by A_* gives

$$A_*\left\{2\frac{\mathcal{P}-\mathcal{P}(1/4)}{x-1/4} + \frac{\mathcal{P}-\mathcal{P}(4)}{x-4}\right\} = L\mathcal{P} + \Lambda_{\mathrm{end}}, \quad \Lambda_{\mathrm{end}} = 8(x-4)\mathcal{P}(1/4) + 4(x-1/4)\mathcal{P}(4). \tag{SSA72}$$

Both endpoint constants remain. Integration by parts using (SSA70) then gives

$$\iota(\mathcal{B}_p) = \Omega_K\{(J_H/6 - \mathcal{G})K_{\mathrm{der}} + \mathcal{K}(L\mathcal{P} + \Lambda_{\mathrm{end}})\}. \tag{SSA73}$$

The jet term in (SSA73) is transformed using (SSA71):

$$\Omega_K(J_H K_{\mathrm{der}}) = [x^{-1}]J_H\mathcal{K}' = -[x^{-1}]\mathcal{K}J_H' = -2\Omega_K(\mathcal{K}\Sigma L). \tag{SSA74}$$

Its contribution to S is therefore $+\mathcal{K}\Sigma L/3$.

The original Green primitive uses $H + 2h$, not H:

$$K_{\mathrm{pull}} = K_0(V), \qquad K_0(z) = \sum_{i=1}^{d} \frac{[z^i]\,\mathrm{qi}(Y^2)}{i^2} z^i,$$

so $A_* K'_{\mathrm{pull}} = HL$, with zero primitive constant. Define

$$\mathcal{G} = \mathcal{P}(H + 2h) + K_{\mathrm{pull}} - \zeta C_M, \quad \zeta = (17 - 4x^p - 4x^{-p})/3,$$

and its cleared source is

$$A_*\mathcal{G}' = \{f\mathcal{P} + \zeta C_B + H - \mathcal{P}\}L - A_*(H + 2h)/(2x). \tag{SSA75}$$

Here C_M has exactly the scale $\mathrm{desc}(t^p M(T_c)/10)$ in (103). Its metric derivative in (104) gives

$$\zeta A_* C_M' = A_*\mathcal{R}(H + 2h) - \zeta C_B L.$$

The product rule now reads

$$\begin{aligned} A_*\mathcal{G}' &= \{A_*\mathcal{R} - A_*/(2x)\}(H + 2h) + \mathcal{P}(f - 1)L + HL - \zeta A_* C_M' \\ &= \{f\mathcal{P} + \zeta C_B + H - \mathcal{P}\}L - A_*(H + 2h)/(2x), \end{aligned}$$

which proves (SSA75), including its constant $2h$. It first gives the derivative formula for the candidate source $\chi_c = f\mathcal{P} + \zeta C_B$. Set $\eta_c = 2\,\mathrm{DD}_{1/4}(\chi_c + H) + \mathrm{DD}_4(\chi_c + H) - H/(2x)$, where $\mathrm{DD}_c(F) = (F - F(c))/(x - c)$. At $1/4, 4$, $f = 1$, $H = 0$, $\zeta = 0$; therefore the divided-difference endpoints of $\chi_c + H = f\mathcal{P} + \zeta C_B + H$ are the two endpoints of $\mathcal{P}$. Clearing A_* gives

$$A_*\eta_c = (\chi_c + H)L + \Lambda_{\mathrm{end}} - A_* H/(2x).$$

Using (SSA75), (SSA72), the same right side is $A_*(\mathcal{G}' + 2\,\mathrm{DD}_{1/4}\,\mathcal{P} + \mathrm{DD}_4\,\mathcal{P} + h/x)$. Since $A_* \neq 0$, cancellation gives

$$\eta_c = \mathcal{G}' + 2\,\mathrm{DD}_{1/4}\,\mathcal{P} + \mathrm{DD}_4\,\mathcal{P} + h/x.$$

We now remove the entire homogeneous ambiguity in the original source. The polynomial division and degree calculation for the original shifted native source in (201)–(207) give the full decomposition (208):

$$\chi_{\mathrm{par}} = \chi_c + f(\alpha_1 x^p + \beta_1 + \gamma_1 x^{-p}).$$

Its two actual endpoints are $q, -q$. Since $f(1/4) = f(4) = 1$, $\zeta(1/4) = \zeta(4) = 0$, and $\mathcal{P}(1/4) = q$, $\mathcal{P}(4) = -q$, they give

$$\alpha_1/4 + \beta_1 + 4\gamma_1 = 0, \qquad 4\alpha_1 + \beta_1 + \gamma_1/4 = 0.$$

To fix the third constant, take the coefficient of x^{p-1} in the actual finite source $\eta_{\mathrm{par}} = \mathrm{WD}(\chi_{\mathrm{par}} + H) - H/(2x)$, where $\mathrm{WD} = 2\,\mathrm{DD}_{1/4} + \mathrm{DD}_4$. Here (186) identifies the original moment cutoff e with this $\chi_{\mathrm{par}} = \chi_N$, and (187) identifies their full finite η sources; thus the following coefficient belongs to the original moment source. The full source remainder calculation in the moment section has support $[-p + 1, N - 1]$, apart from the derivative $J_H'/6$ and the central term c_{res}/x. The first support interval excludes $p - 1$, and $[x^{p-1}]J_H' = p[x^p]J_H = 0$; hence $[x^{p-1}]\eta_{\mathrm{par}} = 0$. This uses the already computed full noncentral source, including its negative slot; it assumes no chosen Green constant.

For any polynomial B, division of $x^{p+i}-a^{p+i}$ by $x-a$ gives

$$[x^{p-1}]\,\mathrm{DD}_a(x^pB)=B(a).$$

For a Laurent carrier with largest exponent $<p$, the same coefficient is zero, because a divided difference lowers a nonnegative largest exponent by one and leaves a negative largest exponent negative. The three phases $\mathcal{P}=x^pA_+ + B_\chi + x^{-p}C_-$ therefore give

$$[x^{p-1}]\,\mathrm{WD}(\mathcal{P})=2A_+(1/4)+A_+(4).$$

From (94), $A_+=-4\{q+\Delta E(s)\}/15$, where $\Delta E(t^2)=E(2t)-E(t)$. Put $q_3=\overline{(3^{p-1}-1)/p}$. The full finite logarithm values give

$$E(1)=-q,\quad E(2)=-3q_3/2,\quad E(1/2)=q-3q_3/4.$$

For example these are the halves of $\ell(1)+\ell(-1)=-2q$, $\ell(2)+\ell(-2)=-3q_3$, and $\ell(1/2)+\ell(-1/2)=2q-3q_3/2$. They follow by reducing $\ell(a)\equiv\{1-a^p-(1-a)^p\}/p \pmod p$ with its rational denominators retained; at $a=-1/2$ the numerator is $(2^p+1-3^p)/2^p$, giving $q-3q_3/2$. Consequently

$$2\Delta E(1/4)+\Delta E(1)=2\{-2q+3q_3/4\}+\{q-3q_3/2\}=-3q,$$

and $2A_+(1/4)+A_+(4)=0$. The coefficient of $\mathcal{G}'$ at $p-1$ is zero by differentiation, and that of h/x is zero by support; thus $[x^{p-1}]\eta_c=0$.

By linearity, $\eta_{\mathrm{par}}-\eta_c=\mathrm{WD}\{f(\alpha_1x^p+\beta_1+\gamma_1x^{-p})\}$. Since $\deg f\le N<p$, the last two phases have zero coefficient at $p-1$, and the first has coefficient $\alpha_1\{2f(1/4)+f(4)\}=3\alpha_1$. Thus $3\alpha_1=0$, so $\alpha_1=0$. The endpoint rows now say $\beta_1+4\gamma_1=0$ and $4\beta_1+\gamma_1=0$. Subtracting the second from four times the first gives $15\gamma_1=0$; hence $\gamma_1=\beta_1=0$, as $3,15$ are units. We have attached the actual source:

$$\chi_{\mathrm{par}}=f\mathcal{P}+\zeta C_B,\qquad \eta_{\mathrm{par}}=\eta_c. \tag{SSA75a}$$

The canonical moment remainder differentiates to $\eta_{\mathrm{par}}-c_{\mathrm{res}}/x-J'_H/6$, by its original primitive definition, which proves (SSA71) for this same source. The first parameter source is $\Sigma=fa-(4x^p-11)C_B$. With $\alpha=1-2x^{-p}/3$, the phase identity

$$\zeta+(4x^p-11)/3=2\alpha$$

shows that (SSA73)–(SSA75) attach (SSA69) to the marked adjoint density

$$\begin{aligned}\mathcal{A}_{\mathrm{mark}}&=\mathcal{K}\left[\{f(a/3-\mathcal{P})-2\alpha C_B-H\}L+\frac{A_*(H+2h)}{2x}\right]-\mathcal{K}\Lambda_{\mathrm{end}}\\&\quad-A_*HQ'_b-A_*\mathcal{M}_{\mathrm{pr}}/3-\frac{5}{24}A_*\mathcal{D}_{I_1},\\\mathcal{D}_{I_1}&=x^{N+p}\mathcal{D}^\vee(\Xi),\\\iota(S)&=\iota(T_{\mathrm{ret}}+G_{\mathrm{KQ}})+\Omega_K(\mathcal{A}_{\mathrm{mark}}).\end{aligned} \tag{SSA76}$$

No endpoint or norm term has been absorbed into an unspecified integration constant.

6.6 The full reciprocal norm and its Euler transport

We spell out the native square defect used by the norm; this avoids replacing the original Ξ by an independently defined quadratic expression. Let $\delta_{\mathrm{sq}}=\bar{\delta}_{\mathbb{Q}}$ be the reduction of the actual rational square reflection quotient in (111), and put $\Sigma=fa-(4x^p-11)C_B$. The rational first Taylor row for the same node square gives

$$\chi_{\mathrm{par}}=\delta_{\mathrm{sq}}+(\Sigma-x^N\Sigma^\vee)/3=f\mathcal{P}+\zeta C_B,\qquad \chi_{\mathrm{par}}+x^N\chi_{\mathrm{par}}^\vee=0. \tag{SSA77}$$

The literal rational Taylor row is

$$F_{\mathrm{full},\mathbb{Q}}(z) = F_{\mathrm{term},\mathbb{Q}}(z) + \frac{p}{6} J^{\mathrm{nat}}_{\mathbb{Q}}(z) + p^2 R_2(z), \quad R_2 \in R[z],$$

where the complete ranges are $0 \le j \le N$ and

$$[z^j]F_{\mathrm{term},\mathbb{Q}} = C_j(-p/6), \qquad [z^j]J^{\mathrm{nat}}_{\mathbb{Q}} = \partial_\tau C_j(-p/6).$$

Here $C_j = A_j/D_j$ is the finite parameter coefficient defined in the moment Taylor calculation. To retain its quotient denominators, put $s_0 = -p/6$, $\varepsilon_p = p/6$, $A_s = A_j(s_0)$, $D_s = D_j(s_0)$, and write the exact polynomial Taylor rows

$$A_j(0) = A_s + \varepsilon_p A_j'(s_0) + \varepsilon_p^2 A_r, \quad D_j(0) = D_s + \varepsilon_p D_j'(s_0) + \varepsilon_p^2 D_r.$$

The polynomial remainders A_r, D_r are integral; both $D_j(0)$ and D_s are units, as their reductions agree and $j! \neq 0$ for $j < p$. Clearing their denominators gives

$$r_j = \frac{A_r D_s^2 - A_s D_r D_s - (A_j'(s_0)D_s - A_s D_j'(s_0))D_j'(s_0) - \varepsilon_p(A_j'(s_0)D_s - A_s D_j'(s_0))D_r}{D_j(0)D_s^2} \in R.$$

Thus the exact coefficient remainder is $\varepsilon_p^2 r_j = p^2 r_j/36$. Take the full weighted reciprocal difference and divide it by p over $\mathbb{Q}$. The remainder becomes pR and reduces to zero. The terminating coefficient polynomial is the same $\mathrm{qi}(Y_1^2)$ already attached to the rational node, and its weighted difference gives δ_{sq}. The first field jet calculation gives $\overline{J^{\mathrm{nat}}_{\mathbb{Q}}} = 2\,\mathrm{qi}(Y\widehat{R})$, including its full range through N. This proves the coefficient $1/3$ in the first equality of (SSA77). Its second equality is the complete three-phase normalization (SSA75a). The last equality is algebraic: a weighted difference $F - x^N F^\vee$ and its weighted inverse have sum zero. These are thus identities of the same first Taylor carrier.

Here is a coefficient proof of the square-defect normalization. Let $X = x^p$, $X^{-1} = x^{-p}$, $\psi = 17 - 4X - 4X^{-1}$, $\phi = 4X^{-1} - 11$, and $B^\# = x^N C_B^\vee$. Then $A_*^p = X\psi$. The inverse of the first source is

$$x^N \Sigma^\vee = f a^\vee - \phi B^\#.$$

Substituting these two source rows into (SSA77) gives

$$\delta_{\mathrm{sq}} + 5C_B/3 = f\{\mathcal{P} - (a - a^\vee)/3\} - \phi(C_B + B^\#)/3. \tag{SSA78}$$

The anti-reciprocal equation in (SSA77) is $f(\mathcal{P} + \mathcal{P}^\vee) + \psi(C_B + B^\#)/3 = 0$. Multiplying (SSA78) by $X\psi$ therefore gives

$$A_*^p(\delta_{\mathrm{sq}} + 5C_B/3) = f\mathcal{N}_{72}, \quad \mathcal{N}_{72} = 2X(3-2X)\mathcal{P} + (4-11X)\mathcal{P}^\vee - A_*^p(a - a^\vee)/3. \tag{SSA79}$$

For the coefficient of $\mathcal{P}^\vee$, the substitution uses $X\phi = 4 - 11X$; the coefficient of $\mathcal{P}$ is $X\psi + X\phi = 2X(3-2X)$.

All derivative-invisible powers have now been displayed. The native finite derivative rows are

$$\mathcal{P}' = \mathcal{R} - \frac{1}{2x}, \quad (\mathcal{P}^\vee)' = -x^{-N}\mathcal{R} + \frac{1}{2x}, \quad A_*\mathcal{R} = \tfrac{1}{2}\psi s^N,$$

$$A_* a' = \tfrac{3}{2}\{4x - 11 - s^N(4X - 11)\}, \quad A_*(a^\vee)' = -\tfrac{3}{2}\{4x^{-1} - 11 - x^{-N}s^N\phi\}. \tag{SSA80}$$

They follow by differentiating the explicitly truncated finite primitives; the last two are the native cubic logarithmic rows. Substituting them in the derivative of (SSA79) gives

$$\begin{aligned} A_*\mathcal{N}_{72}' = {} & 2X(3-2X)\{\tfrac{1}{2}\psi s^N - A_*/(2x)\} \\ & + (4-11X)\{-\tfrac{1}{2}x^{-N}\psi s^N + A_*/(2x)\} \\ & - \tfrac{1}{2}A_*^p\{4x - 11 - s^N(4X-11)\} \end{aligned}$$

$$
\begin{aligned}
&\quad -\tfrac{1}{2}A_*^p\{4x^{-1}-11-x^{-N}s^N\phi\}\\
&=\tfrac{5}{2}A_*^p(1-s^N).
\end{aligned}
\tag{SSA81}
$$

The last equality uses $A_*/x = 17-4x-4x^{-1}$, $A_*^p = X\psi$, and $XX^{-1}=1$. Collecting the s^N terms gives $-5X\psi/2$; the remaining terms give $5X\psi/2$. The finite primitive $D=D_*$ has $A_*D' = 5(1-s^N)/2$. Thus $\mathcal{N}'_{72} = A_*^pD'$. Differentiate (SSA79); since $(A_*^p)'=0$, it follows that

$$
f(\delta_{\mathrm{sq}}+5C_B/3-fD)'-f'(\delta_{\mathrm{sq}}+5C_B/3-fD)=0. \tag{SSA82}
$$

The error polynomial has degree at most $p-2$: $\deg\delta_{\mathrm{sq}}\le p-2$, $\deg C_B\le N+3d-1\le p-2$, $\deg f\le 3d\le N$, and $\deg D\le N-1$. It vanishes at $x=1/4$. Indeed $C_B(1/4)=h/10$, $D(1/4)=q+h/6$, and $f(1/4)=1$. The actual rational square reflection quotient has endpoint

$$
\delta_{\mathbb{Q}}(1/4)=\frac{1-4^{-N}}{p}=\frac{q_2^{\mathbb{Z}}}{1+pq_2^{\mathbb{Z}}},
$$

because $Y_1(1)^2=1$ and $4^N=1+pq_2^{\mathbb{Z}}$. It therefore reduces to q. The error endpoint is $q+(5/3)(h/10)-(q+h/6)=0$. For clarity the elementary flatness argument in characteristic p is as follows. Reduce the fraction error$/f$ to coprime polynomials A/B, each of degree $<p$. Equation (SSA82) gives $A'B-AB'=0$; coprimality implies $A\mid A'$ and $B\mid B'$, so both derivatives are zero. A polynomial of degree $<p$ with derivative zero is constant. Thus error is a constant multiple of f; its endpoint forces that constant to be zero. We have proved

$$
\delta_{\mathrm{sq}}=fD-5C_B/3. \tag{SSA83}
$$

No nonvanishing assumption on f at other points is needed; the single endpoint $f(1/4)=1$ supplies $f\neq 0$.

Combining (SSA77), (SSA83) yields the reflected first source

$$
x^N\Sigma^\vee=f(a-3\mathcal{P}+3D)+\phi C_B. \tag{SSA84}
$$

The same crossed square in (SSA48) has native pullback

$$
\Xi_K=\Xi(V)=2a\Sigma-fa^2. \tag{SSA85}
$$

Indeed expand $\widehat{R}=Y(L_0-h)-t^pn$ before folding: $\widehat{R}^2-t^{2p}n^2=Y^2(L_0-h)^2-2t^pYn(L_0-h)$. The mixed first source is $Y\widehat{R}=Y^2(L_0-h)-t^pYn$; their relation is $\widehat{R}^2-t^{2p}n^2=2(L_0-h)Y\widehat{R}-Y^2(L_0-h)^2$. The cubic transport of t^pYn is exactly $(4x^p-11)C_B$, so this becomes (SSA85). Since $x^Nf^\vee=f$, inversion gives

$$
x^N\Xi_K^\vee=2a^\vee(x^N\Sigma^\vee)-f(a^\vee)^2.
$$

Subtract this equation from (SSA85), insert (SSA84), and collect:

$$
\begin{aligned}
\Xi_K-x^N\Xi_K^\vee&=f\{a^2-2a^\vee(a-3\mathcal{P}+3D)+(a^\vee)^2\}\\
&\quad -2C_B\{(4x^p-11)a+\phi a^\vee\}\\
&=fZ_{52}+2C_BR_{52}.
\end{aligned}
\tag{SSA86}
$$

This proves the two quadratic carriers in the native section from the full original norm.

For a general polynomial G, (SSA66) with its constant restored is

$$
A_*\mathcal{D}^\vee(G)=\{G(V(x^{-1}))-G(0)\}L. \tag{SSA87}
$$

It follows that the original upper projection is

$$
\iota(I_1(\Xi))=\Omega_K\{x^{N+p}(\Xi_K^\vee-\Xi(0))L\}=-\Omega_K\{x^p(\Xi_K-x^N\Xi_K^\vee)L\}. \tag{SSA88}
$$

For the second equality add and subtract $x^p\Xi_K L$. Both $x^p\Xi_K L$ and $x^{N+p}\Xi(0)L$ are polynomials with nonnegative exponents; dividing by A_*, whose origin expansion has nonnegative exponents, gives zero x^{-1} coefficient. This argument removes exactly these two polynomial residues and keeps both signed halves of Ξ_K. By (SSA86) the norm contribution to (SSA76) is therefore

$$-\tfrac{5}{24}\iota(I_1(\Xi)) = \tfrac{5}{24}\Omega_K\{x^p(fZ_{52} + 2C_B R_{52})L\}. \tag{SSA89}$$

Use the native $V_0 = -\mathcal{K}(\mathcal{P} - a/3) + 5x^p Z_{52}/24$. The marked density after (SSA89) is $fV_0 L + \mathcal{N}_0$, where

$$\begin{aligned}\mathcal{N}_0 = \mathcal{K}\{(-2\alpha C_B - H)L + A_*(H + 2h)/(2x)\} - \mathcal{K}\Lambda_{\mathrm{end}} \\ - A_* H Q_b' - A_*\mathcal{M}_{\mathrm{pr}}/3 + 5x^p C_B R_{52} L/12.\end{aligned} \tag{SSA90}$$

Apply the Euler source $A_* H' = (f - 1)L$ without any support cutoff:

$$fV_0 L = V_0 L + A_* V_0 H', \qquad \Omega_K(A_* V_0 H') = -\Omega_K(A_* H V_0').$$

The second identity is the zero coefficient of $(HV_0)'$. Thus

$$\iota(S) = \iota(T_{\mathrm{ret}} + G_{\mathrm{KQ}}) + \Omega_K(V_0 L - A_* H V_0' + \mathcal{N}_0). \tag{SSA91}$$

This step keeps the reciprocal and Frobenius terms of V_0 as Laurent polynomials.

6.7 The universal source, the origin and the mixed collector

Separate the bracket derivative from (SSA90):

$$R_{\mathrm{root}} = C_B B_0 L + H\{-L\mathcal{K} + A_*\mathcal{K}/(2x)\} + hA_*\mathcal{K}/x - \mathcal{K}\Lambda_{\mathrm{end}} - A_*\mathcal{M}_{\mathrm{pr}}/3, \tag{SSA92}$$

where $B_0 = -2\alpha\mathcal{K} + 5x^p R_{52}/12$. Then $\mathcal{N}_0 = R_{\mathrm{root}} - A_* H Q_b'$. The bracket value in (SSA64) is $\iota(G_{\mathrm{KQ}}) = \Omega_K(Q_b L)$. Adding it to (SSA91) gives the universal density

$$D^{\mathrm{orig}}_{\mathrm{univ}} = (V_0 + Q_b)L - A_* H(V_0 + Q_b)' + R_{\mathrm{root}}, \quad \iota(S) = \iota(T_{\mathrm{ret}}) + \Omega_K(D^{\mathrm{orig}}_{\mathrm{univ}}). \tag{SSA93}$$

Expanding its derivative produces $-A_* H Q_b'$. Thus the bracket value has been moved into this density, and its derivative has been retained; neither can be omitted.

The balanced section supplies the explicitly defined S_{origin} in (179), with

$$\Omega_K(S_{\mathrm{origin}}) = \frac{2}{3}(\bar{C}_{\mathbb{Q}} + \bar{B}^{\mathrm{tr}}_{\mathbb{Q}} - \bar{P}^{\mathrm{tr}}_{\mathbb{Q}}) + \frac{2}{3}Q_O(\mathcal{Q}_{\mathrm{pr}}).$$

Put

$$V_{\mathrm{div}} = \iota\left(\frac{2}{3}\overline{V_{\mathrm{bal}}/p} - \frac{103}{1000}h^2\right), \quad D_{\mathrm{full}} = D^{\mathrm{orig}}_{\mathrm{univ}} + S_{\mathrm{origin}}.$$

Equation (SSA93) therefore attaches the original scalar to a whole, independently specified source:

$$\iota(S) = V_{\mathrm{div}} + \Omega_K(D_{\mathrm{full}}). \tag{SSA94}$$

The differential Cartier contraction proving this expression for S_{origin}, including the full proper source, is the coefficient calculation in (179); it applies to this exact $\mathcal{Q}_{\mathrm{pr}}$.

Use the native definitions

$$\delta = \tfrac{2}{5}D_s j + \tfrac{2}{3}De, \qquad V_{\mathrm{mix}} = V_0 + Q_b + \delta, \qquad W_{\mathrm{need}} = \delta L - A_* H\delta' + C_B bL + H(\Sigma_H + s^N\varphi\Phi_H).$$

These are finite Laurent carriers, with all reciprocal terms specified before their residues are taken. Add and subtract $\delta L - A_* H\delta'$ in D_{full}, then add and subtract $C_B bL + H(\Sigma_H + s^N\varphi\Phi_H)$. The exact result is

$$D_{\mathrm{full}} = Q_b L + (S_{\mathrm{origin}} - W_{\mathrm{need}}) + D_{\mathrm{mix}}, \tag{SSA95}$$

where

$$D_{\mathrm{mix}} = (V_0 + \delta)L - A_* H (V_0 + Q_b + \delta)' + R_{\mathrm{root}} + C_B b L + H(\Sigma_H + s^N \varphi \Phi_H). \tag{SSA96}$$

The original origin source $\rho = \Omega_K(S_{\mathrm{origin}} - W_{\mathrm{need}})$, not just its evaluated scalar, now gives

$$\iota(S) = V_{\mathrm{div}} + \rho + \iota(G_{\mathrm{KQ}}) + \Omega_K(D_{\mathrm{mix}}). \tag{SSA97}$$

In particular D_{mix} contains $-A_* H Q_b'$.

We finish by identifying this mixed carrier with the original Green collection used in (209). The low-H definition (183) is

$$\mathcal{H}_{\mathrm{low}} = -A_* H V_0' + H(-L\mathcal{K} + A_*\mathcal{K}/(2x)) + C_B B_0 L - \mathcal{B}_{\mathrm{met}} + A_* \mathcal{C}_H, \quad \mathcal{B}_{\mathrm{met}} = -A_* B_0 C_M' + 3 h s^N B_0. \tag{SSA98}$$

Its equality with the original native H source is (254); the polynomial support argument in (249) is applied only when that source is contracted. For the two endpoint values already determined in (95), (SSA72) gives

$$-\mathcal{K}\Lambda_{\mathrm{end}} = (31q - 4qx)\mathcal{K}. \tag{SSA99}$$

Both endpoint constants have now been substituted: explicitly $\Lambda_{\mathrm{end}} = 8q(x-4) - 4q(x-1/4) = (4qx - 31q)$.

Subtract (SSA98) from (SSA96), using (SSA92). The terms $-A_* H V_0'$, $H(-L\mathcal{K} + A_*\mathcal{K}/(2x))$, and $C_B B_0 L$ cancel literally. The surviving B_0 term is

$$\mathcal{B}_{\mathrm{met}} = -A_* B_0 C_M' - 6 h s^N \alpha \mathcal{K} + 5 h s^N x^p R_{52}/4. \tag{SSA100}$$

Also

$$V_0 L = -L\mathcal{K}(\mathcal{P} - a/3) + 5 x^p Z_{52} L/24.$$

Use (SSA99), $\chi = \chi_0 + h/3$, $\chi_0' = \mathcal{P}' - a'/3$, and the native derivatives to compute

$$2 h A_* \chi_0' = h s^N \{\psi + (4x^p - 11)\} - h A_*/x - h(4x - 11) = 6 h s^N \alpha - h A_*/x + hL/3.$$

Here $\psi = 17 - 4x^p - 4x^{-p}$ and $\psi + (4x^p - 11) = 6\alpha$. Thus $-hL/3 + hA_*/x - 6hs^N\alpha = -2hA_*\chi_0'$. The residual kernel terms in $V_0 L + hA_*\mathcal{K}/x - \mathcal{K}\Lambda_{\mathrm{end}} - 6hs^N\alpha\mathcal{K}$ reduce to the full

$$\mathcal{G}_B = -L\mathcal{K}\chi_0 + (31q - 4qx)\mathcal{K} - 2hA_*\mathcal{K}\chi_0', \tag{SSA101}$$

with the original root χ retained in (SSA98). The $x^p Z_{52}$, $x^p R_{52}$, and De terms give

$$\mathcal{G}_2 = L(5x^p Z_{52}/24 + 2De/3) + s^N(5hx^p R_{52}/4 - 10he/3). \tag{SSA102}$$

The remaining terms after subtracting $N_j, \mathcal{G}_B, \mathcal{G}_2, \mathcal{H}_{\mathrm{low}}, \mathcal{J}_{\mathrm{met}}$ are exactly

$$\begin{aligned} &\delta L - A_* H \delta' - A_* H Q_b' + C_B b L + H(\Sigma_H + s^N \varphi \Phi_H) - A_* \mathcal{C}_H \\ &\quad - A_* B_0 C_M' - A_* \mathcal{M}_{\mathrm{pr}}/3 + 10 h s^N e/3 - 2DeL/3 - N_j - \mathcal{J}_{\mathrm{met}}. \end{aligned} \tag{SSA103}$$

This is the independently derived $\mathcal{E}_{\mathrm{rem}}$ in (210), with its bracket derivative intact. Equations (SSA96)–(SSA103) prove the whole-carrier identity

$$D_{\mathrm{mix}} = N_j + \mathcal{G}_B + \mathcal{G}_2 + \mathcal{H}_{\mathrm{low}} + \mathcal{J}_{\mathrm{met}} + \mathcal{E}_{\mathrm{rem}}. \tag{SSA104}$$

Taking its origin coefficient in (SSA97) proves (209) for the SAME S in (SSA3). The subsequent mixed Green calculation proves the residue of $\mathcal{E}_{\mathrm{rem}}$ is zero; this conclusion is not assumed in the source attachment.

There is one origin correction if (SSA98) is written using χ_0. Define

$$\mathcal{H}_{\mathrm{low}}^{(0)} = \mathcal{H}_{\mathrm{low}} - A_*(h/3)H\mathcal{K}', \qquad \mathcal{E}_{\mathrm{rem}}^{(0)} = \mathcal{E}_{\mathrm{rem}} + A_*(h/3)H\mathcal{K}'. \tag{SSA105}$$

Their sum is unchanged. Since $\deg H \le 3d \le N$ and $H(4) = H(1/4) = 0$, the explicit kernel gives, for $1 \le j \le \deg H$,

$$[x^{-j-1}]\mathcal{K}' = (23 \cdot 4^j - 29 \cdot 4^{-j})/6, \quad [x^{-1}]H\mathcal{K}' = \frac{1}{6}\{23(H(4) - H(0)) - 29(H(1/4) - H(0))\} = H(0). \tag{SSA106}$$

The positive band and the second negative band cannot meet these indices; the first negative band's derivative uses $-j/(p-j) = 1$ in the field. Hence

$$\Omega_K(\mathcal{H}_{\mathrm{low}}^{(0)}) = \Omega_K(\mathcal{H}_{\mathrm{low}}) - \frac{h}{3}H(0), \quad \Omega_K(\mathcal{E}_{\mathrm{rem}}^{(0)}) = \Omega_K(\mathcal{E}_{\mathrm{rem}}) + \frac{h}{3}H(0). \tag{SSA107}$$

Here $H(0) = H_0(4/125)$; the normalization $H_0(0) = 0$ does not set this number to zero. The original form (209) uses (SSA98); the χ_0 form uses the two corrections in (SSA107) once each.

Every fixed scalar denominator in this attachment has prime factors among $2, 3, 5$; its variable integration denominators are positive indices below p. All coefficient maps and origin traces commute with the injective embedding $\mathbb{F}_p \hookrightarrow K$. Thus the native contraction of (209), followed by injectivity, proves $S = P$ in the prime field for the original moment scalar.

The rational connection and balanced trace in the preceding sections attach the source scalar S to the following complete expression:

$$\iota(S) = V_{\mathrm{div}} + \rho + \iota(G_{\mathrm{KQ}}) + \Omega(N_j) + \Omega(\mathcal{G}_B) + \Omega(\mathcal{G}_2) + \Omega(\mathcal{H}_{\mathrm{low}}) + \Omega(\mathcal{J}_{\mathrm{met}}) + \Omega(\mathcal{E}_{\mathrm{rem}}). \tag{209}$$

Here $\iota : \mathbb{F}_p \longrightarrow K$ is the scalar embedding. G_{KQ} is the full reciprocal cubic Euler-bracket residue from the rational source. Its explicit finite reduction is given in (304) below. To prove that the last term in (209) is zero one must use its source. After the two endpoint values in (95), the same remaining density is

$$\begin{aligned}\mathcal{E}_{\mathrm{rem}} = {} & \delta L - A_* H\delta' - A_* H Q_b' + C_B b L + H(\Sigma_H + s^N \varphi \Phi_H) - A_* \mathcal{C}_H \\ & - A_* B_0 C_M' - \frac{A_* \mathcal{M}_{\mathrm{pr}}}{3} + \frac{10h}{3} s^N e - \frac{2}{3} DeL - N_j - \mathcal{J}_{\mathrm{met}}.\end{aligned} \tag{210}$$

Q_b' is differentiated from the independently defined polynomial $B_Q(V(x^{-1}))$. Its value and derivative are kept together. For computing the derivative, the elementary intermediate identities are

$$\begin{aligned}& tR'(t) = \ell(t), \quad \ell'(t) = \sum_{j=0}^{p-2} t^j, \quad \ell(1-t) = \ell(t), \quad \ell(1+t) = \ell(-t), \\ & K_h'(t) = \frac{1}{2}\left(-\frac{\ell(t)}{1-t} + \frac{\ell(-t)}{1+t}\right) - \frac{L_0(t)}{6}\left(-\frac{1}{1-t} + \frac{1}{1+t}\right) + (2h/3 - q)E'(t), \\ & B_h'(t) = -\frac{2}{3}K_h'(t) - \frac{2}{3}(L_0(t) - h)E'(t).\end{aligned} \tag{211}$$

These are cleared rational identities: their displayed denominators can be multiplied out before evaluation, so no endpoint is excluded.

Here is the algebraic certificate for the cancellation, including its load-bearing differential identities:

$$\begin{aligned} H\big(\tfrac{3}{2}s^N b - A_*\delta' - A_* Q_b' \\ + s^N \varphi \Phi_H\big) - A_* \mathcal{C}_H \end{aligned} &= -\frac{2}{5} A_* H D_s j' + \mathcal{E}_H, \\ C_B b L &= \frac{3}{2} H s^N b + 3h s^N b - A_* C_M' b, \\ \begin{aligned} A_* C_M(B_0' + b') - \tfrac{1}{3}A_*\mathcal{M}_{\mathrm{pr}} \\ + H\Sigma_H - \tfrac{2}{5}A_* H D_s j' \end{aligned} &= \mathcal{J}_{\mathrm{met}} - \mathcal{E}_H, \\ \Pi' &= C_M'(B_0 + b) + C_M(B_0' + b'), \end{aligned} \tag{212}$$

$$\delta = \frac{2}{5} D_s j + \frac{2}{3} De,$$
$$N_j = \frac{2}{5} L D_s j - 2h s^N j,$$
$$b = -\frac{2}{3} j - \frac{10}{9} e.$$

The second line is (104) multiplied by $b/2$. The fourth through seventh lines are product differentiation and the literal definitions. For the first and third lines, differentiate the finite odd sum, use $\varphi' = 0$ and $s(s^N)' + s^N/10 = 0$, then use (95). More explicitly, if $F_o(z) = \sum_{i=0}^{N-1} z^i/(2i+1)$, then $F_o + 2zF_o' = \sum_{i=0}^{N-1} z^i$ and $(1-z)(F_o + 2zF_o') = 1 - z^N$. Substitute $z = sx^{-3}(4-11x)^2/25$. The geometric-series numerator is $125x^3 - (x+1)(4-11x)^2 = (4x-1)(4-x)^2$; it keeps both endpoints. The following intermediate rows give the cancellations explicitly. Write $v = a^\vee$, $S = s^N$, $X_N = x^N$, $Y_N = x^{-N}$ and

$$B_c = q - 2h/3 - 2v/3 - a/3 - \mathcal{P}^\vee - X_N(a+h)/3.$$

The odd quotient and the descended bracket derivative are

$$O_q\, y_q = 3 Q_H Y_N O_s,$$
$$H Q_b' = -Q_H Y_N O_s + \frac{1}{9} v y_q - \frac{1}{3}(v - q + 2h/3)(2Hv' + y_q),$$
$$2Hv' + y_q = 3\varphi S Y_N Q_H.$$

Here $O_q = \mathrm{inv}_2(O(t)/t)(V(x^{-1}))$ is defined by the even polynomial $O(t)/t$; its constant term is included. For the second row, differentiating B_h and substituting $z = 1 - t^2$ first gives

$$z B_Q' = -O_q(z)/3 + \{L_{\mathrm{nat}}(z) - h\}/9 - (1-z)^N \{L_{\mathrm{nat}}(z) - q - h/3\}/3,$$

where $L_{\mathrm{nat}} = \mathrm{inv}_2(q + E)$. Multiplication by $H\, d\log V(x^{-1})$ gives the stated row, since $L_{\mathrm{nat}}(V(x^{-1})) = v + h$. The first row is the direct substitution of $O(t)/t = \sum_{i=0}^{N-1} t^{2i}/(2i+1)$. All these equations may be cleared by their nonzero polynomial denominators before extracting a coefficient.

Use $\delta' = (2/5)(D_s' j + D_s j') + (2/3)(D'e + Dv')$. In the first line of (212), the j terms cancel as

$$-HSj - HS(Y_N - 1)j + HY_N O_s = -HSY_N\varphi(e + a/3).$$

The e terms from b and D' combine to $-5HSe/3 - 5H(1-S)e/3 = -5He/3$. Together with Φ_H, the remaining phase is

$$-HSY_N\varphi(e + a/3) + H\varphi S\Phi_H = H\varphi S Y_N B_c.$$

Replace it by $AB_c(2Hv' + y_q)/3$, using $AQ_H = H$. The terms apart from $-(2/5)AHD_s j'$ now become

$$-\frac{5}{3}He - \frac{2}{3}AHDv' + \frac{2}{9}AH\{v - a - X_N(a+h) - 3\mathcal{P}^\vee\}v'$$
$$- \frac{1}{9}A\{a + X_N(a+h) + 3\mathcal{P}^\vee\}y_q = A\mathcal{C}_H + \mathcal{E}_H.$$

This proves the first line of (212).

For the third line put $C_g = 2C_M/3 + 2HD_s/5 = x^{-(N-1)}s^{N+1}P_C$ and

$$Z_{\log} = \frac{-11x^2 - 4x - 8}{2x(x+1)(4-11x)}, \qquad O_{\mathrm{den}} = \frac{3}{2}x^{-N}s^p\varphi^2 - \frac{75}{2}x^{N+1}, \qquad O_{\mathrm{diff}} = \frac{O_{\mathrm{den}}}{A_* s^{N+1}}.$$

For the odd derivative, put $R_j = O_s + S\varphi(e + a/3)$, $w_x = 4 - 11x$, and $L_{\log} = 3(x-4)/(10xsw_x)$. Termwise differentiation of the finite geometric sum gives the exact cleared numerator

$$A_*(sR_j' + R_j/10) = \frac{75}{2}\{Sx^{2-p}w_x^{p-1} - x^{N+1}\}$$

$$+ s^{N+1}\varphi\left[-\frac{3}{2}\{4/x - 11 - x^{-N}S\varphi\} + \frac{1}{2}\{4x - 11 - S(4x^p - 11)\}\right].$$

Use $w_x^p = x^p\varphi$, $s^p = (x^p+1)/5$, and $A_*a' = \frac{3}{2}\{4x - 11 - S(4x^p - 11)\}$. Multiplication by w_x and collection then give

$$A_*(sR_j' + R_j/10)w_x = O_{\mathrm{den}}w_x - S\varphi\,\frac{3(x-4)}{10x}A_* + s^{N+1}\varphi A_*a'w_x/3.$$

Since $sS' + S/10 = 0$, $s^{N+1}j' = sR_j' + R_j/10$. Division by the nonzero polynomials A_*, s, w_x yields $j' = O_{\mathrm{diff}} - \varphi L_{\log} + \varphi a'/3$. Finally $Z_{\log} = L_{\log} + 1/(2x)$, by expanding the common numerator. Thus $j' + \varphi Z_{\log} = O_{\mathrm{diff}} + \varphi(1/(2x) + a'/3)$. Moreover $\Sigma_H = -(2/5)A_*\varphi D_s Z_{\log}$ and $\mathcal{M}_{\mathrm{pr}} = 2C_M\varphi Z_{\log}$. The full conic numerator is

$$\begin{aligned} -A_*C_gO_{\mathrm{diff}} &= -x^{-(N-1)}P_CO_{\mathrm{den}} \\ &= \frac{3}{10}x^2P_C\{125 - (1+x^{-p})(4x^{-p} - 11)^2\} \\ &= \frac{3}{10}x^2P_C(4 - 33x^{-p} + 72x^{-2p} - 16x^{-3p}). \end{aligned}$$

Here $s^p = (x^p+1)/5$ and $2N+1 = p$ account for every phase. Hence

$$-A_*C_gj' + H\Sigma_H - A_*\mathcal{M}_{\mathrm{pr}}/3 = \frac{3}{10}x^2P_C(4 - 33x^{-p} + 72x^{-2p} - 16x^{-3p}) - A_*C_g\varphi(1/(2x) + a'/3).$$

Finally, direct differentiation of B_0 gives

$$C_M(B_0' - 10v'/9) - \frac{2}{3}C_M\varphi(1/(2x) + a'/3) = \mathcal{M}.$$

Substituting $b' = -2j'/3 - 10v'/9$ in the preceding rows cancels the $C_M\varphi$ term, leaving $\mathcal{J}_{\mathrm{met}} - (2/5)A_*HD_s\varphi(1/(2x) + a'/3)$. This is the third line of (212). Also $A_*D_s' = (5/2)s^N(x^{-N} - 1)$ and $A_*(a^\vee)' = -\frac{3}{2}(4/x - 11) + \frac{3}{2}x^{-N}s^N\varphi$, by applying the reciprocal chain rule to (95). These identities, (211), and $R_{52}' = -\varphi(a^\vee)' - (4x^p - 11)a'$ give the first and third lines of (212) after the product rule for Z_{52}. In particular the j' terms in the first and third lines have opposite signs, while the entire correction $\mathcal{E}_H$ is retained. To finish, substitute the first and second lines in (210); its terms involving $\delta L, N_j, DeL$ have sum $2hSj$, while its harmonic terms have sum $3hSb + 10hSe/3 + 2hSj = 0$. The terms with C_M' become $-A_*C_M'(B_0 + b)$. The third line cancels $\mathcal{E}_H, \Sigma_H, \mathcal{M}_{\mathrm{pr}}, \mathcal{J}_{\mathrm{met}}$ and the two j' rows, leaving $-A_*C_M(B_0' + b')$. Thus product differentiation gives

$$\mathcal{E}_{\mathrm{rem}} = -A_*\Pi', \qquad \Omega(\mathcal{E}_{\mathrm{rem}}) = 0. \tag{213}$$

The calculation is an equality of finite Laurent polynomials. It can equivalently be made in $K((x))$, whose full-coefficient inclusion is injective. No value of a remaining collector has been assumed. The conversion $\chi = \chi_0 + h/3$ is used once in deriving (210); its origin correction is $(h/3)H(0)$, with $H(0) = H_0(V(0))$, not $H_0(0)$.

6.8 Finite coefficient contractions

We give the full rows used to contract (209). These are finite-sum calculations; the rules (91)–(93) specify the coefficient operation. The details below also identify the terms which cancel only jointly. For any t define

$$\mathcal{A}_e(t) = -\frac{R(1-t) + R(1+t)}{2}, \qquad G(z) = D_N(4z, 4) - D_N(z, 1) + 2D_N(4z, 1) - 2D_N(z, 1/4).$$

Write $\mathcal{A}_t = S(1, 2t) - S(1/2, t)$ and $\mathcal{B}_t = S(2, 2t) - S(1, t)$ for the two reciprocal triangular rows. Only their difference occurs below. Put

$$\mathcal{Q}_w = G(3/11) + 4h\big(E(2w) - E(w)\big) + 6(q - C)q_{11}.$$

The q_{11} term is part of the conic origin; it is not discarded.

The primitive part of the evaluated $\mathcal{Z}$ in (148) is

$$H_{\mathrm{PJ}} = -\frac{22H_P}{15\epsilon} + \frac{8\{2J_\Delta(a_*) + J_\Delta(b_*)\}}{15\epsilon} - \frac{69\{I_0(a_*) - I_0(b_*)\}}{5\epsilon}. \tag{214}$$

Here I_z is the unit primitive in (128), $I_z(y) = (2\epsilon/15)\mathsf{H}(z,y)$, and H_P, J_Δ are the explicit combinations in the finite-trace calculation. Substituting their definitions in (214) gives the first row below. The primitive rows use the ordered basis

$$(\mathsf{H}(a_*,a_*), \mathsf{H}(b_*,a_*), \mathsf{H}(a_*,b_*), \mathsf{H}(b_*,b_*), \mathsf{H}(0,a_*), \mathsf{H}(0,b_*)).$$

They are

	aa	ba	ab	bb	$0a$	$0b$
H_{PJ}	$208/225$	$368/225$	$-43/225$	$-38/225$	$-22/5$	$11/5$
H_j	$-232/225$	$-76/45$	$67/225$	$2/9$	$176/75$	$-11/75.$

(215)

Every triangular inner sum in (99) includes $j = i$. Expanding those sums and changing the order of summation gives

$$H_{\mathrm{PJ}} + H_j = \frac{77}{75}(\mathcal{B}_u - \mathcal{A}_u) - \frac{8}{25}(q - 3r/4)^2. \tag{216}$$

One useful check on this expansion is that the intervening seven rational rows have sum $3S(1,1) - 2S(1/2,1/2) + S(1,2) - S(1/2,1) - 2S(2,1) + 2S(1,1/2) - S(2,2) = 12(q-3r/4)^2$. They follow by pairing i with $N+1-i$ and using the full logarithm reflection formulas of Section 3.1. Indeed the sum of the two coefficient rows is $(-8,-4,8,4,-154,154)/75$. Substitution of (101) cancels $A(u), B(u), D_5$ with coefficients $8-8, 4-4, 154-154$ respectively, and leaves

$$\frac{77}{75}(\mathcal{B}_u - \mathcal{A}_u) - \frac{2}{75}\{3S(1,1) - 2S(1/2,1/2) + S(1,2) - S(1/2,1) - 2S(2,1) + 2S(1,1/2) - S(2,2)\}.$$

The seven values (70) give $12(q-3r/4)^2$ for the brace, proving (216) with its exact coefficient.

The complete mixed residue is

$$\begin{aligned}\Omega(N_j) = {} & \frac{11}{15}\mathcal{Q}_w - \frac{11}{10}G(1/5) + H_j + \frac{509}{25}q^2 + \frac{316}{25}hq + \frac{197}{375}h^2 - \frac{62}{25}qC - \frac{149}{75}C^2 \\ & - \frac{88}{15}hC - \frac{44}{5}(q-C)k - \frac{11}{10}hk.\end{aligned} \tag{217}$$

Section 6.9 gives the full integration-by-parts derivation of this row from the original quotient. The following exact window identities also display its infinity correction. Let $D_e = -3D$, $D_a = D_e(a_*)$, $D_b = D_e(b_*)$, $D_0 = D_e(0) = C$ and $\kappa = -2D_b - D_a + 3D_0$. Define

$$\begin{aligned}\Lambda_+ &= -\frac{2}{15}(2H_{D_e}(a_*,x) + H_{D_e}(b_*,x)) - \frac{4h}{15}(D_e - D_0), \\ \Lambda_- &= -\frac{2}{15}(2H_{D_e}(b_*,x)^\vee + H_{D_e}(a_*,x)^\vee - 3H_{D_e}(0,x)^\vee), \\ \Lambda &= \Lambda_+ + \Lambda_- - (\Lambda_+ + \Lambda_-)(-1).\end{aligned}$$

The support is $[-N+2, N-1]$, and

$$A_*\Lambda' = \frac{2}{5}LD_s - 2hs^N - \lambda_c, \quad \lambda_c = -\frac{2\kappa}{15}\frac{A_*}{x} + \frac{16h}{15}(x-4) - \frac{4h}{15}(x-a_*).$$

Thus $\Lambda(-1) = 0$ cancels the branch factor of its divided difference. Integrate $j\Lambda'$ by parts, using (92); the odd derivative is the geometric series specified after (212). The product contributions reduce to

$$T = \Omega(LD_eD_e^\vee) = -\frac{9}{40}h^2 + D_b^2 + D_a^2/2 - 3D_0^2/2, \quad U = \Omega(D_e s^N x^{-N}) = \frac{D_b - D_a}{15} - \frac{h}{50}.$$

Here $D_a+D_b = h/2$, $D_b = 3q+h$ by (94). The required reciprocal corrections are, with $\lambda_0 = -2\kappa/15$,

$$\begin{aligned}
[x^{-1}](D_e\Lambda') &= -2T/15 - \lambda_0 D_0,\\
\Omega(s^N\Lambda) &= \Omega(\Lambda) + 4T/225 + 2\lambda_0 D_0/15,\\
\Omega(s^N\Lambda^\vee) &= \Omega(\Lambda^\vee) - 4T/225 - 4D_aD_b/75 - 4hU/15 - 4qD_0/75.
\end{aligned}$$

Each is obtained by writing $(x-c)H'_{D_e} = D_e - D_e(c)$ and applying (93). They fix the six coefficients in the second row of (215). Here are the complete reciprocal windows. Write

$$\begin{aligned}
a &= x^pU_0 + V_0 + c_0,\\
U_0 &= 12D/5 - 4q/5,\\
V_0 &= -33D/5 + (2/5)(4\ell(x/4) - \ell(4x)/4),\\
c_0 &= 3k/2 - 11q/5 - h,
\end{aligned}$$

and $c_D = 4c_0/3 - q + h$, $F_0 = V_0^\vee + V_0/3 + c_D$. With $R_{\mathrm{mix}} = O_s + s^N\varphi(e + a/3)$, multiplication of the four phases gives

$$R_{\mathrm{mix}} = O_s + s^N\{4x^{-2p}U_0^\vee + x^{-p}(4F_0 - 11U_0^\vee) + 4U_0/3 - 11F_0 - 11x^pU_0/3\}.$$

Keep the window

$$R_{\mathrm{win}} = O_s + s^N\{-11V_0^\vee + \varphi V_0/3 + 4U_0/3 - 11c_D\}.$$

Their difference is exactly $s^N\{4x^{-2p}U_0^\vee + x^{-p}(4V_0^\vee + 4c_D - 11U_0^\vee) - 11x^pU_0/3\}$. Put

$$\widetilde{C} = \frac{2}{5}Ls\beta_D - 2hx^{N-1}, \quad \widehat{U} = x^{N-1}U_0^\vee, \quad \widehat{V} = x^{p-1}V_0^\vee.$$

These are polynomials, $\deg\widetilde{C} \le N-1$, and $N_j = x^{-(N-1)}\widetilde{C}R_{\mathrm{mix}}$. Define

$$\begin{aligned}
P_2 &= 4\widetilde{C}s^N\widehat{U},\\
P_1 &= \widetilde{C}s^N(4\widehat{V} + 4c_Dx^{p-1} - 11x^{N+1}\widehat{U}),\\
P_{\mathrm{pos}} &= -11x^{N+2}\widetilde{C}s^NU_0/3.
\end{aligned}$$

Then

$$N_j - x^{-(N-1)}\widetilde{C}R_{\mathrm{win}} = x^{-6N}P_2 + x^{-5N}P_1 + P_{\mathrm{pos}}. \tag{218}$$

The bounds are $\deg P_2 \le 3N-2 < 6N$ and $\deg P_1 \le 4N-1 < 5N$; P_{pos} has no negative exponents. At the two endpoints, $\widetilde{C}(a_*) = -16h$, $\widetilde{C}(b_*) = -h/4$, and $s(a_*)^N = s(b_*)^N = 1$. Since $a_*^N = b_*^N = 1$, all four phases $a_*^{-6N}, b_*^{-6N}, a_*^{-5N}, b_*^{-5N}$ equal 1. The full numerator values are

	a_*	b_*
P_2	$-256hU_0(b_*)$	$-hU_0(a_*)/4$
P_1	$-16h(4V_0(b_*) + 4c_D - 11U_0(b_*))$	$-h(4V_0(a_*) + 4c_D - 11U_0(a_*))/4.$

The endpoint equations $U_0(a_*)/4 + V_0(a_*) + c_0 = -h$ and $4U_0(b_*) + V_0(b_*) + c_0 = -h$ now give, by (93),

$$\begin{aligned}
\Omega(x^{-6N}P_2 + x^{-5N}P_1) &= \frac{h}{15}(-176U_0(b_*) + 11U_0(a_*)/4 + 21c_0 - 63q)\\
&= \frac{h}{15}(1083q/5 - 3627r/20 + 63k/2),
\end{aligned}$$

where $U_0(a_*) = (4q-3r)/5$, $U_0(b_*) = -2(4q-3r)/5$ and $h = -2q - 3r/2$. The whole calculation in Section 6.9 uses R_{mix} and proves (217), including its infinity phases and original conic q_{11} term. The endpoint correction just displayed compares that source with R_{win}; it is already included in the whole result and is not added a second time.

6.9 Product projections and the complete odd, even and infinity rows

Work over a field K of characteristic $p \geq 19$, with p prime. Put $N = (p-1)/2$, $a_* = 1/4$, $b_* = 4$, $s = (x+1)/5$, $A_* = (4-x)(4x-1)$, and $L = 33 - 12x$. Write $F^\vee(x) = F(x^{-1})$, $F_+ = [F]_{\geq 0}$, and $\operatorname{Res} F = [x^{-1}]F$ for an expansion at zero. Thus $\Omega(F) = \operatorname{Res}(F/A_*)$. All expansions below exist: $A_*(0) = -4$, $s(0) = 1/5$, and $4 - 11x$ has nonzero constant term. The only integer factors in the displayed rational denominators are $2, 3, 5, 11$, which are units under the stated prime bound. For a Laurent series bounded below, $\operatorname{Res} F' = 0$, since its x^{-1} coefficient is $0 \cdot F_0$. The product rule therefore proves every integration-by-parts identity used below. Reciprocals of rational functions are formed before expansion at zero; a reflected infinite expansion is never assumed to be a Laurent series bounded below.

Use the native objects already defined above, with the following abbreviations:

$$q = q_2, \quad r = q_3, \quad k = q_5, \quad j_{11} = q_{11}, \quad h = -2q - \frac{3}{2}r, \quad C = E(2u) - E(u), \quad u^2 = \frac{1}{5}.$$

The polynomial $D = D_\star$, its symmetric Laurent polynomial $D_s = D + D^\vee$, and $D_e = -3D$ satisfy

$$A_* D' = \frac{5}{2}(1 - s^N), \qquad D_e(0) = C, \quad D_e(a_*) = -3q - \frac{h}{2}, \quad D_e(b_*) = 3q + h.$$

Set $D_a = D_e(a_*)$, $D_b = D_e(b_*)$. For $\mathsf{H}(c, y) = H_{D_e}(c, y)$, as defined in (99), use the ordered rows

$$H_{aa}, H_{ba}, H_{ab}, H_{bb}, H_{0a}, H_{0b} = \mathsf{H}(a_*, a_*), \mathsf{H}(b_*, a_*), \mathsf{H}(a_*, b_*), \mathsf{H}(b_*, b_*), \mathsf{H}(0, a_*), \mathsf{H}(0, b_*).$$

Whole and strict projections. For $c \neq 0$, the geometric expansion at zero gives

$$\operatorname{Res} \frac{F}{x - c} = -\sum_{i \geq 1} F_{-i} c^{-i} = F_+(c) - F(c). \tag{219}$$

Here F is finite, so the sum is finite. Coefficientwise,

$$F = F_+ + (F^\vee)_+^\vee - F_0.$$

Since $L/A_* = 2/(x - a_*) + 1/(x - b_*)$, this implies

$$\Omega(LF) = -(F^\vee)_+(a_*) - 2(F^\vee)_+(b_*) + 3F_0. \tag{220}$$

This includes every negative coefficient and the origin correction.

The branch factor of (105) is

$$x^{N-1} D_s = s^{N+1} \beta_D, \qquad \deg \beta_D \leq N - 3.$$

Define its actual square polynomial

$$B_{\mathrm{sq}} = \frac{1}{5} s\, x^3 \beta_D^2.$$

Then $x^3 \mid B_{\mathrm{sq}}$ and $\deg B_{\mathrm{sq}} \leq p - 3$. Using $2N + 2 = p + 1$ and $s^p = (x^p + 1)/5$, square the branch identity:

$$D_s^2 = x^{3-p} s^{p+1} \beta_D^2 = (1 + x^{-p}) B_{\mathrm{sq}}. \tag{221}$$

Its nonnegative part is B_{sq}, because every exponent of $x^{-p} B_{\mathrm{sq}}$ is negative. Its constant coefficient is zero. For $F = DD^\vee$, therefore,

$$0 = 2D(0)^2 + 2F_0, \qquad B_{\mathrm{sq}} = D^2 + 2F_+ + D(0)^2.$$

Consequently the complete nonnegative part and the strict positive part are

$$F_+ = \frac{1}{2}(B_{\mathrm{sq}} - D^2 - D(0)^2), \qquad [F]_{>0} = \frac{1}{2}(B_{\mathrm{sq}} - D^2 + D(0)^2). \tag{222}$$

At the two endpoints $a_*^N = b_*^N = 1$, $s(a_*) = a_*$, $s(b_*) = 1$. Evaluation of the branch identity gives $\beta_D(a_*) = 16(D(a_*) + D(b_*))$ and $\beta_D(b_*) = (D(a_*) + D(b_*))/4$; hence

$$B_{\mathrm{sq}}(a_*) = \frac{1}{5}(D(a_*) + D(b_*))^2, \qquad B_{\mathrm{sq}}(b_*) = \frac{4}{5}(D(a_*) + D(b_*))^2.$$

For the literal even polynomial $D_e = -3D$, define $B_e = 9B_{\mathrm{sq}}$. The same branch calculation is therefore

$$\begin{aligned}
(D_e + D_e^\vee)^2 &= (1 + x^{-p})B_e, \qquad [D_e D_e^\vee]_0 = -C^2, \\
[D_e D_e^\vee]_{\geq 0} &= \frac{1}{2}(B_e - D_e^2 - C^2), \qquad [D_e D_e^\vee]_{>0} = \frac{1}{2}(B_e - D_e^2 + C^2), \\
B_e(a_*) &= \frac{1}{5}(D_a + D_b)^2 = \frac{h^2}{20}, \qquad B_e(b_*) = \frac{4}{5}(D_a + D_b)^2 = \frac{h^2}{5}.
\end{aligned} \tag{223}$$

For the strict endpoint functional $\mathcal{T}_0(F) = 2[F]_{>0}(b_*) + [F]_{>0}(a_*)$, this proves

$$\mathcal{T}_0(DD^\vee) = \frac{9}{10}(D(a_*) + D(b_*))^2 - \frac{1}{2}D(a_*)^2 - D(b_*)^2 + \frac{3}{2}D(0)^2. \tag{224}$$

Using (220), reciprocal symmetry of F, and (222), gives the product row

$$\begin{aligned}
T &:= \Omega(LD_eD_e^\vee) \\
&= 9\left[-\frac{9}{10}(D(a_*) + D(b_*))^2 + \frac{1}{2}D(a_*)^2 + D(b_*)^2 - \frac{3}{2}D(0)^2\right] \\
&= -\frac{9}{40}h^2 + \frac{1}{2}D_a^2 + D_b^2 - \frac{3}{2}C^2.
\end{aligned} \tag{225}$$

In the last line $D(a_*) + D(b_*) = -h/6$.

The other strict triangle used in the native coefficient rows follows from the literal log convolution, including its diagonal. For any polynomial $W = \sum_{i=0}^{p-1} W_i x^i$, put

$$P_{W,c}(x) = \sum_{i=1}^{p-1} W_i \sum_{j=1}^{i} \frac{c^j x^{i-j}}{j}.$$

Multiplying W by $\ell(c/x) = \sum_{j=1}^{p-1} c^j x^{-j}/j$ shows, term by term, that $[W\ell(c/x)]_{\geq 0} = P_{W,c}$: an exponent is nonnegative precisely when $j \leq i$. Moreover $P_{W,c}(z) = H_W(z, c)$. Let

$$f_{\mathrm{mix}} = \frac{2}{5}\left(4\ell(x/4) - \frac{1}{4}\ell(4x)\right).$$

Then the reflected complete nonpositive triangle of $f_{\mathrm{mix}}D^\vee$ is the polynomial

$$[(f_{\mathrm{mix}}D^\vee)^\vee]_{\geq 0}(z) = \frac{2}{15}\left(\frac{1}{4}\mathsf{H}(z, b_*) - 4\mathsf{H}(z, a_*)\right). \tag{226}$$

Indeed $D_e = -3D$ changes each of the two ordered primitives by -3, giving exactly the displayed factor and signs. Subtracting this entire nonpositive part from the whole endpoint evaluation,

$$\begin{aligned}
\mathcal{T}_0(f_{\mathrm{mix}}D^\vee) &= 2f_{\mathrm{mix}}(b_*)D(a_*) + f_{\mathrm{mix}}(a_*)D(b_*) \\
&\quad - \frac{2}{15}\left[\frac{1}{4}(2H_{ab} + H_{bb}) - 4(2H_{aa} + H_{ba})\right].
\end{aligned} \tag{227}$$

No diagonal or constant is removed separately. The two native bands of (98) are

$$a = x^p U_0 + f_{\mathrm{mix}} - \frac{33}{5}D + c_0, \qquad U_0 = \frac{12}{5}D - \frac{4}{5}q, \qquad c_0 = \frac{3}{2}k - \frac{11}{5}q - h.$$

All exponents of $x^p U_0 D^\vee$ are positive, because $\deg D \leq N - 1 < p$. The term $c_0 D^\vee$ has no strict positive exponent. Thus the exact second product projection is

$$\begin{aligned}
\mathcal{T}_0(aD^\vee) &= 2b_* U_0(b_*)D(a_*) + a_* U_0(a_*)D(b_*) \\
&\quad + \mathcal{T}_0(f_{\mathrm{mix}}D^\vee) - \frac{33}{5}\mathcal{T}_0(DD^\vee).
\end{aligned} \tag{228}$$

Equations (224)–(228) supply the whole triangle values used by the native G_2 row.

The shifted product and the two weighted primitive rows. The branch factor also computes the metric residue without a prime-class restriction. Put $n = N - 1$. From $s^N D_s = x^{-n} s^p \beta_D$,

$$s^N D_s = \frac{1}{5}(x^{p-n}\beta_D + x^{-n}\beta_D).$$

The first term is polynomial. For $\deg P \le n$,

$$\Omega(x^{-n}P) = \frac{a_*^n P(b_*) - b_*^n P(a_*)}{15}.$$

This is the full negative-coefficient geometric sum; it includes the coefficient of degree n, whose two endpoint contributions cancel. Use $a_*^n = 4$, $b_*^n = 1/4$ and the two values of β_D above:

$$\Omega(s^N D_s) = \frac{1}{75}\left[4\frac{D(a_*) + D(b_*)}{4} - \frac{1}{4}\,16(D(a_*) + D(b_*))\right] = \frac{h}{150}.$$

For $F = s^N D_e^\vee$, its polynomial complement $s^N D_e$ has zero Ω, so $\Omega(F) = -h/50$. The inversion identity $\Omega(F^\vee) = \Omega(F) + (F(a_*) - F(b_*))/15$ then yields

$$U := \Omega(D_e s^N x^{-N}) = \frac{D_b - D_a}{15} - \frac{h}{50}. \tag{229}$$

We record the primitive and its normalization explicitly. Set

$$\begin{aligned}
\Lambda_+(x) &= -\frac{2}{15}\big(2\mathsf{H}(a_*, x) + \mathsf{H}(b_*, x)\big) - \frac{4h}{15}(D_e(x) - C),\\
\Lambda_-(x) &= -\frac{2}{15}\big(2\mathsf{H}(b_*, x)^\vee + \mathsf{H}(a_*, x)^\vee - 3\mathsf{H}(0, x)^\vee\big),\\
\nu &= (\Lambda_+ + \Lambda_-)(-1), \qquad \Lambda = \Lambda_+ + \Lambda_- - \nu.
\end{aligned}$$

Thus $\Lambda(-1) = 0$, its support is contained in $[-N+2, N-1]$, $\Lambda_+(0) = 0$, and Λ_- is strictly negative. The improved lower bound follows because the degree $N-1$ terms of the three primitive polynomials cancel with weights $2 + 1 - 3 = 0$. Indeed the coefficient of x^{N-1} in $\mathsf{H}(c, x)$ is $(D_e)_{N-1}/(N-1)$, independently of c; all higher coefficients are zero because $\deg D_e \le N - 1$. Let

$$\kappa = -2D_b - D_a + 3C, \quad \lambda_0 = -\frac{2}{15}\kappa, \qquad \lambda_c = \lambda_0 A_*/x + \frac{16h}{15}(x - 4) - \frac{4h}{15}(x - a_*).$$

The source $(x - c)\partial_x \mathsf{H}(c, x) = D_e(x) - D_e(c)$ proves the primitive equation directly. For its positive part,

$$A_*\Lambda_+' = \frac{8}{15}\left[2(x - b_*)(D_e - D_a) + (x - a_*)(D_e - D_b)\right] + 2h(1 - s^N).$$

For a reciprocal row, $\partial_x \mathsf{H}(c, x)^\vee = -(D_e^\vee - D_e(c))/(x(1 - cx))$. Thus

$$A_*\Lambda_-' = -\frac{2}{15}\left[2(4/x - 1)(D_e^\vee - D_b) + (4/x - 16)(D_e^\vee - D_a) + 3(A_*/x)(D_e^\vee - C)\right].$$

The coefficient of D_e in the first line is $-2L/15$; that of $D_e^\vee$ in the second is also $-2L/15$, since $2(4/x - 1) + (4/x - 16) + 3A_*/x = L$. The remaining coefficients of $x, 1, x^{-1}$ are, respectively,

$$\frac{-16D_a - 8D_b - 24C}{15}, \quad \frac{32D_a - 2D_b + 102C + 30h}{15}, \quad \frac{8D_a + 16D_b - 24C}{15}.$$

Using $D_a + D_b = h/2$ and $\kappa = -2D_b - D_a + 3C$, these become $(-8\kappa - 12h)/15$, $(34\kappa + 63h)/15$, $-8\kappa/15$, the three coefficients of $-\lambda_c$. Since $D_e = -3D$, the resulting equation and partial fractions are

$$A_*\Lambda' = \frac{2}{5}LD_s - 2hs^N - \lambda_c, \qquad \frac{\lambda_c}{A_*} = \frac{\lambda_0}{x} - \frac{4h}{15(x - a_*)} + \frac{h}{15(x - b_*)}. \tag{230}$$

In particular $\lambda_0 = 2(q-C)/5 + h/5$ and $\lambda_0 - h/5 = 2(q-C)/5$. The source $A_* D'_e = -15(1-s^N)/2$ and $\operatorname{Res}(D_e F)' = 0$ give, for every finite F,

$$\Omega(s^N F) = \Omega(F) - \frac{2}{15}\operatorname{Res}(D_e F'). \tag{231}$$

Multiplying (230) by D_e/A_*, the polynomial products contribute zero Ω, and $D_e D_s = -D_e(D_e + D_e^\vee)/3$; therefore

$$\operatorname{Res}(D_e \Lambda') = -\frac{2}{15}T - \lambda_0 C, \qquad \Omega(s^N \Lambda) = \Omega(\Lambda) + \frac{4}{225}T + \frac{2\lambda_0 C}{15}.$$

For the reflected primitive, differentiation and $A_*^\vee = A_*/x^2$ give

$$A_*(\Lambda^\vee)' = -\frac{2}{5}L^\vee D_s + 2hx^{-N}s^N + \lambda_c^\vee.$$

The three residues needed here are

$$\Omega(L^\vee D_e^2) = 3C^2, \quad \Omega(L^\vee D_e D_e^\vee) = T + 3D_a D_b, \quad \Omega(D_e \lambda_c^\vee) = (\lambda_0 - h/5)C.$$

For the first, $L^\vee = 33 - 12/x$ and $\Omega(x^{-1}D_e^2) = -C^2/4$. For the second apply inversion to $L D_e D_e^\vee$; $L(a_*) = 30$, $L(b_*) = -15$, so the endpoint difference divided by 15 is $3D_aD_b$. For the third, the only negative term of $\lambda_c^\vee$ is $(-4\lambda_0 + 4h/5)x^{-1}$, giving the stated residue. Consequently

$$\operatorname{Res}(D_e(\Lambda^\vee)') = \frac{2}{15}(3C^2 + T + 3D_aD_b) + 2hU + (\lambda_0 - h/5)C.$$

Equation (231), followed by $\lambda_0 - h/5 = 2(q-C)/5$, cancels its two C^2 terms:

$$\Omega(s^N \Lambda^\vee) = \Omega(\Lambda^\vee) - \frac{4}{225}T - \frac{4}{75}D_aD_b - \frac{4h}{15}U - \frac{4}{75}qC. \tag{232}$$

Directly evaluating the negative primitive polynomial at a_*, b_* in the formula for Ω, and then the reflected positive polynomial, gives

$$\begin{aligned}
\Omega(\Lambda) &= -\frac{2}{225}(2H_{ba} + H_{aa} - 3H_{0a} - 2H_{bb} - H_{ab} + 3H_{0b}),\\
\Omega(\Lambda^\vee) &= -\frac{2}{225}(2H_{aa} + H_{ba} - 2H_{ab} - H_{bb}) - \frac{4h}{225}(D_a - D_b).
\end{aligned} \tag{233}$$

The constant ν contributes no Ω; these evaluations thus retain the chosen normalization without introducing an arbitrary constant. Substitute (225) and (229) into (232), and multiply by 6:

$$\begin{aligned}
6\Omega(s^N\Lambda^\vee) = &-\frac{4}{75}(2H_{aa} + H_{ba} - 2H_{ab} - H_{bb}) + \frac{7}{125}h^2 - \frac{4}{75}D_a^2 - \frac{8}{75}D_b^2\\
&- \frac{8}{25}D_aD_b + \frac{4}{25}C^2 - \frac{8}{25}qC.
\end{aligned} \tag{234}$$

Indeed $-8h(D_a - D_b)/75$ from the reflected origin cancels $-8h(D_b - D_a)/75$ from U; the remaining square terms are $3h^2/125 + 4h^2/125 = 7h^2/125$.

The genuine odd source and its reflected residue. Define

$$\begin{aligned}
w &= 4 - 11x, \quad \phi = 4x^{-p} - 11,\\
z_o &= \frac{s x^{-3} w^2}{25}, \quad F_o(z) = \sum_{i=0}^{N-1} \frac{z^i}{2i+1},\\
O_s &= x^{N-1} w F_o(z_o), \quad Q_o = \frac{O_s}{s^N}.
\end{aligned}$$

These are the full odd cutoff, including $i = N-1$; all $2i+1 < p$. The finite geometric sum proves

$$(1-z)(F_o + 2zF_o') = 1 - z^N.$$

The literal cubic derivatives and complements are

$$z_o' = \frac{3x^{-4}(x-4)w}{125}, \quad 1 - z_o = \frac{(x-4)^2(4x-1)}{125x^3}, \quad (x+1)w^2 + (x-4)^2(4x-1) = 125x^3.$$

The identity $a_*^N = b_*^N = 1$ used throughout follows from $4^N = 2^{p-1} = 1$, so it is valid in both congruence classes of p. Since $N = -1/2$ in K, differentiation of O_s yields

$$sO_s' + \frac{1}{10}O_s = \frac{3}{10}x^{N-2}(x-4)(F_o + 2z_oF_o').$$

Multiply by A_*, substitute the finite geometric identity and the displayed $1 - z_o$, to obtain

$$A_*\left(sO_s' + \frac{1}{10}O_s\right) = \frac{75}{2}x^{N+1}(z_o^N - 1).$$

Also $(s^N)' = -s^{N-1}/10$. The quotient rule therefore applies with exactly the same numerator. Put

$$g = \frac{3(x-4)}{10xsw} = \frac{1}{2(x+1)} + \frac{1}{x-4/11} - \frac{3}{2x}, \qquad \rho_o = \frac{\frac{3}{2}x^{-N}s^p\phi^2 - \frac{75}{2}x^{N+1}}{s^{N+1}A_*}.$$

For the reciprocal native source,

$$A_*(a^\vee)' = -\frac{3}{2}(4/x - 11 - x^{-N}s^N\phi).$$

Use $z_o^N = s^N x^{1-N}\phi/w$ (because $w^p = 4 - 11x^p = x^p\phi$) and the cubic complement above in the quotient numerator. More explicitly, multiplying the proposed derivative by $s^{N+1}A_*$ gives

$$\begin{aligned} s^{N+1}A_*\{\rho_o - \phi g - \phi(a^\vee)'\} &= \frac{3}{2}x^{-N}(s^p - s^{2N+1})\phi^2 - \frac{75}{2}x^{N+1} \\ &\quad + s^{N+1}\phi\left[\frac{3}{2}(4/x - 11) - A_*g\right] \\ &= -\frac{75}{2}x^{N+1} + s^{N+1}\phi\,\frac{3\{(x+1)w^2 + (x-4)^2(4x-1)\}}{10xsw} \\ &= \frac{75}{2}(s^Nx^2\phi/w - x^{N+1}) = \frac{75}{2}x^{N+1}(z_o^N - 1). \end{aligned}$$

Here the first term vanished because $2N+1 = p$. Cancellation of the nonzero denominators proves

$$Q_o' = \rho_o - \phi\big(g + (a^\vee)'\big). \tag{235}$$

This step retains both Frobenius phases of ρ_o.

Let

$$K_j = \frac{\frac{2}{5}LD_s - 2hs^N}{A_*} = \Lambda' + \frac{\lambda_c}{A_*}, \quad \widetilde{C} = \frac{2}{5}Ls\beta_D - 2hx^{N-1}, \quad N_o = O_sx^{-(N-1)}\widetilde{C}.$$

The branch identity gives $Q_oK_j = N_o/A_*$. Define the odd residue

$$\mathcal{O}_{\mathrm{res}} = \mathrm{Res}\left(Q_o\frac{\lambda_c}{A_*} - \Lambda\rho_o\right).$$

Integration by parts of ΛQ_o, using (235), gives

$$\mathcal{O}_{\mathrm{res}} = \Omega(N_o) - \mathrm{Res}(\Lambda\phi g) - \mathrm{Res}(\Lambda\phi(a^\vee)'). \tag{236}$$

These latter two carriers are finite. For the first, $\Lambda(-1)=0$ makes $\Lambda/(x+1)$ a Laurent polynomial, and

$$\Lambda\phi g=\frac{3}{2}\,\frac{\Lambda}{x+1}\,x^{-p-1}w^{p-1}(x-4).$$

The other carrier is the product of finite Laurent polynomials.

For the reflected differential, use $\operatorname{Res}(-x^{-2}F^{\vee})=-\operatorname{Res}F$, proved by sending an exponent i to $-i-2$. The reflected quotient is $Q_o^{\infty}=Q_o^{\vee}$, the reflected density $\rho_o^{\infty}=-x^{-2}\rho_o^{\vee}$, and the reflected pole part $\lambda^{\infty}=-\lambda_c^{\vee}/A_*$. Define $g^{\infty}=-x^{-2}g^{\vee}$ and

$$K_j^{\infty}=-x^{-2}K_j^{\vee}=-\frac{(\frac{2}{5}LD_s-2hs^N)^{\vee}}{A_*}=(\Lambda^{\vee})'+\lambda^{\infty}.$$

The reciprocal chain rule applied to (235) gives

$$(Q_o^{\infty})'=\rho_o^{\infty}-\phi^{\vee}(g^{\infty}+a'),\qquad Q_o^{\infty}K_j^{\infty}=-N_o^{\vee}/A_*.$$

Integration by parts of $\Lambda^{\vee}Q_o^{\infty}$ therefore gives

$$\operatorname{Res}(Q_o^{\infty}\lambda^{\infty}-\Lambda^{\vee}\rho_o^{\infty})=-\Omega(N_o^{\vee})-\operatorname{Res}(\Lambda^{\vee}\phi^{\vee}g^{\infty})-\operatorname{Res}(\Lambda^{\vee}\phi^{\vee}a').$$

The last two products are, respectively, $-x^{-2}(\Lambda\phi g)^{\vee}$ and $-x^{-2}(\Lambda\phi(a^{\vee})')^{\vee}$. They are finite carriers, so the coefficient reflection identity applies to each and yields

$$\mathcal{O}_{\mathrm{res}}^{\infty}:=\operatorname{Res}(Q_o^{\infty}\lambda^{\infty}-\Lambda^{\vee}\rho_o^{\infty})=-\Omega(N_o^{\vee})+\operatorname{Res}(\Lambda\phi g)+\operatorname{Res}(\Lambda\phi(a^{\vee})').$$

The original endpoints are

$$O_s(a_*)=5q,\quad O_s(b_*)=-10q,\quad \widetilde{C}(a_*)=-16h,\quad \widetilde{C}(b_*)=-h/4.$$

To obtain the odd values, $z_o(a_*)=z_o(b_*)=1$ and

$$F_o(1)=\frac{\ell(1)-\ell(-1)}{2}=q,\qquad a_*^{N-1}w(a_*)=4\cdot\frac{5}{4}=5,\qquad b_*^{N-1}w(b_*)=\frac{1}{4}(-40)=-10.$$

Here $\ell(1)=0$ and $\ell(-1)=-2q$ are the rational finite-log values proved above. For the adjoint endpoints, the branch values give

$$\widetilde{C}(a_*)=\frac{2}{5}\cdot 30\cdot\frac{1}{4}\cdot 16(-h/6)-2h\cdot 4=-16h,$$
$$\widetilde{C}(b_*)=\frac{2}{5}(-15)\cdot\frac{(-h/6)}{4}-2h\cdot\frac{1}{4}=-h/4.$$

Using $a_*^{N-1}=4$, $b_*^{N-1}=1/4$, one gets $N_o(a_*)=-20hq$, $N_o(b_*)=10hq$. The inversion balance of Ω now proves

$$\mathcal{O}_{\mathrm{res}}+\mathcal{O}_{\mathrm{res}}^{\infty}=2hq. \tag{237}$$

The reflected quotient is regular at zero. In fact

$$P_{\infty}=x^NO_s^{\vee}=(4x-11)F_o\left(\frac{s(4x-11)^2}{25}\right),\qquad Q_o^{\infty}=P_{\infty}/s^N.$$

Its actual constant term is

$$I=5^N[x^N]O_s=-11\cdot 5^NF_o(121/125).$$

For the reflected pole source, direct use of (230) gives

$$\lambda^{\infty}=\frac{\lambda_{\infty}}{x}+\frac{4h(x+1)}{5A_*},\qquad \lambda_{\infty}=\frac{2}{15}\kappa+\frac{h}{5}=-\frac{2}{5}(q-C).$$

Indeed $\lambda_c^\vee = \lambda_0 A_*/x + 4h/(5x) - 21h/5$; the displayed decomposition follows from $A_* + 4x(x+1) = 21x - 4$. Its second term is regular, so $\mathrm{Res}(Q_o^\infty \lambda^\infty) = -2(q-C)I/5$. The reflected odd density, with both phases still present, is

$$\begin{aligned}\rho_o^\infty &= -\frac{x^{N+1}\{\frac{3}{2}x^{N-p}s^p(4x^p-11)^2 - \frac{75}{2}x^{-N-1}\}}{s^{N+1}A_*} \\ &= \frac{75s^N}{2s^pA_*} - \frac{3s^N(4x^p-11)^2}{2A_*}.\end{aligned}$$

The Laurent series $s^N\Lambda^\vee/A_*$ has no exponent below $-(N-1)$. Since $s^p = (1+x^p)/5$, all positive terms in $(1+x^p)^{-1}$ shift its lower bound above -1. Thus the first residue is $(375/2)\Omega(s^N\Lambda^\vee)$. The x^p, x^{2p} terms in the second residue also vanish by this same bound, leaving its constant 121. Therefore

$$\mathrm{Res}(\Lambda^\vee \rho_o^\infty) = \left(\frac{375}{2} - \frac{363}{2}\right)\Omega(s^N\Lambda^\vee) = 6\Omega(s^N\Lambda^\vee).$$

This explains both the denominator cutoff and the phase cutoff at infinity. Equation (237) becomes

$$\mathcal{O}_{\mathrm{res}} = 2hq + \frac{2}{5}(q-C)I + 6\Omega(s^N\Lambda^\vee). \tag{238}$$

The odd origin dictionary, for either fifth root. Let $t_0 = -11u/5$, so $t_0^2 = 121/125$, and let $O(t) = \sum_{i=0}^{N-1} t^{2i+1}/(2i+1) = tF_o(t^2)$. The full odd/even decomposition of the finite log, together with finite entropy reflection, gives

$$t^pO(t) = \frac{1}{2}\ell(t^2) - E(t).$$

Equivalently this follows by splitting all indices $1, \dots, p-1$ into $2i$ and $2i+1$ in the finite definitions of E and ℓ; there is no truncated extra tail. Since $u^p = u(1/5)^N$,

$$I = \frac{5}{u^p}O(t_0), \qquad u^pt_0^p = -\frac{11}{25}, \qquad I = -\frac{125}{11}\left(\frac{1}{2}\ell(121/125) - E(t_0)\right).$$

The two finite-log evaluations used here are

$$\ell(121/125) = 3k - \frac{8}{125}q - \frac{242}{125}j_{11}, \qquad E(t_0) = \frac{11}{5}C - \frac{16}{5}q + \frac{3}{2}k.$$

For the first identity the integral entropy polynomial $(1 - X^p - (1-X)^p)/p$ reduces to $\ell(X)$: its coefficient of X^i, $1 \le i < p$, reduces to $1/i$ by $\binom{p}{i}/p = \binom{p-1}{i-1}/i$. At $X = a/b$, where $p \nmid ab(b-a)$, its value reduces to

$$\ell(a/b) = \frac{bq_b - aq_a - (b-a)q_{b-a}}{b}.$$

This follows by writing the integral numerator as $b^p - a^p - (b-a)^p = p[bQ_b - aQ_a - (b-a)Q_{b-a}]$ before reduction. The exact product identity $Q_{mn} = Q_m + Q_n + pQ_mQ_n$ gives $q_{125} = 3k$, $q_{121} = 2j_{11}$, $q_4 = 2q$; substitution of $a = 121, b = 125$ proves the first displayed value. No inequality $125 < p$ is used: only the factors $5, 11, 2$ need be units. For the second value, $4u^3 - 3u = t_0$ and $5u^2 - 1 = 0$. The native descent is $a(5t^2-1) = q - h + E(4t^3 - 3t)$. Evaluate (98) at zero, with $D(0) = -C/3$ and $f_{\mathrm{mix}}(0) = 0$:

$$a_0 = \frac{11}{5}C + \frac{3}{2}k - \frac{11}{5}q - h, \qquad E(t_0) = a_0 - q + h = \frac{11}{5}C - \frac{16}{5}q + \frac{3}{2}k.$$

Substituting without cancelling a root phase prematurely gives

$$\begin{aligned}I &= -\frac{125}{11}\left[\frac{3}{2}k - \frac{4}{125}q - \frac{121}{125}j_{11} - \frac{11}{5}C + \frac{16}{5}q - \frac{3}{2}k\right] \\ &= 25C - 36q + 11j_{11}.\end{aligned} \tag{239}$$

Since I is a coefficient of the original O_s, this also proves the independence of the two choices of u.

Even pole rows and the native derivative row. The actual mixed quotient is

$$R_j = O_s + s^N\phi(e + a/3), \qquad j = R_j/s^N = Q_o + \phi(e + a/3), \qquad e = a^\vee - (q - h).$$

The whole branch gives the finite numerator

$$N_j = x^{-(N-1)}\widetilde{C}R_j = \left(\frac{2}{5}LD_s - 2hs^N\right) j,$$

so its coefficient residue is exactly $\mathrm{Res}(jK_j)$. Here $\phi' = 0$, so (235) cancels $e' = (a^\vee)'$ and gives $j' = \rho_o - \phi g + \phi a'/3$. Integration by parts of Λj now proves the whole reduction

$$\begin{aligned} J_{\mathrm{raw}} := \Omega(N_j) = \mathrm{Res}(jK_j) &= \mathrm{Res}\left(j\frac{\lambda_c}{A_*} - \Lambda j'\right) \\ &= \mathcal{O}_{\mathrm{res}} + \mathrm{Res}(\Lambda\phi g) + \Omega(\phi e\lambda_c) + \frac{1}{3}\Omega(\phi a\lambda_c) - \frac{1}{3}\mathrm{Res}(\Lambda\phi a'). \end{aligned} \tag{240}$$

The first equality uses the actual branch carrier N_j, so this computes the full residue, including the deep windows of (218). No separate window correction is added to (240).

The partial fraction formula (219) gives for every finite F

$$\Omega(F\lambda_c) = \lambda_0 F_0 - \frac{4h}{15}(F_+(a_*) - F(a_*)) + \frac{h}{15}(F_+(b_*) - F(b_*)).$$

Put $a_0 = a(0)$. The reciprocal even carrier $F = \phi e$ has only nonpositive exponents, and

$$F_0 = -11(a_0 - q + h), \qquad F(a_*) = -5q, \qquad F(b_*) = 10q.$$

Hence

$$\Omega(\phi e\lambda_c) = -2hq - \frac{22}{5}(q - C)(a_0 - q + h). \tag{241}$$

For the native carrier $F = \phi a$, write $a = x^pU_0 + B_0$, with $B_0 = f_{\mathrm{mix}} - 33D/5 + c_0$. The entire nonnegative part, including its p-slot, is $F_+ = 4U_0 - 11a$. Its constant is

$$F_0 = -\frac{16}{5}(C + q) - 11a_0.$$

Let $\tau = (4q - 3r)/5$. The two low-band endpoint values are

$$B_0(a_*) = -h - \tau/4, \qquad B_0(b_*) = -h + 8\tau.$$

Therefore

$$F_+(a_*) - F(a_*) = -4a_*^{-p}B_0(a_*) = 16h + 4\tau, \qquad F_+(b_*) - F(b_*) = -4b_*^{-p}B_0(b_*) = h - 8\tau.$$

Substituting gives

$$\frac{1}{3}\Omega(\phi a\lambda_c) = \frac{\lambda_0}{3}\left[-\frac{16}{5}(C + q) - 11a_0\right] - \frac{7}{5}h^2 - \frac{32}{75}hq + \frac{8}{25}hr. \tag{242}$$

For the logarithmic carrier use $F = \Lambda\phi$. Since Λ_- is strictly negative and the whole support of Λ is below p,

$$F_+ = -11(\Lambda_+ - \nu), \qquad F_0 = -11\Lambda_0 = 11\nu.$$

At $x = -1$, $F = 0$ because $\Lambda(-1) = 0$; at $x = 4/11$, $F = 0$ because $\phi(4/11) = 0$. The three partial fractions of g thus give

$$\mathrm{Res}(\Lambda\phi g) = -11\Lambda_+(4/11) - \frac{11}{2}\Lambda_+(-1);$$

the normalization constants cancel as $11\nu+(11/2)\nu-(3/2)(11\nu)=0$. The moving primitive identity of (100) is

$$2\mathsf{H}(a_*,v)+\mathsf{H}(b_*,v)=\frac{1}{2}\left[G((1+v)/5)-G(1/5)\right].$$

Use $D_e(-1)=0$, $G(0)=0$, and the two coordinates $(1-1)/5=0$, $(1+4/11)/5=3/11$; one obtains

$$\operatorname{Res}(\Lambda\phi g)=\frac{11}{15}G(3/11)+\frac{44h}{15}D_e(4/11)-\frac{11}{10}G(1/5)-\frac{22}{5}hC. \tag{243}$$

The finite entropy coordinate formula for D_e gives $D_e(4/11)=C_w=E(2w_0)-E(w_0)$, where $w_0^2=3/11$. It holds for either choice of w_0, because the left side is the fixed polynomial value at $4/11$.

For the last derivative row, the full native source is

$$A_*a'=\frac{3}{2}\left(4x-11-s^N(4x^p-11)\right).$$

The product of the two Frobenius factors is $\phi(4x^p-11)=137-44x^p-44x^{-p}$. Consequently

$$A_*\left(-\frac{1}{3}\Lambda\phi a'\right)=\frac{1}{6}L\Lambda\phi+\frac{137}{2}s^N\Lambda-22x^ps^N\Lambda-22x^{-p}s^N\Lambda. \tag{244}$$

The support of $s^N\Lambda$ is contained in $[-N+2,p-2]$; thus its x^p shift is polynomial, and its negative p-shift has full residue

$$\Omega(x^{-p}s^N\Lambda)=\frac{-4\Lambda(a_*)+\Lambda(b_*)/4}{15}.$$

This follows from the shifted-polynomial geometric formula after writing $s^N\Lambda=x^{-N+2}P$, with $\deg P\le 3N-3$, and using the endpoint powers in the resulting shift $p+N-2$. The two proper-end pole evaluations are

$$\begin{aligned}(\Lambda\phi)_+(a_*)-(\Lambda\phi)(a_*)&=-16\Lambda_+(a_*)-5\Lambda_-(a_*)+16\nu,\\ (\Lambda\phi)_+(b_*)-(\Lambda\phi)(b_*)&=-\Lambda_+(b_*)+10\Lambda_-(b_*)+\nu.\end{aligned}$$

Insert these in (219) and (244). The terms in ν cancel, leaving

$$\begin{aligned}R_{\mathrm{der}}:=-\frac{1}{3}\operatorname{Res}(\Lambda\phi a')=\frac{137}{2}\Omega(s^N\Lambda)+\frac{8}{15}(\Lambda_+(a_*)-\Lambda_+(b_*))\\ +\frac{21}{5}\Lambda_-(a_*)+\frac{13}{10}\Lambda_-(b_*).\end{aligned} \tag{245}$$

For clarity, the coefficients of the six primitive rows in the four terms on the right are the following integers divided by 225:

	H_{aa}	H_{ba}	H_{ab}	H_{bb}	H_{0a}	H_{0b}
$(137/2)\Omega(\Lambda)$	-137	-274	137	274	411	-411
$(8/15)(\Lambda_+(a_*)-\Lambda_+(b_*))$	-32	-16	32	16	0	0
$(21/5)\Lambda_-(a_*)$	0	0	-126	-252	0	378
$(13/10)\Lambda_-(b_*)$	-39	-78	0	0	117	0
sum	-208	-368	43	38	528	-33

The nonprimitive terms are $-32h(D_a-D_b)/225$ from the positive endpoints and $274T/225+137\lambda_0C/15$ from the unit row. Thus

$$R_{\mathrm{der}}=H_{\mathrm{even}}-\frac{32h}{225}(D_a-D_b)+\frac{274}{225}T+\frac{137}{15}\lambda_0C, \tag{246}$$

where the completely specified row is

$$H_{\mathrm{even}}=\frac{-208H_{aa}-368H_{ba}+43H_{ab}+38H_{bb}}{225}+\frac{176}{75}H_{0a}-\frac{11}{75}H_{0b}.$$

Final coefficient collection. Substitute (238), (234), (241), (242), (243), and (246) in (240). This first gives the exact intermediate packet

$$\begin{aligned} J_{\mathrm{raw}} &= \mathcal{O}_{\mathrm{res}} + \frac{11}{15}G(3/11) + \frac{44h}{15}C_w - \frac{11}{10}G(1/5) + H_{\mathrm{even}} \\ &\quad - 2hq - \frac{22}{5}hC - \frac{22}{5}(q-C)(a_0 - q + h) \\ &\quad + \frac{\lambda_0}{3}\left[-\frac{16}{5}(C+q) - 11a_0\right] - \frac{7}{5}h^2 - \frac{32}{75}hq + \frac{8}{25}hr \\ &\quad - \frac{32h}{225}(D_a - D_b) + \frac{274}{225}T + \frac{137}{15}\lambda_0 C. \end{aligned} \tag{247}$$

The odd H-row is $-4(2H_{aa} + H_{ba} - 2H_{ab} - H_{bb})/75$. Adding it to H_{even} produces

$$H_j = -\frac{232}{225}H_{aa} - \frac{76}{45}H_{ba} + \frac{67}{225}H_{ab} + \frac{2}{9}H_{bb} + \frac{176}{75}H_{0a} - \frac{11}{75}H_{0b}.$$

For the ordinary coefficients make the substitutions

$$D_a = -3q - h/2, \quad D_b = 3q + h, \quad a_0 = \frac{11}{5}C - \frac{11}{5}q + \frac{3}{2}k - h,$$
$$\lambda_0 = \frac{2}{5}(q-C) + \frac{h}{5}, \quad r = -\frac{2}{3}h - \frac{4}{3}q, \quad I = 25C - 36q + 11j_{11},$$
$$T = -\frac{9}{40}h^2 + \frac{1}{2}D_a^2 + D_b^2 - \frac{3}{2}C^2.$$

After excluding the j_{11} term, the five groups have the following coefficients. Each row is the expansion of the displayed group, so every quadratic cancellation is explicit:

	q^2	hq	h^2	qC	C^2	hC
odd	$-324/25$	$66/25$	$12/125$	$602/25$	$-246/25$	0
log	0	0	0	0	0	$-22/5$
reciprocal	$352/25$	-2	0	$-594/25$	$242/25$	0
native	$14/5$	$151/75$	$-22/25$	$-484/75$	$274/75$	$-247/75$
derivative	$411/25$	$749/75$	$491/375$	$274/75$	$-137/25$	$137/75$
sum	$509/25$	$316/25$	$197/375$	$-62/25$	$-149/75$	$-88/15$

Here the ordinary odd group is

$$2hq + \frac{2}{5}(q-C)(25C - 36q) + \frac{7}{125}h^2 - \frac{4}{75}D_a^2 - \frac{8}{75}D_b^2 - \frac{8}{25}D_aD_b + \frac{4}{25}C^2 - \frac{8}{25}qC;$$

the other four groups are, in order, $-22hC/5$, the right side of (241), the right side of (242), and $-32h(D_a - D_b)/225 + 274T/225 + 137\lambda_0 C/15$. Their remaining coefficients are

	qk	Ck	hk
odd	0	0	0
log	0	0	0
reciprocal	$-33/5$	$33/5$	0
native	$-11/5$	$11/5$	$-11/10$
derivative	0	0	0
sum	$-44/5$	$44/5$	$-11/10$

The odd term $2(q-C)I/5$ contributes $22(q-C)j_{11}/5$. Define the full conic row $Q_w = G(3/11) + 4hC_w + 6(q-C)j_{11}$. Its three coefficients are $11/15$, $44/15$, and $22/5$, exactly those already present. Therefore the whole residue is

$$\begin{aligned} J_{\mathrm{raw}} &= \frac{11}{15}Q_w - \frac{11}{10}G(1/5) + H_j + \frac{509}{25}q^2 + \frac{316}{25}hq + \frac{197}{375}h^2 \\ &\quad - \frac{62}{25}qC - \frac{149}{75}C^2 - \frac{88}{15}hC - \frac{44}{5}(q-C)k - \frac{11}{10}hk. \end{aligned} \tag{248}$$

This is the residue in (217), computed from its actual quotient and full finite branch. The support estimates above justify all suppressed infinity phases; no Hasse-nonzero hypothesis, missing endpoint, unspecified primitive constant or root-choice premise occurs.

6.10 The whole low-H attachment and its regular remainder

The density (183) uses the original slot $\chi = \mathcal{P} - a/3$. Its entire attachment is

$$\mathcal{H}_{\mathrm{low}} = A_* \left(\mathscr{R} + \frac{5}{24} x^p a \, l_y \right), \qquad l_y = LQ_H, \tag{249}$$

where $\mathscr{R}$ is the complete source in (257). We prove this identity before extracting its residue. In this subsection put

$$\begin{aligned} y = H, \quad Q = Q_H, \quad z_y = y/x, \quad q_y = y_q, \\ G = x^p, \quad X_p = x^{-p}, \quad X_n = x^N, \quad S = s^N, \\ B_p = 4G - 11, \quad \phi = 4X_p - 11, \end{aligned}$$

and write $u = a$, $v = a^\vee$, $D_v = D^\vee$, $w = D + D_v$, $c = q - h$. Thus $GX_p = 1$ and $G' = X_p' = 0$. The derivative of X_n is retained whenever it occurs. All rational constants below have denominators with prime factors only $2, 3, 5$, which are units for $p \geq 19$.

Define the densities, each with its original y multiplier,

$$d_u = yu', \quad d_v = yv', \quad d_P = y\mathcal{P}', \quad d_D = yD', \quad d_{D_v} = yD_v'.$$

The complete cleared derivatives (95) and their reciprocal give

$$\begin{aligned} d_D &= \frac{5}{2}(1 - S)Q, & d_{D_v} &= \frac{5}{2}(X_n^{-1}S - 1)Q, \\ d_u &= \frac{3}{5}B_p d_D - \frac{1}{2}l_y - \frac{3}{2}B_p Q, & d_v &= \frac{3}{5}\phi d_{D_v} - \frac{1}{2}q_y + \frac{3}{2}\phi Q. \end{aligned} \tag{250}$$

For the reciprocal equations use $A_*^\vee = x^{-2}A_*$, $s^\vee = x^{-1}s$, and $(F^\vee)' = -x^{-2}(F')^\vee$. The positive d_u equation follows on substituting $d_D = 5(1 - S)Q/2$: its right side is $-3B_pSQ/2 - l_y/2$, and $L = -3(4x - 11)$. For the negative equation, the origin term in q_y is essential. Since $A_* = -4x^2 + 17x - 4$,

$$q_y = (18 - 12x)Q - 3A_*Q/x = 3(4x^{-1} - 11)Q.$$

The right side of its equation therefore becomes $3\phi X_n^{-1}SQ/2 - q_y/2$, exactly the reciprocal derivative. In particular the two shifted equations are

$$2d_u + l_y = -3B_pSQ, \qquad X_n(2d_v + q_y) = 3\phi SQ, \qquad SQ = Q - \frac{2}{5}d_D. \tag{251}$$

The full source bands also imply

$$\begin{aligned} \mathcal{P} - D &= (u - v)/3 + \phi w/5, \\ \mathcal{P}^\vee - D_v &= (v - u)/3 + B_p w/5, \\ d_P - d_D &= (d_u - d_v)/3 + \phi(d_D + d_{D_v})/5. \end{aligned} \tag{252}$$

For completeness, let $M = \ell(x/4) - \ell(4x)$ and $\kappa = 3k/2 - 11q/5 - h$. The actual bands already printed give

$$v = \frac{3}{5}\phi D_v + \frac{2}{5}X_p M + \kappa - \frac{4}{5}qX_p, \qquad \mathcal{P} - u/3 - D = \frac{1}{5}\phi D - \frac{2}{15}X_p M - \frac{\kappa}{3} + \frac{4}{15}qX_p.$$

Subtracting $v/3 - \phi D_v/5$ proves the first equation in (252), including its origin constant. Full inversion proves the second, and differentiation proves the third because $\phi' = 0$.

The kernel and metric branch cancel before collection. The cleared metric source (104) gives

$$C_B L + A_* C_M' - 3hS = \frac{3}{2} Sy, \qquad \frac{C_B B_0 L - \mathcal{B}_{\rm met}}{A_*} = \frac{3}{2} B_0 SQ.$$

Indeed its C_B terms cancel because $L + 3(4x - 11) = 0$, and its constant branch cancels $3hS$. No value of H is divided by. Differentiate $V_0 = -\mathcal{K}\chi + 5GZ_{52}/24$, using $G' = 0$. Then (183) yields

$$\mathcal{H}_{\rm low}/A_* = \chi y \mathcal{K}' + \mathcal{K}Q\{A_*\chi' + A_*/(2x) - L\} - \frac{5}{24} G y Z_{52}' + \frac{3}{2} B_0 SQ + \mathcal{C}_H.$$

The factor in braces has the full value

$$\begin{aligned} A_*\chi' + A_*/(2x) - L &= \frac{1}{2} S(17 - 4G - 4X_p) - \frac{1}{2}(4x - 11 - SB_p) - L \\ &= S(3 - 2X_p) - \frac{1}{2}(4x - 11) - L \\ &= S(3 - 2X_p) - \frac{5}{6} L. \end{aligned}$$

Thus it alone is not $-5L/6$. Its branch cancels against the kernel part of the actual $B_0 = -2(1 - 2X_p/3)\mathcal{K} + 5GR_{52}/12$:

$$\frac{3}{2} B_0 SQ = -(3 - 2X_p) S\mathcal{K}Q + \frac{5}{8} GSR_{52}Q.$$

Consequently

$$\mathcal{H}_{\rm low}/A_* = \chi y \mathcal{K}' - \frac{5}{6} \mathcal{K} l_y - \frac{5}{24} G y Z_{52}' + \frac{5}{8} GSR_{52}Q + \mathcal{C}_H. \tag{253}$$

The actual product derivative and its second branch. Define the entire derivative block

$$\begin{aligned} \Xi = {} & \frac{5}{12} G\{v d_u + (u - (1 + X_n)v) d_v\} + \frac{5}{24} G(u l_y - X_n v q_y) \\ & - \frac{5}{4} G\{v(d_P - d_D) + (\mathcal{P} - D) d_v\}. \end{aligned}$$

The product rule for $Z_{52} = (u - v)^2 + 6v(\mathcal{P} - D)$ gives

$$y Z_{52}' = 2(u - v)(d_u - d_v) + 6\{d_v(\mathcal{P} - D) + v(d_P - d_D)\}.$$

Subtracting Ξ from $-5GyZ_{52}'/24$ cancels the last two product terms. The remaining difference is

$$\begin{aligned} & -\frac{5}{12} G u\, d_u + \frac{5}{12} G X_n v\, d_v - \frac{5}{24} G u l_y + \frac{5}{24} G X_n v q_y \\ & = -\frac{5}{24} G u(2d_u + l_y) + \frac{5}{24} G v X_n (2d_v + q_y) \\ & = \frac{5}{8} GS(B_p u + \phi v) Q = -\frac{5}{8} GSR_{52}Q, \end{aligned}$$

where both equations of (251) were used. This cancels the second branch in (253), proving the whole native attachment

$$\mathcal{H}_{\rm low}/A_* = \chi y \mathcal{K}' - \frac{5}{6} \mathcal{K} l_y + \Xi + \mathcal{C}_H. \tag{254}$$

Its right side is the original scalar-assembly low-H source, with every derivative and reciprocal retained.

Reduction to the expanded rows with the own-variable term retained. Substitute the first and third equations of (252) in Ξ. Its two d_u coefficients cancel as $5Gv/12 - 5Gv/12 = 0$. The coefficients of d_v are

$$\frac{5}{12}G\{u - (1+X_n)v + v - u + v\} = \frac{5}{12}G(1-X_n)v.$$

Therefore

$$\Xi = \frac{5}{12}G(1-X_n)vd_v + \frac{5}{24}G(ul_y - X_nvq_y) - \frac{1}{4}G\phi\{v(d_D + d_{D_v}) + wd_v\}.$$

Now expand the six original terms of $\mathcal{C}_H$ in (180), using the second equation of (252). Its d_v coefficient is

$$\begin{aligned} &-\frac{2}{3}\{w + (v-u)/3 + B_pw/5\} + \frac{2}{9}\{v - u - X_n(u+h)\} \\ &\quad = -\frac{2}{15}(B_p+5)w - \frac{2}{9}X_n(u+h), \end{aligned}$$

and its q_y coefficient is

$$-\frac{1}{9}\{u + X_n(u+h)\} - \frac{1}{3}\{D_v + (v-u)/3 + B_pw/5\} = -v/9 - D_v/3 - B_pw/15 - X_n(u+h)/9.$$

The terms involving X_n in $\Xi + \mathcal{C}_H$ thus combine to

$$-\left(\frac{5}{24}Gv + \frac{u+h}{9}\right)X_n(2d_v + q_y) = -3\phi\left(\frac{5}{24}Gv + \frac{u+h}{9}\right)\left(Q - \frac{2}{5}d_D\right).$$

Put

$$t_0 = 133G - 12, \quad \tau = \frac{5}{12}Gv + \frac{t_0}{60}w, \quad b_0 = \frac{5}{24}Gv + \frac{u+h}{9}.$$

The coefficient of d_v before its density substitution is exactly τ, since

$$\frac{1}{4}G\phi + \frac{2}{15}(B_p+5) = \frac{12 - 133G}{60} = -\frac{t_0}{60}.$$

The whole remaining expression is now

$$\begin{aligned} \Xi + \mathcal{C}_H = &\frac{5}{24}Gul_y + \tau d_v - \frac{1}{4}G\phi v(d_D + d_{D_v}) - \frac{5}{3}(v-c)Q \\ &- (v/9 + D_v/3 + B_pw/15)q_y - \frac{1}{5}\phi wz_y - \frac{2}{15}\phi wd_u - 3\phi b_0(Q - 2d_D/5). \end{aligned}$$

Substitute the two equations for d_u, d_v in (250). The resulting coefficients are listed with their complete cancellations:

$$\begin{aligned} [d_D]: &\quad -G\phi v/4 - 2\phi B_pw/25 + 6\phi b_0/5 = \frac{2\phi}{15}(u + h - 3B_pw/5), \\ [d_{D_v}]: &\quad 3\phi\tau/5 - G\phi v/4 = \phi t_0 w/100, \\ [q_y]: &\quad -\tau/2 - v/9 - D_v/3 - B_pw/15 \\ &\quad = -\frac{15G+8}{72}v + \left(\frac{5}{6} - \frac{11}{8}G\right)w - D_v/3 \\ &\quad = -\frac{15G+8}{72}v + D/3 + G\phi w/8, \\ [l_y]: &\quad \phi w/15, \qquad [z_y]: \quad -\phi w/5, \\ [Q]: &\quad 3\phi\tau/2 - 5(v-c)/3 + \phi B_pw/5 - 3\phi b_0 \\ &\quad = -\frac{5}{3}(v-c) - \frac{1}{3}\phi(u+h) + \frac{1}{8}\phi(33G - 20)w. \end{aligned}$$

In the q_y row, $t_0 + 8B_p = 165G - 100$ and $G\phi = 4 - 11G$. In the Q row the two $5G\phi v/8$ terms cancel, and $(t_0 + 8B_p)/40 = (33G - 20)/8$. These are exactly the six coefficient rows of (257); the remaining term is $5Gul_y/24$. Together with (254), this proves (249) as an identity of Laurent polynomials. The divisions by A_* above may be performed in $K(x)$; every displayed cleared expression is Laurent polynomial, and $A_* \neq 0$, so multiplication by A_* is injective.

Uniform support and the scalar assembly. The actual positive carrier is the polynomial

$$a = q - h + \sum_{i=1}^{N} \frac{(1-V)^i}{2i}.$$

Thus $\deg a \le 3N$, and the actual polynomial density satisfies $\deg l_y \le 3d-1$. The exact remainder

$$R_{\mathrm{reg}} = \frac{5}{24} x^p a\, l_y \tag{255}$$

has support contained in

$$[p,\, p+3N+3d-1] = [p,\, 3p-3-\delta], \qquad \delta = N-3d \in \{0,2\}.$$

In particular $[x^{-1}]R_{\mathrm{reg}} = 0$, without an endpoint or origin assignment. By (92),

$$\Omega(\mathcal{H}_{\mathrm{low}}) = \operatorname{Res}\mathscr{R} + \operatorname{Res} R_{\mathrm{reg}} = \frac{8}{75} J_H(1) + \frac{137}{5} J_s = \frac{97}{375} h^2.$$

The original scalar assembly therefore uses the whole density $A_*(\mathscr{R} + R_{\mathrm{reg}})$. Replacing it by $A_*\mathscr{R}$ is justified only after this zero-residue proof. The joint metric remains unchanged, and its collection is

$$\Omega(\mathcal{H}_{\mathrm{low}} + \mathcal{J}_{\mathrm{met}}) = \left(\frac{97}{375} - \frac{1213}{750}\right) h^2 = -\frac{1019}{750} h^2.$$

If the low-H slot is instead written with $\chi_0 = \chi - h/3$, its whole density subtracts $A_*(h/3)y\mathcal{K}'$. The same R_{reg} remains. The kernel-origin calculation in (259), with its constant factor $h/3$, gives $\operatorname{Res}((h/3)y\mathcal{K}') = (h/3)y_0$. Hence this single conversion changes the collected scalar to $-1019h^2/750 - hy_0/3$; it does not replace the pulled-back origin $y_0 = H(0)$ by the own-variable value $H_0(0) = 0$.

6.11 The individual low-H source rows

We expand the coefficient calculation underlying the first row of (268). Throughout $p \ge 19$, $N = (p-1)/2$, $d = \lfloor p/6 \rfloor$, and $H_0(0) = 0$, $\deg H_0 \le d$. Thus $6d < p$, $3d \le N$, and $\delta = N - 3d$ is 0 or 2. Write $\operatorname{Res} F = [x^{-1}]F$. By (249), the original density (183) satisfies $\mathcal{H}_{\mathrm{low}}/A_* = \mathscr{R} + R_{\mathrm{reg}}$. The polynomial remainder (255) has zero residue. The following calculation evaluates the complete $\mathscr{R}$, so $\Omega(\mathcal{H}_{\mathrm{low}}) = \operatorname{Res}\mathscr{R}$.

Finite densities and whole source decomposition. For this calculation set

$$\begin{aligned}
&y = H_0(V), && y_0 = y(0), && Q = \frac{y}{A_*} = \frac{4x-1}{125}(H_0/z)(V),\\
&z_y = y/x, && l_y = (33-12x)Q, && q_y = (18-12x)Q - 3z_y,\\
&J_Q = (18-12x)Q - 3(y-y_0)/x, && Q_s = Qs^N, && J_0 = \operatorname{Res}(D_s Q), \qquad J_s = \operatorname{Res}(D_s Q_s).
\end{aligned}$$

The letter z in H_0/z is its own polynomial variable: $H_0/z = \sum_{i=1}^{d} (H_0)_i z^{i-1}$. Hence Q, J_Q, l_y are polynomials, and $q_y = J_Q - 3y_0/x$. Also

$$q_y - l_y + 3z_y = -15Q, \qquad yD' = \frac{5}{2}(Q - Q_s). \tag{256}$$

The first equality subtracts the displayed affine factors. The second multiplies $A_*D' = 5(1-s^N)/2$ by the actual polynomial quotient Q.

Put $G = x^p$, $X_p = x^{-p}$, $B_p = 4G - 11$, $\phi = 4X_p - 11$, $D_v = D^\vee$, $w = D + D_v$, $\mathcal{L}_a = \ell(x/4)$, $\mathcal{L}_b = \ell(4x)$, $M = \mathcal{L}_a - \mathcal{L}_b$, and $L_{\log} = 4\mathcal{L}_a - \mathcal{L}_b/4$. Keep κ, h, q as independent constants during the algebra. The two affine sources and their constant parts are

$$u_c = \kappa - \frac{4qG}{5}, \quad v_c = \kappa - \frac{4qX_p}{5}, \quad c = q - h, \quad u = \frac{3}{5}B_pD + \frac{2}{5}L_{\log} + u_c, \quad v = \frac{3}{5}\phi D_v + \frac{2}{5}X_pM + v_c.$$

For the actual native source $u = a$, $v = a^\vee$ and $\kappa = 3k/2 - 11q/5 - h$. Indeed the whole two-band formula for a, printed in (271)'s derivation, gives $u = a$. Full logarithm reversal $j \mapsto p - j$ gives the Laurent identities $\mathcal{L}_a^\vee = -X_p\mathcal{L}_b/4$, $\mathcal{L}_b^\vee = -4X_p\mathcal{L}_a$; therefore $L_{\log}^\vee = X_pM$, proving $v = a^\vee$. The actual source slot is

$$C_{\text{slot}} = \frac{4X_p - 6}{5}D - \frac{2}{15}X_pM + \frac{4qX_p}{15} - \frac{\kappa}{3} = \mathcal{P} - a/3.$$

This follows by the same full-band cancellations as (271), before its final subtraction of $h/3$.

With $R_D = yD'$, $R_v = yD'_v$, the original collected source is

$$\begin{aligned}\mathscr{R} = {} & C_{\text{slot}}y\mathcal{K}' - \frac{5}{6}\mathcal{K}l_y + \frac{2\phi}{15}(u + h - 3B_pw/5)R_D + \frac{\phi(133G - 12)}{100}wR_v \\ & - \frac{15G + 8}{72}vq_y + \frac{1}{3}Dq_y + \frac{1}{8}G\phi wq_y + \frac{1}{15}\phi wl_y - \frac{1}{5}\phi wz_y \\ & + \left\{-\frac{5}{3}(v - c) - \frac{1}{3}\phi(u + h) + \frac{1}{8}\phi(33G - 20)w\right\}Q. \end{aligned} \tag{257}$$

Its seven entire groups are

$$\begin{aligned}\mathscr{R}_C &= C_{\text{slot}}y\mathcal{K}' - \frac{5}{6}\mathcal{K}l_y, \\ \mathscr{R}_{\log} &= \frac{4}{75}\phi L_{\log}R_D - \frac{15 + 8X_p}{180}Mq_y - \frac{2}{3}X_pMQ - \frac{2}{15}\phi L_{\log}Q, \\ \mathscr{R}_D &= \left(\frac{5}{6} - \frac{11}{8}G\right)Dq_y + \left(\frac{11}{15} - \frac{4}{15}X_p\right)D_vq_y + \frac{1}{15}\phi wl_y - \frac{1}{5}\phi wz_y \\ &\quad + \frac{\phi(133G - 12)}{40}DQ + \frac{\phi(33G - 28)}{8}D_vQ, \\ \mathscr{R}_2 &= -\frac{2}{25}\phi B_pD_vR_D + \frac{\phi(133G - 12)}{100}wR_v, \\ \mathscr{R}_{\text{reg,D}} &= \frac{2}{15}\phi(u_c + h)R_D, \\ \mathscr{R}_{\text{reg,Q}} &= \left\{-\frac{5}{3}(v_c - c) - \frac{1}{3}\phi(u_c + h)\right\}Q, \\ \mathscr{R}_{\text{const}} &= -\frac{15G + 8}{72}v_cq_y. \end{aligned}$$

Here is the algebra proving their sum equals $\mathscr{R}$. The R_D coefficient expands to

$$\frac{2\phi}{15}\left(-\frac{3}{5}B_pD_v + \frac{2}{5}L_{\log} + u_c + h\right) = -\frac{2}{25}\phi B_pD_v + \frac{4}{75}\phi L_{\log} + \frac{2}{15}\phi(u_c + h).$$

The R_v, l_y, z_y coefficients are already identical. The nonconstant Q coefficients are respectively

$$D: \quad \phi\left(-\frac{B_p}{5} + \frac{33G - 20}{8}\right) = \frac{\phi(133G - 12)}{40}, \qquad D_v: \quad \phi\left(-1 + \frac{33G - 20}{8}\right) = \frac{\phi(33G - 28)}{8},$$

and the logarithms contribute $-2X_pM/3 - 2\phi L_{\log}/15$. After removing v_c, the q_y coefficients in the original source are

$$D: \quad \frac{1}{3} + \frac{G\phi}{8}, \quad D_v: \quad -\frac{\phi(15G + 8)}{120} + \frac{G\phi}{8} = -\frac{\phi}{15}, \quad M: \quad -\frac{(15G + 8)X_p}{180}.$$

Subtract the corresponding coefficients in the seven-group sum. The differences are $(GX_p - 1)D/2$, 0, and $-(GX_p - 1)M/12$. Thus the full residual is

$$\mathscr{R} - \sum \mathscr{R}_\bullet = \frac{GX_p - 1}{12}(6D - M)q_y = 0.$$

Only the exact Laurent equality $GX_p = 1$ is used here.

Degree windows and the two kernel origins. The precise bounds needed for coefficient selection are

$$\begin{aligned}\deg y \le 3d = N-\delta, \quad \deg Q \le 3d-2,\\ \deg l_y, \deg J_Q \le 3d-1, \quad \deg D \le N-1,\\ \deg(yD') \le 3d+N-2 = p-3-\delta.\end{aligned}$$

In particular Q, QD have degree $< p-2$, and every polynomial used in an ordinary integral below has degree $< p-1$. The Laurent densities q_y, z_y, l_y, Q have support in $[-1, N-1]$. Their products with D have support in $[-1, p-3]$, and with D_v in $[-N, N-1]$. Consequently multiplying any of these densities or products by G or X_p gives zero residue: it asks for coefficient $-p-1$ or $p-1$, respectively, outside the specified support.

Differentiating the complete finite logarithms gives

$$A_*\mathcal{L}_a' = 4(x-a_*)(1-x^{p-1}), \qquad A_*\mathcal{L}_b' = 4(x-b_*)(1-x^{p-1}).$$

For example $(1-x/4)\mathcal{L}_a' = (1/4)(1-(x/4)^{p-1})$; since $4^{p-1}=1$ and $A_* = (4-x)(4x-1)$, the first identity follows. The second is the same geometric identity with scale 4. The derivatives of G, X_p, X_p^2 are zero. Therefore

$$\mathfrak{B} := A_*\mathcal{K}' = (1-x^{p-1})\{20x + 15/2 + (113x/2+4)X_p + (1953x/8 - 113/2)X_p^2\}. \tag{258}$$

The three affine coefficients follow from the literal kernel weights:

band	x coefficient	constant
0	$(140-20)/6 = 20$	$(-35+80)/6 = 15/2$
1	$(368-29)/6 = 113/2$	$(-92+116)/6 = 4$
2	$(1472-29/4)/6 = 1953/8$	$(-368+29)/6 = -113/2$

Its full twelve-term expansion has exponents and coefficients

n	1	0	$1-p$	$-p$	$1-2p$	$-2p$
c_n	20	15/2	113/2	4	1953/8	−113/2
n	p	$p-1$	0	−1	$-p$	$-p-1$
c_n	−20	−15/2	−113/2	−4	−1953/8	113/2

The two occurrences at 0 and $-p$ remain separate until coefficient selection. For a polynomial F of degree $< p-2$, the only term contributing to $\mathrm{Res}(\mathfrak{B}F)$ is the exponent -1, giving $-4F(0)$. After multiplication by X_p, only the original exponent $p-1$ contributes, giving $-15F(0)/2$. Every other term asks for a negative coefficient or for a coefficient of index at least $p-2$. Hence

$$\mathrm{Res}\{(\alpha X_p + \beta)\mathfrak{B}F\} = (-15\alpha/2 - 4\beta)F(0).$$

Since $A_*(0) = -4$, the actual identity $y = A_*Q$ gives $Q(0) = -y_0/4$. The D and constant parts of $\mathscr{R}_C$, before its logarithms, are thus

$$\begin{aligned}\mathrm{Res}\{y(4X_p/5 - 6/5)D\mathcal{K}'\} &= (-6+24/5)Q(0)D(0) = \frac{3}{10}y_0D(0),\\ \mathrm{Res}\{y(4qX_p/15 - \kappa/3)\mathcal{K}'\} &= (-2q+4\kappa/3)Q(0) = (q/2-\kappa/3)y_0.\end{aligned} \tag{259}$$

Finite integration and the three logarithm groups. For a polynomial F of degree $< p-1$, define

$$I_F(c) = \sum_{j\ge 0}\frac{F_j c^{j+1}}{j+1}, \qquad O_F(c) = \sum_{j=1}^{p-1}\frac{F_j c^j}{j}.$$

The first sum only has $j+1<p$, so all its denominators are units, and its polynomial has derivative F and value zero at zero. Define

$$P_H(z)=\sum_{i=1}^{d}\frac{(H_0)_i z^i}{i},\qquad P=P_H(V(0)),\qquad T_O=\frac{1}{4}O_y(b_*)-4O_y(a_*).$$

Since $P_H'=H_0/z$, $l_y=P_H(V)'$. This uses $V'/V=(33-12x)/A_*$, verified by differentiating its displayed cubic factors. Its degree is $<p-1$, and

$$I_{l_y}(c)=P_H(V(c))-P,\qquad I_{l_y}(a_*)=I_{l_y}(b_*)=-P.$$

For $c\neq 0$, let $L_c=\ell(x/c)$. Direct coefficient reversal gives

$$[x^{p-1}](L_cF)=-c^{-p}I_F(c),\qquad [x^p](L_cF)=-c^{-p}O_F(c).\tag{260}$$

For the first equality its term with F_j has log index $p-1-j$, inverse denominator $-1/(j+1)$, and power $c^{-(p-1-j)}=c^{-p}c^{j+1}$. For the second the log index is $p-j$, its inverse is $-1/j$, and $j=0$ contributes nothing because the log cutoff excludes p. The product has degree $<2p-2$, so coefficients at or above $2p-1$ are zero. Also the coefficients of L_c at $0,p$ are zero. Thus the two original simple-pole terms $\mathrm{Res}(x^{-1}L_c)$ and $\mathrm{Res}(X_px^{-1}L_c)$ are both zero; no assignment of their multiplying origin scalar is required.

The logarithmic derivative part of $\mathscr{R}_C$ is $-2X_pMy\mathcal{K}'/15$. Let

$$F_0=(20x+15/2)Q,\quad F_1=(113x/2+4)Q,\quad F_2=(1953x/8-113/2)Q,\quad Z=2l_y-(y-y_0)/x.$$

These three row polynomials have degree at most $3d-1<p-1$. The entire derivative product is

$$y\mathcal{K}'=(1-x^{p-1})(F_0+X_pF_1+X_p^2F_2).$$

For each L_c, the six products in $X_pL_c(1-x^{p-1})(F_0+X_pF_1+X_p^2F_2)$ ask respectively for coefficients

row	F_0	$-F_0$	F_1	$-F_1$	F_2	$-F_2$
index	$p-1$	0	$2p-1$	p	$3p-1$	$2p$

in L_cF_i. By (260), only the first and fourth survive, giving $c^{-p}\{O_{F_1}(c)-I_{F_0}(c)\}$. The exact row identity is

$$xZ=F_1-xF_0-F_1(0).$$

To check it, $2xl_y-y=(-20x^2+49x+4)Q=F_1-xF_0$, and $F_1(0)=4Q(0)=-y_0$. Integrating the positive coefficients after division by x gives $O_{F_1}(c)-I_{F_0}(c)=I_Z(c)$. Finally

$$I_Z(c)=2(P_H(V(c))-P)-O_y(c).$$

Thus the derivative-log contribution is

$$\begin{aligned}\mathrm{Res}(-2X_pMy\mathcal{K}'/15)&=\frac{8}{15}I_Z(a_*)-\frac{1}{30}I_Z(b_*)\\&=-P+\frac{2}{15}T_O.\end{aligned}\tag{261}$$

The factors $8/15,-1/30$ retain $a_*^{-p}=4,b_*^{-p}=1/4$.

For the other cubic log term, the whole three-band kernel gives

$$\mathrm{Res}(\mathcal{K}F)=-\frac{23}{6}I_F(b_*)+\frac{29}{6}I_F(a_*).$$

Its regular bands give zero; its X_p^2 bands ask for coefficient $2p-1$ and give zero by degree. Its X_p bands give exactly $(92/6)(-I_F(b_*)/4)$ and $-(29/24)(-4I_F(a_*))$ by (260). For $F=l_y$, both endpoint integrals are $-P$, so

$$\mathrm{Res}(-5\mathcal{K}l_y/6)=\frac{5}{6}P.\tag{262}$$

The four terms of $\mathscr{R}_{\log}$ contribute, respectively,

$$
\begin{aligned}
&-\frac{16}{75}\{I_{R_D}(b_*) - I_{R_D}(a_*)\},\\
&\frac{2}{45}\{I_{J_Q}(b_*)/4 - 4I_{J_Q}(a_*)\},\\
&\frac{2}{3}\{I_Q(b_*)/4 - 4I_Q(a_*)\},\\
&\frac{8}{15}\{I_Q(b_*) - I_Q(a_*)\}.
\end{aligned}
$$

Indeed the first and fourth use

$$\mathrm{Res}(\phi L_{\log} F) = 4\{4(-I_F(b_*)/4) - (1/4)(-4I_F(a_*))\} = -4\{I_F(b_*) - I_F(a_*)\}.$$

The second uses only its $8X_p$ phase, with coefficient $-8/180 = -2/45$; the 15 phase is polynomial. Its $3y_0/x$ subtraction has the zero log indices $0, p$ just proved. The third uses the X_pM difference directly.

The first and fourth contributions cancel their unweighted integral by $R_D = 5(Q - Q_s)/2$:

$$-\frac{16}{75}\Delta I_{R_D} + \frac{8}{15}\Delta I_Q = -\frac{8}{15}(\Delta I_Q - \Delta I_{Q_s}) + \frac{8}{15}\Delta I_Q = \frac{8}{15}\Delta I_{Q_s},$$

where $\Delta I_F = I_F(b_*) - I_F(a_*)$. The polynomial primitive already used for the branch is

$$I_s = \mathrm{desc}\{(2/15)t^p J_H(4t^3 - 3t)\}, \qquad I_s' = Q_s.$$

Its degree is at most $N + 3d - 1 \le p - 2$. Hence $I_{Q_s} = I_s - I_s(0)$: the difference has derivative zero, degree $< p$, and value zero at zero. At $x = b_*, a_*$, choose respectively $t = 1, 1/2$; the cubic arguments are $1, -1$, and $t^p = 1, 1/2$. Oddness of J_H gives

$$I_s(b_*) = \frac{2}{15}J_H(1), \qquad I_s(a_*) = -\frac{1}{15}J_H(1), \qquad \Delta I_{Q_s} = \frac{1}{5}J_H(1).$$

This gives the first contribution $8J_H(1)/75$. For the other two, (256) gives

$$J_Q + 15Q = l_y - 3(y - y_0)/x, \qquad I_{J_Q+15Q}(c) = P_H(V(c)) - P - 3O_y(c).$$

Thus their sum, retaining both endpoint primitives, is

$$
\begin{aligned}
\frac{2}{45}\{I_{J_Q+15Q}(b_*)/4 - 4I_{J_Q+15Q}(a_*)\} &= \frac{2}{45}\{15P/4 - 3T_O\}\\
&= \frac{1}{6}P - \frac{2}{15}T_O.
\end{aligned}
$$

We have proved the exact remaining-log row

$$\mathrm{Res}\,\mathscr{R}_{\log} = \frac{8}{75}J_H(1) + \frac{1}{6}P - \frac{2}{15}T_O. \tag{263}$$

The three logarithm groups now cancel both auxiliary primitives:

$$\left(-P + \frac{2}{15}T_O\right) + \frac{5}{6}P + \left(\frac{8}{75}J_H(1) + \frac{1}{6}P - \frac{2}{15}T_O\right) = \frac{8}{75}J_H(1).$$

The linear D and scalar origin rows. Expand the complete G, X_p factors of $\mathscr{R}_D$. Its phase-zero part is

$$\frac{5}{6}Dq_y + \frac{11}{15}D_vq_y - \frac{11}{15}wl_y + \frac{11}{5}wz_y + \frac{664}{40}DQ + \frac{440}{8}D_vQ.$$

Its remaining terms, all retained before the support test, are

$$-\frac{11}{8}GDq_y - \frac{4}{15}X_pD_vq_y + \frac{4}{15}X_pwl_y - \frac{4}{5}X_pwz_y - \frac{48X_p + 1463G}{40}DQ - \frac{112X_p + 363G}{8}D_vQ.$$

Here $664 = 4 \cdot 133 + 11 \cdot 12$ and $440 = 4 \cdot 33 + 11 \cdot 28$ use $GX_p = 1$. Every remaining term has zero residue by the degree windows above. The polynomial products Dl_y, DQ have zero residue; $\operatorname{Res}(Dz_y) = y_0D(0)$, $\operatorname{Res}(Dq_y) = -3y_0D(0)$. Their origin contribution is

$$\left(-\frac{5}{2} + \frac{11}{5}\right) y_0D(0) = -\frac{3}{10}y_0D(0).$$

For the reciprocal products use the full identity $q_y = l_y - 3z_y - 15Q$. Their sum is

$$\frac{11}{15}\operatorname{Res}(D_vq_y) - \frac{11}{15}\operatorname{Res}(D_vl_y) + \frac{11}{5}\operatorname{Res}(D_vz_y) + 55\operatorname{Res}(D_vQ) = 44J_0.$$

Here $\operatorname{Res}(DQ) = 0$ identifies $\operatorname{Res}(D_vQ) = J_0$. Therefore

$$\operatorname{Res}\mathscr{R}_D = -\frac{3}{10}y_0D(0) + 44J_0. \tag{264}$$

For the scalar log row, retain the actual bilinear phases:

$$\mathscr{R}_{\text{const}} = -\frac{1}{72}\{15\kappa G + 8\kappa - 12q - (32q/5)X_p\}q_y.$$

The G, X_p terms vanish by their support, while $\operatorname{Res} q_y = -3y_0$. Hence

$$\operatorname{Res}\mathscr{R}_{\text{const}} = \frac{8\kappa - 12q}{24}y_0 = (\kappa/3 - q/2)y_0. \tag{265}$$

Both origins in (259) have now been cancelled by actual source contributions.

The complete quadratic branch. The two mixed products $F_1 = yD_vD'$, $F_2 = ywD'_v$ have supports contained, respectively, in

$$[-(N-1),\, 3d+N-2], \qquad [-(2N-1),\, 3d+N-3].$$

Both are contained in $[-(p-2), p-2]$. Thus both G and X_p multiples of both products have zero residue. Expanding the actual phase coefficients gives

$$-\frac{2}{25}\phi B_p = -\frac{274}{25} + \frac{88}{25}(X_p + G), \qquad \frac{1}{100}\phi(133G - 12) = \frac{166}{25} - \frac{12}{25}X_p - \frac{1463}{100}G.$$

Hence the surviving quadratic source is

$$-\frac{274}{25}yD_vD' + \frac{166}{25}ywD'_v = \frac{166}{25}yww' - \frac{88}{5}ywD' + \frac{274}{25}yDD'.$$

This uses $D'_v = w' - D'$, $D_v = w - D$, and $(166 + 274)/25 = 88/5$. The last term is polynomial, so has zero residue. The first two cleared derivatives give

$$ywD' = \frac{5}{2}w(Q - Q_s), \qquad yww' = \frac{5}{2}wQ_s(x^{-N} - 1).$$

For the second, use the full reciprocal derivative $A_*w' = 5s^N(x^{-N}-1)/2$. Its reciprocal term has zero residue for a concrete degree reason. The previously derived branch factor is

$$w = x^{-(N-1)}s^{N+1}\beta_D, \qquad \deg\beta_D \le N-3.$$

Since $s^p = (1+x^p)/5$, the entire reciprocal product becomes

$$Qws^N x^{-N} = \frac{1}{5}(x^2 + x^{2-p})Q\beta_D.$$

Its coefficient at -1 asks for coefficients of $Q\beta_D$ at -3 and $p-3$. The first is zero because it is a polynomial; the second is zero because

$$\deg(Q\beta_D) \le 3d - 2 + N - 3 = p - 6 - \delta < p - 3.$$

No reciprocal phase was dropped before this identity. We have therefore proved

$$\operatorname{Res}(yww') = -\frac{5}{2}J_s, \qquad \operatorname{Res}(ywD') = \frac{5}{2}(J_0 - J_s),$$

and consequently

$$\begin{aligned}\operatorname{Res}\mathscr{R}_2 &= \frac{166}{25}\left(-\frac{5}{2}J_s\right) - \frac{88}{5}\frac{5}{2}(J_0 - J_s)\\ &= \left(-\frac{83}{5} + 44\right)J_s - 44J_0 = \frac{137}{5}J_s - 44J_0.\end{aligned} \tag{266}$$

The remaining regular rows and total. The two remaining groups multiply the polynomials R_D and Q by affine combinations of $1, G, X_p$. For the bilinear product,

$$(\alpha+\beta G)(\gamma+\nu X_p) = (\alpha\gamma+\beta\nu) + \beta\gamma G + \alpha\nu X_p.$$

The unshifted polynomial has zero residue, the G phase asks for a negative coefficient, and the X_p phase asks for coefficient $p-1$. Both $\deg R_D, \deg Q < p-1$, so all are zero. This proves $\operatorname{Res}\mathscr{R}_{\mathrm{reg,D}} = \operatorname{Res}\mathscr{R}_{\mathrm{reg,Q}} = 0$ without specializing κ, h, q. The original own-variable term $(5/24)G\,U(V)l_y$, for any polynomial U, is also polynomial with positive shift and has zero residue.

The complete contribution table, before any cancellation, is

source	residue
$y(4X_p/5 - 6/5)D\mathcal{K}'$	$3y_0D(0)/10$
$y(4qX_p/15 - \kappa/3)\mathcal{K}'$	$(q/2 - \kappa/3)y_0$
$-2X_pMy\mathcal{K}'/15$	$-P + 2T_O/15$
$-5\mathcal{K}l_y/6$	$5P/6$
$\mathscr{R}_{\log}$	$8J_H(1)/75 + P/6 - 2T_O/15$
$\mathscr{R}_D$	$-3y_0D(0)/10 + 44J_0$
$\mathscr{R}_2$	$137J_s/5 - 44J_0$
$\mathscr{R}_{\mathrm{const}}$	$(\kappa/3 - q/2)y_0$
$\mathscr{R}_{\mathrm{reg,D}},\ \mathscr{R}_{\mathrm{reg,Q}},\ (5/24)G\,U(V)l_y$	0

The $y_0D(0)$, y_0, P, T_O, and J_0 columns cancel separately. It follows for the full original source that

$$\operatorname{Res}\mathscr{R} = \frac{8}{75}J_H(1) + \frac{137}{5}J_s. \tag{267}$$

The preceding branch and Gauss calculations give $J_s = -J_H(1)/75$ and $J_H(1) = -h^2$. Thus

$$\operatorname{Res}\mathscr{R} = \left(\frac{8}{75} - \frac{137}{375}\right)J_H(1) = -\frac{97}{375}J_H(1) = \frac{97}{375}h^2,$$

with all original source and origin terms included.

The low-H and metric rows are

$$\Omega(\mathcal{H}_{\mathrm{low}}) = \frac{97}{375}h^2, \qquad \Omega(\mathcal{J}_{\mathrm{met}}) = -\frac{1213}{750}h^2. \tag{268}$$

For the first row, define the odd unit primitive $J_H(t) = \int_0^t H_0(1-z^2)/(1-z^2)\,dz$, where the integrand means $\sum_{i=1}^{d}(H_0)_i(1-z^2)^{i-1}$. All degrees are below p, so this is an ordinary polynomial primitive. The kernel terms, the logarithm terms and the two linear origin terms contract jointly to $8J_H(1)/75$; the quadratic branch contributes $137J_s/5$. The two regular sources have zero origin. The branch contraction is $J_s = -J_H(1)/75$: pull back $x = z^2$, use $t^2 = (z^2+1)/5$, and integrate the full D_s derivative by parts. The primitive becomes $(2/15)t^pJ_H(4t^3-3t)$. After subtraction of $J_H(1)t^p(4t^3-3t)$ the Euler correction is an exact polynomial primitive. The remaining two-pole trace is $J_H(1)/5$, and the substitution $x = z^2$ gives the factor 2. Thus $8J_H(1)/75 + 137J_s/5 = -97J_H(1)/375$. Here is the full conic justification for that branch contraction. The density and its primitive are

$$Q_s = \frac{4x-1}{125}\frac{H_0(V)}{V}s^N, \qquad I_s = \mathrm{desc}\{(2/15)t^pJ_H(T(t))\}, \qquad I_s' = Q_s, \qquad J_s = [x^{-1}](D_sQ_s).$$

Differentiate the primitive under $x = 5t^2-1$: $d(t^p)/dt = 0$, $T' = 3(4t^2-1)$ and $J_H'(T) = H_0(1-T^2)/(1-T^2)$; after $1-T^2 = (1-t^2)(4t^2-1)^2$ the result is $Q_s\,dx/dt$. Pulling back $x = z^2$ and taking the coefficient of z^{-1} in the product derivative gives

$$[z^{-1}]\left(D_s(z^2)'I_s(z^2)\right) + 2J_s = 0.$$

Put $t^2 = (z^2+1)/5$. The cleared derivative is

$$A_*(z^2)D_s(z^2)' = 5z\,t^{p-1}(z^{-(p-1)}-1).$$

Let $\Delta(t) = J_H(T(t)) - J_H(1)T(t)$. It is odd, has degree at most $6d-3$, and vanishes at $1,-1,1/2,-1/2$, since T has values $1,-1,-1,1$ there. These four values are distinct for $p \geq 19$, so

$$\Delta(t) = (1-t^2)(4t^2-1)P_\Delta(t), \qquad P_\Delta(-t) = -P_\Delta(t), \qquad \deg P_\Delta \leq 6d-7 \leq p-8.$$

The identity $A_*(z^2) = 25(1-t^2)(4t^2-1)$ and $5t\,dt = z\,dz$ transform the full correction differential to

$$t^p\,dD_s(z^2)\,\Delta(t) = \left\{\left(\frac{t^2}{z}\right)^p z - (t^2)^p\right\}P_\Delta(t)\,dt.$$

Neither reciprocal phase has been suppressed. For the first use the rational conic parameter $t = \eta(r_0+r_0^{-1})/2$, $z = (r_0-r_0^{-1})/2$, where $5\eta^2 = 1$. At $r_0 = 1,-1$, respectively, $z = 0$ and $dz/dr_0 = 1$; the inverse local coordinate therefore exists. For an odd polynomial P of degree $m \leq p-8$ the finite Laurent differential $P(t)z\,dt/dr_0$ has support in $[-m-3, m+1]$, and only even exponents. Its coefficient at -1 is zero. For every surviving exponent j, $|j+1| < p$ and $j+1 \neq 0$. It consequently has the explicit primitive $\sum_j F_j r_0^{j+1}/(j+1)$. Multiplication by a pth power has derivative zero and preserves the zero residue. For the second phase, the odd polynomial $P_\Delta(t)\,dt$ has a polynomial primitive: all integration denominators are at most $p-7$. Its pth-power multiplier again preserves zero residue. Residues are preserved by either inverse local coordinate because its leading coefficient is a unit. Thus the entire Δ correction contributes zero.

The remaining trace is computed without a primitive assumption. In the original z coordinate,

$$t^pD_s(z^2)'T(t) = \frac{z(1+z^{2p})(4z^2-11)}{5A_*(z^2)}(z^{-(p-1)}-1).$$

Only z^{2-p} times the even unit series can contribute at z^{-1}. The trace is therefore

$$\frac{1}{5}[x^{N-1}]\frac{4x-11}{A_*(x)} = \frac{1}{5}[x^{N-1}]\left\{-\frac{2}{3(x-1/4)} - \frac{1}{3(x-4)}\right\} = \frac{1}{5}\{(2/3)4^N + (1/3)4^{-N}\} = \frac{1}{5}.$$

Here $4^N = 2^{p-1} = 1$ in the prime field. It follows that the IBP equation is $(2/15)J_H(1)/5 + 2J_s = 0$, proving $J_s = -J_H(1)/75$ for either value of η and either conic point. To evaluate it, expanding the primitive gives

$$J_H(1) = \sum_{i=1}^{d} \frac{\bar{b}_i}{i} \sum_{j=0}^{i-1} \frac{(-1)^j \binom{i-1}{j}}{2j+1} = \frac{1}{2} \sum_{i=1}^{d} \frac{(-d)_i (d+1)_i}{(i!)^2 i^2} = -h^2.$$

For the middle equality put $\beta_i = \sum_{j=0}^{i-1} (-1)^j \binom{i-1}{j} / (2j+1)$. Use the finite polynomial functional sending z^m to $1/(m+1)$; all degrees here are at most $2d-2 < p-1$, so these divisions are units. Applied to the derivative of $z(1-z^2)^i$, for $1 \le i < d$, the functional gives its endpoint difference, which is zero. Hence

$$0 = \beta_{i+1} - 2i(\beta_i - \beta_{i+1}), \qquad \beta_1 = 1, \qquad \beta_i = \frac{(i-1)!}{2(1/2)_i}.$$

The coefficient recurrence for $\bar{b}_i$ gives, in both $p = 6d+1$ and $p = 6d+5$,

$$\bar{b}_i = \frac{(-d)_i (d+1)_i}{(i!)^2} \frac{(1/2)_i}{i!}, \qquad \frac{\bar{b}_i}{i} \beta_i = \frac{(-d)_i (d+1)_i}{2(i!)^2 i^2}.$$

This proves the displayed middle equality term by term. For the last finite identity, subtract the degree-$d-1$ sum from the degree-d sum using the adjacent factorial ratio; the difference is $-4h_{d-1}/d - 2/d^2$. Its value at $d = 0$ is zero, proving the value $-2h_d^2$ for the sum before the factor $1/2$. An explicit partial-fraction proof supplies both harmonic sums used here. For $A_i = (-1)^i \binom{d}{i} \binom{d+i}{i}$, cancellation of the simple poles at $z = -i$ and comparison of the values at infinity give

$$\prod_{j=1}^{d} \frac{j-z}{j+z} = (-1)^d - \sum_{i=1}^{d} \frac{i A_i}{z+i}.$$

At $z = -i$ the residue on the left is $(-1)^{i-1}(d+i)!/(i!(i-1)!(d-i)!) = -iA_i$. The constant, linear and quadratic coefficients at $z = 0$ are $1, -2h_d, 2h_d^2$; on the right the latter two are $\sum_i A_i/i$ and $-\sum_i A_i/i^2$. Thus $\sum_i A_i/i = -2h_d$ and $\sum_i A_i/i^2 = -2h_d^2$. Every i and every $j+z$ constant at the expansion point is a unit. This proves the first row of (268).

For the second, the three conic phases in $\mathcal{J}_{\mathrm{met}}$ give $-6h^2/125$ by (93). Indeed $H(a_*) = H(b_*) = 0$ and $P_M(0) = M(1)$ give $P_C(a_*) = -8M(1)/15$ and $P_C(b_*) = M(1)/60$. Since $\deg(x^2 P_C) \le p-2$, the three complete windows have values

$$\Omega(x^{-jp} x^2 P_C) = \frac{M(1)}{15} \{(4/15)4^{-j} + (1/30)4^j\} \quad (j = 1, 2, 3).$$

Their respective braces, after multiplication by $-33, 72, -16$, are $-33/5, 198/5, -171/5$. Their sum is $-6/5$; multiplication by $3/10$ therefore gives $-3M(1)/125 = -6h^2/125$. The unshifted phase is a polynomial and contributes zero. The metric kernel and its two Euler bands give

$$[x^{-1}]\mathcal{M} = \frac{1199}{450}(2C_M)(a_*) - \frac{583}{225}(2C_M)(b_*) = -\frac{1177}{1500} M(1).$$

Their origin value $(2C_M)(0)$ cancels only after the bands are added. The endpoint values are $(2C_M)(a_*) = -M(1)/10$ and $(2C_M)(b_*) = M(1)/5$. Finally The exact finite expression for this same M is

$$M = -2 \sum_{j=0}^{d-1} (Y_{j+1} n_{j+1} + Y_j n_j)/(j+1).$$

Here is its polynomial attachment. Writing $q_t = 1 - t^2$, the four adjacent derivative identities are

$$q_t Y_j' = (j+1)(tY_j - Y_{j+1}), \qquad q_t Y_{j+1}' = (j+1)(Y_j - tY_{j+1}),$$
$$q_t n_j' = (j+1)(tn_j - n_{j+1}) - Y_j, \qquad q_t n_{j+1}' = (j+1)(n_j - tn_{j+1}) - Y_{j+1}.$$

They follow from the coefficient recurrence at $j = 0$ and induction using the two adjacent recurrences: differentiating the recurrence at $j+1$ gives its new upper derivative after division by $j+2$. Every $j+2 \le d$ is a unit. Multiplying and adding these four rows gives

$$q_t(Y_{j+1}^2 + Y_j^2)' = -2(j+1)t(Y_{j+1}^2 - Y_j^2),$$
$$q_t(Y_{j+1}n_{j+1} + Y_j n_j)' = -2(j+1)t(Y_{j+1}n_{j+1} - Y_j n_j) - Y_{j+1}^2 - Y_j^2.$$

For $H_\Sigma = \sum_{j=0}^{d-1}(Y_{j+1}^2 + Y_j^2 - 2)/(j+1)$, the first row telescopes to $q_t H_\Sigma' = -2t(Y_d^2 - 1)$. The finite definition of H_0 and (102) give exactly $q_t\{H_0(1-t^2)\}' = -2t(Y_d^2 - 1)$. Both polynomials have degree at most $2d < p$ and vanish at 1; cancelling q_t and then the unit derivative indices proves $H_\Sigma = H_0(1-t^2)$. Summing the second row, multiplied by $-2/(j+1)$, gives

$$q_t\left\{-2\sum_{j=0}^{d-1}\frac{Y_{j+1}n_{j+1} + Y_j n_j}{j+1}\right\}' = 4tY_d n_d + 2H_\Sigma + 4h = U.$$

Its degree is less than p and its origin is zero by oddness. It therefore equals the primitive M previously defined. Evaluating at 1 gives $M(1) = 2h^2$. More explicitly, $n_j(1) = -h_j$ and $Y_j(1) = 1$ give

$$M(1) = 2\sum_{j=0}^{d-1}\frac{h_{j+1} + h_j}{j+1} = 2\sum_{j=0}^{d-1}(h_{j+1}^2 - h_j^2) = 2h_d^2.$$

Consequently $-6h^2/125 - 1177h^2/750 = -1213h^2/750$. The sufficient degree bound used for P_C is $p-4$. No stronger top-degree cancellation is required. For completeness the metric contraction before its endpoint evaluation is as follows. Put $m = 2C_M = s^{N+1}g$, where $g = (4x-11)P_M(V)/25$. Then $\deg g \le N-2$, $\deg m \le p-2$, $g(a_*) = 4m(a_*)$, $g(b_*) = m(b_*)$ and $ms^N = (x^p+1)g/5$. With $m_a = m(a_*)$, $m_b = m(b_*)$, $m_0 = m(0)$, the cleared two Euler derivatives are

$$U_m = \frac{3}{2}\{m(4x-11) + 11g/5 + 7x^p g/5 - 4x^{2p}g/5\} = mA_* a',$$
$$V_m = -\frac{3}{2}(4m/x - 11m) + \frac{3}{10}\{-7x^{-N}g + 4x^{-N-p}g - 11x^{p-N}g\} = mA_*(a^\vee)'.$$

These are (95) with $ms^N = (x^p+1)g/5$ substituted. Applying (93) separately to their nonpolynomial bands gives

$\Omega(x^{-p}U_m)$	$(24m_a + 9m_b)/50$
$\Omega(V_m)$	$3m_0/2 - 18m_a/25 - 3m_b/25$
$\Omega(x^pV_m)$	$2(m_b - 4m_a)/25$
$[x^{-1}]\{-\varphi m/(6x)\}$	$11m_0/6.$

For the first row only $x^{-p}\{m(4x-11) + 11g/5\}$ survives. For the second the three negative shifts are $1, N, N+p$; their values use $a_*^N = b_*^N = 1$, $g(a_*) = 4m_a$, $g(b_*) = m_b$. After multiplication by x^p, precisely the $-N-p$ band survives in the third row. In the last row $m_p = 0$ by $\deg m \le p-2$. All positive-shift summands are polynomials and have $\Omega = 0$. The Euler and origin contraction is therefore

$$-\frac{4}{9}\frac{24m_a + 9m_b}{50} - \frac{25}{18}\left(\frac{3}{2}m_0 - \frac{18}{25}m_a - \frac{3}{25}m_b\right) + \frac{55}{24}\frac{2}{25}(m_b - 4m_a) + \frac{11}{6}m_0 = -\frac{1}{4}m_0 + \frac{4}{75}m_a + \frac{27}{100}m_b.$$

The terms with positive phase in U_m have zero residue, so the coefficient $4/9$ is the full negative-phase coefficient. For the first metric kernel no general residue theorem is needed: write $m = \sum_{i=0}^{p-2} m_i x^i$ in (96). The two products contributing at x^{-1} have logarithm index $p-i$, $1 \le i \le p-2$. Their respective sums are

$$\frac{29}{6}(m_a - m_0) - \frac{23}{6}(m_b - m_0), \qquad -\frac{20}{9}(m_a - m_0) + \frac{35}{36}(m_b - m_0).$$

Thus $[x^{-1}]\{-m(1-2x^{-p}/3)\mathcal{K}'\} = 47m_a/18 - 103m_b/36 + m_0/4$. Adding the Euler row cancels m_0 and gives $1199m_a/450 - 583m_b/225$ as asserted. The half-scale $C_M = m/2$ in (107) remains unchanged.

The end products are

$$\begin{aligned} G_B := \Omega(\mathcal{G}_B) = & -\frac{1}{20}G(1/5) + \frac{1}{10}R(-2) + \frac{7}{15}R(16) \\ & + \frac{313}{3}q^2 - \frac{27}{10}qr - 13qk - \frac{15}{2}rk \\ & - \frac{15}{4}k^2 + \frac{2}{5}qC + \frac{3}{10}rC, \end{aligned} \tag{269}$$

$$\begin{aligned} G_2 := \Omega(\mathcal{G}_2) = & \frac{5}{12}G(1/5) - \frac{5}{6}R(-2) - \frac{1}{20}R(16) \\ & - \frac{193}{50}q^2 - \frac{9}{100}qr - \frac{177}{800}r^2 \\ & + 4qk + \frac{11}{15}C^2 - \frac{10}{3}qC - \frac{5}{2}rC. \end{aligned} \tag{270}$$

6.12 The complete end-product calculation for G_B

We prove (269) using the full bands in (96). Retain the notation $a_* = 1/4$, $b_* = 4$, $q = q_2$, $r = q_3$, $k = q_5$, $h = -2q - 3r/2$. Put

$$\mathcal{L}_a(x) = \ell(x/4), \qquad \mathcal{L}_b(x) = \ell(4x), \qquad E_D(x) = -3D(x), \qquad \lambda = \ell(16) = 15(r+k) - 64q.$$

All the equalities below are in K, of characteristic $p \geq 19$. Every integer denominator occurring below has only the prime factors $2, 3, 5$, so is a unit. Each finite summation denominator lies between 1 and $p-1$. The arguments $16, 1/16$ differ from $0, 1$: their relevant differences have numerator 15. All rational arguments are fixed by the p-power map.

The two source bands and their actual endpoints. The entire source is

$$\begin{aligned} \chi_0 &= W_0 + x^{-p}W_1, \\ W_0 &= -\frac{6}{5}D + \frac{11q}{15} - \frac{k}{2} = \frac{2}{5}E_D + c_0, \qquad c_0 = \frac{11q}{15} - \frac{k}{2}, \\ W_1 &= \frac{4}{5}D - \frac{2}{15}(\mathcal{L}_a - \mathcal{L}_b) + \frac{4q}{15} = -\frac{4}{15}E_D - \frac{2}{15}(\mathcal{L}_a - \mathcal{L}_b) + \frac{4q}{15}. \end{aligned} \tag{271}$$

Here is the coefficient cancellation attaching these bands to the defined $\chi_0 = \mathcal{P} - (a+h)/3$. The recipe for W in (94) gives $W/\epsilon = -2D/5 - 2C/15$. Its substitution in the three coefficients of $\mathcal{P}$ gives

$$\mathcal{P} = x^p\left(\frac{4}{5}D - \frac{4q}{15}\right) - \frac{17}{5}D + \frac{2}{15}(4\mathcal{L}_a - \mathcal{L}_b/4) + x^{-p}\left(\frac{4}{5}D + \frac{4q}{15} + \frac{2}{15}(\mathcal{L}_b - \mathcal{L}_a)\right).$$

The whole cubic identity for a is

$$a = x^p\left(\frac{12}{5}D - \frac{4q}{5}\right) - \frac{33}{5}D + \frac{2}{5}(4\mathcal{L}_a - \mathcal{L}_b/4) + \frac{3k}{2} - \frac{11q}{5} - h.$$

Thus its x^p band cancels exactly in $\mathcal{P} - a/3$; the regular D coefficient becomes $-17/5 + 11/5 = -6/5$, the regular logarithm coefficient becomes $2/15 - (2/5)/3 = 0$, and its constant after subtracting $h/3$ is $-k/2 + 11q/15$. Its negative band is unchanged. This proves (271), including its coefficient at $-p$. The degree bounds are

$$\deg E_D = \deg D \leq N-1, \quad \deg W_0 \leq N-1, \quad \deg W_1 \leq p-1, \quad \deg K_i \leq p-1 \quad (i = 0, 1, 2).$$

The last term in the defining sum for D vanishes because $4^N = 2^{p-1} = 1$.

The half-length logarithms already evaluated in the finite consumer give $L_N(1) = -2q$, $L_N(4) = -3r$, $L_N(1/4) = 2q - 3r/2$. Since $s(a_*) = 1/4$, $s(b_*) = 1$, the defining sum for D yields

$$D(a_*) = \frac{L_N(1/4) - L_N(1)}{6} = \frac{2q}{3} - \frac{r}{4} = q + \frac{h}{6}, \qquad D(b_*) = \frac{L_N(1) - L_N(4)}{6} = -\frac{q}{3} + \frac{r}{2} = -q - \frac{h}{3}.$$

At zero the same sum and the fifth entropy identity give $D(0) = -C/3$. Consequently

$$E_D(a_*) = -3q - \frac{h}{2}, \qquad E_D(b_*) = 3q + h, \qquad E_D(0) = C. \tag{272}$$

Logarithm inversion and $\ell(1) = 0$ give the complete endpoint table

	a_*	b_*
$\mathcal{L}_a$	$-\lambda/16$	0
$\mathcal{L}_b$	0	λ
W_0	$-7q/15 - h/5 - k/2$	$29q/15 + 2h/5 - k/2$
W_1	$16q/15 + 2h/15 + \lambda/120$	$-8q/15 - 4h/15 + 2\lambda/15$

In particular, without assigning a residue in advance,

$$W_0(a_*) + 4W_1(a_*) = \frac{19q}{5} + \frac{h}{3} - \frac{k}{2} + \frac{\lambda}{30} = q,$$
$$W_0(b_*) + \frac{1}{4}W_1(b_*) = \frac{9q}{5} + \frac{h}{3} - \frac{k}{2} + \frac{\lambda}{30} = -q.$$

Both last equalities use the exact relation $\lambda + 84q + 10h - 15k = 0$, which follows by inserting $h = -2q - 3r/2$ into $\lambda = 15(r + k) - 64q$. The phase factors here are $a_*^{-p} = 4$, $b_*^{-p} = 1/4$.

Full regular parts and the inclusive p slot. For a polynomial $F = \sum F_j x^j$, define

$$F^{<p} = \sum_{0 \le j < p} F_j x^j, \qquad F^{\ge p} = \sum_{j \ge p} F_j x^j, \qquad I_p(F) = \sum_{j \ge 0} F_{p+j} x^j.$$

Thus $F^{\ge p} = x^p I_p(F)$, and $I_p(F) = F_p + \sum_{j \ge 1} F_{p+j} x^j$. No constant F_p is removed. For a Laurent polynomial U, write $U_+ = \sum_{j \ge 0} U_j x^j$. At $c \neq 0$, the Laurent polynomial $\Delta_c U = (U(x) - U(c))/(x - c)$ satisfies

$$[x^{-1}]\Delta_c U = U_+(c) - U(c). \tag{273}$$

For x^n with $n \ge 0$ the divided difference is a polynomial. For $n = -m < 0$,

$$\frac{x^{-m} - c^{-m}}{x - c} = -\sum_{t=0}^{m-1} x^{t-m} c^{-t-1},$$

whose coefficient at -1 is $-c^{-m}$. This proves (273) coefficient by coefficient.

Put $F = \mathcal{K}\chi_0$. Its four entire phases are

$$F = K_0 W_0 + x^{-p}(K_1 W_0 + K_0 W_1) + x^{-2p}(K_2 W_0 + K_1 W_1) + x^{-3p} K_2 W_1.$$

The numerators of the last two terms have degree $< 2p$ and $< 3p$, respectively. Hence their regular parts are zero, while

$$\mathcal{K}_+ = K_0, \qquad F_+ = K_0 W_0 + I_p(K_1 W_0 + K_0 W_1).$$

This retains the constant term of I_p. Directly from (91), for every integer $j < 0$,

$$\Omega(Lx^j) = -2a_*^j - b_*^j, \qquad \Omega((31q - 4qx)x^j) = -2qa_*^j + qb_*^j.$$

For $j \geq 0$ both sides vanish when interpreted as the coefficients of the divided differences. Thus

$$\Omega(LF) = 2[x^{-1}]\Delta_{a_*}F + [x^{-1}]\Delta_{b_*}F, \quad \Omega((31q - 4qx)\mathcal{K}) = 2q[x^{-1}]\Delta_{a_*}\mathcal{K} - q[x^{-1}]\Delta_{b_*}\mathcal{K}.$$

Since $\chi_0(a_*) = q$, $\chi_0(b_*) = -q$, equation (273) cancels the full endpoint values:

$$\begin{aligned}\mathcal{D}_B &:= -\Omega(L\mathcal{K}\chi_0) + \Omega((31q - 4qx)\mathcal{K}) \\ &= -2F_+(a_*) - F_+(b_*) + 2qK_0(a_*) - qK_0(b_*) \\ &= 2a_*^{-p}\{K_0W_1 - (K_1W_0 + K_0W_1)^{\geq p}\}(a_*) \\ &\quad + b_*^{-p}\{K_0W_1 - (K_1W_0 + K_0W_1)^{\geq p}\}(b_*).\end{aligned}$$

Here, for example, $q - W_0(a_*) = a_*^{-p}W_1(a_*)$ and $I_p(J)(a_*) = a_*^{-p}J^{\geq p}(a_*)$. The constant part of W_0 produces no high tail because $\deg K_1 < p$. We have therefore proved

$$\mathcal{D}_B = 8\mathcal{T}(a_*) + \frac{1}{4}\mathcal{T}(b_*), \qquad \mathcal{T} = (K_0W_1)^{<p} - \frac{2}{5}(K_1E_D)^{\geq p}. \tag{274}$$

Reversing the full E_D-logarithm tail. For $\deg Q < p$, $c \neq 0$, expand a product before evaluating:

$$(Q\ell(cx))^{\geq p}(z) = \sum_{i=1}^{p-1} Q_i \sum_{j=p-i}^{p-1} \frac{c^j z^{i+j}}{j}.$$

Replace j by $p - t$. The limits become $1 \leq t \leq i$ and $1/(p-t) = -1/t$, giving

$$(Q\ell(cx))^{\geq p}(z) = -(cz)^p \sum_{i=1}^{p-1} Q_i \sum_{t=1}^{i} \frac{z^{i-t}c^{-t}}{t} = -(cz)^p H_Q(z, c^{-1}). \tag{275}$$

The term $t = i$ is exactly the original coefficient at degree p; it contributes $-c^pQ_ic^{-i}/i$ to that coefficient. The constant Q_0 contributes no high term because $\ell(cx)$ has degree $p - 1$. At the rational endpoints $(cz)^p = cz$. With $\mathsf{H} = H_{E_D}$, (275) gives all four tails:

	a_*	b_*
$(E_D\mathcal{L}_a)^{\geq p}$	$-\mathsf{H}(a_*, b_*)/16$	$-\mathsf{H}(b_*, b_*)$
$(E_D\mathcal{L}_b)^{\geq p}$	$-\mathsf{H}(a_*, a_*)$	$-16\mathsf{H}(b_*, a_*)$

(E-tail)

The strict logarithm shuffle. Use the full triangular sum already defined in the finite consumer and put

$$T_2(v, w) = D_{p-1}(v, w) + D_{p-1}(w, v) - R(v) - R(w).$$

Removing the $k = j$ diagonals and reversing $k \mapsto j - k$ in the second off-diagonal sum gives

$$T_2(v, w) = \sum_{j=1}^{p-1} \sum_{i=1}^{j-1} \frac{v^i w^{j-i}}{i(j-i)}.$$

Indeed its combined denominator is

$$\frac{1}{ij} + \frac{1}{(j-i)j} = \frac{1}{i(j-i)}.$$

The rectangular product $\ell(cx)\ell(dx)$, truncated at total degree $< p$, consequently satisfies

$$(\ell(cx)\ell(dx))^{<p}(z) = T_2(cz, dz). \tag{276}$$

All denominators in this triangle are units.

For a rational $t \neq 0, 1$, reflection and inversion in (66) give

$$tR(t) + (1-t)R(1-t) = -\ell(t)^2/2,$$

$$R(t) + t(t-1)R((t-1)/t) = -\ell(t)^2/2.$$

The second equality is reflection at t^{-1}, multiplied by t^2, using $R(t^{-1}) = R(t)/t$, $\ell(t^{-1}) = -\ell(t)/t$. Subtract the first equality from the second and divide by $t-1$; this proves

$$tR((t-1)/t) = R(t) - R(1-t). \tag{277}$$

In the full homogeneous identity (67), use

$$H_2(-t, 1-t) = -t^p R((t-1)/t).$$

This follows by $H_2(a,b) = b^p R(a/b)$ and inversion at $-t/(1-t)$; p is odd, so $(-t)^p = -t^p$. The homogeneous diagonal $H_2(v,v)$ is zero: for $v \neq 0$ it is $v^p R(1) = 0$, and for $v = 0$ every summand is zero. Substitution now gives

$$D_{p-1}(t,t) = tR((t-1)/t), \quad D_{p-1}(1,t) = 2tR((t-1)/t), \quad D_{p-1}(t,1) = R(1-t).$$

Together with (277), these prove the two exact shuffles

$$T_2(t,t) = -2R(1-t), \qquad T_2(1,t) = R(t) - R(1-t).$$

Also $T_2(1,1) = 0$ by $R(0) = R(1) = 0$, and $T_2(v,w) = T_2(w,v)$ from its definition.

Write $R_{16} = R(16)$. Inversion gives $R(1/16) = R_{16}/16$. Reflection at $1/16$ and at 16 respectively gives

$$\frac{R_{16}}{256} + \frac{15}{16}R(15/16) = -\frac{\lambda^2}{512}, \qquad 16R_{16} - 15R(-15) = -\frac{\lambda^2}{2}.$$

Solving these two equations yields

$$R(15/16) = -\frac{\lambda^2}{480} - \frac{R_{16}}{240}, \qquad R(-15) = \frac{\lambda^2}{30} + \frac{16R_{16}}{15}.$$

Thus every required strict low value is explicitly

$$\begin{aligned}
(\mathcal{L}_a^2)^{<p}(a_*) &= T_2(1/16, 1/16) = \frac{\lambda^2}{240} + \frac{R_{16}}{120}, \\
(\mathcal{L}_a\mathcal{L}_b)^{<p}(a_*) &= T_2(1/16, 1) = \frac{\lambda^2}{480} + \frac{R_{16}}{15}, \\
(\mathcal{L}_a\mathcal{L}_b)^{<p}(b_*) &= T_2(1, 16) = -\frac{\lambda^2}{30} - \frac{R_{16}}{15}, \\
(\mathcal{L}_b^2)^{<p}(b_*) &= T_2(16, 16) = -\frac{\lambda^2}{15} - \frac{32R_{16}}{15}.
\end{aligned} \tag{278}$$

The other two square values are $(\mathcal{L}_a^2)^{<p}(b_*) = (\mathcal{L}_b^2)^{<p}(a_*) = T_2(1,1) = 0$.

The three original product blocks. Multiply $K_0 = (35\mathcal{L}_a - 5\mathcal{L}_b)/6$ by the W_1 in (271). The scalar products

$$\frac{1}{6}\left(-\frac{4}{15}\right) = -\frac{2}{45}, \quad \frac{1}{6}\left(-\frac{2}{15}\right) = -\frac{1}{45}, \quad \frac{1}{6}\frac{4q}{15} = \frac{2q}{45}$$

give the exact decomposition

$$\begin{aligned}
K_0 W_1 &= J_E + J_{\log} + J_{\lin}, \\
J_E &= -\frac{2}{45}(35\mathcal{L}_a - 5\mathcal{L}_b)E_D,
\end{aligned}$$

$$J_{\log} = -\frac{1}{45}(35\mathcal{L}_a^2 - 40\mathcal{L}_a\mathcal{L}_b + 5\mathcal{L}_b^2),$$
$$J_{\lin} = \frac{2q}{45}(35\mathcal{L}_a - 5\mathcal{L}_b).$$

The middle polynomial uses $(35\mathcal{L}_a - 5\mathcal{L}_b)(\mathcal{L}_a - \mathcal{L}_b) = 35\mathcal{L}_a^2 - 40\mathcal{L}_a\mathcal{L}_b + 5\mathcal{L}_b^2$. For the E_D part of $\mathcal{T}$, subtract both entire high products:

$$\begin{aligned}\mathcal{T}_E &= J_E^{<p} - \frac{2}{5}(K_1 E_D)^{\geq p} \\ &= -\frac{14}{9}E_D\mathcal{L}_a + \frac{2}{9}E_D\mathcal{L}_b - \frac{206}{45}(E_D\mathcal{L}_a)^{\geq p} + \frac{47}{180}(E_D\mathcal{L}_b)^{\geq p}.\end{aligned}$$

The high coefficients are

$$\frac{14}{9} - \frac{92}{15} = -\frac{206}{45}, \qquad -\frac{2}{9} + \frac{29}{60} = \frac{47}{180}.$$

Using the endpoint logs and the four inclusive tails gives

$$8\mathcal{T}_E(a_*) = \frac{7\lambda E_D(a_*)}{9} + \frac{103}{45}\mathsf{H}(a_*, b_*) - \frac{94}{45}\mathsf{H}(a_*, a_*),$$
$$\frac{1}{4}\mathcal{T}_E(b_*) = \frac{\lambda E_D(b_*)}{18} + \frac{103}{90}\mathsf{H}(b_*, b_*) - \frac{47}{45}\mathsf{H}(b_*, a_*).$$

This is the inclusive p-slot contraction, with no primitive origin term discarded.

For the logarithm block, substitute all six strict low values:

$$J_{\log}^{<p}(a_*) = -\frac{1}{45}\left\{35\left(\frac{\lambda^2}{240} + \frac{R_{16}}{120}\right) - 40\left(\frac{\lambda^2}{480} + \frac{R_{16}}{15}\right)\right\} = -\frac{\lambda^2}{720} + \frac{19R_{16}}{360},$$
$$J_{\log}^{<p}(b_*) = -\frac{1}{45}\left\{-40\left(-\frac{\lambda^2}{30} - \frac{R_{16}}{15}\right) + 5\left(-\frac{\lambda^2}{15} - \frac{32R_{16}}{15}\right)\right\} = -\frac{\lambda^2}{45} + \frac{8R_{16}}{45}.$$

Multiplying by $8, 1/4$ gives

$$8J_{\log}^{<p}(a_*) + \frac{1}{4}J_{\log}^{<p}(b_*) = -\left(\frac{1}{90} + \frac{1}{180}\right)\lambda^2 + \left(\frac{19}{45} + \frac{2}{45}\right)R_{16} = -\frac{\lambda^2}{60} + \frac{7}{15}R_{16}.$$

The linear block has degree $< p$, so its high part is zero. Its actual logs give

$$J_{\lin}(a_*) = -\frac{7q\lambda}{72}, \qquad J_{\lin}(b_*) = -\frac{2q\lambda}{9}, \qquad 8J_{\lin}(a_*) + \frac{1}{4}J_{\lin}(b_*) = -\frac{5}{6}q\lambda.$$

We have proved

$$\begin{aligned}\mathcal{D}_B = &-\frac{94}{45}\mathsf{H}(a_*, a_*) - \frac{47}{45}\mathsf{H}(b_*, a_*) + \frac{103}{45}\mathsf{H}(a_*, b_*) + \frac{103}{90}\mathsf{H}(b_*, b_*) \\ &+ \frac{\lambda}{18}\{14E_D(a_*) + E_D(b_*)\} - \frac{\lambda^2}{60} + \frac{7}{15}R_{16} - \frac{5}{6}q\lambda. \qquad (279)\end{aligned}$$

The four moving primitives. Substitute the first four rows of (101) into the first line of (279). The complete expression is

$$\begin{aligned}&-\frac{97}{60}S(1,1) + \frac{47}{45}S(1/2,1/2) - \frac{47}{90}S(1,2) + \frac{47}{90}S(1/2,1) \\ &\quad + \frac{103}{90}S(2,1) - \frac{103}{90}S(1,1/2) + \frac{103}{180}S(2,2) - \frac{1}{10}A(u) - \frac{1}{20}B(u).\end{aligned}$$

For example the $A(u)$ coefficient is $94/90 - 103/90 = -1/10$, and the $B(u)$ coefficient is $47/90 - 103/180 = -1/20$. Since $u^2 = 1/5$, the defining triangular sums give $G(1/5) = B(u) + 2A(u)$; these two terms therefore give $-G(1/5)/20$. For the seven rational rows in (70), expand the two

occurrences of $(2q-3r)^2$. Multiplication by the displayed coefficients gives the following complete table; its columns are coefficients of $R(-2), q^2, qr, r^2$:

$S(1,1)$	0	$-97/30$	0	0
$S(1/2,1/2)$	$-47/135$	$47/135$	$-47/45$	$47/60$
$S(1,2)$	$-47/135$	$47/135$	$94/45$	$-47/30$
$S(1/2,1)$	$-47/135$	$-376/135$	$94/45$	$-47/120$
$S(2,1)$	$-206/135$	$206/135$	$103/45$	$-103/60$
$S(1,1/2)$	$103/135$	$-103/135$	$103/45$	$-103/60$
$S(2,2)$	$103/54$	$-103/54$	$103/45$	$103/120$
sum	$1/10$	$-97/15$	10	$-15/4$

Hence the four-H part is exactly

$$-\frac{1}{20}G(1/5)+\frac{1}{10}R(-2)-\frac{97}{15}q^2+10qr-\frac{15}{4}r^2. \tag{280}$$

The derivative coefficient, including its invisible slots. Let $U=\sum U_j x^j$ be supported in $[-p,2p-1]$. Define its full endpoint weight

$$b(j)=\frac{29}{6}a_*^j-\frac{23}{6}b_*^j+2\,\mathbf{1}_{j<0}(-2a_*^j-b_*^j).$$

The literal coefficients of the three logarithm bands in $\mathcal{K}$, before any reduction of their cutoffs, are

$$\mathcal{K}_n=\frac{1}{6}\left\{\begin{array}{l}\dfrac{35a_*^n-5b_*^n}{n}\mathbf{1}_{1\le n<p}\\ +\dfrac{92a_*^{n+p}-(29/4)b_*^{n+p}}{n+p}\mathbf{1}_{1\le n+p<p}\\ +\dfrac{368a_*^{n+2p}-(29/16)b_*^{n+2p}}{n+2p}\mathbf{1}_{1\le n+2p<p}\end{array}\right\}.$$

An inactive summand is defined to be zero, without division. For active summands $n+tp$ is a unit. The substitutions $a_*^p=a_*$, $b_*^p=b_*$, $a_*^{-j}=b_*^j$, $b_*^{-j}=a_*^j$ give, at $n=-j$,

j	active band	$j\mathcal{K}_{-j}$
$-p<j<0$	0	$(5a_*^j-35b_*^j)/6=b(j)$
$0<j<p$	1	$(29a_*^j-23b_*^j)/6=b(j)$
$p<j<2p$	2	$(29a_*^j-23b_*^j)/6=b(j)$

For example the band 1 numerator is $92a_*a_*^{-j}-(29/4)b_*b_*^{-j}=23b_*^j-29a_*^j$, while its denominator is $-j$. Band 2 has $368a_*^2=23$ and $(29/16)b_*^2=29$, giving the same result. At $j=0,-p,p$ all log cutoffs are inactive and $j=0$ in K; the derivative of the corresponding monomial is zero. Their full weights, however, are

$$b(0)=1,\qquad b(-p)=\frac{5}{6}\cdot 4-\frac{35}{6}\cdot\frac{1}{4}=\frac{15}{8},\qquad b(p)=\frac{29}{6}\cdot\frac{1}{4}-\frac{23}{6}\cdot 4=-\frac{113}{8}.$$

Summing $[x^{-1}](\mathcal{K}(U_jx^j)')=jU_j\mathcal{K}_{-j}$ therefore proves, with all three slots retained,

$$\begin{aligned}[x^{-1}](\mathcal{K}U')=&\frac{29}{6}U(a_*)-\frac{23}{6}U(b_*)+2\{-2U_-(a_*)-U_-(b_*)\}\\ &-U_0-\frac{15}{8}U_{-p}+\frac{113}{8}U_p,\end{aligned} \tag{281}$$

where $U_-=\sum_{j<0}U_jx^j$. This coefficient identity also holds over any characteristic-p extension field: all coefficient and denominator operations commute with the injective homomorphism from its prime field.

For the whole $U = \chi_0$ in (271), the support is $[-p, N-1]$, $U_- = x^{-p}W_1$, and the three coefficients are

$$U_0 = -\frac{6}{5}D(0) + \frac{11q}{15} - \frac{k}{2}, \qquad U_{-p} = \frac{4}{5}D(0) + \frac{4q}{15}, \qquad U_p = 0.$$

Using the actual endpoint values proved above, the first line of (281) is

$$\begin{aligned}\frac{26q}{3} - 16W_1(a_*) - \frac{1}{2}W_1(b_*) &= \left(\frac{26}{3} - \frac{256}{15} + \frac{4}{15}\right)q \\ &\quad + \left(-\frac{32}{15} + \frac{2}{15}\right)h + \left(-\frac{2}{15} - \frac{1}{15}\right)\lambda \\ &= -\frac{122}{15}q - 2h - \frac{\lambda}{5}.\end{aligned}$$

Subtracting the $0, -p$ slots gives

$$\begin{aligned}[x^{-1}](\mathcal{K}\chi_0') &= -\frac{3}{10}D(0) - \frac{281}{30}q - 2h + \frac{k}{2} - \frac{\lambda}{5} \\ &= \frac{C}{10} + \frac{223}{30}q - \frac{5k}{2}.\end{aligned}$$

The last substitution uses $D(0) = -C/3$, $-2h = 4q + 3r$, and $-\lambda/5 = 64q/5 - 3r - 3k$. In particular the r coefficient is $3 - 3 = 0$, and the q coefficient is $-281/30 + 4 + 64/5 = 223/30$.

The final coefficient collection. The definition in (181) and $\Omega(A_*F) = [x^{-1}]F$ give

$$G_B = \mathcal{D}_B - 2h[x^{-1}](\mathcal{K}\chi_0').$$

By (272), $14E_D(a_*) + E_D(b_*) = -39q - 6h$. Insert (280) and the derivative value into (279). Before expanding either h or λ, this is

$$\begin{aligned}G_B = &-\frac{G(1/5)}{20} + \frac{R(-2)}{10} + \frac{7}{15}R_{16} - \frac{97}{15}q^2 + 10qr - \frac{15}{4}r^2 \\ &+ \left(-3q - \frac{h}{3}\right)\lambda - \frac{\lambda^2}{60} - \frac{h}{5}C - \frac{223}{15}qh + 5hk.\end{aligned}$$

Here the $q\lambda$ coefficient is $-39/18 - 5/6 = -3$, and the $h\lambda$ coefficient is $-6/18 = -1/3$. Substitute $-3q - h/3 = -7q/3 + r/2$ and $\lambda = -64q + 15r + 15k$. The complete scalar expansion is

	q^2	qr	qk	r^2	rk	k^2	qC	rC
$-97q^2/15 + 10qr - 15r^2/4$	$-97/15$	10	0	$-15/4$	0	0	0	0
$(-7q/3 + r/2)\lambda$	448/3	-67	-35	15/2	15/2	0	0	0
$-\lambda^2/60$	$-1024/15$	32	32	$-15/4$	$-15/2$	$-15/4$	0	0
$-223qh/15$	446/15	223/10	0	0	0	0	0	0
$5hk$	0	0	-10	0	$-15/2$	0	0	0
$-hC/5$	0	0	0	0	0	0	2/5	3/10
sum	313/3	$-27/10$	-13	0	$-15/2$	$-15/4$	2/5	3/10

For example the quadratic coefficient is $(-97-1024+446)/15+448/3 = 313/3$, and the r^2 coefficient is $-15/4 + 15/2 - 15/4 = 0$. Thus the displayed value (269) follows:

$$G_B = -\frac{G(1/5)}{20} + \frac{R(-2)}{10} + \frac{7}{15}R(16) + \frac{313}{3}q^2 - \frac{27}{10}qr - 13qk - \frac{15}{2}rk - \frac{15}{4}k^2 + \frac{2}{5}qC + \frac{3}{10}rC.$$

If the full original origin term $[x^{-1}]\{\mathcal{K}(r_0 - h)x^{-1}\}$ is retained, it equals $(r_0 - h)\mathcal{K}_0 = 0$, because every one of the three log bands has coefficient zero at its excluded endpoints $0, p, 2p$. No assumption on $r_0 - h$ is used.

6.13 The full five-tail calculation for G_2

We first expand the five tails giving G_2. For a finite Laurent polynomial define

$$T_j(F) = \sum_{m>jp} F_m\{2\,4^{m-jp} + 4^{-(m-jp)}\}.$$

There is no upper cutoff, and the slot $m = jp$ is excluded. Directly from this definition,

$$T_0(F) = 2F(4) + F(1/4) - 3F_0 \quad (F \in K[x]), \qquad T_j(x^{jp}F) = T_0(F).$$

The reciprocity rule (92), applied coefficient by coefficient to $L = 33 - 12x$, gives $\Omega(Lx^{jp}F^\vee) = -T_j(F)$. To verify its weight, a monomial $F = x^m$ contributes $-(2\,4^{m-jp} + 4^{-(m-jp)})$ if $m > jp$ and zero otherwise: substitute the two coefficients $33x^{jp-m} - 12x^{jp-m+1}$ in (91). This proves the rule on every full signed support.

The original Z_{52} source has the identity

$$Z_{52} = a^2 - (a^\vee)^2 + \frac{6}{5}\varphi a^\vee(D + D^\vee).$$

Multiplication by $5x^p/24$, using $x^p\varphi = 4 - 11x^p$, leaves, after removing the polynomial $L(5x^p a^2/24 - 2D(q-h)/3)$,

$$L\left\{\left(\frac{5}{3} - \frac{11}{4}x^p\right)a^\vee D + \left(1 - \frac{11}{4}x^p\right)a^\vee D^\vee - \frac{5}{24}x^p(a^\vee)^2\right\}.$$

The polynomial removed has $\Omega = 0$. The original R_{52} source similarly gives

$$\mu = -h\Omega\left(s^N(25/3 - 55x^p/4)a^\vee\right);$$

the omitted summand here is exactly $s^N\{-5hx^p(4x^p - 11)a/4 + 10h(q-h)/3\}$, a polynomial. Thus

$$G_2 = -\frac{5}{3}T_0(aD^\vee) - T_0(aD) + \frac{11}{4}\{T_1(aD^\vee) + T_1(aD)\} + \frac{5}{24}T_1(a^2) + \mu. \tag{282}$$

For all five tails use the actual two bands, not an arbitrary polynomial with the same endpoints:

$$a = x^p B_n + L_n, \quad B_n = 12D/5 - 4q/5, \quad L_n = c_0 - 33D/5 + F_m,$$
$$F_m = \frac{2}{5}(4\ell(x/4) - \ell(4x)/4), \quad c_0 = 3k/2 - 11q/5 - h.$$

Here $\deg B_n \le N - 1$, $\deg L_n < p$, and $\tau = (4q - 3r)/5$ gives

	$1/4$	4	0
D	$q + h/6$	$-q - h/3$	$-C/3$
B_n	τ	-2τ	$-4(C+q)/5$
L_n	$-h - \tau/4$	$-h + 8\tau$	a_0

$$a_0 = 11C/5 - q/5 + 3(r+k)/2.$$

Let $X_m = T_1(F_m D)$, retained until the final cancellation. Expanding the bands and their products gives the following full rows:

$$\begin{aligned}
T_0(aD) &= -3hr/4 + a_0 C,\\
T_1(aD^\vee) &= -14q^2/25 + 39qr/25 - 27r^2/50 + 2C^2/5,\\
T_1(aD) &= 8q^2/5 - 3qr + 27r^2/20 - 4C(C+q)/5 + X_m,\\
T_1(a^2) &= 6h\tau - 129\tau^2/4 + 24(C+q)a_0/5 - 66X_m/5 - 6R(16)/25,\\
T_0(aD^\vee) &= R(-2)/2 - G(1/5)/4 + 77q^2/50 + 54qr/25 - 63r^2/200 - 3qk/2 - 11C^2/10.
\end{aligned} \tag{283}$$

The third row is $2B_n(4)D(4) + B_n(1/4)D(1/4) - 3B_n(0)D(0) + X_m$. For the fourth row, expand $(x^pB_n + L_n)^2 = x^{2p}B_n^2 + 2x^pB_nL_n + L_n^2$. The surviving upper terms are $8B_n(4)^2 + B_n(1/4)^2/4 +$

$2T_0(B_nL_n)$ and $T_1(L_n^2) = -66X_m/5 - 6R(16)/25$. The latter equality retains both logarithm products; terms involving $(c_0 - 33D/5)^2$ have degree at most $2N-2 < p$, and $2c_0F_m$ has degree at most $p-1$, so their strict tails are zero. To evaluate the remaining square tail, put $\Pi_p(F)(z) = \sum_{m>p} F_m z^{m-p}$. In a product of two full logarithms reflect both indices by $(i,j) \mapsto (p-i, p-j)$. The condition $i+j > p$ becomes $(p-i)+(p-j) < p$, and the strict diagonal $i+j=p$ stays excluded. The two denominator signs cancel. Since the scale constants $1/4$ and 4 lie in the prime field, this gives

$$\Pi_p(F_m^2)(z) = \frac{4}{25} z^p \{T_2((4z)^{-1}, (4z)^{-1}) + T_2(4/z, 4/z) - 2T_2((4z)^{-1}, 4/z)\}.$$

The complete strict shuffle T_2 is defined and proved in Section 6.12. At $z=4$ its arguments are $1/16, 1$, so $T_2(t,t) = -2R(1-t)$, $T_2(1,t) = R(t) - R(1-t)$ and $T_2(1,1) = 0$ cancel both copies of $R(15/16)$. At $z = 1/4$ the arguments are $1, 16$, and the same equations cancel both copies of $R(-15)$. Consequently

$$\Pi_p(F_m^2)(4) = -\frac{32}{25} R(1/16), \qquad \Pi_p(F_m^2)(1/4) = -\frac{2}{25} R(16), \qquad T_1(F_m^2) = -\frac{6}{25} R(16),$$

where the last equality uses $R(1/16) = R(16)/16$. Expanding L_n^2 now gives its stated row, including the uncancelled mixed coefficient $-66X_m/5$.

For the reciprocal second row the complete projection is $T_1(aD^\vee) = (12/5)T_0(DD^\vee)$, since $\deg L_n < p$. The full reciprocal triangle is

$$T_0(DD^\vee) = \frac{9}{10}(D(1/4) + D(4))^2 - D(1/4)^2/2 - D(4)^2 + 3D(0)^2/2.$$

For the last row its nonpositive triangle gives

$$\begin{aligned} T_0(aD^\vee) &= \frac{33}{5}\left\{\frac{3}{5}(D(1/4) + D(4))^2 - D(1/4)^2 - D(4)^2/2 - C^2/6\right\} \\ &\quad + (11q/5 - 3k/2)q - \mathcal{H}_\triangle, \\ \mathcal{H}_\triangle &= \frac{2}{15}\left\{\frac{1}{4}(2\mathsf{H}(1/4,4) + \mathsf{H}(4,4)) - 4(2\mathsf{H}(1/4,1/4) + \mathsf{H}(4,1/4))\right\} \\ &= -R(-2)/2 - 11q^2/5 + 9qr/5 - 27r^2/40 + G(1/5)/4. \end{aligned}$$

Here is the complete projection argument for both reciprocal triangles. For a Laurent polynomial $F = \sum_m F_m x^m$ put $F_+ = \sum_{m>0} F_m x^m$ and $F_-^{\mathrm{rev}} = \sum_{m\le 0} F_m x^{-m}$. Then $F = F_+ + (F_-^{\mathrm{rev}})^\vee$, including the $m=0$ slot, and

$$T_0(F) = 2F(4) + F(1/4) - 2F_-^{\mathrm{rev}}(1/4) - F_-^{\mathrm{rev}}(4).$$

Define the actual branch square

$$B_D = \frac{1}{5} s x^3 \beta_D^2, \qquad \deg B_D \le p-3, \qquad (D + D^\vee)^2 = (1 + x^{-p}) B_D.$$

The last identity follows directly from (105): $x^{-2N+2}s^{2N+2}\beta_D^2 = x^{3-p}s^{p+1}\beta_D^2$ and $s^p = (x^p+1)/5$. Since B_D has zero origin and degree less than p, the strict positive part of the right side is exactly B_D, and its origin is zero. Expanding the left side as $D^2 + (D^2)^\vee + 2DD^\vee$ therefore gives

$$(DD^\vee)_0 = -D(0)^2, \quad (DD^\vee)_+ = \tfrac{1}{2}(B_D - D^2 + D(0)^2), \quad (DD^\vee)_-^{\mathrm{rev}} = \tfrac{1}{2}(B_D - D^2 - D(0)^2).$$

For this calculation set $d_a = D(a_*)$, $d_b = D(b_*)$; the earlier D_a, D_b remain endpoints of $D_e = -3D$. The finite definition gives $d_a = \{E(1/2) - E(1)\}/3 = q + h/6$ and $d_b = \{E(1) - E(2)\}/3 = -q - h/3$, using $E(1/2) = q - 3r/4$, $E(1) = -q$, $E(2) = -3r/2$ and $h = -2q - 3r/2$. Evaluation of the branch

square at the two endpoints gives $B_D(1/4) = (d_a + d_b)^2/5$ and $B_D(4) = 4(d_a + d_b)^2/5$, since the factors $1 + x^{-p}$ there are 5 and 5/4. Thus the strict positive weighted value is

$$\frac{1}{2}\{\tfrac{9}{5}(d_a + d_b)^2 - 2d_b^2 - d_a^2 + 3D(0)^2\},$$

and the complete nonpositive weighted value is

$$\frac{1}{2}\{\tfrac{6}{5}(d_a + d_b)^2 - 2d_a^2 - d_b^2 - 3D(0)^2\}.$$

These are the two displayed quadratic triangles.

For the mixed triangle use the full convolution, with its diagonal:

$$(D^\vee \ell(\alpha x))^{\mathrm{rev}}_-(z) = \sum_{i=1}^{N-1} D_i \sum_{j=1}^{i} \frac{\alpha^j z^{i-j}}{j} = -\frac{1}{3}\mathsf{H}(z, \alpha).$$

Every $1 \le j \le i \le N-1 < p$ is a unit, so this is a finite polynomial identity; the constant coefficient of D contributes no term. Consequently, for $F_m = (2/5)(4\ell(x/4) - \ell(4x)/4)$,

$$(D^\vee F_m)^{\mathrm{rev}}_-(z) = \frac{2}{15}\{\mathsf{H}(z, 4)/4 - 4\mathsf{H}(z, 1/4)\}.$$

The nonpositive reversal of $aD^\vee$ is precisely

$$c_0 D - \frac{33}{5}(DD^\vee)^{\mathrm{rev}}_- + (D^\vee F_m)^{\mathrm{rev}}_-.$$

When forming $(aD^\vee)^{\mathrm{rev}}_-$, the shifted summand of $(aD^\vee)^\vee$ is $x^{-p} D B_n^\vee$. Its largest exponent is at most $N - 1 - p < 0$, so its nonnegative truncation is zero. The complete product has weighted endpoint value $-h(2d_a + d_b)$. Since $2d_a + d_b = q$, $-h - c_0 = 11q/5 - 3k/2$ and $D(0)^2 = C^2/9$, subtracting the just computed nonpositive reversal gives exactly the displayed formula for $T_0(aD^\vee)$ and the full mixed correction $\mathcal{H}_\triangle$. The moving-H endpoint formulas (101) and the seven rational rows (70) give $\mathcal{H}_\triangle$; all six coefficients, including the two zero endpoints, are retained.

The metric uses the actual differential of D:

$$\Omega(s^N F) = \Omega(F) - \frac{2}{5}[x^{-1}](FD').$$

Its two derivative pairs and poles are

$$[x^{-1}](a^\vee D') = -11q/50 + 21r/100 + C/2, \qquad [x^{-1}](x^p a^\vee D') = 12q/25 - 6r/25,$$
$$\Omega(a^\vee) = 0, \qquad \Omega(x^p a^\vee) = \tau/5.$$

For example $[x^{-1}](D^\vee D') = q/5 - r/10$ follows by the following whole branch calculation. Equation (105) gives $\beta_D(a_*) = 16(d_a + d_b)$ and $\beta_D(b_*) = (d_a + d_b)/4$. Since $s^N D_s = (x^p + 1)x^{-(N-1)}\beta_D/5$ and $\deg \beta_D \le N - 3$, (93) gives $\Omega(s^N D_s) = -(d_a + d_b)/25$. Multiplying $A_* D' = 5(1 - s^N)/2$ by D_s and taking Ω gives

$$[x^{-1}](D^\vee D') = \frac{5}{2}\left\{\frac{d_a - d_b}{15} + \frac{d_a + d_b}{25}\right\} = \frac{4d_a - d_b}{15} = \frac{q}{5} - \frac{r}{10}.$$

The polynomial product DD' has coefficient zero at -1. This also evaluates the diagonal sum $\sum_{i=1}^{N-1} i D_i^2$. The reciprocal logarithm pair is $-\{D(4) - D(0)\}/4 + 4\{D(1/4) - D(0)\}$. Inserting $L_n = c_0 - 33D/5 + F_m$ gives the first derivative pair; the negative phase has no such coefficient because $\deg D < p$. Inserting $B_n = 12D/5 - 4q/5$ gives the second pair. Consequently

$$\mu = -88hq/75 + 37hr/100 + 5hC/3.$$

Substitution in (282) cancels X_m with coefficient $11/4 - (5/24)(66/5) = 0$. The remaining substitutions of τ, a_0, h give (270). No high logarithm tail has been assigned a value before its coefficient cancels.

6.14 The complete finite cubic primitive and its two weighted rows

Let $p \geq 19$ be prime, let K be a field of characteristic p, and put $N = (p-1)/2$. Write $q = q_2$, $r = q_3$ for the images of the integer Fermat quotients and $h = -2q - 3r/2$. All denominators below are units: the summation indices lie in $1, \ldots, p-1$, and the remaining denominators have prime factors only $2, 3, 5$. Define

$$\ell(x) = \sum_{j=1}^{p-1} \frac{x^j}{j}, \qquad R(x) = \sum_{j=1}^{p-1} \frac{x^j}{j^2}, \qquad E(x) = \frac{\ell(x) + \ell(-x)}{2}, \quad O(x) = \frac{\ell(x) - \ell(-x)}{2}.$$

Every formula containing an indeterminate is an identity of polynomials over K, not merely an identity of functions on $\mathbb{F}_p$.

6.14.1 The full triangle, its diagonal and the rectangle

For $n < p$ define

$$D_n(x,y) = \sum_{j=1}^{n} \sum_{k=1}^{j} \frac{x^k y^{j-k}}{kj}, \qquad \mathscr{D}(x,y) = D_{p-1}(x,y), \qquad S(x,y) = D_N(x^2, y^2).$$

The $k = j$ term is x^j/j^2; hence

$$\begin{aligned} T_2(x,y) &= \mathscr{D}(x,y) + \mathscr{D}(y,x) - R(x) - R(y) \\ &= \sum_{\substack{i,j \geq 1 \\ i+j<p}} \frac{x^i y^j}{ij}. \end{aligned}$$

Indeed, at $i + j = m < p$ the two off-diagonal terms have coefficient $1/(im) + 1/(jm) = 1/(ij)$. The diagonal is subtracted explicitly. Put

$$H_2(x,y) = \sum_{k=1}^{p-1} \frac{x^k y^{p-k}}{k^2}.$$

For $x \neq 0$, the involution $(j,k) \mapsto (p-k, p-j)$ preserves $1 \leq k \leq j \leq p-1$. Its reflected denominator is $(p-j)(p-k) = jk$ in K, and its numerator is $x^{p-j} y^{j-k}$. Thus

$$\mathscr{D}(x,y) = x^p \mathscr{D}(x^{-1}, y/x). \tag{284}$$

For $y \neq 0$, another substitution, $l = p - j$, gives

$$y^p \mathscr{D}(x/y, 1/y) = -\sum_{\substack{k,l \geq 1 \\ k+l \leq p}} \frac{x^k y^l}{kl} = -T_2(x,y) + H_2(x,y).$$

Here $1/(kj) = -1/(kl)$; on $k + l = p$ this is $1/k^2$. This is the entire boundary row, including its nonzero terms.

Define the two sign packets

$$\begin{aligned} N_o(x,y) &= \mathscr{D}(x,y) - \mathscr{D}(-x,y) - \mathscr{D}(x,-y) + \mathscr{D}(-x,-y), \\ M(x,y) &= \mathscr{D}(x,y) + \mathscr{D}(-x,y) - \mathscr{D}(x,-y) - \mathscr{D}(-x,-y). \end{aligned}$$

In the sum of all four signs, only even $j = 2a$, even $k = 2b$ remain. Their sign factor 4 cancels the denominator $4ab$, giving

$$\mathscr{D}(x,y) + \mathscr{D}(-x,y) + \mathscr{D}(x,-y) + \mathscr{D}(-x,-y) = S(x,y).$$

The diagonal $a = b$ remains. Since p is odd, $H_2(-x,-y) = -H_2(x,y)$, so the sum of its four signs is zero. Adding the four second-reciprocity equations therefore gives

$$y^p M(x/y, 1/y) = -S(x,y) - S(y,x) + 2\{R(x) + R(-x) + R(y) + R(-y)\}.$$

Replace (x,y) by $(x/y, 1/y)$ and multiply by y^p to obtain

$$M(x,y) = y^p \left[2\{R(x/y) + R(-x/y) + R(1/y) + R(-1/y)\} - S(x/y, 1/y) - S(1/y, x/y)\right]. \quad (285)$$

Applying (284) to the four signs also gives $N_o(x,y) = x^p N_o(x^{-1}, y/x)$.

For $xy \neq 0$, split the product $\ell(x)\ell(y)$ into the three regions $i+j < p$, $i+j = p$, $i+j > p$. The first region is $T_2(x,y)$. The middle is $-H_2(x,y)$ because $p - i = -i$. In the third, replace (i,j) by $(p-i, p-j)$; its value is $x^p y^p T_2(x^{-1}, y^{-1})$. Thus

$$\ell(x)\ell(y) = T_2(x,y) - H_2(x,y) + x^p y^p T_2(x^{-1}, y^{-1}).$$

Add this identity at the four signs. The middle packets cancel by joint negation; the low packet is the four-sign S packet above. The high packets, transported with the first reciprocity, give

$$4E(x)E(y) = S(x,y) + S(y,x) - 2\{R(x) + R(-x) + R(y) + R(-y)\} + y^p N_o(x, x/y) + x^p N_o(y, y/x). \quad (286)$$

No individual middle row has been assigned zero.

6.14.2 The original F_b, F_d contraction

Define

$$\begin{aligned}
A(t) &= S(2t,1) - S(t,1/2), & B(t) &= S(2t,2) - S(t,1),\\
\mathcal{A}_t &= S(1,2t) - S(1/2,t), & \mathcal{B}_t &= S(2,2t) - S(1,t),\\
\chi(t) &= S(t,t) + S(2t,2t) - S(2t,t) - S(t,2t),\\
F_b(t) &= -N_o(t,1) + 2N_o(t,1/2) + N_o(2t,2) - 2N_o(2t,1),\\
F_d(t) &= -M(t,1) + 2M(t,1/2) + M(2t,2)/2 - M(2t,1).
\end{aligned}$$

Let $X = t^p$ and $P_R(t) = R(t) + R(-t)$. For $t \neq 0$, (286) at $(t,t), (2t,2t), (2t,t)$ reads

$$\begin{aligned}
4E(t)^2 &= 2S(t,t) - 4P_R(t) + 2XN_o(t,1),\\
4E(2t)^2 &= 2S(2t,2t) - 4P_R(2t) + 4XN_o(2t,1),\\
4E(2t)E(t) &= S(2t,t) + S(t,2t) - 2P_R(2t) - 2P_R(t)\\
&\quad + XN_o(2t,2) + 2XN_o(t,1/2).
\end{aligned}$$

The first plus the second minus twice the third cancels both P_R terms and gives $4(E(2t) - E(t))^2 = 2\chi(t) - 2XF_b(t)$. Consequently

$$t^p F_b(t) = \chi(t) - 2(E(2t) - E(t))^2. \quad (287)$$

For F_d, (285) gives the following four whole rows, including their diagonal contributions:

$$\begin{array}{r|l}
-M(t,1) & -2P_R(t) - 2P_R(1) + S(t,1) + S(1,t)\\
2M(t,1/2) & 2P_R(2t) + 2P_R(2) - S(2t,2) - S(2,2t)\\
M(2t,2)/2 & 2P_R(t) + 2P_R(1/2) - S(t,1/2) - S(1/2,t)\\
-M(2t,1) & -2P_R(2t) - 2P_R(1) + S(2t,1) + S(1,2t).
\end{array}$$

The variable P_R terms cancel. The S terms are exactly $A - B + \mathcal{A}_t - \mathcal{B}_t$. Inversion and the rational values of (66) give

$$R(1) = R(-1) = 0, \quad R(2) = -q^2, \quad R(1/2) = -q^2/2, \quad R(-1/2) = -R(-2)/2.$$

Thus the constant row is $2\{P_R(2) + P_R(1/2) - 2P_R(1)\} = R(-2) - 3q^2$, and

$$F_d(t) = A(t) - B(t) + \mathcal{A}_t - \mathcal{B}_t + R(-2) - 3q^2. \quad (288)$$

6.14.3 The endpoint normalization from the same triangle

For $a \in K$ define the polynomial

$$Z(a,t) = \frac{\mathscr{D}(at,-a) - \mathscr{D}(at,a)}{2} + \mathscr{D}(at,a/2) - \mathscr{D}(at,-a/2).$$

Expanding its four terms and collecting the sign packets defined above gives, coefficient by coefficient,

$$\begin{aligned} F_b &= 2Z(1,t) + 2Z(-1,t) - 2Z(2,t) - 2Z(-2,t), \\ F_d &= 2Z(1,t) - 2Z(-1,t) - Z(2,t) + Z(-2,t). \end{aligned}$$

For example $Z(1,t) + Z(-1,t) = -N_o(t,1)/2 + N_o(t,1/2)$; $Z(2,t) + Z(-2,t) = -N_o(2t,2)/2 + N_o(2t,1)$. Likewise $Z(1,t) - Z(-1,t) = -M(t,1)/2 + M(t,1/2)$ and $Z(2,t) - Z(-2,t) = -M(2t,2)/2 + M(2t,1)$. Put

$$\begin{aligned} F_e(t) &= \{Z(1,t) + Z(-1,t)\}/2 + Z(2,t) + Z(-2,t), \\ M_E(t) &= (8q - 6r)E(t) + (4q - 3r)E(2t), \\ M_O(t) &= -3rO(t) + 3rO(2t)/2, \\ V_{\mathrm{cub}}(t) &= t^{2p}F_b(t) + t^pF_d(t) + F_e(t) + t^pM_E(t) + M_O(t), \\ K_J &= V_{\mathrm{cub}}(1). \end{aligned}$$

The definition of K_J fixes the endpoint before its value is evaluated.

Here is the full rational evaluation. Write $s = R(-2)$. The reflection equation (66), the logarithm values $\ell(3) = 2q - 3r$, $\ell(4) = -8q + 3r$, and inversion give

$$\begin{aligned} 3R(3) - 2s &= -(2q - 3r)^2/2, \\ 4R(4) - 3R(-3) &= -(-8q + 3r)^2/2, \qquad R(4) = -6q^2 - 2s, \\ -R(-1/2)/2 + 3R(3/2)/2 &= -(q - 3r/2)^2/2, \\ 4R(4/3)/3 - R(-1/3)/3 &= -(-8q/3 + r)^2/2. \end{aligned}$$

In the third equation $R(-1/2) = -s/2$; in the fourth $R(-1/3) = -R(-3)/3$. Solving these linear equations gives

$$\begin{aligned} R(3) &= 2s/3 - 2q^2/3 + 2qr - 3r^2/2, \\ R(-3) &= 8q^2/3 - 8s/3 - 8qr + 3r^2/2, \\ R(3/2) &= -s/6 - q^2/3 + qr - 3r^2/4, \\ R(4/3) &= -(8q - 3r)^2/24 - R(-3)/12, \qquad R(2/3) = 2R(3/2)/3. \end{aligned}$$

For $y(1-y) \neq 0$ in the prime field, the full homogeneous triangle identity (67) specializes to

$$\mathscr{D}(x,y) = (1-y)R\left(\frac{x-y}{1-y}\right) - (1-y)R\left(\frac{-y}{1-y}\right) + yR\left(\frac{y-x}{y}\right).$$

In particular its $x = 1$ and $x = y$ rows are respectively $2yR((y-1)/y)$ and $yR((y-1)/y)$: use $R(0) = R(1) = 0$ and $R(z) = zR(1/z)$ at $z = -y/(1-y)$. At $y = 1$, the homogeneous identity itself gives $\mathscr{D}(x,1) = R(1-x)$, since the two homogeneous terms with second argument zero vanish. At $y = -1$ it gives $\mathscr{D}(x,-1) = 2R((1+x)/2) + q^2 - R(1+x)$. Therefore the actual terms in the three Z endpoints are

a	$\mathscr{D}(a,-a)$	$\mathscr{D}(a,a)$	$\mathscr{D}(a,a/2)$	$\mathscr{D}(a,-a/2)$
1	$2q^2$	0	0	$-R(3)$
-1	$-q^2$	q^2	$-(R(3) + R(-3))/2$	$(R(3) + R(-3))/2$
2	$3R(4/3) - 3R(2/3) + 2q^2$	$-q^2$	0	$2R(3/2) + q^2 - R(3)$.

All arguments and scalar factors in this table are from the preceding homogeneous formula; no endpoint with zero denominator is used. Thus

$$\begin{aligned} Z(1,1) &= q^2 + R(3) = 2s/3 + q^2/3 + 2qr - 3r^2/2, \\ Z(-1,1) &= -q^2 - R(3) - R(-3) = 2s - 3q^2 + 6qr, \\ Z(2,1) &= 3R(4/3)/2 - 3R(3/2) + q^2/2 + R(3) \\ &= 3s/2 - 7q^2/2 + 3qr. \end{aligned}$$

The endpoint logarithms are

$$E(1) = -q, \quad O(1) = q, \quad E(2) = -3r/2, \quad O(2) = -2q + 3r/2.$$

The coefficients of $Z(1,1), Z(-1,1), Z(2,1), Z(-2,1)$ in $F_b(1) + F_d(1) + F_e(1)$ are $9/2, 1/2, -2, 0$. Moreover $M_E(1) = -8q^2 + 9r^2/2$ and $M_O(1) = -6qr + 9r^2/4$. Consequently

$$K_J = \frac{9}{2}Z(1,1) + \frac{1}{2}Z(-1,1) - 2Z(2,1) - 8q^2 - 6qr + \frac{27}{4}r^2.$$

The coefficients of s, q^2, qr, r^2 after substitution are

$$3 + 1 - 3 = 1, \quad 3/2 - 3/2 + 7 - 8 = -1, \quad 9 + 3 - 6 - 6 = 0, \quad -27/4 + 27/4 = 0.$$

Hence

$$K_J = R(-2) - q^2. \tag{289}$$

6.14.4 The cleared cubic derivative and its Frobenius kernel

Put $a = 1 - t^2$, $b = 4t^2 - 1$ and $X = t^p$. Termwise differentiation of the full triangle gives

$$(x - y)\partial_x \mathscr{D}(x, y) = \ell(x) - \ell(y) : \quad (x - y)\sum_{k=1}^{j} x^{k-1}y^{j-k} = x^j - y^j.$$

For a nonzero scalar c this gives $(t - d)\partial_t \mathscr{D}(ct, cd) = \ell(ct) - \ell(cd)$. Apply it to the four terms of $Z(c, t)$. Their logarithm coefficient is

$$\frac{1}{2(t+1)} - \frac{1}{2(t-1)} + \frac{1}{t - 1/2} - \frac{1}{t + 1/2} = \frac{1}{a} + \frac{4}{b} = \frac{3}{ab}.$$

Clearing the denominator gives the polynomial identity

$$\begin{aligned} abZ(c,t)' &= 3\ell(ct) - Z_R(c,t), \\ Z_R(c,t) &= \frac{b}{2}\{(1+t)\ell(c) + (1-t)\ell(-c)\} \\ &\quad + a\{(4t+2)\ell(c/2) - (4t-2)\ell(-c/2)\}. \end{aligned} \tag{290}$$

The rational logarithms are

$$\ell(1) = 0, \quad \ell(-1) = \ell(2) = -2q, \quad \ell(-2) = 2q - 3r, \quad \ell(1/2) = q, \quad \ell(-1/2) = q - 3r/2.$$

Thus the four corrections are explicitly

c	$Z_R(c,t)$
1	$-q(1-t)b + a\{4q + (6t-3)r\}$
-1	$-q(1+t)b + a\{4q - (6t+3)r\}$
2	$q\{-2tb + (8t-4)a\} - 3r(1-t)b/2$
-2	$q\{2tb - (8t+4)a\} - 3r(1+t)b/2.$

Let

$$A_1 = 2X^2 + 2X + 1/2, \quad A_{-1} = 2X^2 - 2X + 1/2, \quad A_2 = -2X^2 - X + 1, \quad A_{-2} = -2X^2 + X + 1.$$

The definition of V_{cub} is $\sum_{c=\pm1,\pm2} A_c Z(c,t) + XM_E + M_O$. The pair sums and differences of the displayed corrections are

$$\begin{aligned} Z_R(1) + Z_R(-1) &= q(10 - 16t^2) - 6ra, \\ Z_R(1) - Z_R(-1) &= 2qtb + 12rta, \\ Z_R(2) + Z_R(-2) &= -8qa - 3rb, \\ Z_R(2) - Z_R(-2) &= qt(20 - 32t^2) + 3rtb. \end{aligned}$$

Multiply these by $2X^2 + 1/2$, $2X$, $1 - 2X^2$, $-X$ respectively. Their sum, with $3q(1 - 4X^2)$, is

$$\begin{aligned} 3q(1 - 4X^2) + \sum_c A_c Z_R(c,t) = {} & 24q(1 - 2t^2)(X^2 - Xt) \\ & + 9r\{2(2t^2 - 1)X^2 + (3 - 4t^2)Xt - t^2\}. \end{aligned}$$

The four basic derivative rows are

$$aE'(t) = t - X, \quad bE(2t)' = -4(t - X), \quad aO'(t) = 1 - t^{p-1}, \quad bO(2t)' = -2(1 - t^{p-1}).$$

They follow from $\ell'(t) = \sum_{j=0}^{p-2} t^j$ and the two signs, using $2^p = 2$, $2^{p-1} = 1$. In particular

$$\begin{aligned} abM'_E &= 6(4q - 3r)(2t^2 - 1)(t - X), \\ abM'_O &= -9rt^2(1 - t^{p-1}) = 9r(Xt - t^2). \end{aligned}$$

Multiplying the first by X and adding the second gives exactly the preceding correction polynomial. Hence

$$ab(XM'_E + M'_O) = 3q(1 - 4X^2) + \sum_c A_c Z_R(c,t). \tag{291}$$

Because $X' = 0$, (290) and (291) imply

$$abV'_{\mathrm{cub}} = 3\left[q(1 - 4X^2) + \sum_c A_c \ell(ct)\right] = 3\{q + E(T_c(t))\}, \qquad T_c(t) = 4t^3 - 3t.$$

The last equality is the four-log form of the complete cubic logarithm identity proved in Section 4.1; its coefficients are exactly the four A_c displayed here.

Define $L(t) = q + E(t)$ and

$$J_L(t) = \frac{R((1+t)/2) - R((1-t)/2)}{2}.$$

Since $xR'(x) = \ell(x)$, logarithm reflection and $\ell((1+t)/2) = q + E(t)$ give

$$(1 - t^2)J'_L(t) = L(t).$$

This is a cleared polynomial identity, valid also at $t = \pm1$. The cubic identities $1 - T_c(t)^2 = ab^2$ and $T'_c(t) = 3b$ therefore give $ab(J_L \circ T_c)' = 3L(T_c)$. Subtract the preceding source for V_{cub} and cancel the nonzero polynomial ab: the difference has zero derivative.

Each $Z(c,t)$ has degree at most $p - 1$, since its triangle powers have $k \le j \le p - 1$. Thus $\deg V_{\mathrm{cub}} \le 3p - 1$, and $\deg(J_L \circ T_c) \le 3p - 3$. A polynomial of degree $< 3p$ with zero derivative has only the coefficients of $1, t^p, t^{2p}$: at any other exponent j, its derivative coefficient is j times its coefficient and j is nonzero in K. Moreover $Z(c,-t) = -Z(-c,t)$. Consequently F_b, F_e, M_O are odd, F_d, M_E are even, and both V_{cub} and $J_L \circ T_c$ are odd. Since p is odd and 2 is a unit, the constant and t^{2p} coefficients of their difference vanish. Its t^p coefficient is its value at $t = 1$. Here $T_c(1) = 1$, $J_L(1) = \{R(1) - R(0)\}/2 = 0$, and $V_{\mathrm{cub}}(1) = K_J$. We obtain the fully normalized identity

$$J_L(T_c(t)) = t^{2p}F_b + t^pF_d + F_e + t^pM_E + M_O - t^pK_J. \tag{292}$$

The last term is forced by the original endpoint.

6.14.5 The complete high-p band and origin row

Put

$$U_J(t) = t^p F_b(t) + F_d(t) + M_E(t) - K_J, \qquad V_J(t) = t^p\{F_e(t) + M_O(t)\}.$$

Multiplication of (292) by t^p gives

$$t^p J_L(T_c(t)) = t^{2p} U_J(t) + V_J(t). \tag{293}$$

In particular, at $u^2 = 1/5$, one has $u^{2p} = 1/5$ and $u^p J_L(T_c(u)) = U_J(u)/5 + V_J(u)$.

To identify the actual coefficient row, rather than merely this evaluation, put $H_j = \sum_{k=1}^{j} 1/k$ and $b_j = (q + H_j/2)/(2j+1)$, $0 \le j < N$. The harmonic value $H_N = -2q$ gives

$$(1-t^2)\left(\sum_{j=0}^{N-1} b_j t^{2j+1}\right)' = q + \sum_{j=1}^{N} \frac{t^{2j}}{2j} = L(t).$$

Indeed, at t^{2j} the coefficient difference is $(q + H_j/2) - (q + H_{j-1}/2) = 1/(2j)$; the top coefficient uses $q + H_N/2 = 0$. This polynomial and J_L have degree $< p$, the same derivative after cancellation of $1 - t^2$, and value zero at 0. Their difference is constant and zero. Thus $J_L(t) = \sum_{j=0}^{N-1} b_j t^{2j+1}$ with this full cutoff.

Define the actual polynomial in the original coordinate by

$$\mathscr{J}(x) = \sum_{j=0}^{N-1} \frac{b_j}{5^{2j+1}} \left(\frac{x+1}{5}\right)^{N+j+1} (4x-11)^{2j+1}. \tag{294}$$

Its degree is at most $N + 3(N-1) + 2 = 2p - 3$. Under $x = 5t^2 - 1$, each summand is $b_j t^{p+2j+1}(4t^2 - 3)^{2j+1}$, so

$$\mathscr{J}(5t^2 - 1) = t^p J_L(T_c(t)) = t^{2p} U_J(t) + V_J(t).$$

More generally if $t^2 = (x+1)/5$, its value is $t^p J_L(t(4x-11)/5)$. At $x = 4$, take $t = 1$; at $x = 1/4$, take $t = 1/2$. The arguments of J_L are $1, -1$, respectively, and $J_L(-1) = 0$ as well. Hence both original endpoints of $\mathscr{J}$ are zero. At $x = 0$, take $t = u$ and write $T_0 = -11u/5$; then $T_c(u) = T_0$ and $\mathscr{J}(0) = u^p J_L(T_0)$. This carrier differs from the parameter-jet Euler primitive H_J in the moment calculation.

Split the polynomial, keeping the coefficient at p, as

$$\mathscr{J}(x) = H_{\mathrm{lo}}(x) + x^p H_{\mathrm{hi}}(x), \qquad H_{\mathrm{lo}} = \sum_{j=0}^{p-1} [x^j]\mathscr{J}\, x^j, \quad H_{\mathrm{hi}} = \sum_{j=0}^{p-1} [x^{p+j}]\mathscr{J}\, x^j.$$

Both bands have degree $< p$; the split is exact because $\deg \mathscr{J} < 2p$. Substitute $x = 5t^2 - 1$ and use $(5t^2-1)^p = 5t^{2p} - 1$. Comparison with (293) gives

$$t^{2p}\{5H_{\mathrm{hi}}(5t^2-1) - U_J(t)\} + \{H_{\mathrm{lo}}(5t^2-1) - H_{\mathrm{hi}}(5t^2-1) - V_J(t)\} = 0.$$

The second brace has degree $< 2p$: its first two terms have degree at most $2p-2$, and V_J has degree at most $2p-1$. Its coefficients cannot cancel any coefficient of the first brace multiplied by t^{2p}. Consequently both braces vanish. At $t = u$, where $5u^2 - 1 = 0$, the first gives the actual high coefficient and the full evaluation gives the actual origin:

$$[x^p]\mathscr{J} = \frac{U_J(u)}{5}, \qquad \mathscr{J}(0) = \frac{U_J(u)}{5} + V_J(u). \tag{295}$$

The original finite endpoint functional is therefore

$$-\frac{22}{15}\mathscr{J}(0) + \frac{77}{15}[x^p]\mathscr{J} = \frac{11}{15}U_J(u) - \frac{22}{15}V_J(u). \tag{296}$$

Using (287), (288) and (289) in its definition gives, for every $t \neq 0$,

$$U_J(t) = A(t) - B(t) + \chi(t) + \mathcal{A}_t - \mathcal{B}_t - 2\{E(2t) - E(t)\}^2 + (4q - 3r)\{2E(t) + E(2t)\} - 2q^2. \quad (297)$$

The constant is $-3q^2-(-q^2) = -2q^2$; the two $R(-2)$ terms cancel. At $t = u$, with $C = E(2u)-E(u)$ and $\mathcal{L} = A(u) - B(u) + \chi(u)$, this is exactly the original row (300). If desired its even-log alphabet follows directly from (77): $E(u) = C/5 - 6q/5 + q_5/2$ and $2E(u) + E(2u) = 8C/5 - 18q/5 + 3q_5/2$.

6.14.6 The actual canceled bracket and the V_J cancellation

Let $\mathcal{A}_e(t) = -\{R(1 - t) + R(1 + t)\}/2$ and

$$S_2(t) = R((1 - t)/2) + R((1 + t)/2).$$

The actual triangular source for the bracket is

$$R_D(t) = \{\mathscr{D}(t, 1) - \mathscr{D}(t, -1) - \mathscr{D}(-t, -1) + \mathscr{D}(-t, 1)\}/4.$$

Use the two whole polynomial $y = 1, -1$ rows evaluated above. The four terms have sum $2\{R(1 - t) + R(1 + t) - S_2(t) - q^2\}$, so $R_D = \{R(1 - t) + R(1 + t) - S_2 - q^2\}/2$. The original source $R_D - qL + q^2 + S_2/3 + (2c/3)L$ consequently equals

$$K_c(t) = \frac{R(1 - t) + R(1 + t) + q^2}{2} - \frac{S_2(t)}{6} + \left(\frac{2c}{3} - q\right) L(t).$$

Reflection (66) at $(1 + t)/2$ is

$$\frac{1 + t^p}{2} R((1 + t)/2) + \frac{1 - t^p}{2} R((1 - t)/2) = -\frac{L(t)^2}{2}.$$

Thus $S_2(t) = -L(t)^2 - 2t^p J_L(t)$ and, by substitution,

$$K_c(t) = -\mathcal{A}_e(t) + \frac{q^2}{2} + \frac{L(t)^2}{6} + \frac{t^p J_L(t)}{3} + \left(\frac{2c}{3} - q\right) L(t). \quad (298)$$

The endpoints are zero before any descent: at $t = \pm 1$, $L(t) = 0$, $S_2(t) = R(0) + R(1) = 0$, and the remaining row is $(R(0) + R(2) + q^2)/2 = 0$. Both L and K_c are even polynomials. Hence

$$B_T(t) = -\frac{2}{3} K_h(t) - \frac{(L(t) - h)^2}{3} + \frac{h^2}{3}$$

has a unique polynomial descent $B_Q(z)$ satisfying $B_Q(1 - t^2) = B_T(t)$. Explicitly take the even coefficients of $B_T(t) = \sum b_{2j} t^{2j}$ and set $B_Q(z) = \sum b_{2j}(1 - z)^j$. Its endpoint is $B_Q(0) = B_T(1) = 0$.

For any polynomial P with $P(0) = 0$, write $P(z) = zP_1(z)$. The original rational Q coordinate and its pole multiplier are

$$Q(x) = \frac{(4 - x)^2(4x - 1)}{125x^3}, \qquad \frac{1}{x - 4} + \frac{8}{4x - 1}.$$

This is precisely the native bracket density: with $A_* = (4 - x)(4x - 1)$ and $L = 33 - 12x$,

$$\frac{1}{x - 4} + \frac{8}{4x - 1} = \frac{-(4x - 1) + 8(4 - x)}{(4 - x)(4x - 1)} = \frac{L}{A_*}.$$

Thus the coefficient computed below is $\Omega(LB_Q(Q(x)))$, where $\Omega(F) = [x^{-1}](F/A_*)$. Put $y = x^{-1}$. Direct multiplication, keeping the entire density, gives

$$Q(x) = \frac{4 - 33y + 72y^2 - 16y^3}{125} = Q_{\mathrm{rev}}(y),$$

$$Q(x)\left(\frac{1}{x-4}+\frac{8}{4x-1}\right)=\frac{12y-81y^2+132y^3}{125}=F_{\mathrm{rev}}(y).$$

Thus the canceled polynomial density is $P_1(Q_{\mathrm{rev}}(y))F_{\mathrm{rev}}(y)$. Since $F_{\mathrm{rev}}(0)=0$, $[y]F_{\mathrm{rev}}=12/125$ and $Q_{\mathrm{rev}}(0)=4/125$, its full coefficient of y is

$$\frac{12}{125}P_1(4/125)=3P(4/125).$$

This cancels only the proven factor z, with $4/125$ a unit. For $P=B_Q$ use $1-T_0^2=4/125$, which follows from $T_0=-11u/5$ and $u^2=1/5$. The actual bracket coefficient is therefore

$$\iota(G_{\mathrm{KQ}})=3B_Q(4/125)=-2K_h(T_0)-a_0^2+h^2,\qquad a_0=L(T_0)-h.$$

Insert (298) and $L(T_0)=a_0+h$. The square and linear terms collect as follows:

$$-\frac{(a_0+h)^2}{3}-a_0^2-2\left(\frac{2h}{3}-q\right)(a_0+h)+h^2=-\frac{4}{3}a_0^2+2(q-h)a_0-\frac{2}{3}h^2+2qh.$$

Because $T_0^p=-11u^p/5$, the retained primitive term is $-2T_0^pJ_L(T_0)/3=22u^pJ_L(T_0)/15$. By (295) this proves

$$\iota(G_{\mathrm{KQ}})=2\mathcal{A}_e(T_0)-\frac{4}{3}a_0^2+2(q-h)a_0-\frac{2}{3}h^2+2qh-q^2+\frac{22}{75}U_J+\frac{22}{15}V_J.$$

Adding the actual endpoint functional (296) cancels V_J with coefficient $22/15-22/15=0$. The U_J coefficient is $22/75+11/15=77/75$. Consequently the complete bracket contraction is

$$\iota(G_{\mathrm{KQ}})+\frac{11}{15}U_J-\frac{22}{15}V_J=2\mathcal{A}_e(T_0)-\frac{4}{3}a_0^2+2(q-h)a_0-\frac{2}{3}h^2+2qh-q^2+\frac{77}{75}U_J.\tag{299}$$

This is the source-selected formula (304); its two weighted rows are the origin and full high-p coefficient of the same polynomial (294).

6.15 Collection to the corrected finite consumer

The weighted cubic rows in this collection are finite triangular sums. Put $\mathscr{D}=D_{p-1}$ and define

$$\begin{aligned}
Z(r_0,t)&=\frac{\mathscr{D}(r_0t,-r_0)-\mathscr{D}(r_0t,r_0)}{2}+\mathscr{D}(r_0t,r_0/2)-\mathscr{D}(r_0t,-r_0/2),\\
F_b&=2Z(1,t)+2Z(-1,t)-2Z(2,t)-2Z(-2,t),\\
F_d&=2Z(1,t)-2Z(-1,t)-Z(2,t)+Z(-2,t),\\
F_e&=\frac{Z(1,t)+Z(-1,t)}{2}+Z(2,t)+Z(-2,t),\\
M_E&=(8q-6r)E(t)+(4q-3r)E(2t),\\
M_O&=-3rO(t)+\frac{3}{2}rO(2t),\\
K_J&=R(-2)-q^2,\\
U_J(t)&=t^pF_b+F_d+M_E-K_J,\qquad V_J(t)=t^p(F_e+M_O).
\end{aligned}$$

In the formulas below U_J,V_J mean their values at $t=u$. These definitions contain the full $p-1$ cutoff, including the triangle diagonal. For reference their endpoint rows are

$$\begin{array}{c|c}
Z(1,1) & 2R(-2)/3+q^2/3+2qr-3r^2/2\\
Z(-1,1) & 2R(-2)-3q^2+6qr\\
Z(2,1) & 3R(-2)/2-7q^2/2+3qr.
\end{array}$$

Substitution in $9Z(1,1)/2 + Z(-1,1)/2 - 2Z(2,1) - 8q^2 - 6qr + 27r^2/4$ gives K_J: the coefficients of $R(-2), q^2, qr, r^2$ are $1, -1, 0, 0$ respectively. Changing the triangular indices (i,j) to $(p-j, p-i)$ proves

$$\mathscr{D}(x,y) = x^p \mathscr{D}(x^{-1}, y/x) \quad (x \neq 0).$$

Both indices remain in $1, \dots, p-1$, and the factors in the denominator change by two minus signs. Pairing the two full logarithm indices gives $t^p F_b = \chi(t) - 2(E(2t) - E(t))^2$. The same four substitutions in F_d give $F_d = A(t) - B(t) + \mathcal{A}_t - \mathcal{B}_t + R(-2) - 3q^2$. Consequently

$$U_J = \mathcal{L} + \mathcal{A}_u - \mathcal{B}_u - 2C^2 + (4q - 3r)\{2E(u) + E(2u)\} - 2q^2. \tag{300}$$

The reciprocal primitive $J_L(t) = \{R((1+t)/2) - R((1-t)/2)\}/2$ has the exact cubic pullback

$$J_L(4t^3 - 3t) = t^{2p} F_b + t^p F_d + F_e + t^p M_E + M_O - t^p K_J.$$

The identities above follow by substituting the full triangular definitions and the finite reflection identity; the homogeneous triangle formula in Section 3.1 retains the zero endpoint. Multiplication by t^p gives $t^p J_L(4t^3 - 3t) = t^{2p} U_J + t^p(F_e + M_O)$, using the definition of U_J. At $u^2 = 1/5$, the pulled-back value is $U_J/5 + V_J$, since $u^{2p} = (u^2)^p = 1/5$. Its complete high-p coefficient is $U_J/5$. These are the two distinct weighted rows; they cannot be replaced by an assumed value of the primitive.

The evaluated remainder $\mathcal{U}$ in (139) is

$$\begin{aligned}\mathcal{U} = {} & \frac{7}{15}G(1/5) + \frac{17}{15}R(4) - \frac{1}{25}R(16) + \frac{11}{10}\{D_N(4/5,0) - D_N(1/5,0)\} \\ & + \frac{451}{75}q^2 + \frac{31}{5}qr + \frac{9}{10}r^2 + \frac{51}{5}qk - \frac{3}{2}rk + \left(\frac{104}{5}q - \frac{44}{5}k\right)C - \frac{22}{5}q\sum_{j=1}^{N}\frac{(4/5)^j}{2j}.\end{aligned} \tag{301}$$

Use the explicitly evaluated $\mathcal{Z}$ in (148) and the balanced moment row $\mathcal{V}$ in (154). The whole-source identity (175) gives $V_{\mathrm{div}} + \rho = \mathcal{V} + \mathcal{U} + \mathcal{Z} + 287h^2/600$. With (217), their full sum is

$$\begin{aligned}B_2 := {} & \mathcal{V} + \mathcal{U} + \mathcal{Z} + \iota(G_{\mathrm{KQ}}) + \Omega(N_j) \\ = {} & \frac{77}{75}\mathcal{L} + \frac{22}{5}(K_e(2u) - K_e(u)) - \frac{11}{30}G(1/5) - \frac{5056}{375}q^2 + \frac{204}{25}qr + \frac{104}{25}qk - \frac{153}{25}rk \\ & - \frac{399}{500}r^2 + \frac{3}{4}k^2 + \frac{5434}{375}qC - \frac{341}{125}rC - \frac{319}{75}C^2 - \frac{11}{15}R(-2) - \frac{11}{150}R(16).\end{aligned} \tag{302}$$

Here the genuine conic packet cancels with coefficient $-11/15 + 11/15$, rather than being declared zero. For transparency the remaining transformations in this collection are

$$\begin{aligned}D_N(4/5,0) - D_N(1/5,0) &= 4\{K_e(2u) - K_e(u) - \mathcal{A}_e(2u) + \mathcal{A}_e(u)\}, \\ \sum_{j=1}^{N}\frac{(4/5)^j}{2j} &= 6C/5 - 6q/5 + k/2, \qquad R(4) = -6q^2 - 2R(-2), \\ \ell(16) &= 15(r+k) - 64q.\end{aligned} \tag{303}$$

For the first identity, setting the second argument to zero retains exactly the $k = j$ terms:

$$D_N(t^2, 0) = \sum_{j=1}^{N}\frac{t^{2j}}{j^2} = 2\{R(t) + R(-t)\} = 4\{K_e(t) - \mathcal{A}_e(t)\}.$$

Subtract this equality at $t = 2u$ and $t = u$. The logarithm sum in the second identity is $E(2u)$; substituting $E(2u) = E(u) + C$ and (77) gives $6C/5 - 6q/5 + k/2$. Reflection at -1 gives $R(2) = -q^2$, and quadratic distribution at 2 gives $R(4) = 6R(2) - 2R(-2)$. Finally, integer entropy at 16 gives

$$\ell(16) = \frac{\overline{1 - 16^p + 15^p}}{p} = -16\overline{Q_{16}} + 15\overline{Q_{15}} = -64q + 15(r+k).$$

Here $16^{p-1} = (1 + pQ_2)^4$ and $15^{p-1} = (1 + pQ_3)(1 + pQ_5)$ justify the last reduction. The two weighted J rows, say U_J, V_J, occur with coefficients $11/15, -22/15$ in the normalized Z expression. Their whole Euler-bracket combination is

$$\iota(G_{\mathrm{KQ}}) + \frac{11}{15}U_J - \frac{22}{15}V_J = 2\mathcal{A}_e(T_0) - \frac{4}{3}a_0^2 + 2(q-h)a_0 - \frac{2}{3}h^2 + 2qh - q^2 + \frac{77}{75}U_J,$$
$$T_0 = -11u/5, \qquad a_0 = E(T_0) + q - h. \tag{304}$$

Thus V_J cancels with its actual coefficient. Combining the ordered rows (216), the identities (303), and $h = -2q - 3r/2$ gives (302) by collecting the nine finite alphabet coefficients. Here is the full coefficient collection. Column A is the balanced product row together with $23R(-2)/15 + 4G(1/5)/15$; column B is (301) after (303). Column C contains the non-H part of the normalized Z row and (304), apart from $77U_J/75$. Column D contains the contracted moving-H and J_{raw} rows and $77U_J/75$ after (300). The $\mathcal{A}_e(2u) - \mathcal{A}_e(u)$ and $\mathcal{A}_e(T_0)$ coefficients cancel before this table.

	A	B	C	D	sum
q^2	304/25	337/75	−254/25	−2497/125	−5056/375
qr	−7	31/5	66/5	−106/25	204/25
qk	−12	8	43/5	−11/25	104/25
rk	3/2	−3/2	−63/20	−297/100	−153/25
r^2	−3/20	9/10	−51/20	501/500	−399/500
k^2	15/4	0	−3	0	3/4
qC	−2618/75	388/25	1354/75	1978/125	5434/375
rC	−22/5	0	−11/5	484/125	−341/125
kC	44/5	−44/5	−44/5	44/5	0
C^2	37/25	0	−127/75	−101/25	−319/75
$G(1/5)$	4/15	7/15	0	−11/10	−11/30
$R(-2)$	23/15	−34/15	0	0	−11/15
$R(16)$	0	−1/25	−1/30	0	−11/150
$K_e(2u) - K_e(u)$	0	22/5	0	0	22/5
$\mathcal{L}$	0	0	0	77/75	77/75

It is important here that these are the full rows of the original source, including the proper norm, rather than a restriction of its support.

The explicit target scalar, obtained by reducing its integer Fermat quotients, is

$$P = \frac{304}{3}q^2 - 11qr - 11qk - 9rk - 3r^2 - 3k^2 + \frac{5}{12}R(16). \tag{305}$$

In (209), insert (213), (268), and the balanced collection (302). Since $97/375 - 1213/750 = -1019/750$,

$$\iota(S) - P = B_2 + \frac{287}{600}h^2 + G_B + G_2 - \frac{1019}{750}h^2 - P.$$

Substitution of (269), (270) and (305), followed by the polynomial identity $h = -2q - 3r/2$, gives exactly F in (89). In particular the coefficient of q^2 is $-2233/125$. By (90), $\iota(S - P) = 0$. The final coefficient additions, with the four harmonic substitutions $h^2 = 4q^2 + 6qr + 9r^2/4$, are recorded

explicitly below. The columns are respectively B_2, $287h^2/600$, G_B, G_2, $-1019h^2/750$, and $-P$:

	B_2	$287h^2/600$	G_B	G_2	$-1019h^2/750$	$-P$	sum
q^2	$-5056/375$	$287/150$	$313/3$	$-193/50$	$-2038/375$	$-304/3$	$-2233/125$
qr	$204/25$	$287/100$	$-27/10$	$-9/100$	$-1019/125$	11	$1386/125$
qk	$104/25$	0	-13	4	0	11	$154/25$
rk	$-153/25$	0	$-15/2$	0	0	9	$-231/50$
r^2	$-399/500$	$861/800$	0	$-177/800$	$-3057/1000$	3	0
k^2	$3/4$	0	$-15/4$	0	0	3	0
qC	$5434/375$	0	$2/5$	$-10/3$	0	0	$4334/375$
rC	$-341/125$	0	$3/10$	$-5/2$	0	0	$-616/125$
C^2	$-319/75$	0	0	$11/15$	0	0	$-88/25$
$G(1/5)$	$-11/30$	0	$-1/20$	$5/12$	0	0	0
$R(-2)$	$-11/15$	0	$1/10$	$-5/6$	0	0	$-22/15$
$R(16)$	$-11/150$	0	$7/15$	$-1/20$	0	$-5/12$	$-11/150$

The coefficients $77/75$ of $\mathcal{L}$ and $22/5$ of $K_e(2u) - K_e(u)$ pass unchanged from B_2 to F. The embedding ι is injective: a field homomorphism preserving 1 has zero kernel. Thus the same original prime-field scalar satisfies

$$S = P \quad \text{in } \mathbb{F}_p. \tag{306}$$

Neither conic root imposes a prime class or a sign condition. The proof uses global nonzero polynomials for cancellation, never a nonzero specialized Hasse value.

7 Completion of the all-prime congruence

Proof of Theorem 1. For $p \geq 19$, the scalar in the actual rational moment is $S = P$ by (306). Its identification with that moment is the equivalence (56); all divisions by p in its construction occur in $\mathbb{Z}_{(p)}$ before reduction. Therefore

$$M_p \equiv \frac{1728^p}{45000} \pmod{p^3}.$$

The literal adapter (42) and the factorial normalization (10) give

$$W_p \equiv 3375p^2 c_p M_p \equiv 3375p^2 \frac{120}{1728^p} \frac{1728^p}{45000} = 9p^2 \pmod{p^5}.$$

To verify the precision of the second replacement, write $c_p = 120/1728^p + p^3 a$ and $M_p = 1728^p/45000 + p^3 b$ with $a, b \in \mathbb{Z}_{(p)}$. The error in their product is

$$p^3 \left(a\, 1728^p/45000 + b\, 120/1728^p\right) + p^6 ab.$$

Its coefficient factors are integral since $p \geq 19$ does not divide 1728 or 45000. Multiplication by $3375p^2$ gives an error in $p^5\mathbb{Z}_{(p)}$. The exact rational simplification is $3375 \cdot 120 = 405000 = 9 \cdot 45000$.

If $7 \leq p < 19$ is prime, trial division by $2, 3, 5$ leaves exactly $7, 11, 13, 17$. Section 1.8 supplies the exact integer certificates for these four primes. Their denominator $10^{6(p-1)}$ is prime to p, so those certificates give the same congruence in $\mathbb{Z}_{(p)}$. The two cases exhaust every prime $p \geq 7$. □